All recorded signal plots

Every recorded signal and every matrix component is retained.

Wheel order: front left, front right, rear left, rear right. Multi-signal plots have explicit legends.

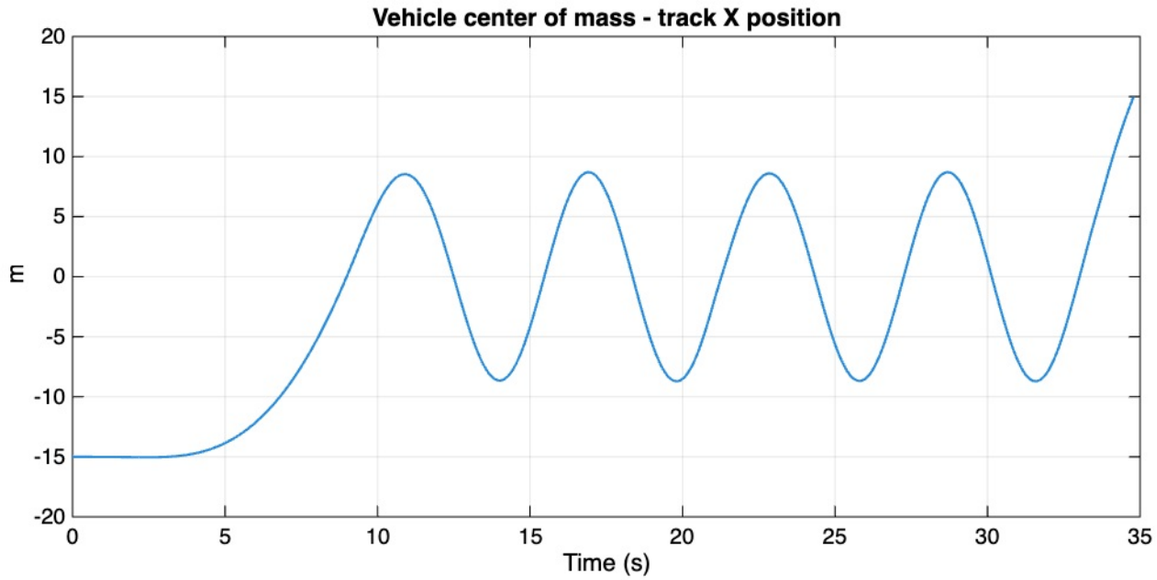
Track coordinates are earth-fixed; vehicle coordinates rotate with the car; tire signals use their tire frame.

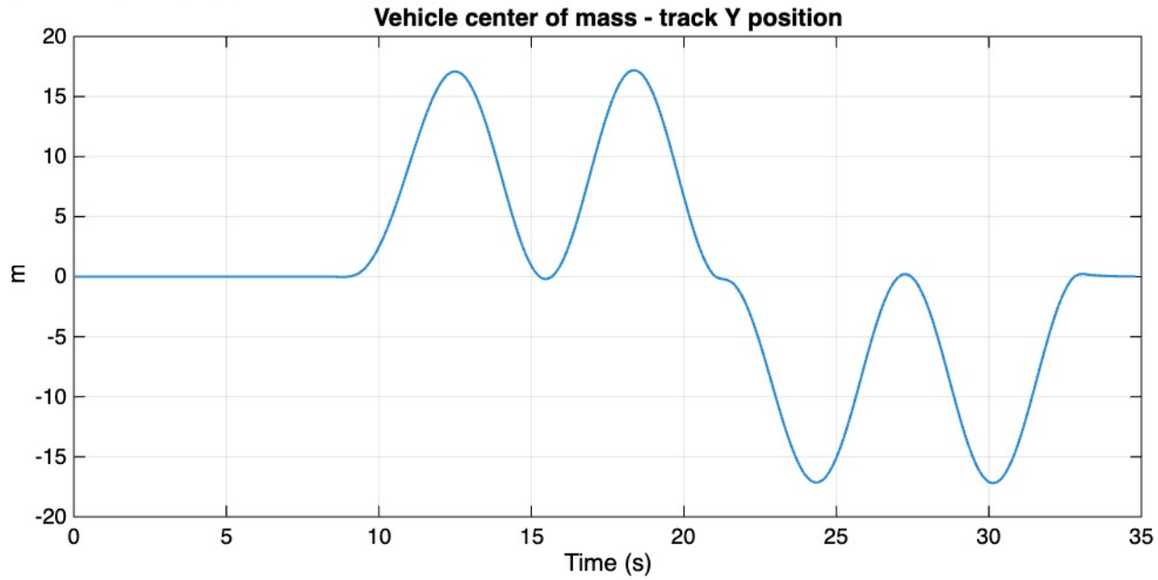
Values and coordinate signs are unchanged. Angular units are preserved; acceleration in g is labeled explicitly.

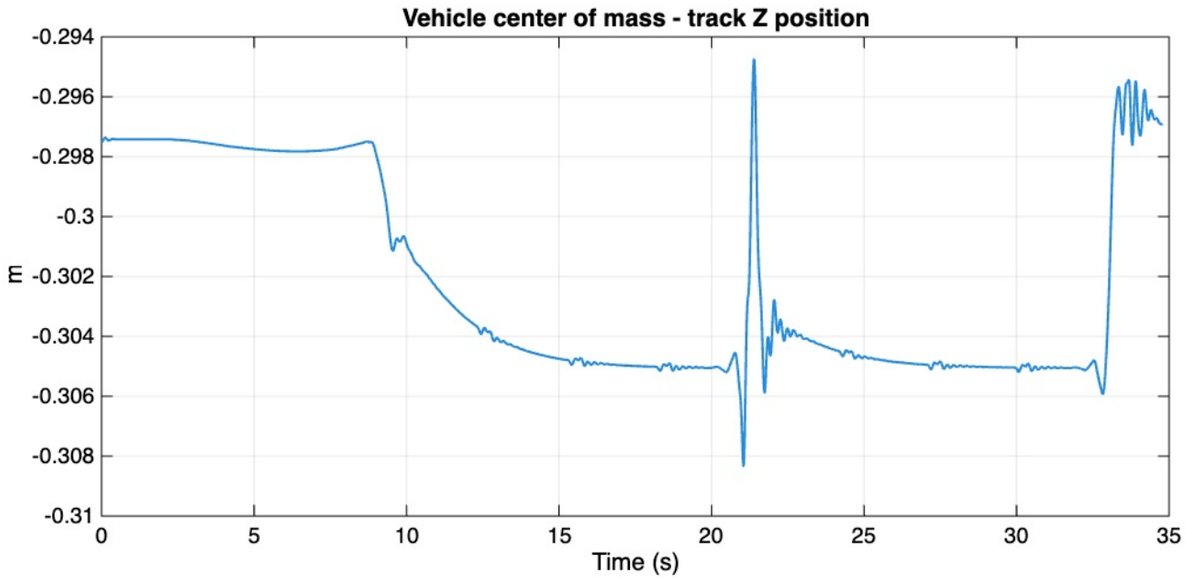
Zero body tire inputs are not zero tire forces: the current model routes active forces through suspension.

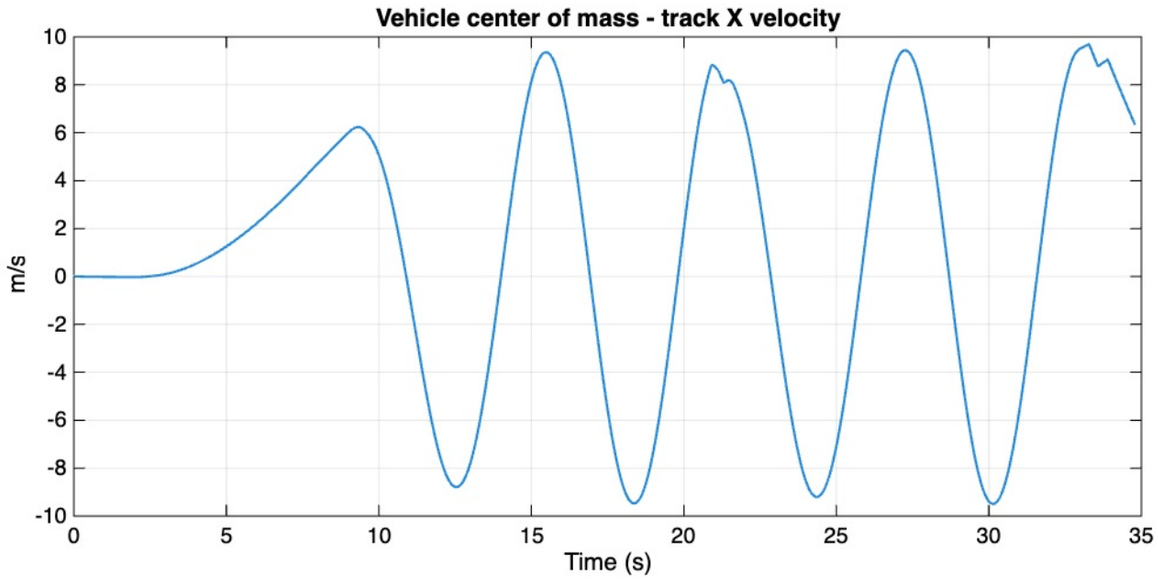
The companion CSV index maps each readable title to the original recorded signal and PDF page.

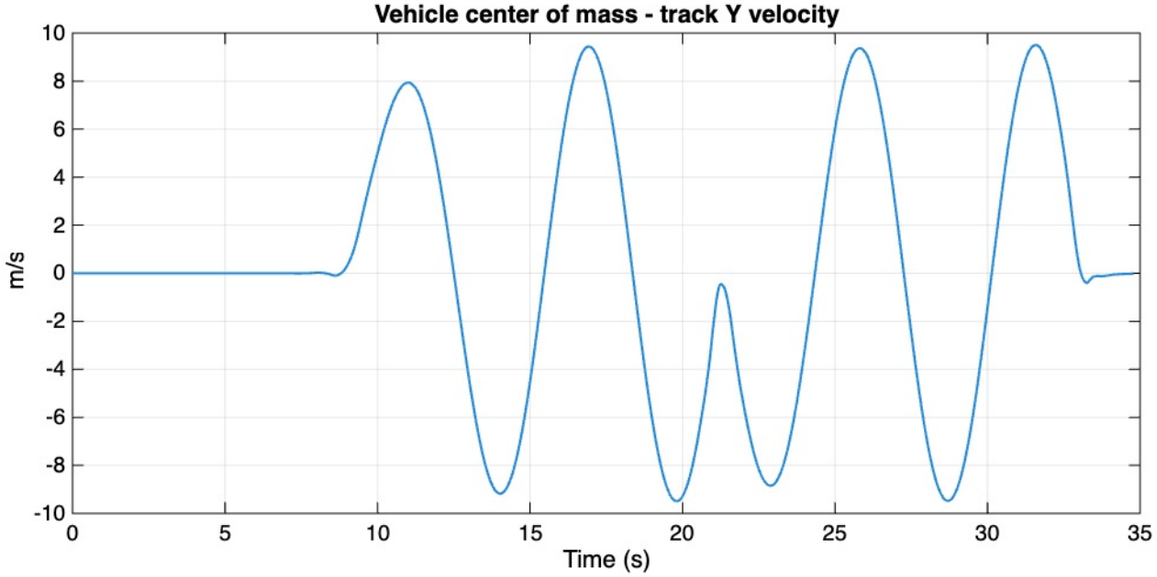
All recorded signal plots - plot page 1

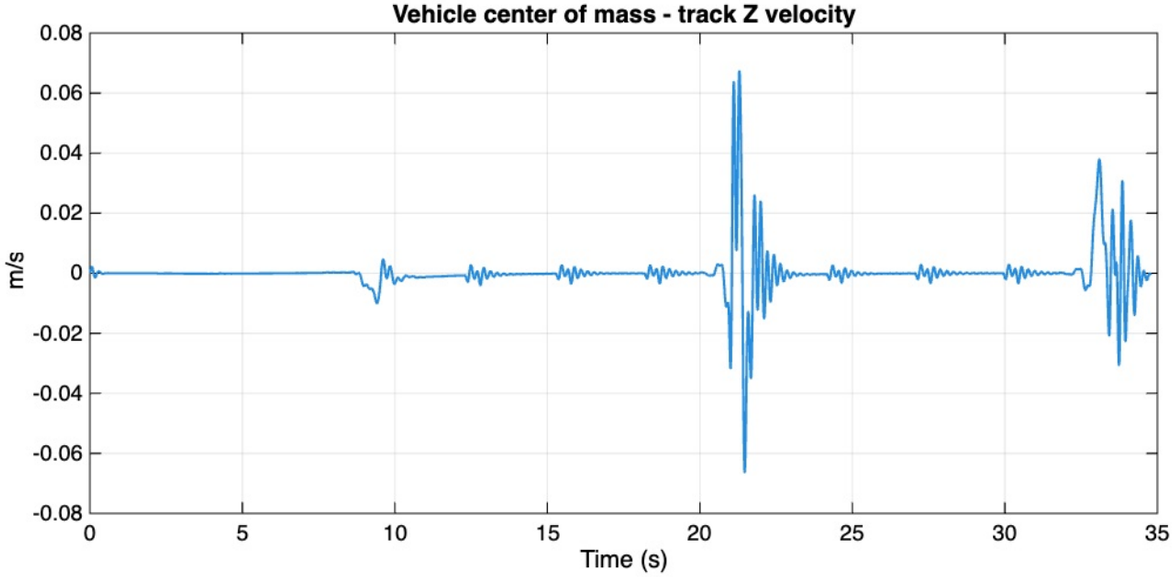




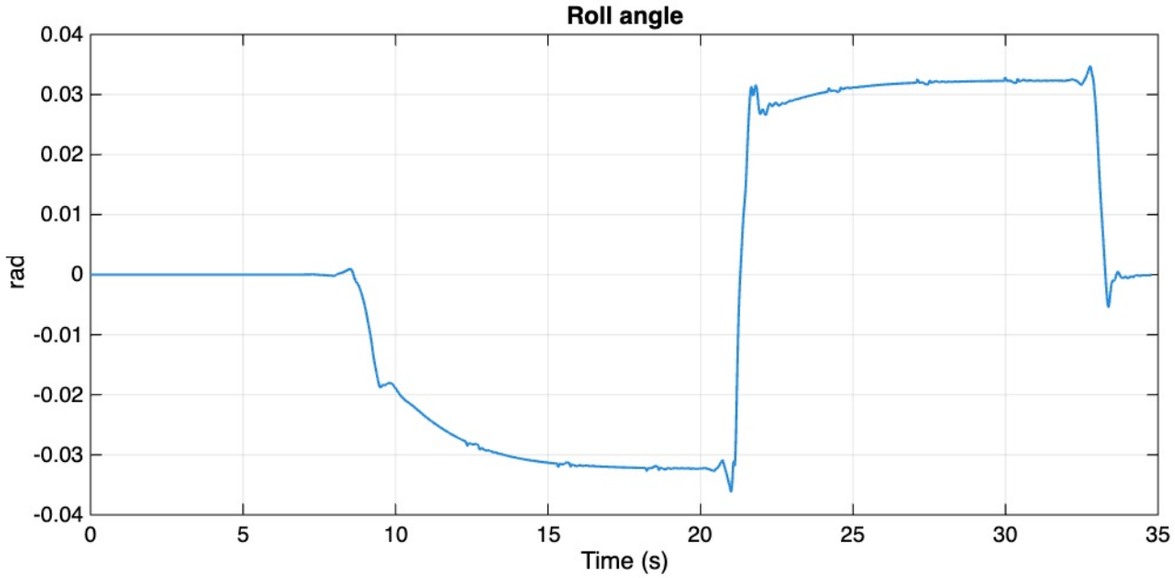


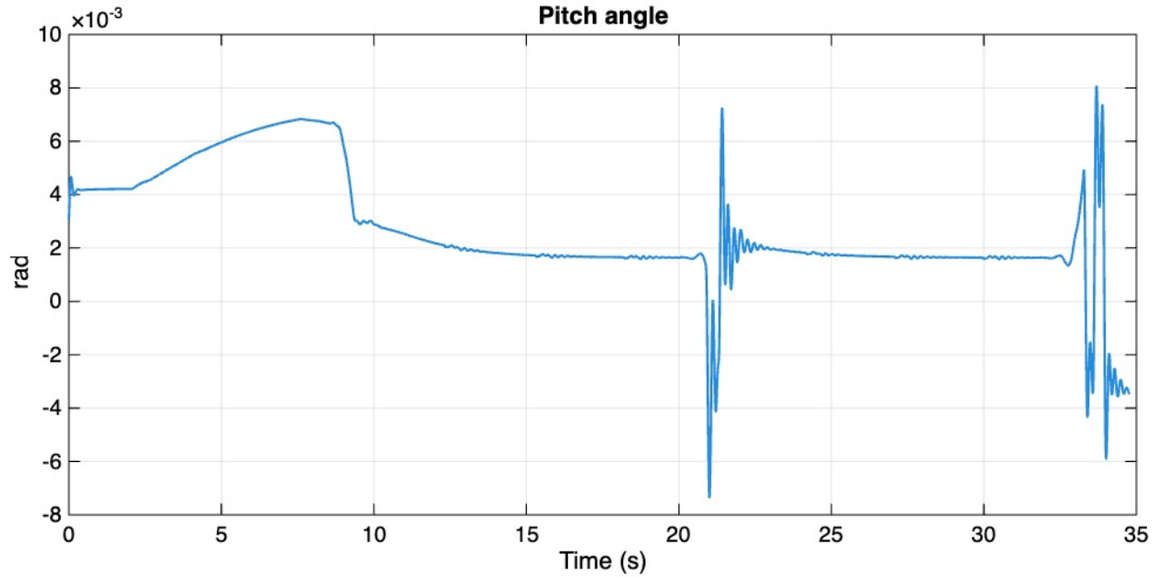


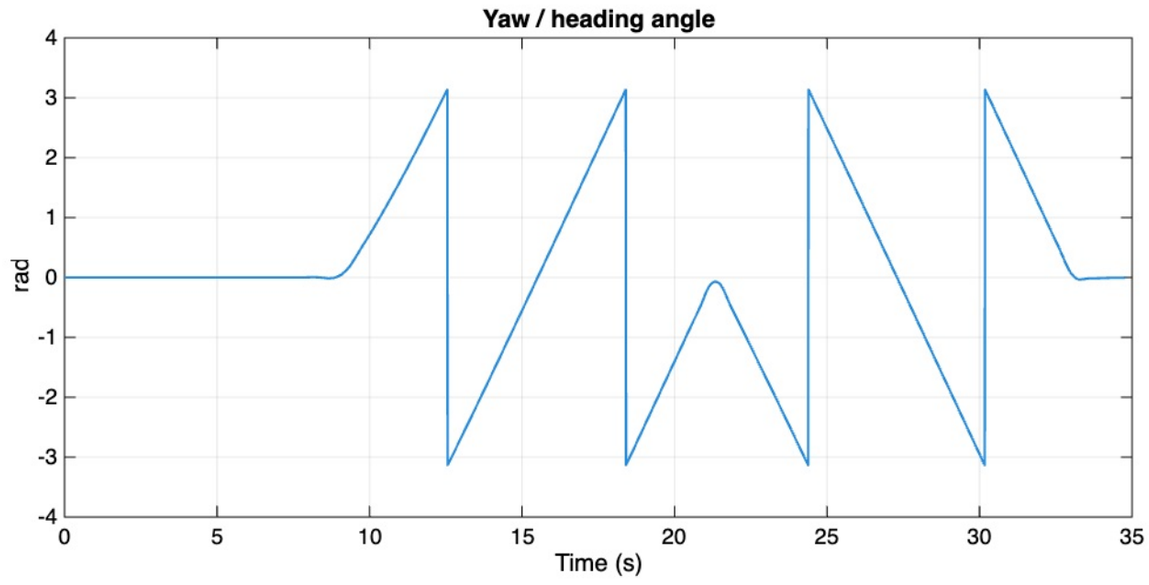


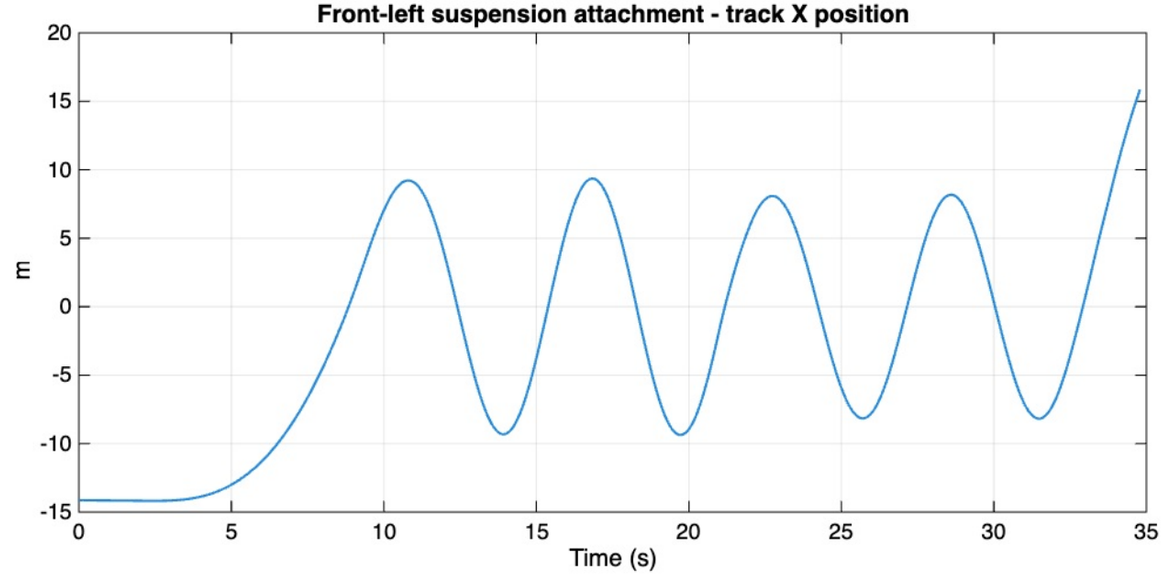


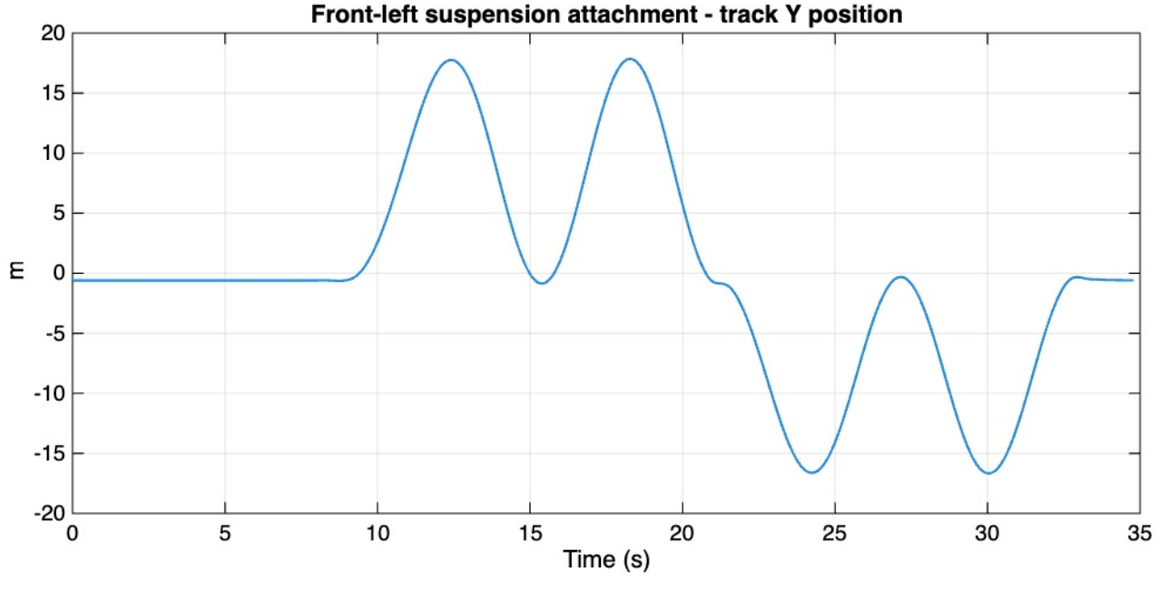
All recorded signal plots - plot page 2

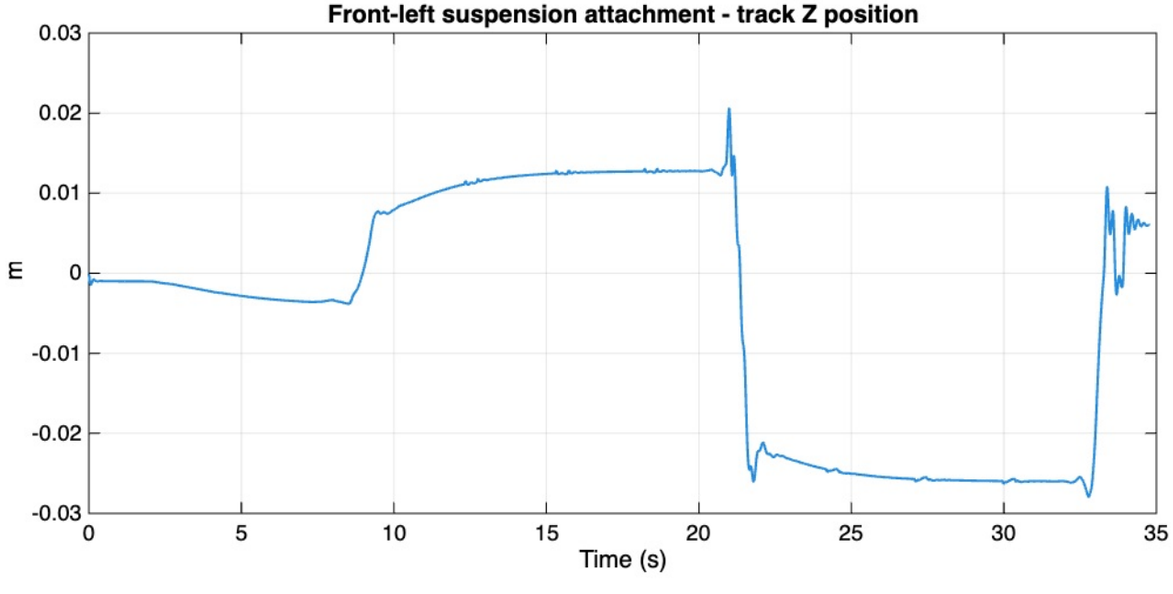


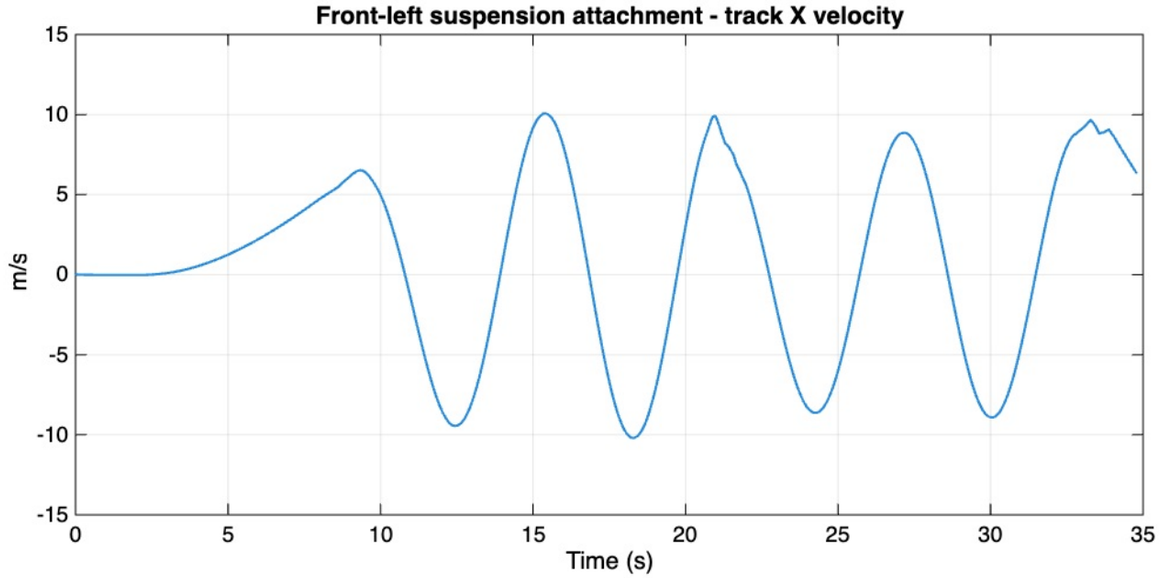


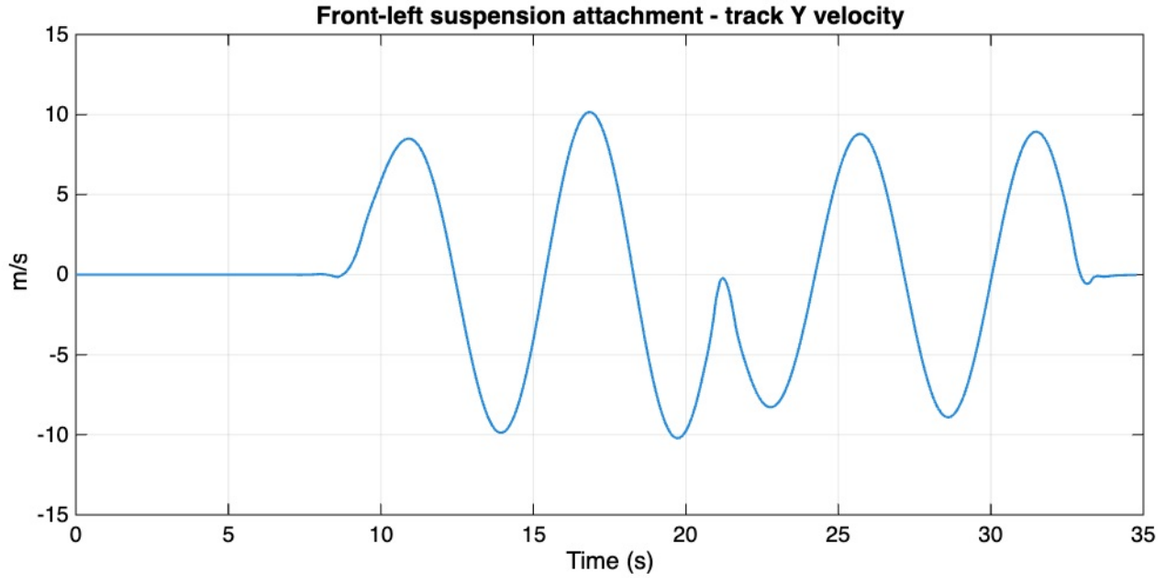


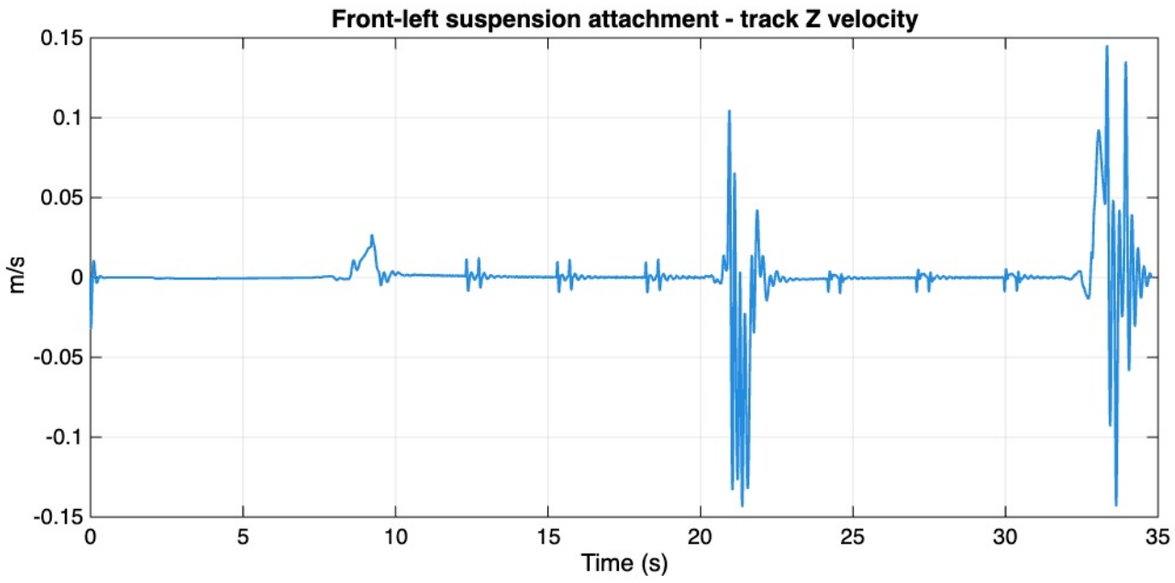


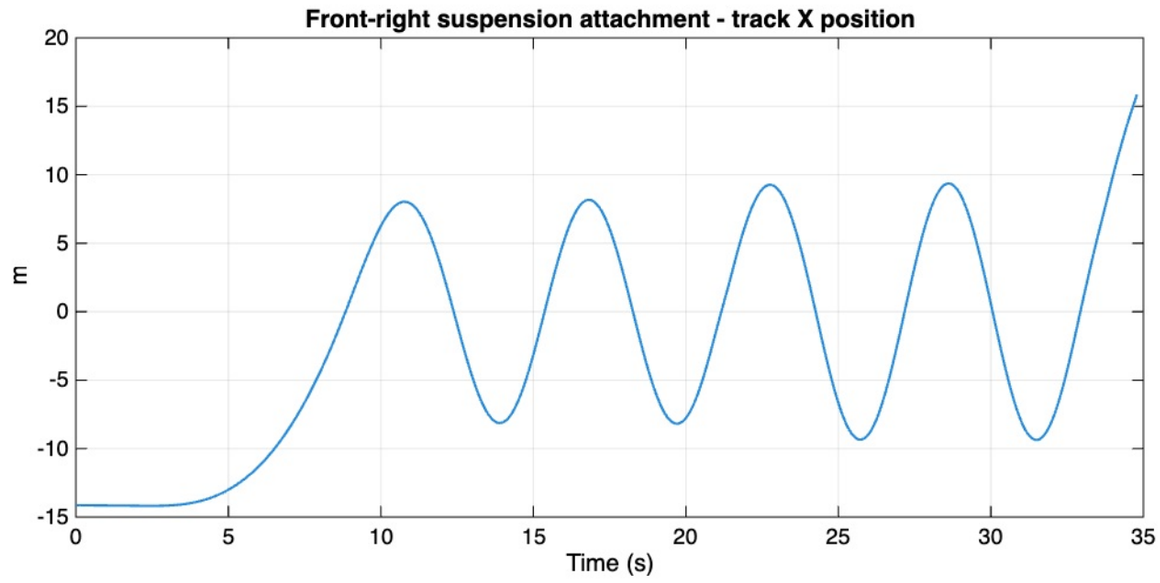


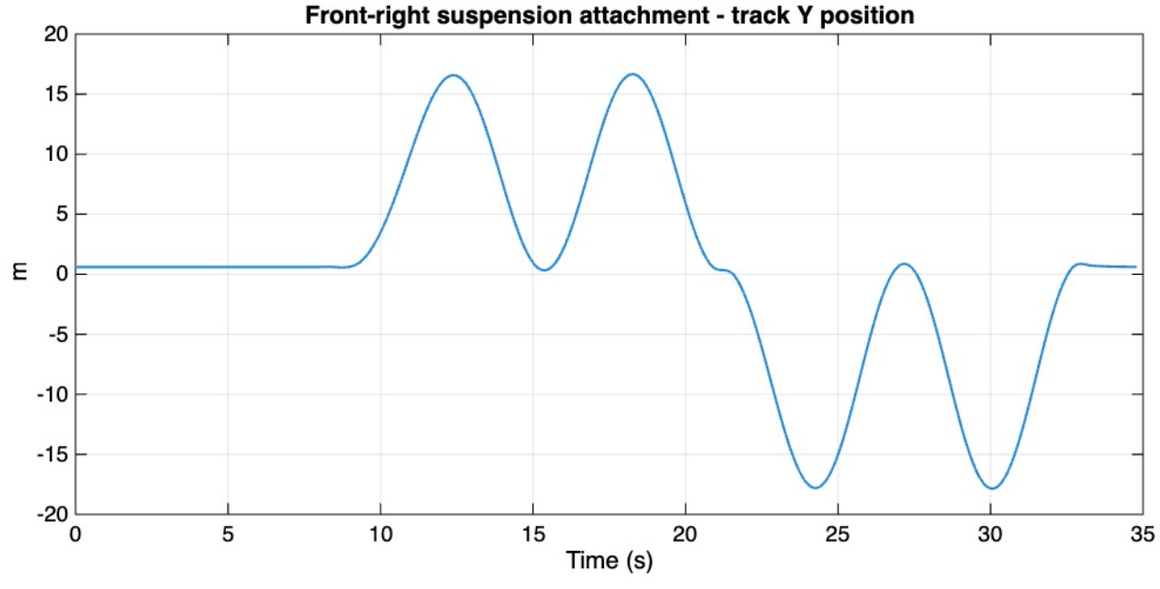


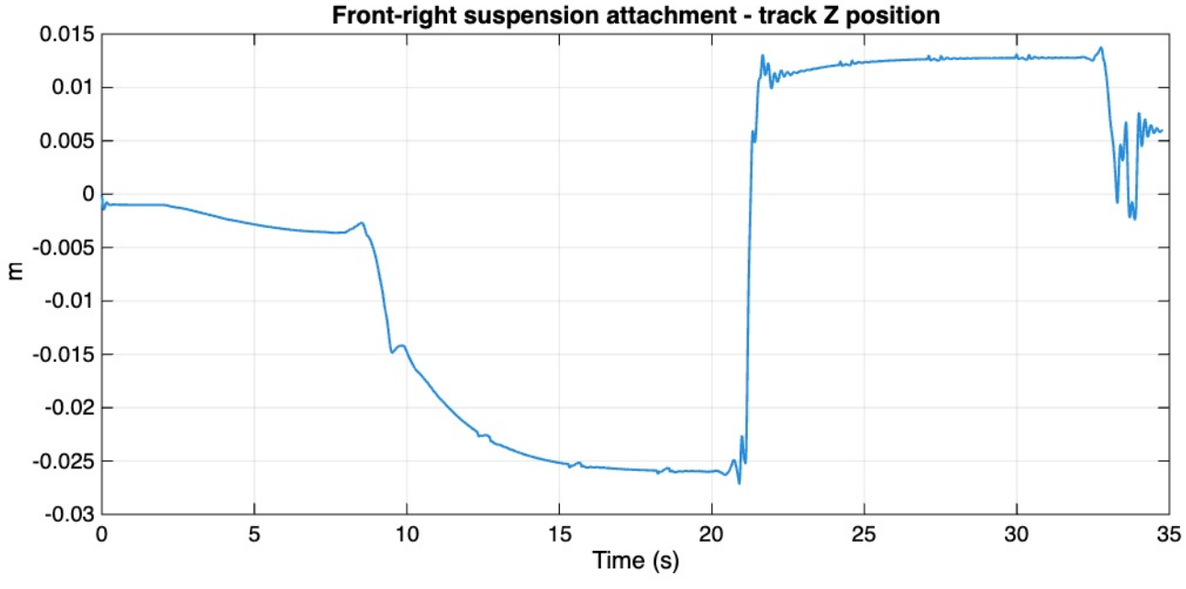

Front-left suspension attachment - track X velocity
m/s
Time (s)


Front-left suspension attachment - track Y velocity
m/s
Time (s)

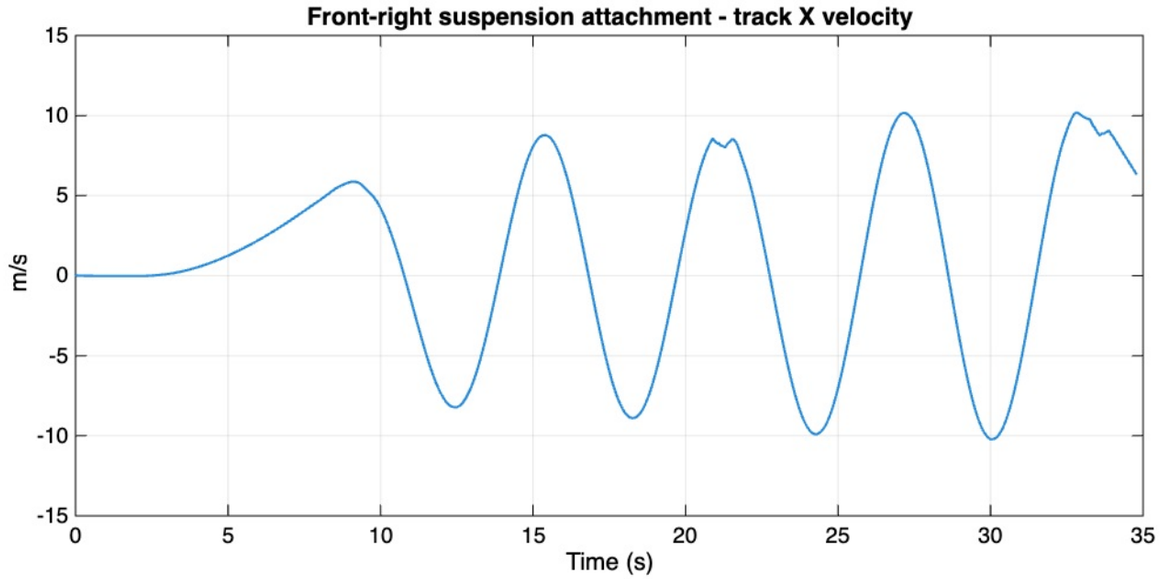

Front-left suspension attachment - track Z velocity
m/s
Time (s)

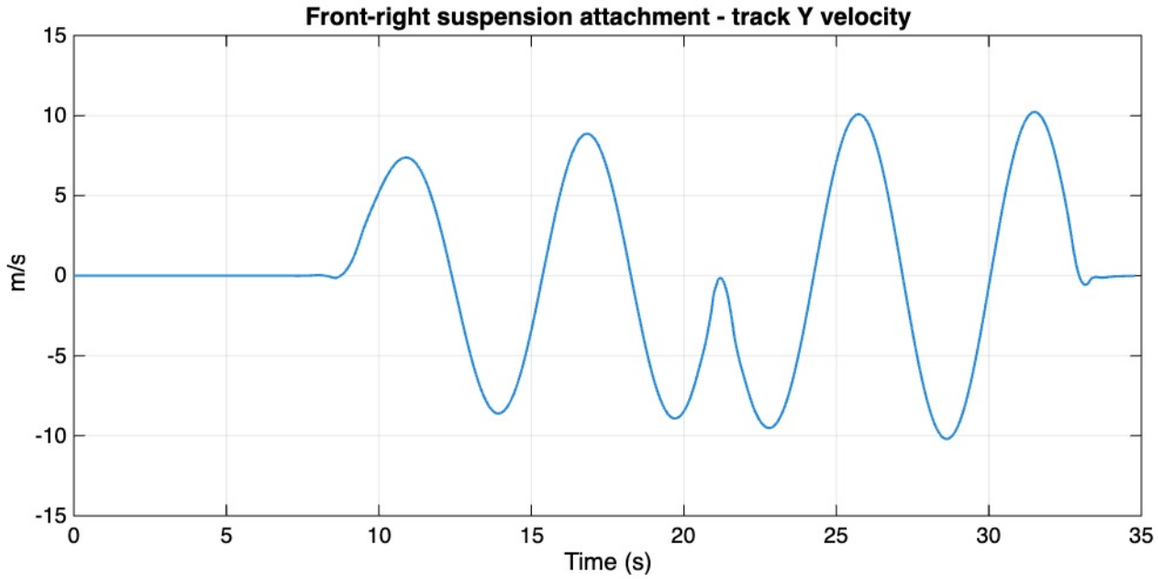

Front-right suspension attachment - track X position
m
Time (s)

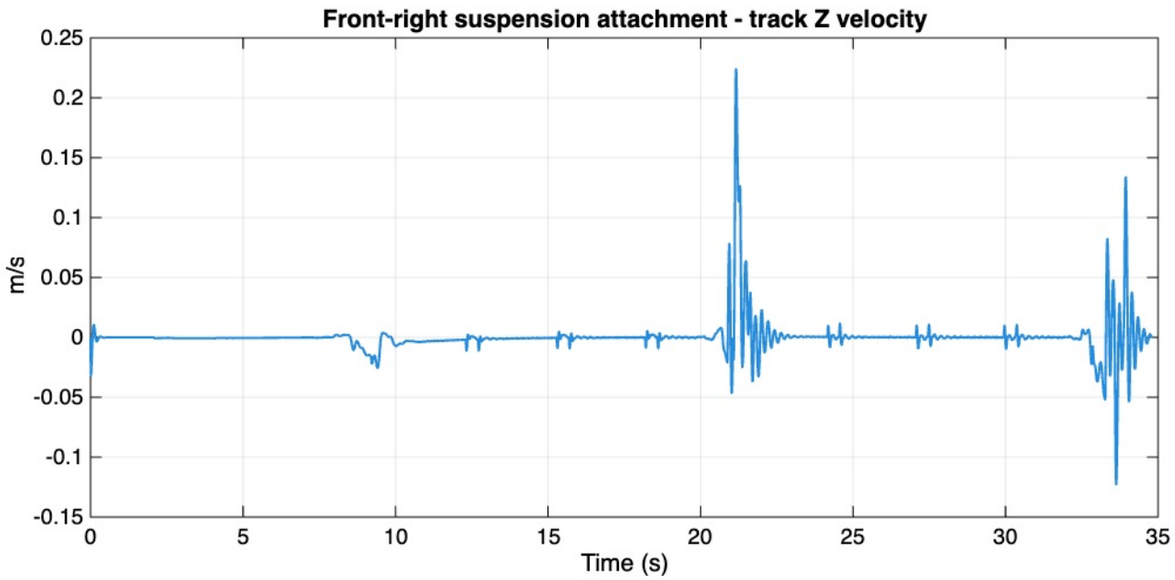

Front-right suspension attachment - track Y position
m
Time (s)

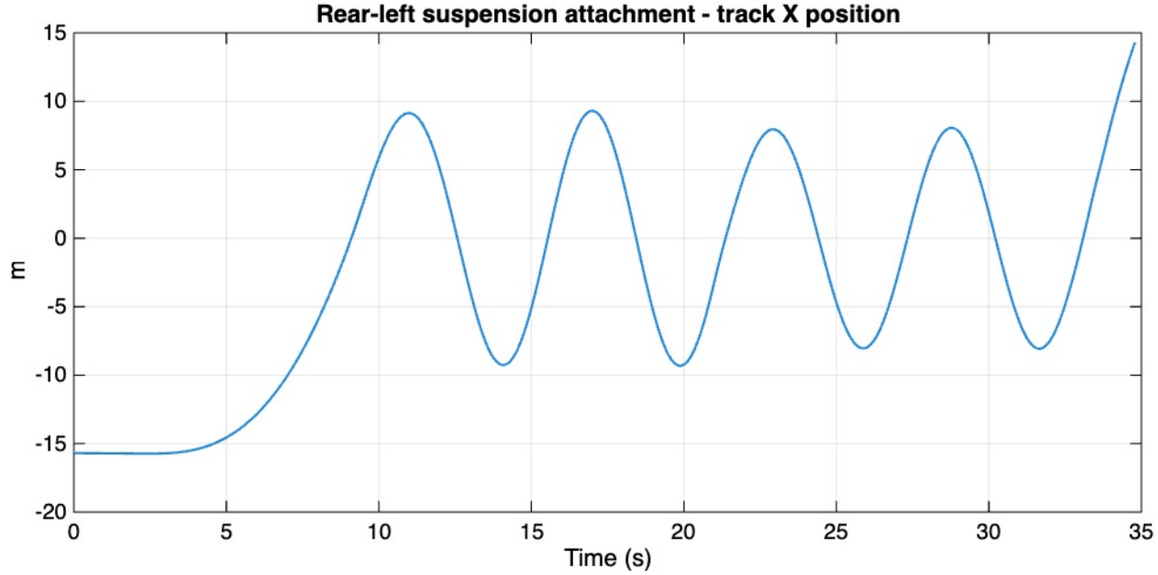

Front-right suspension attachment - track Z position
m
Time (s)

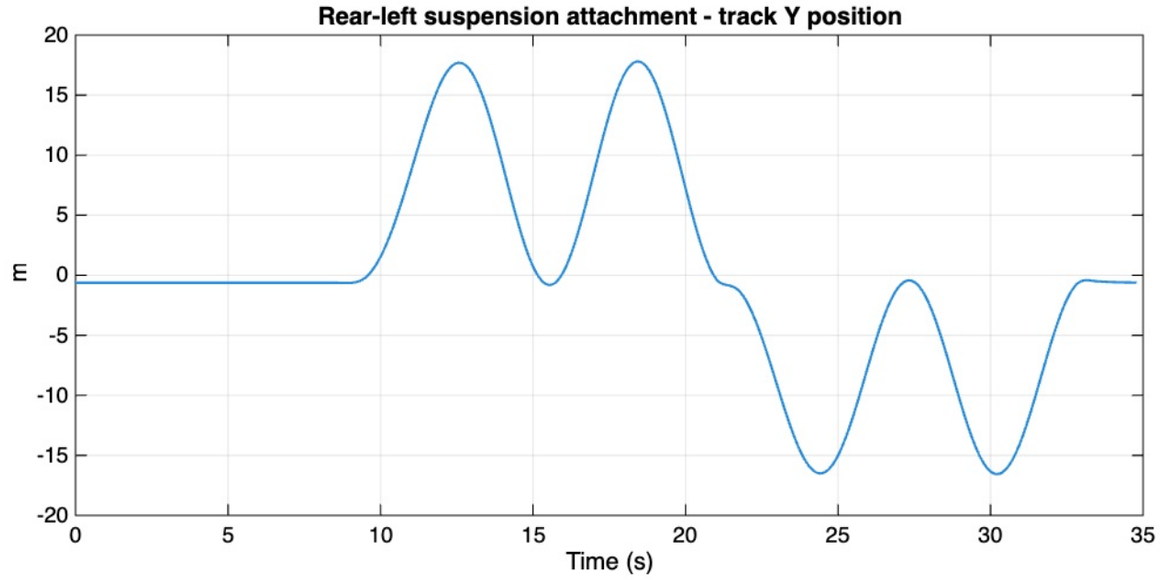
All recorded signal plots - plot page 4

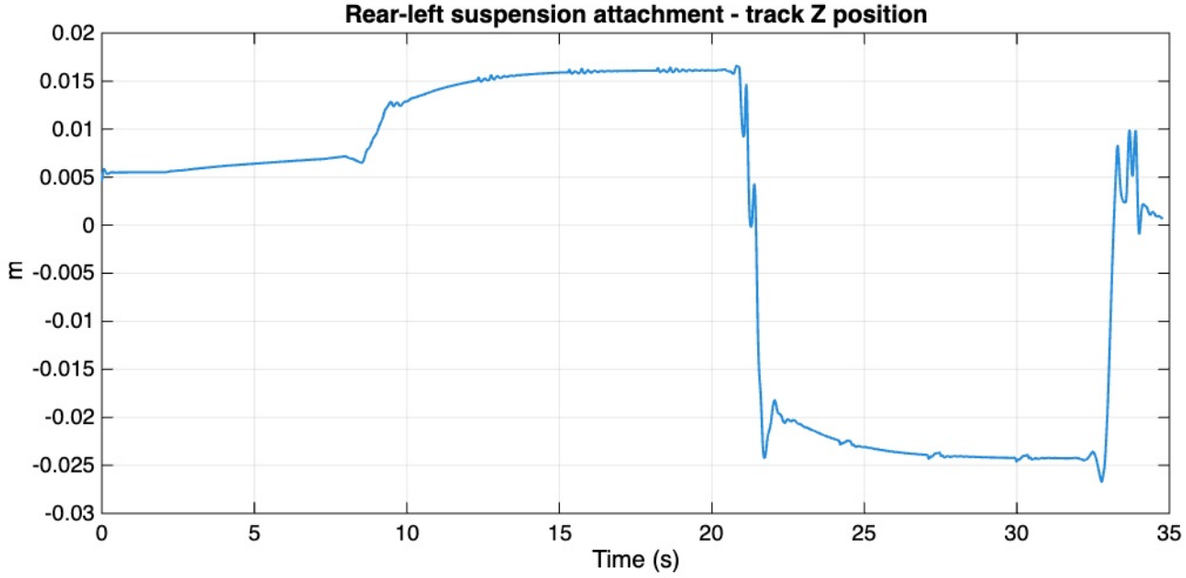




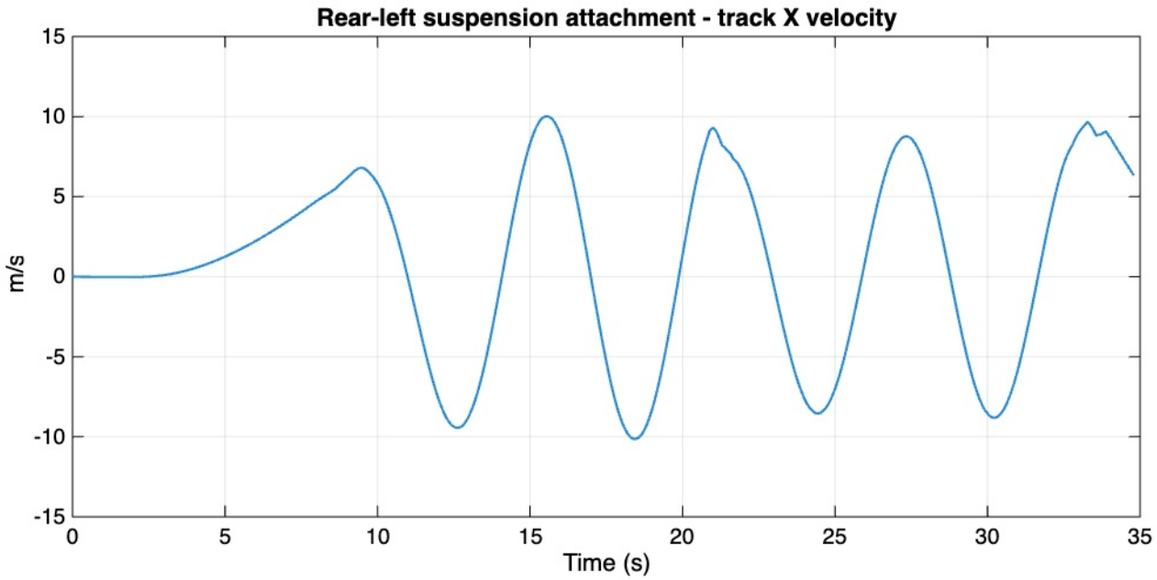


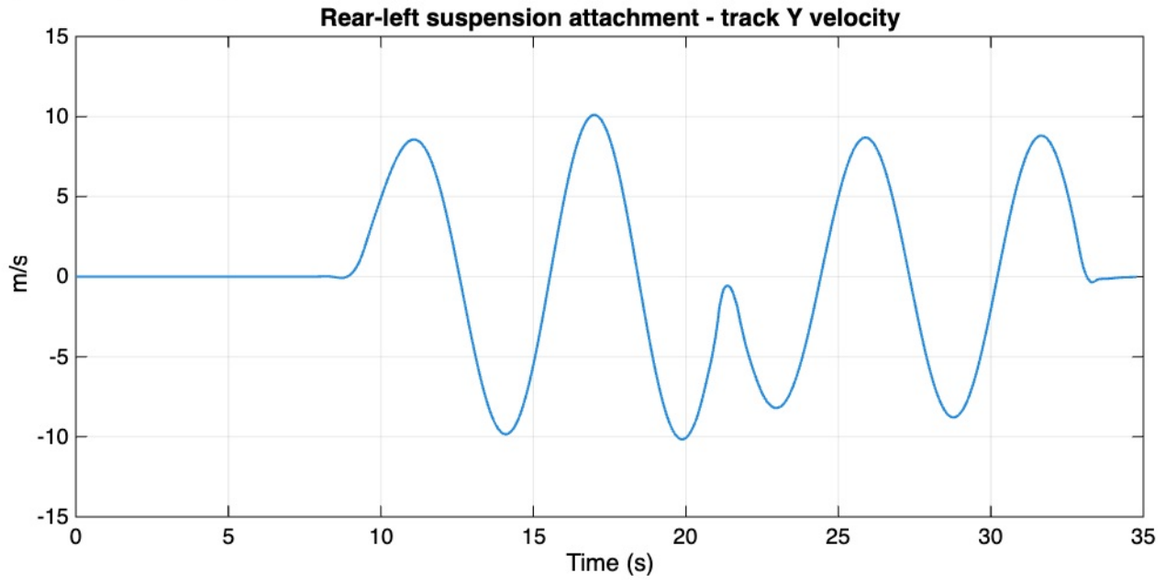


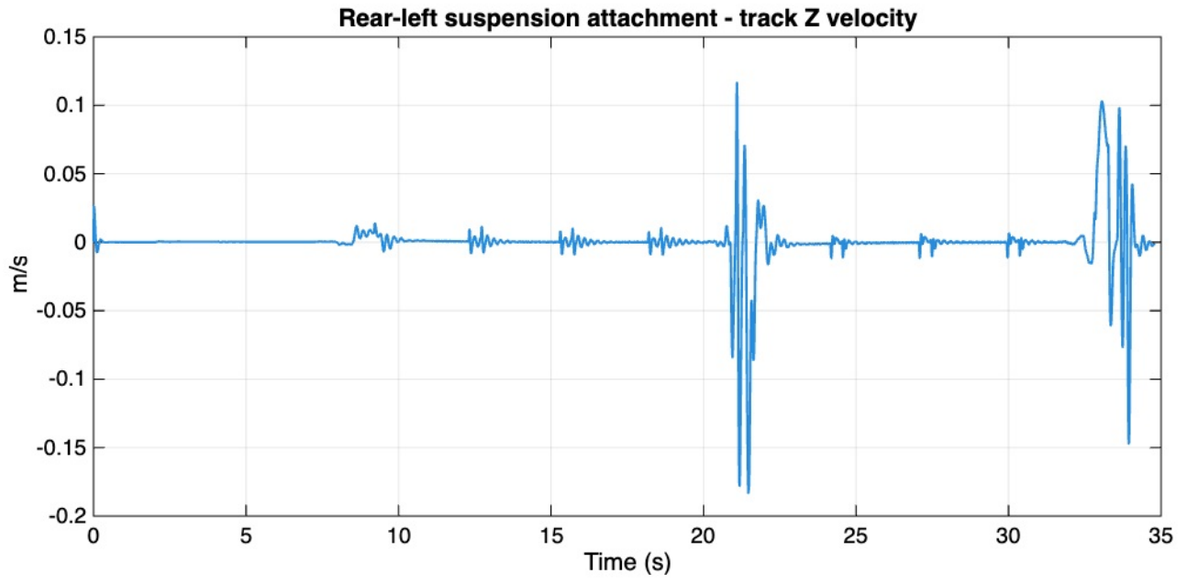


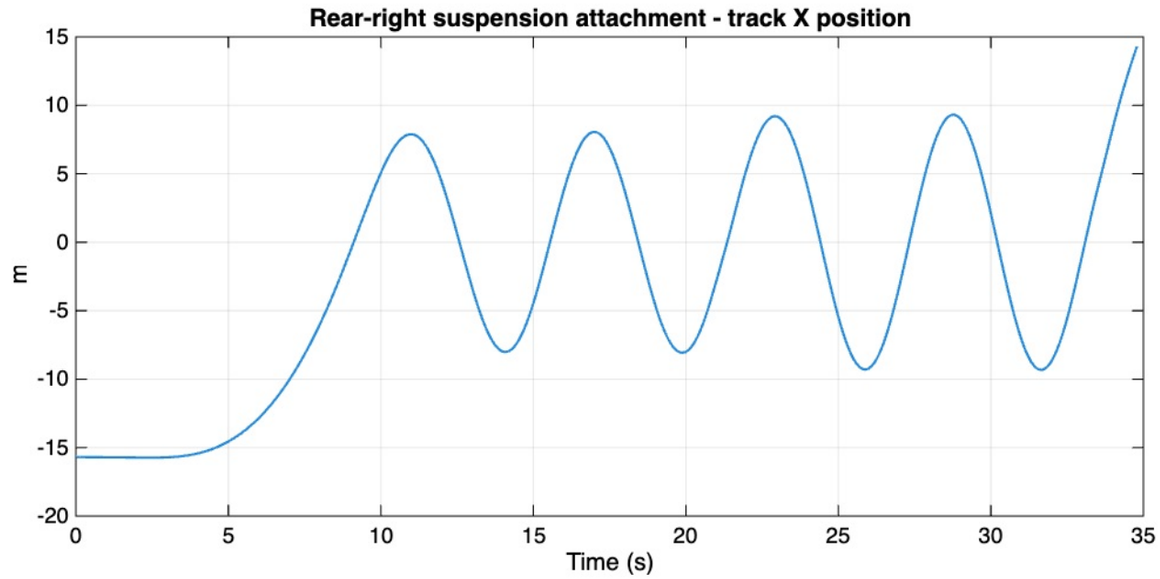


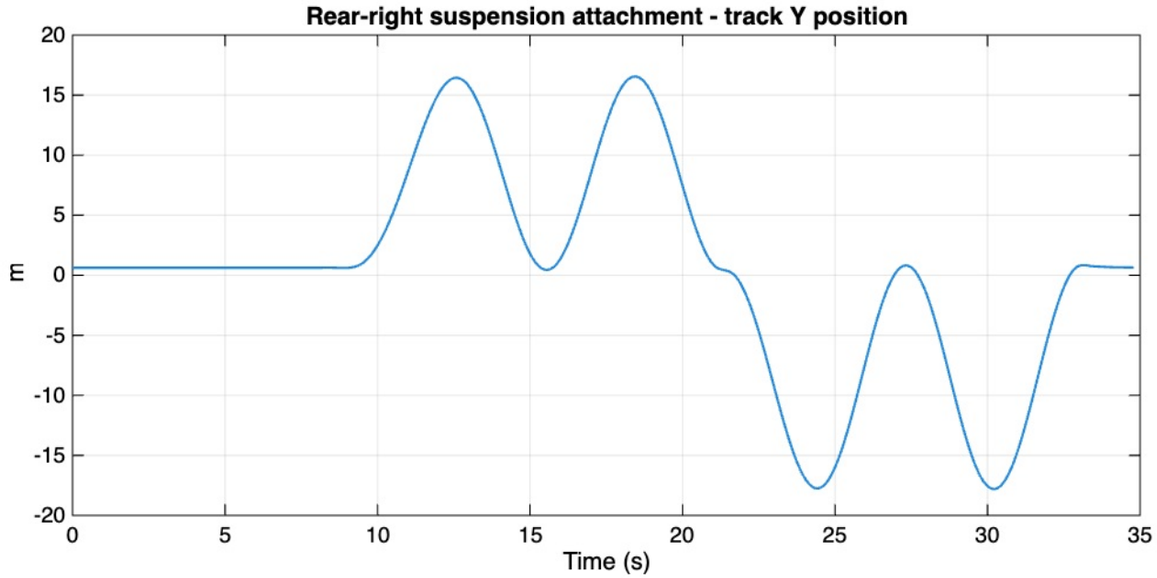
All recorded signal plots - plot page 5

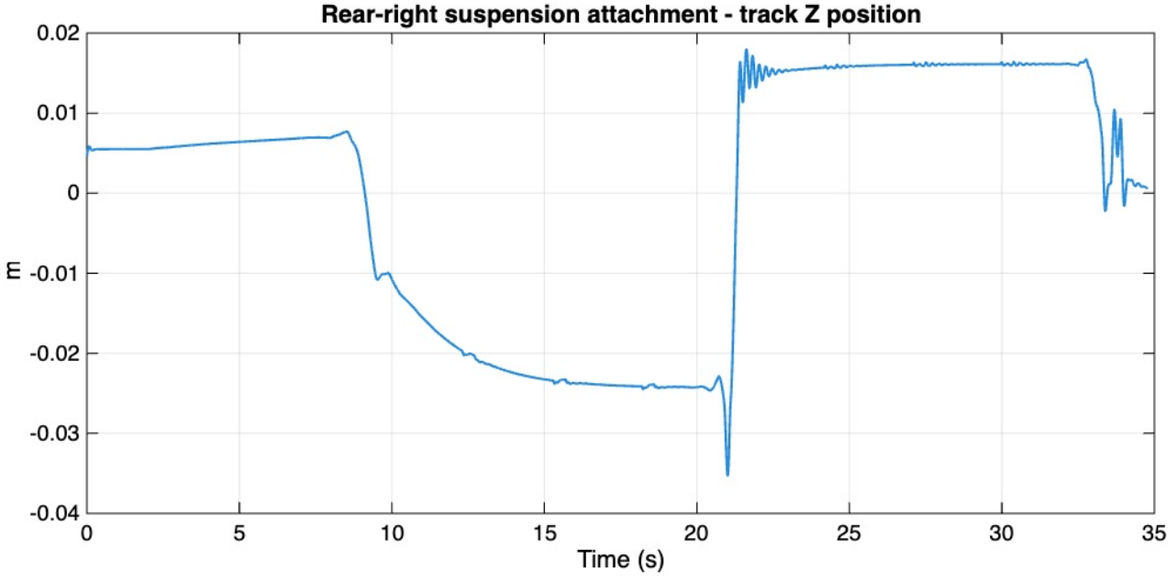




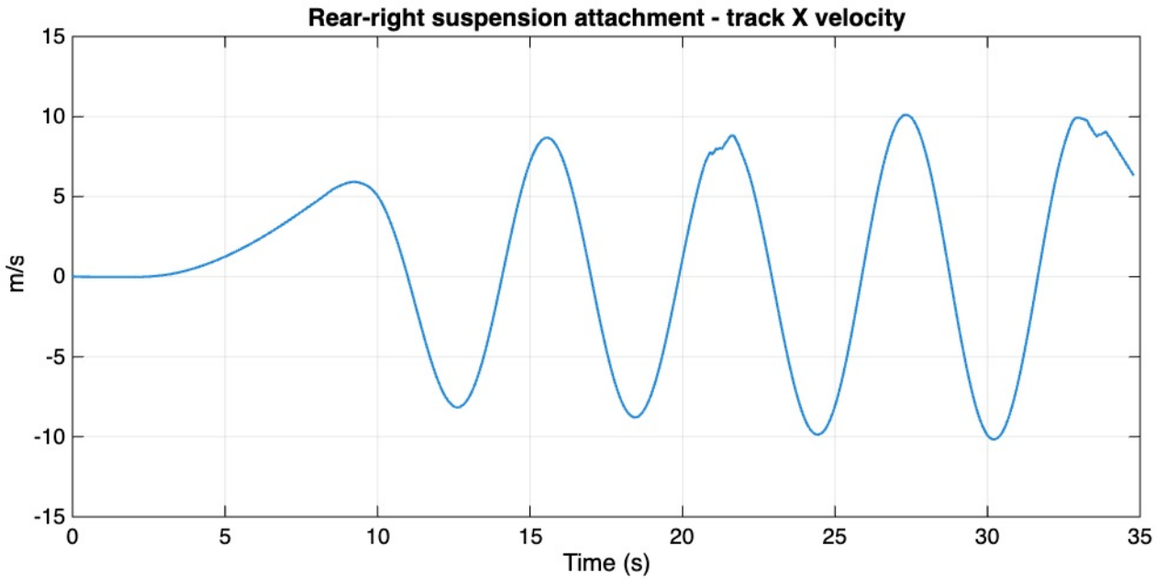


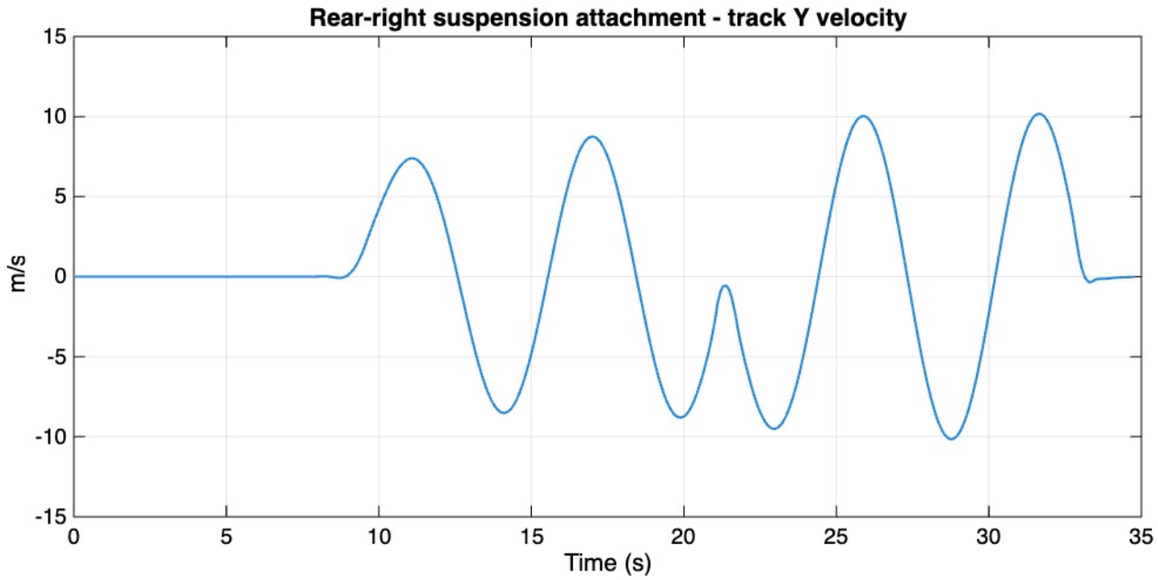


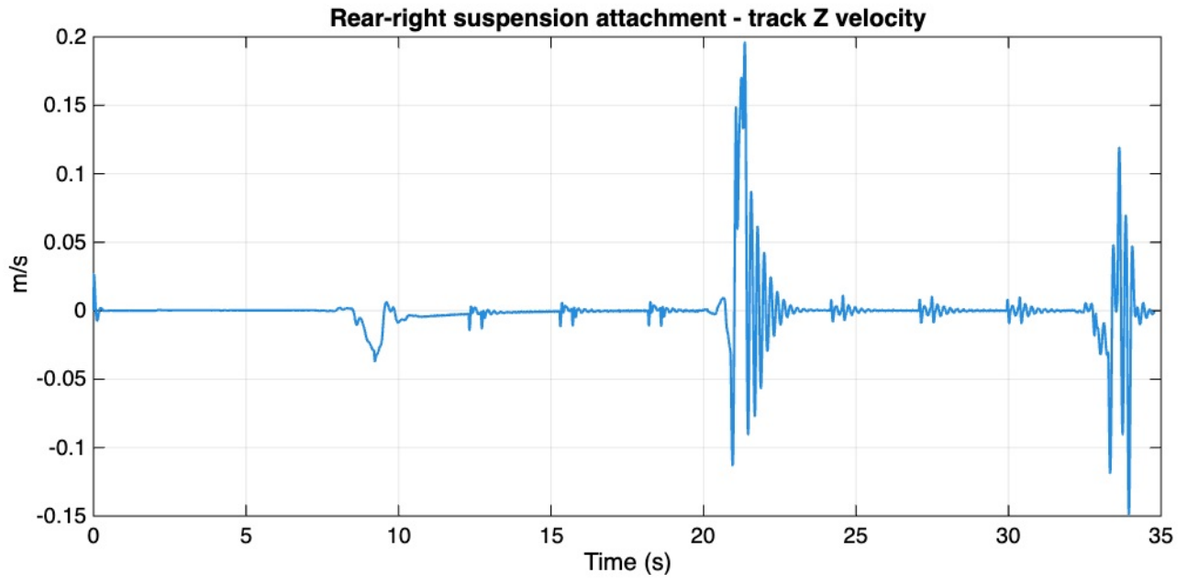


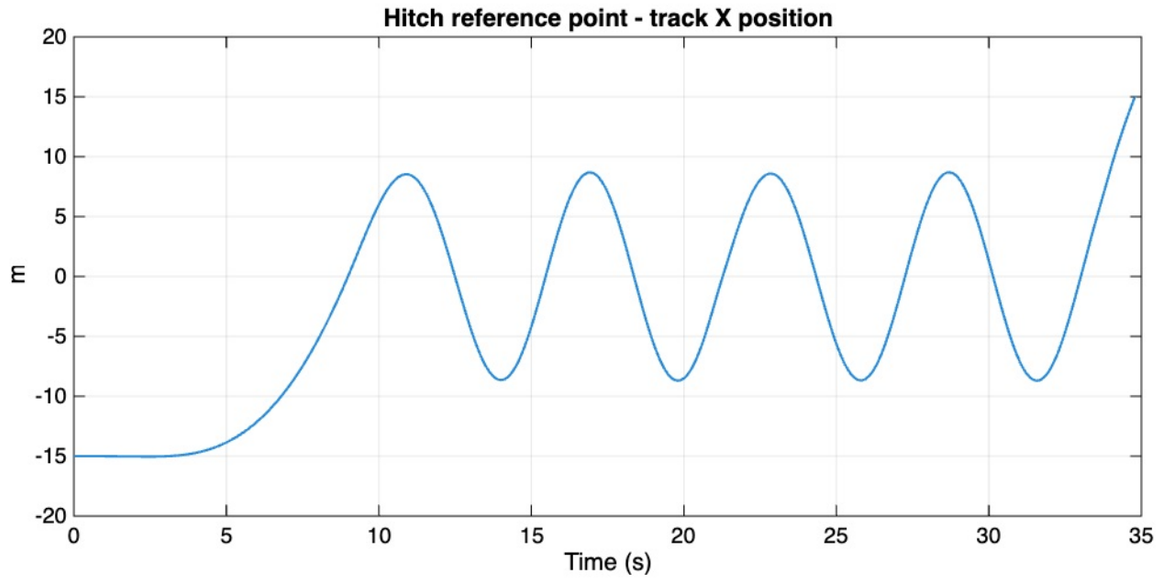


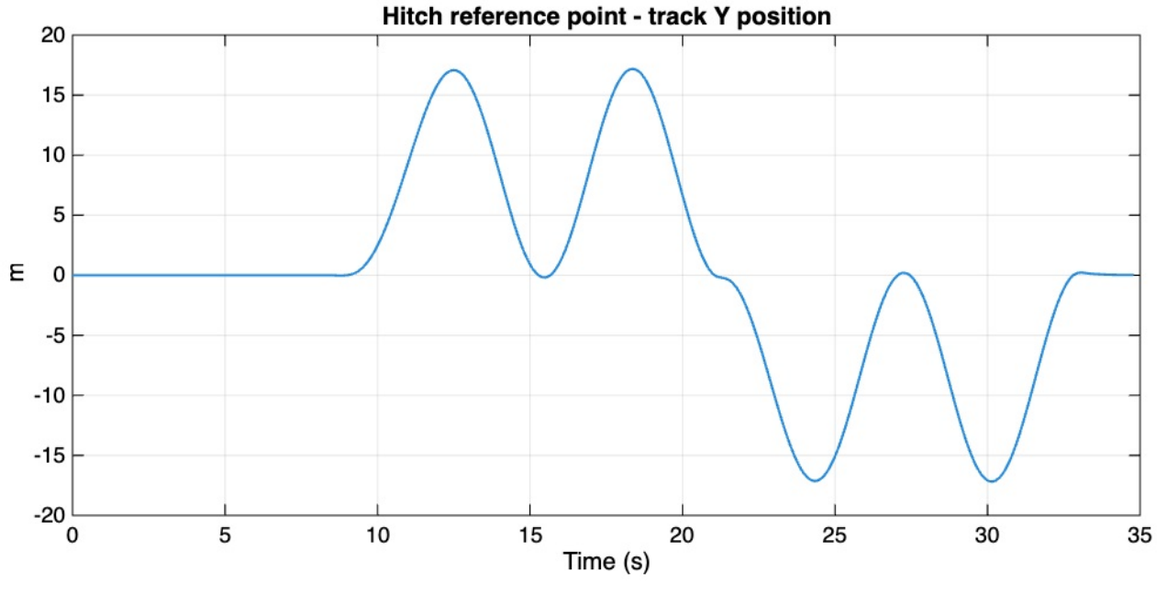
All recorded signal plots - plot page 6

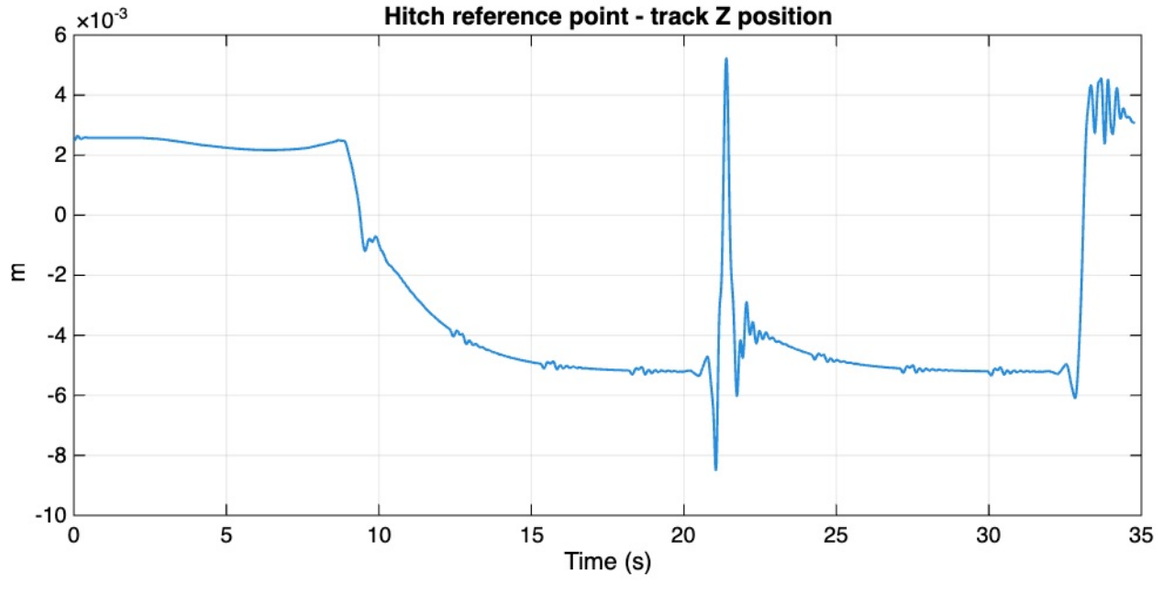


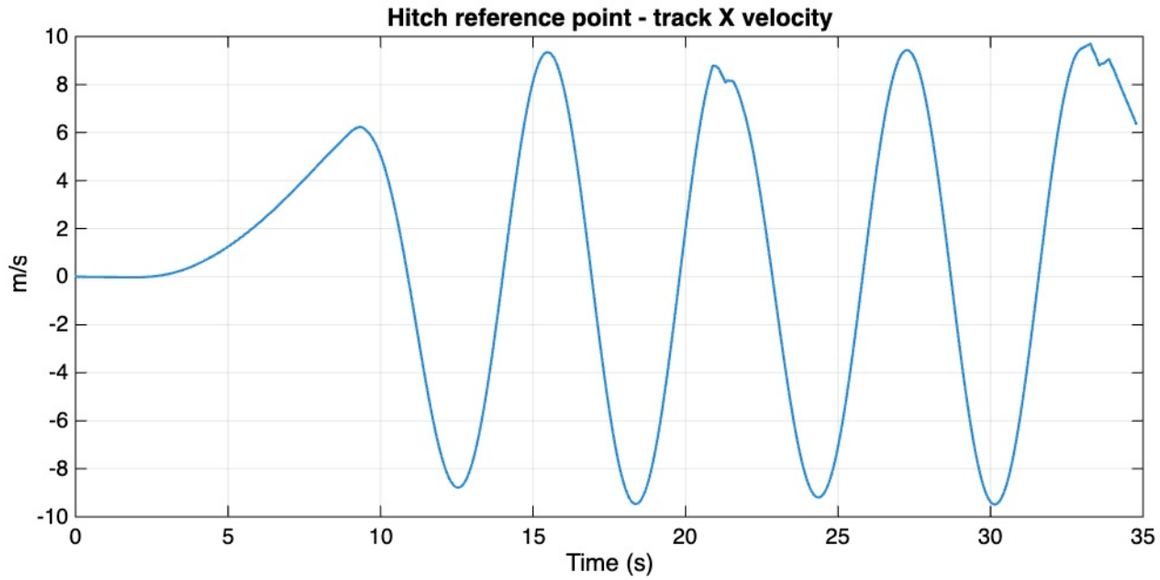


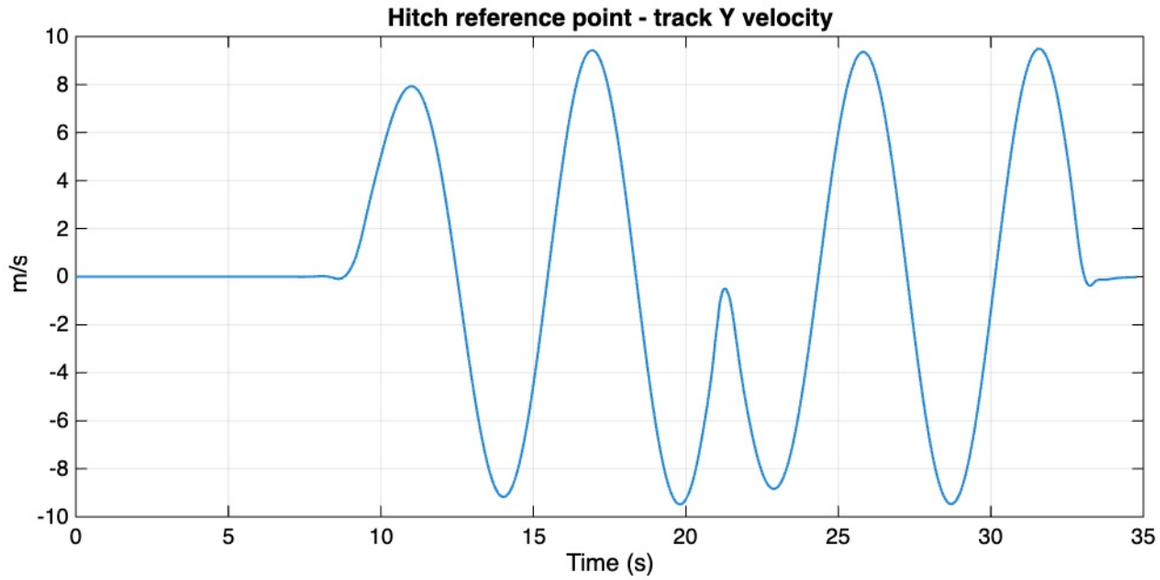


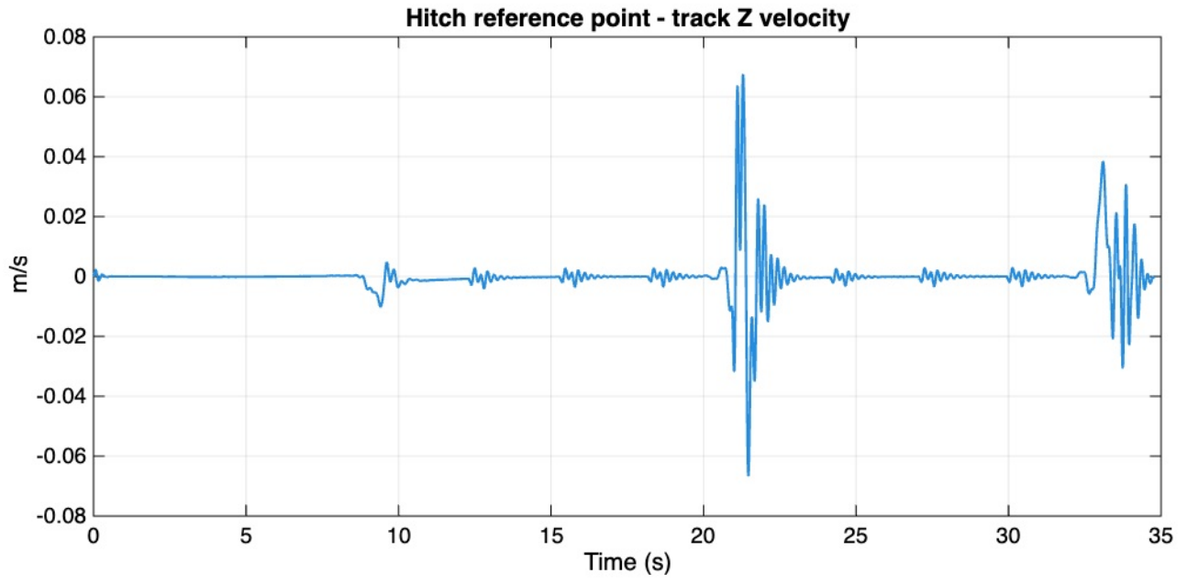


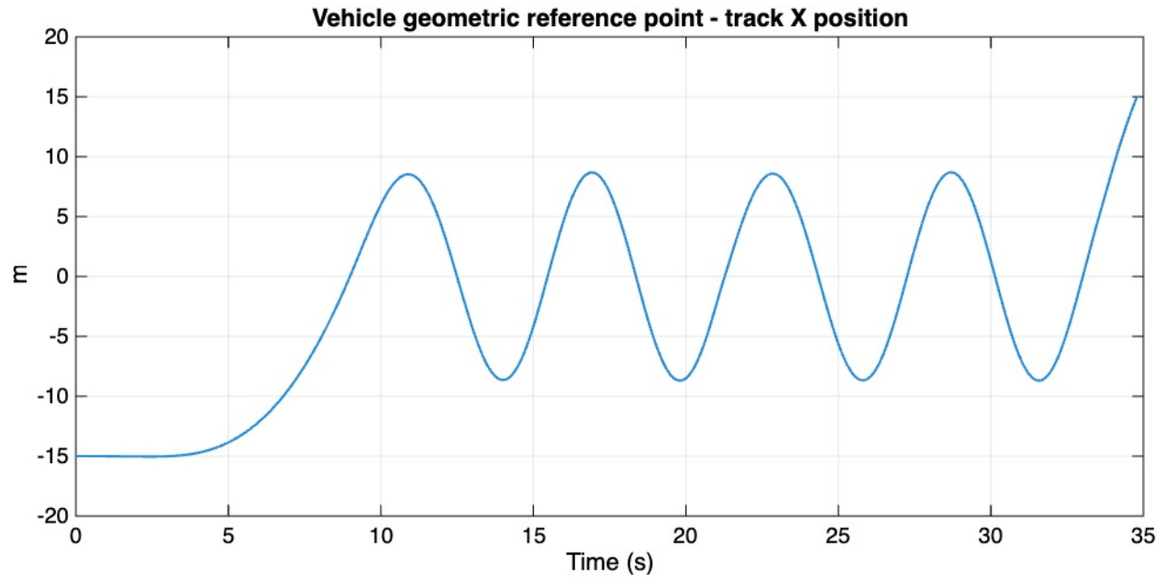


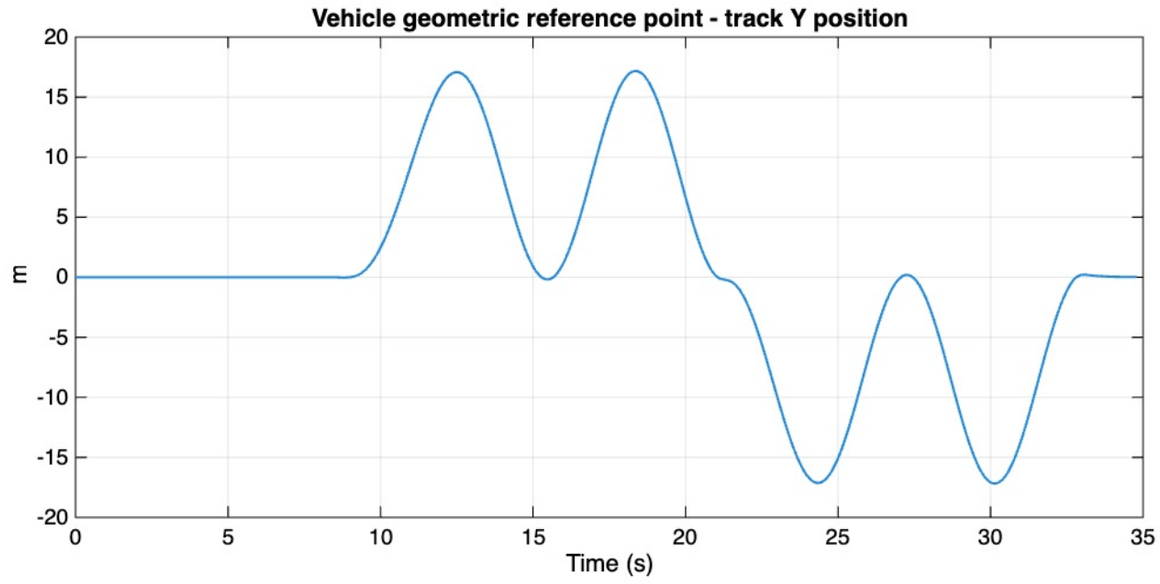


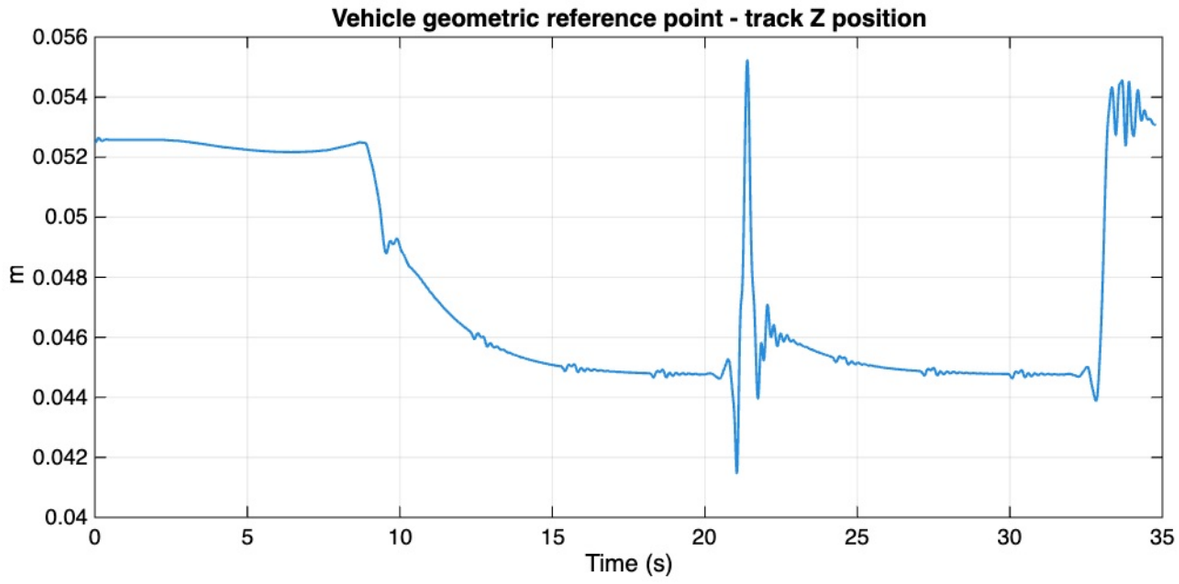

Hitch reference point - track X velocity
m/s
Time (s)


Hitch reference point - track Y velocity
m/s
Time (s)

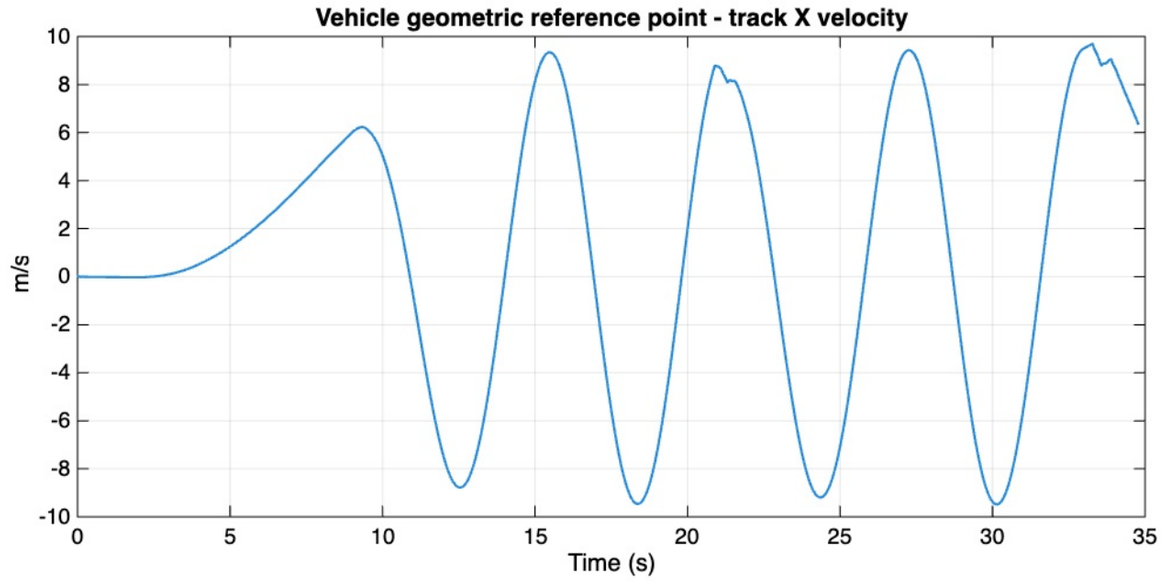

Hitch reference point - track Z velocity
m/s
Time (s)

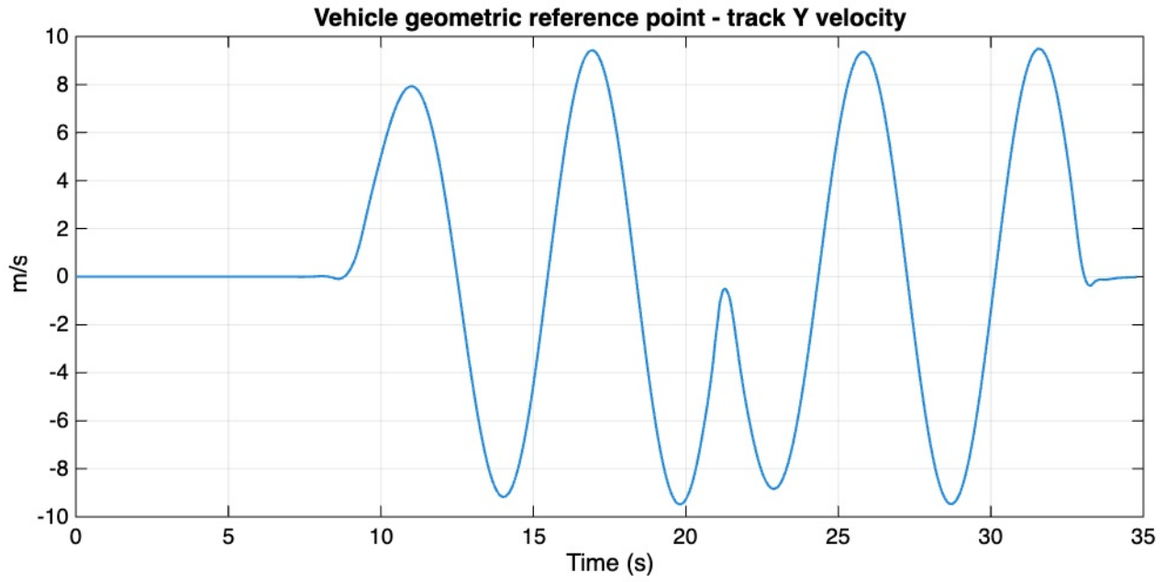

Vehicle geometric reference point - track X position
m
Time (s)

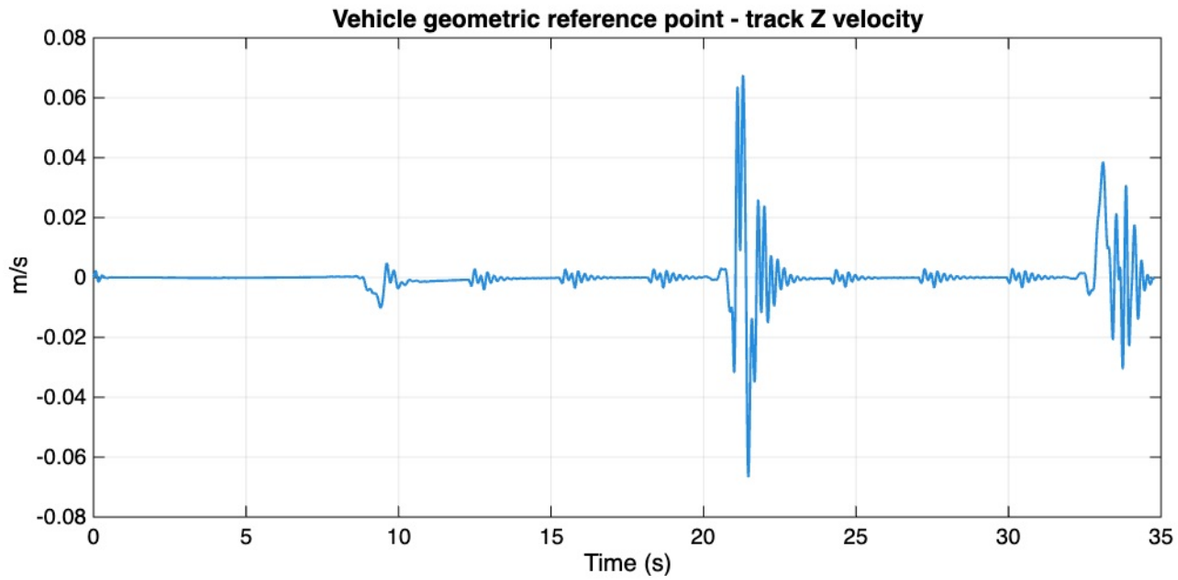

Vehicle geometric reference point - track Y position
m
Time (s)

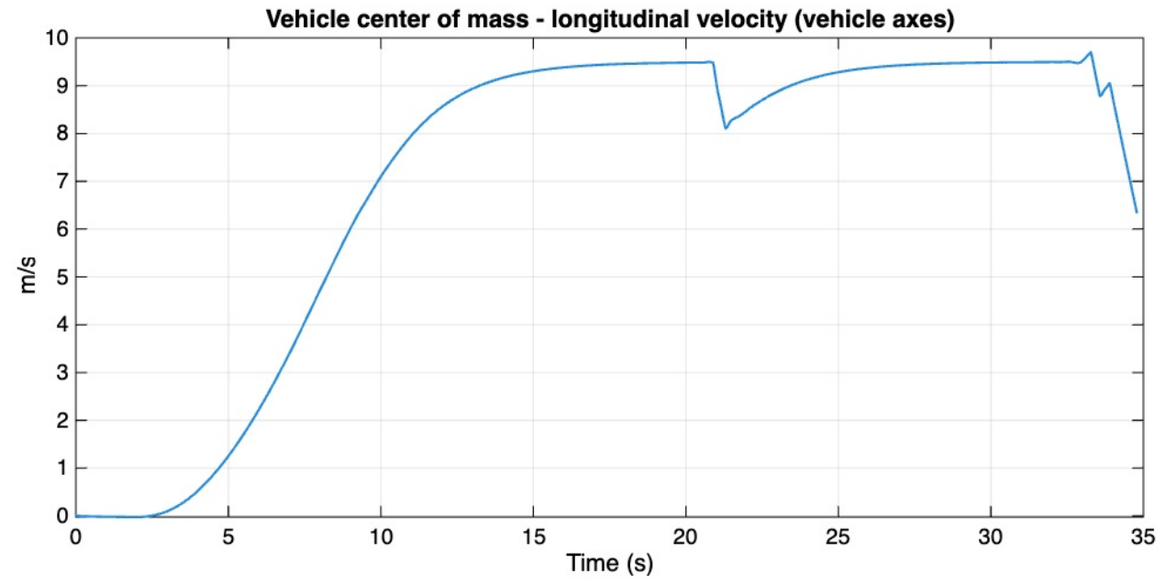

Vehicle geometric reference point - track Z position
m
Time (s)

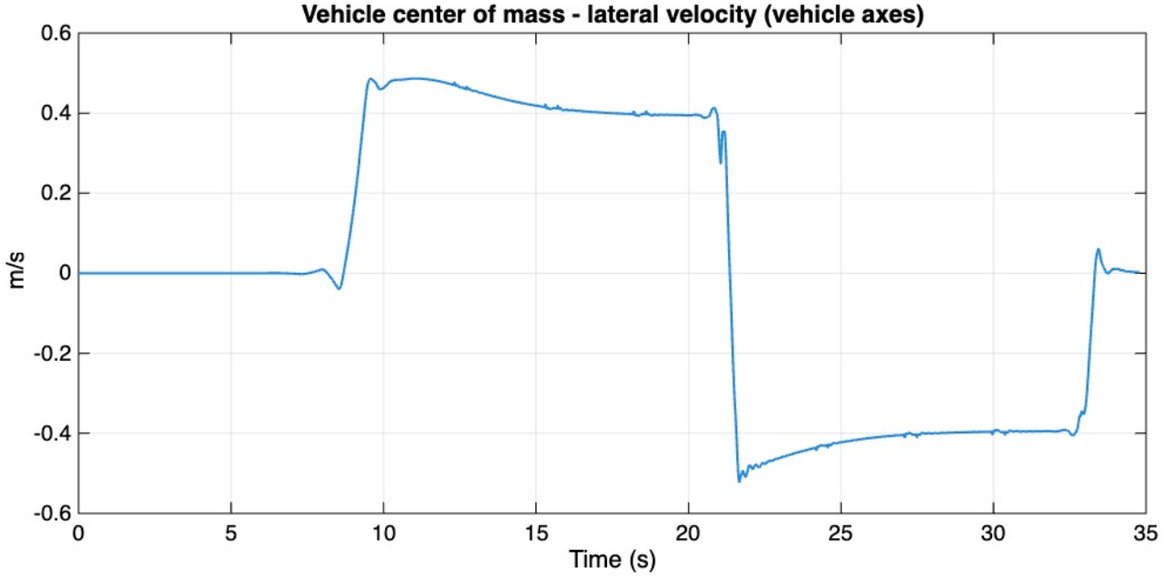
All recorded signal plots - plot page 8

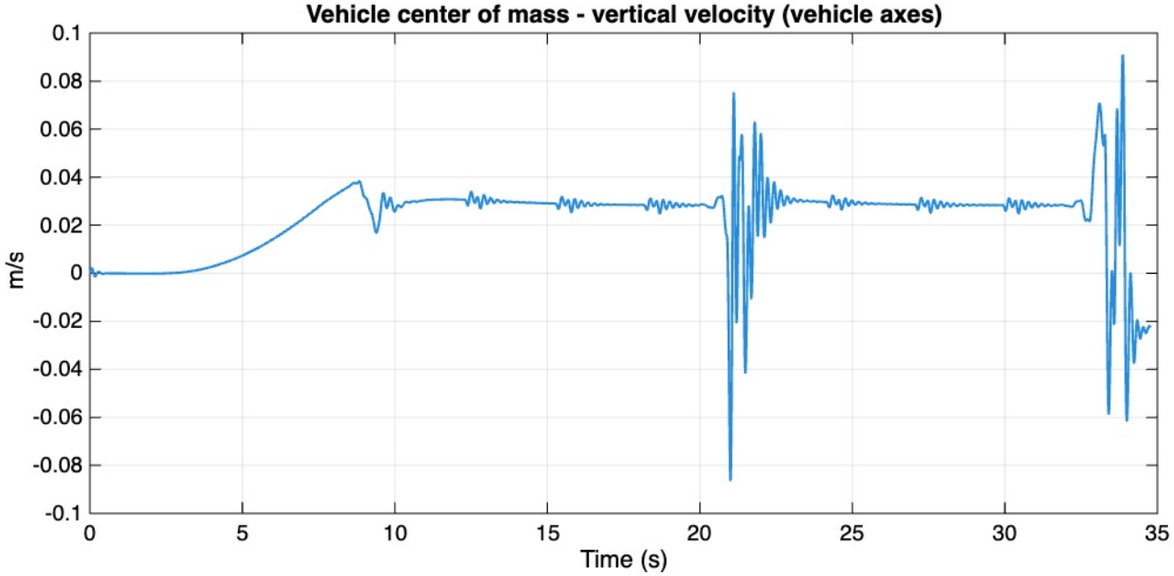




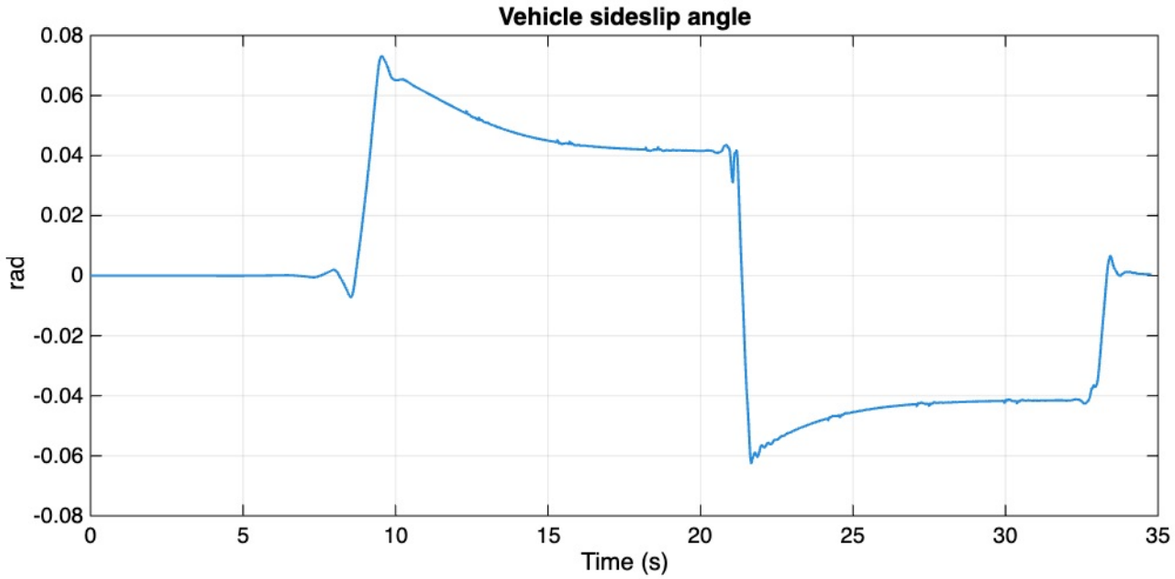


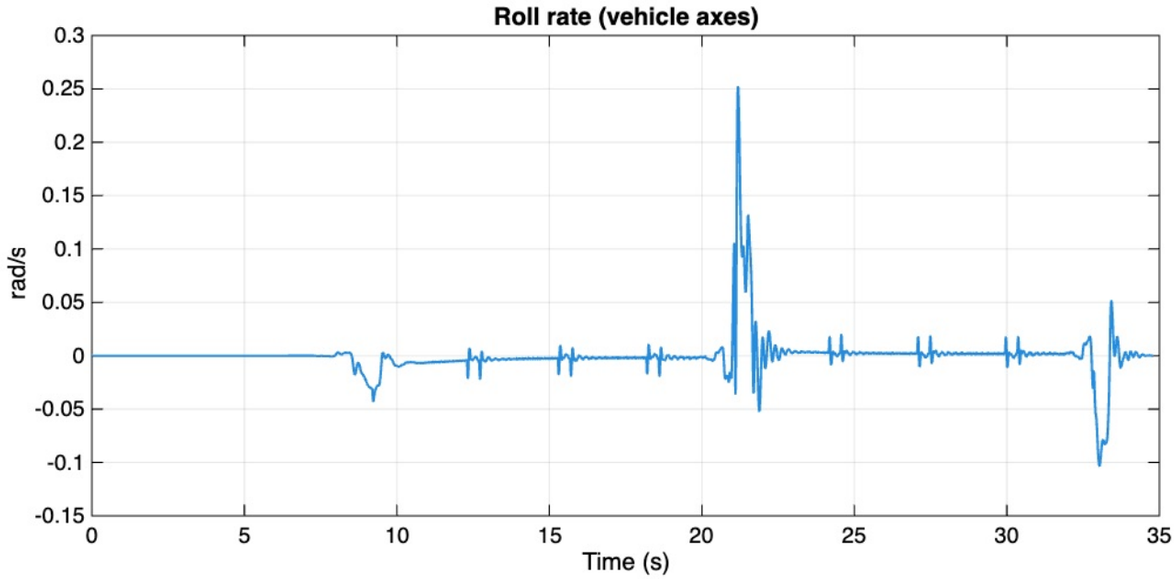


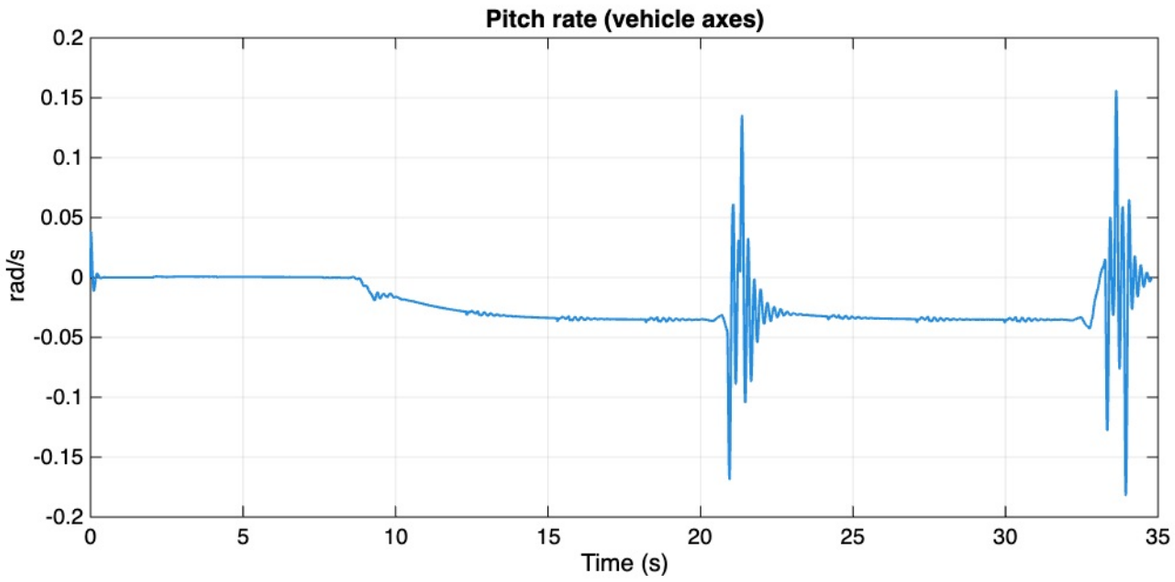


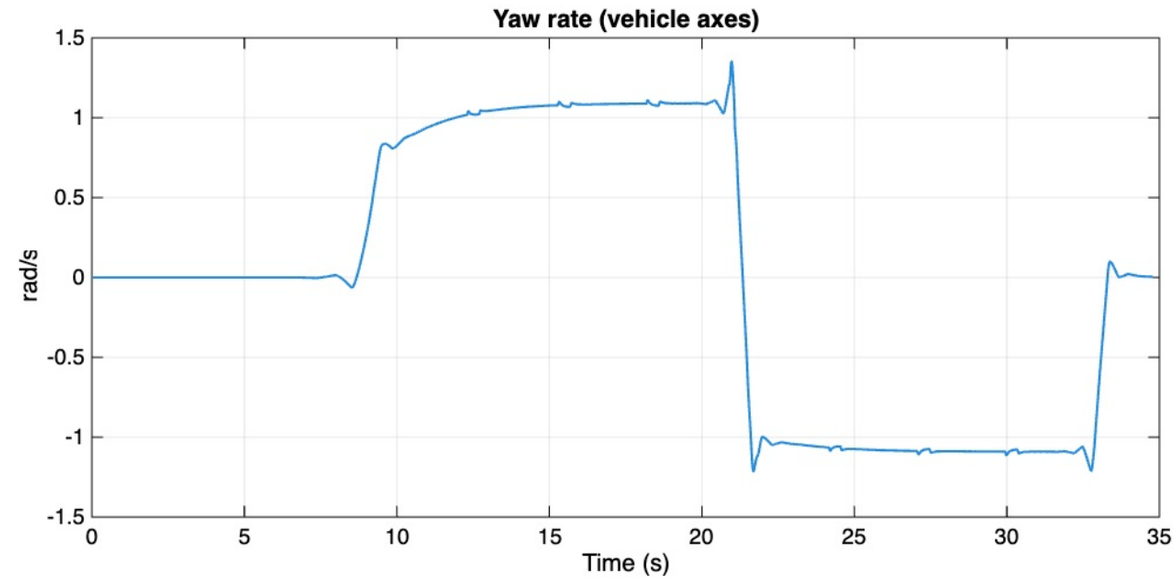


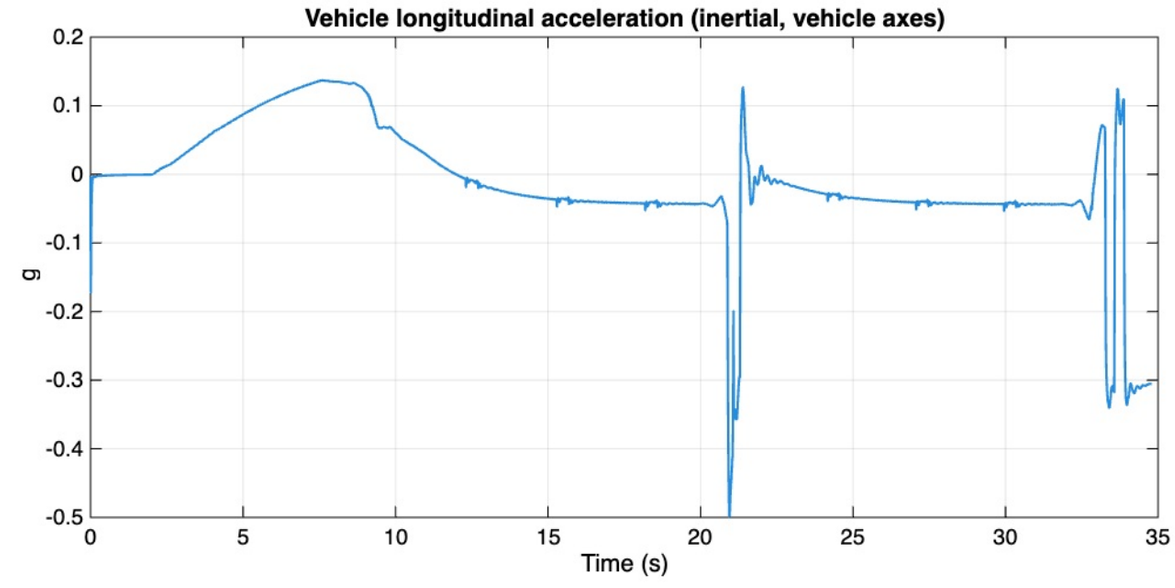
All recorded signal plots - plot page 9

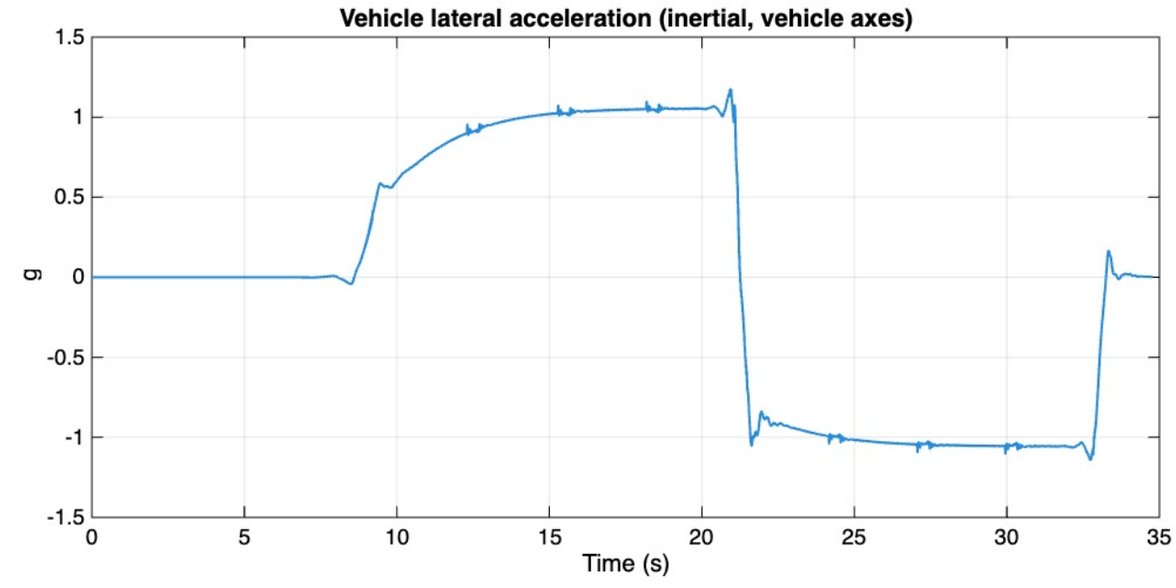


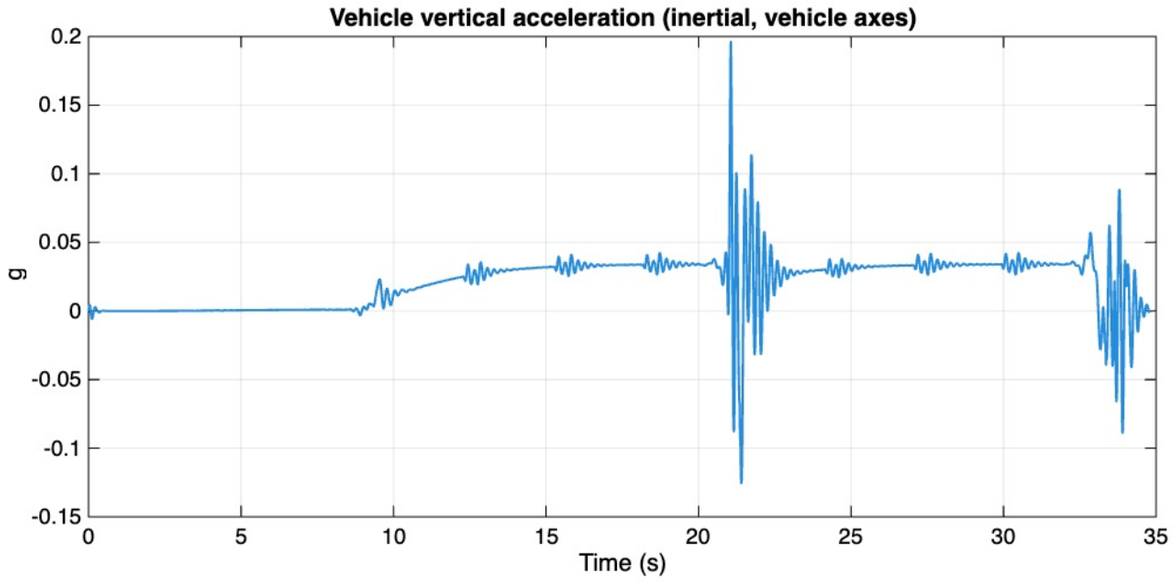


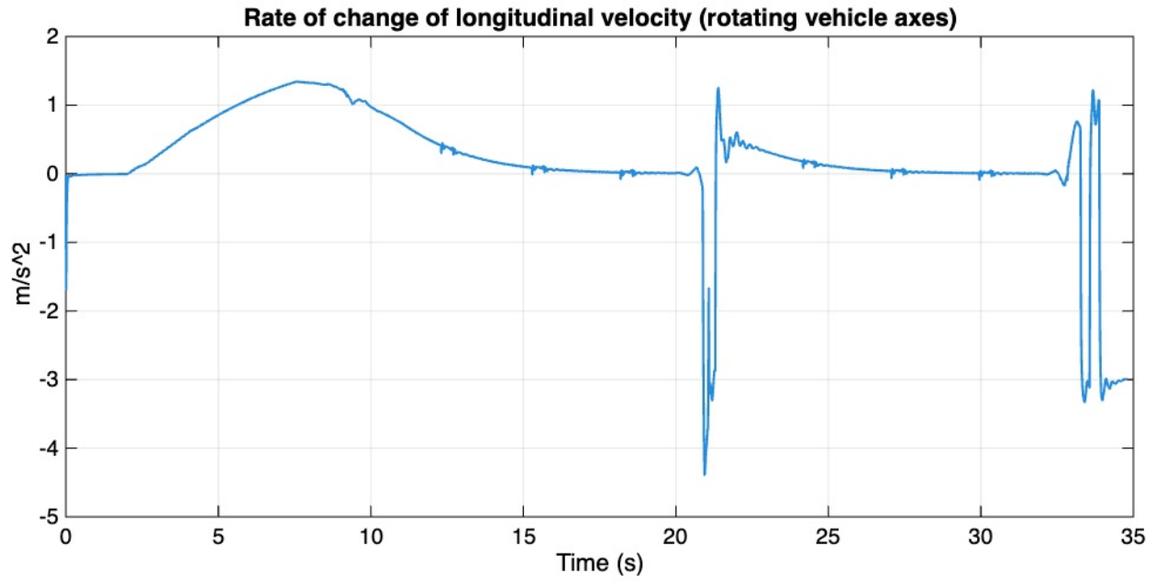


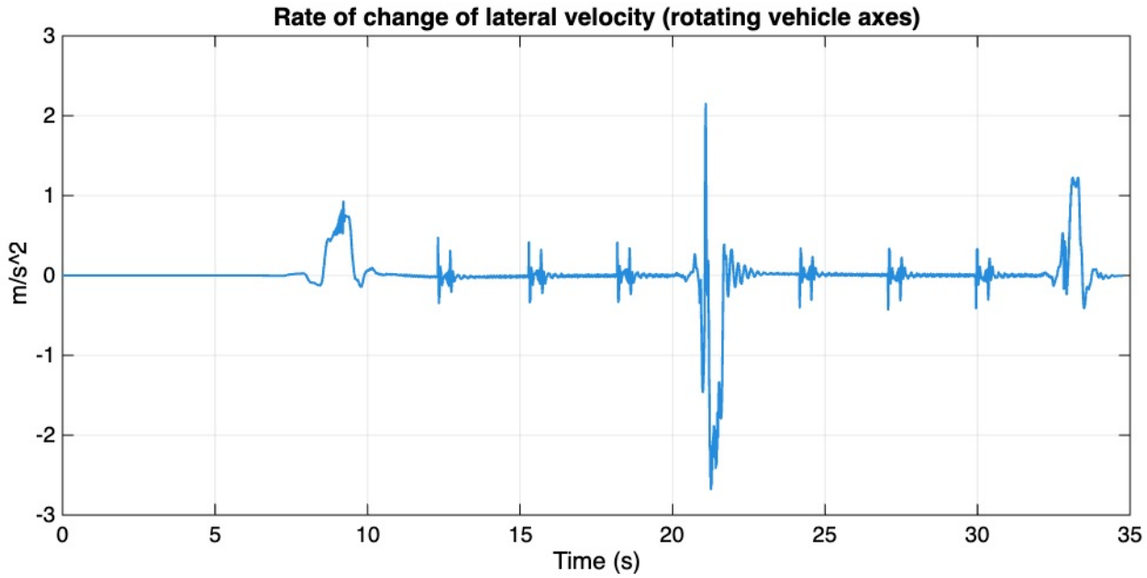


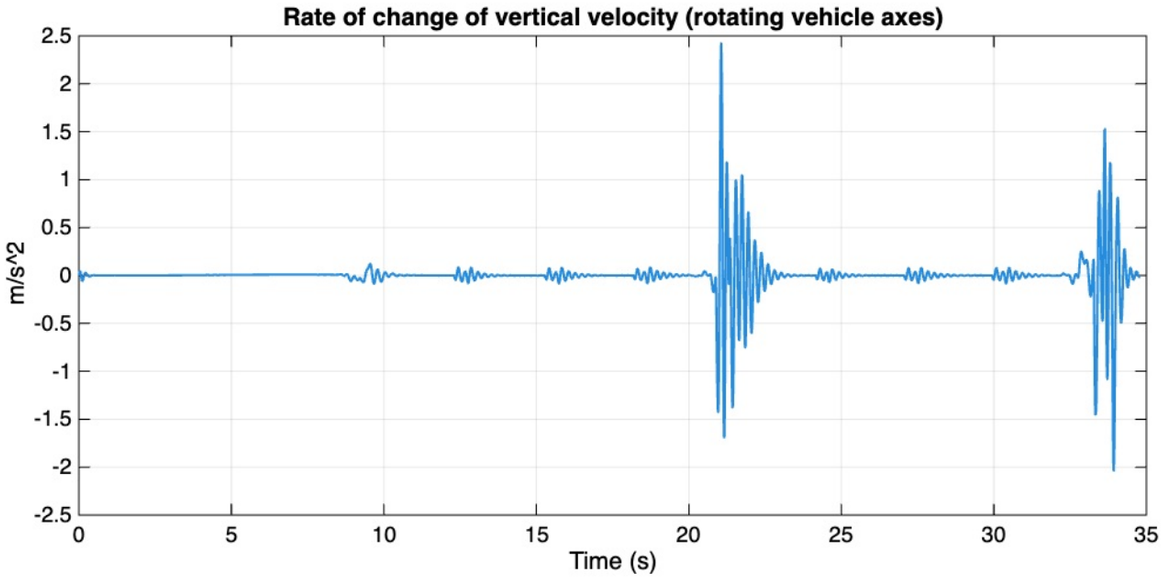


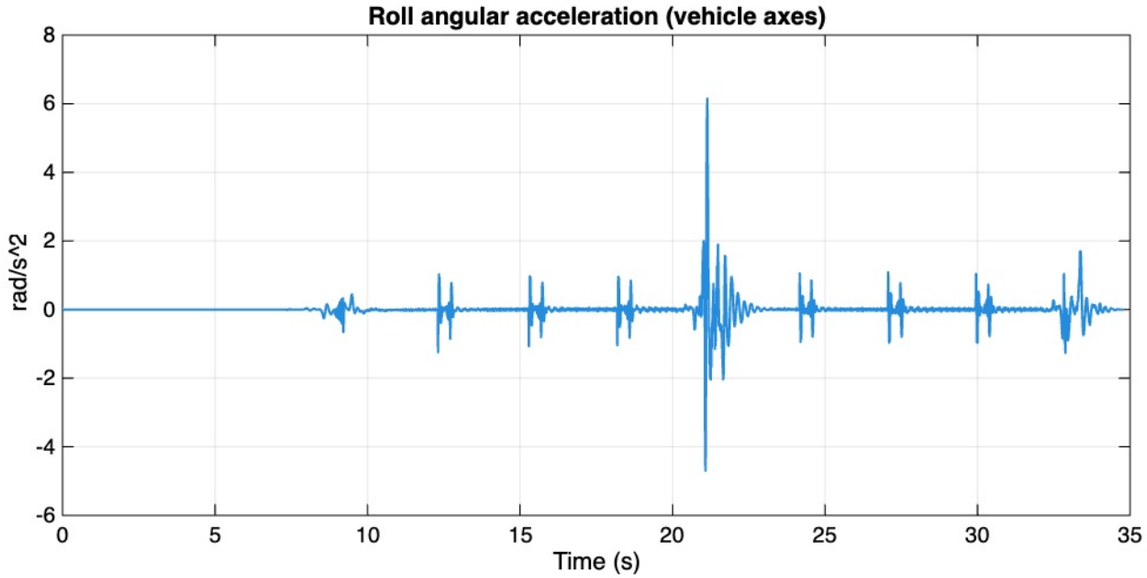


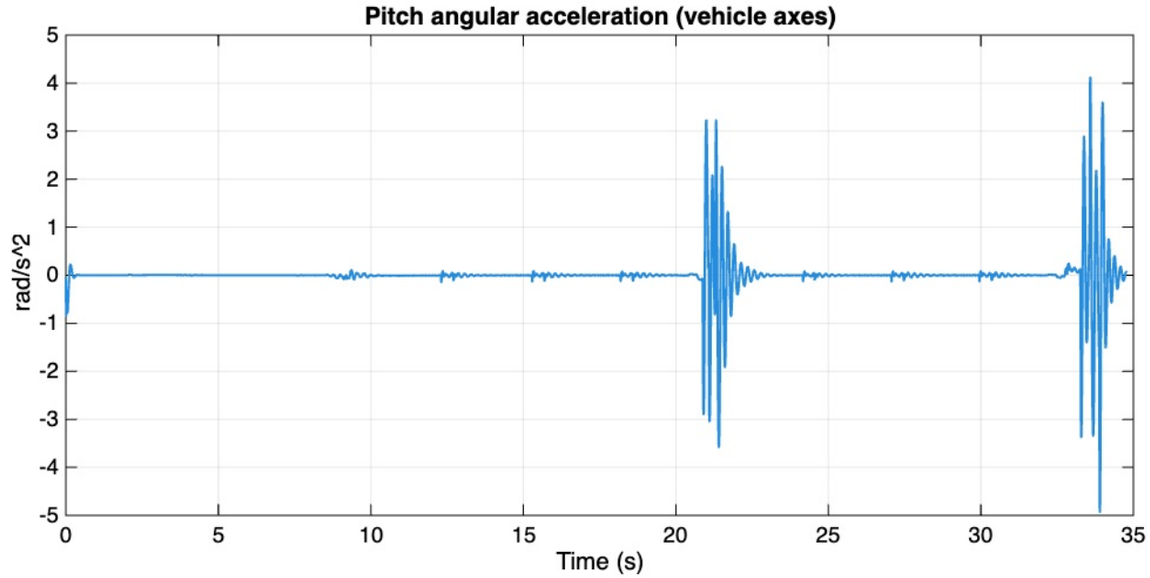

Vehicle vertical acceleration (inertial, vehicle axes)
g
Time (s)

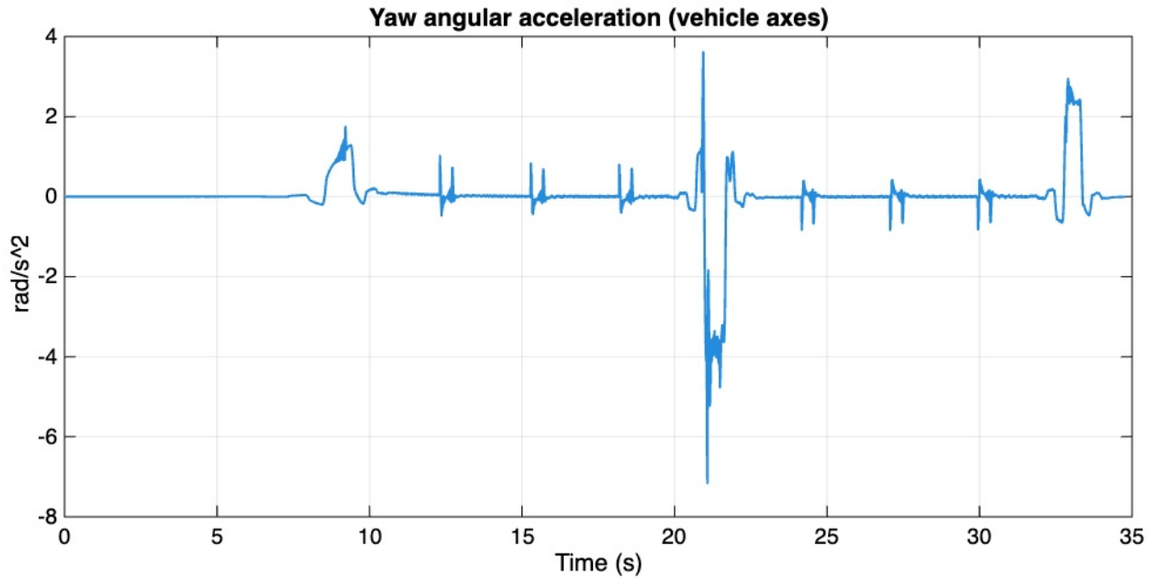

Rate of change of longitudinal velocity (rotating vehicle axes)
m/s^2
Time (s)

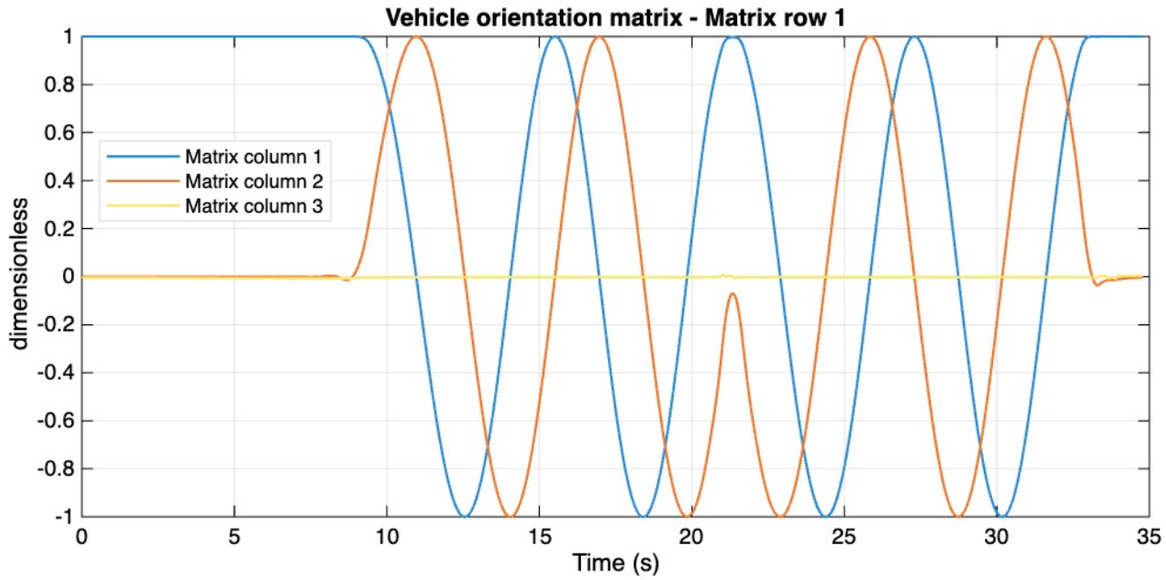

Rate of change of lateral velocity (rotating vehicle axes)
m/s^2
Time (s)

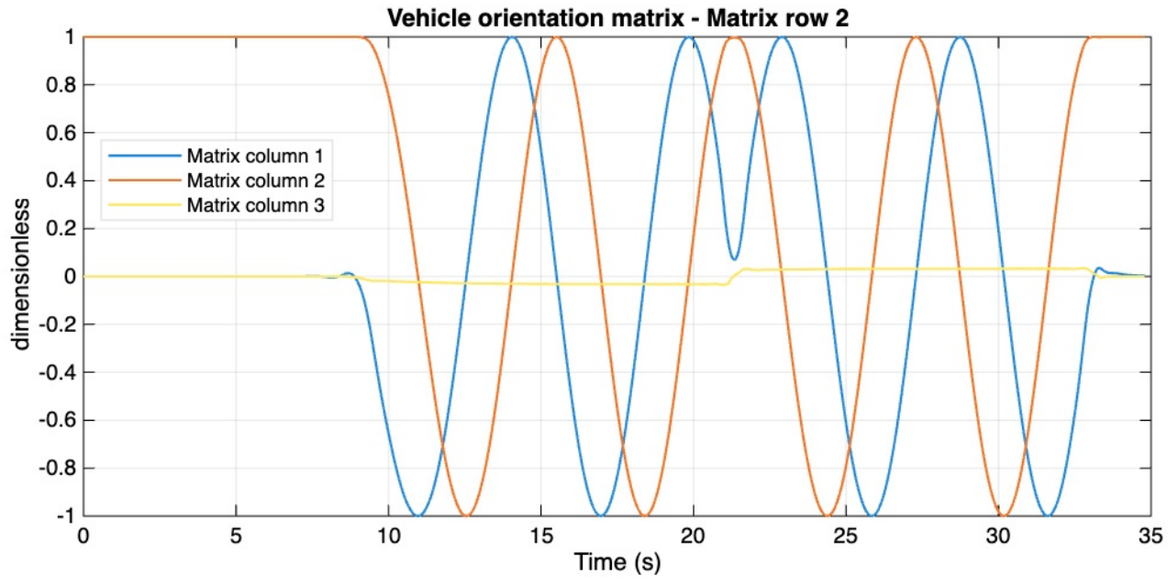

Rate of change of vertical velocity (rotating vehicle axes)
m/s^2
Time (s)

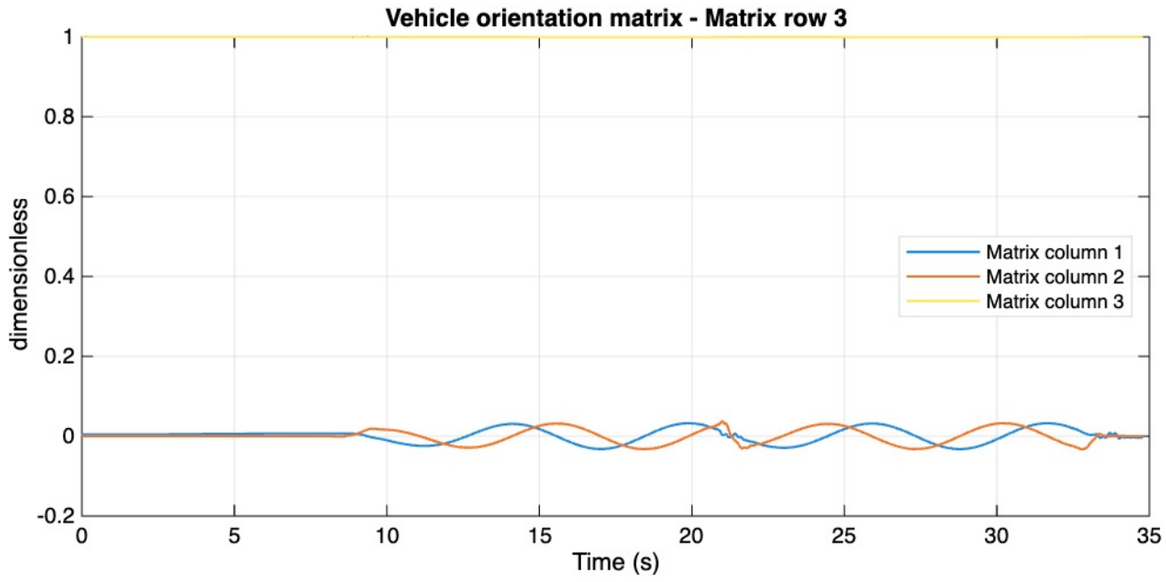

Roll angular acceleration (vehicle axes)
rad/s^2
Time (s)

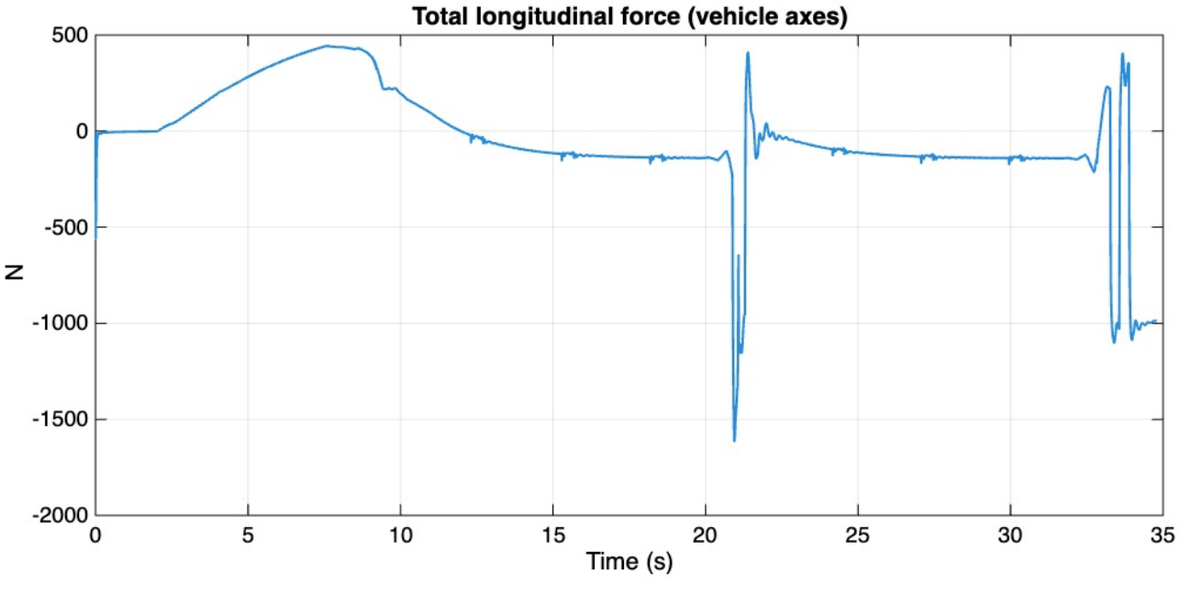

Pitch angular acceleration (vehicle axes)
rad/s^2
Time (s)

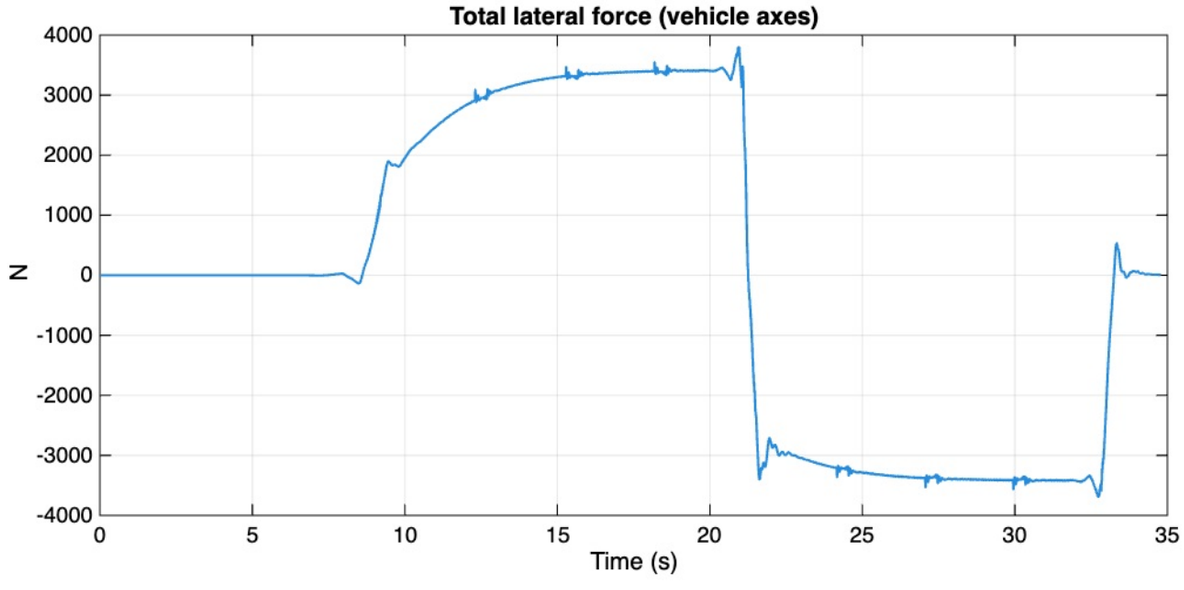

Yaw angular acceleration (vehicle axes)
rad/s^2
Time (s)

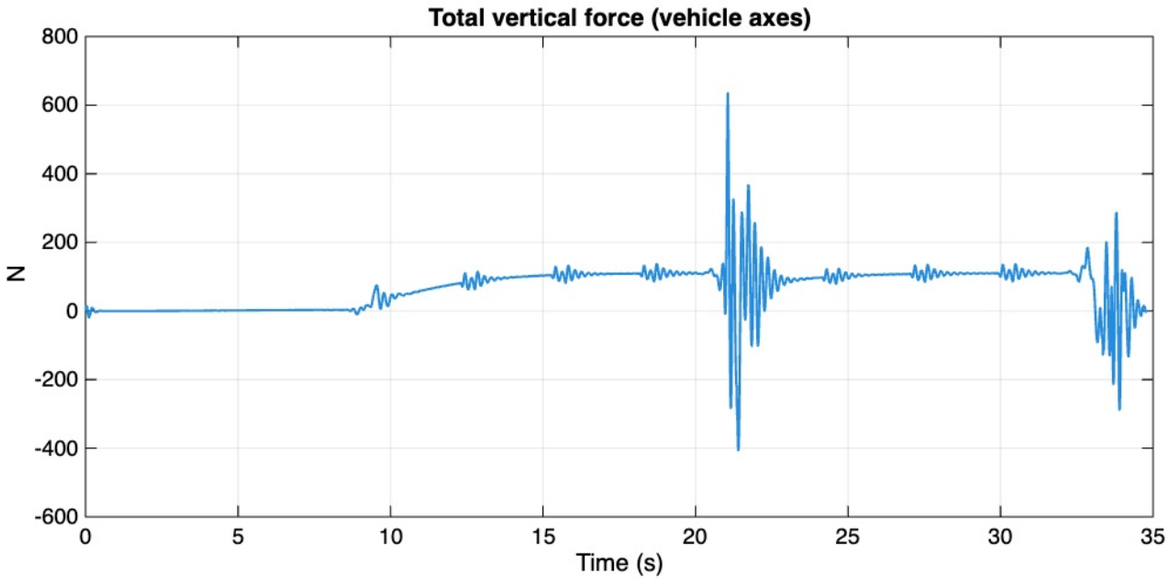

Vehicle orientation matrix - Matrix row 1
Matrix column 1
Matrix column 2
Matrix column 3
dimensionless
Time (s)

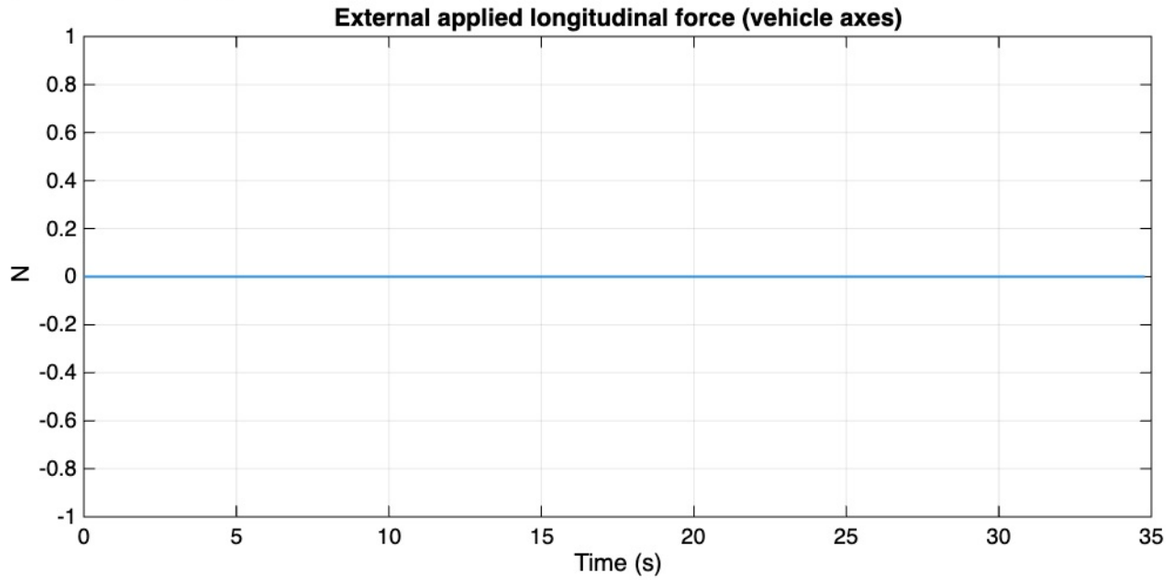

Vehicle orientation matrix - Matrix row 2
Matrix column 1
Matrix column 2
Matrix column 3
dimensionless
Time (s)

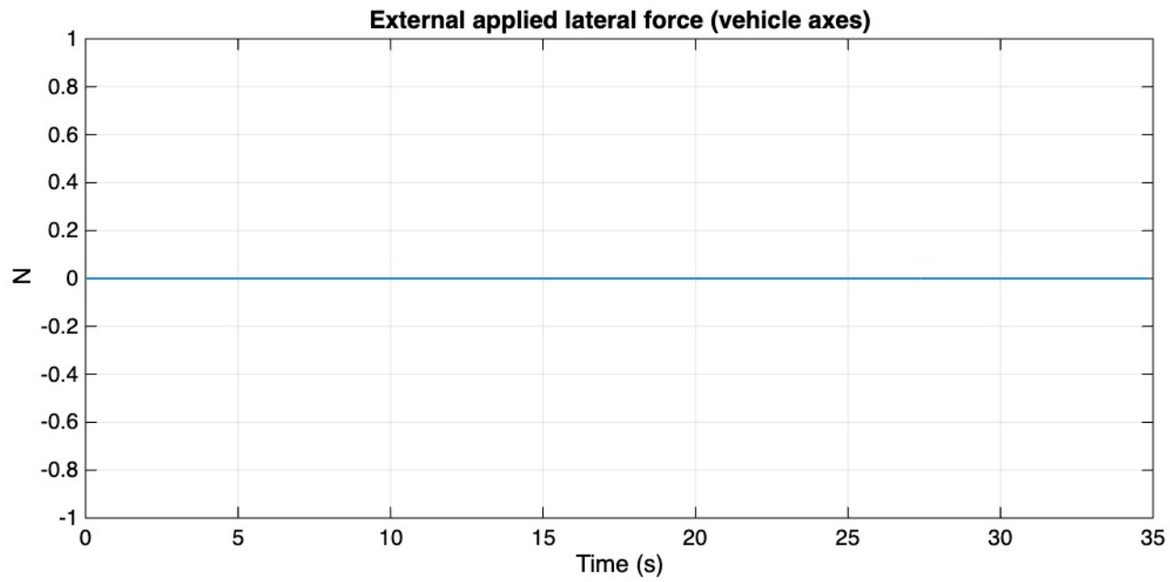

Vehicle orientation matrix - Matrix row 3
Matrix column 1
Matrix column 2
Matrix column 3
dimensionless
Time (s)

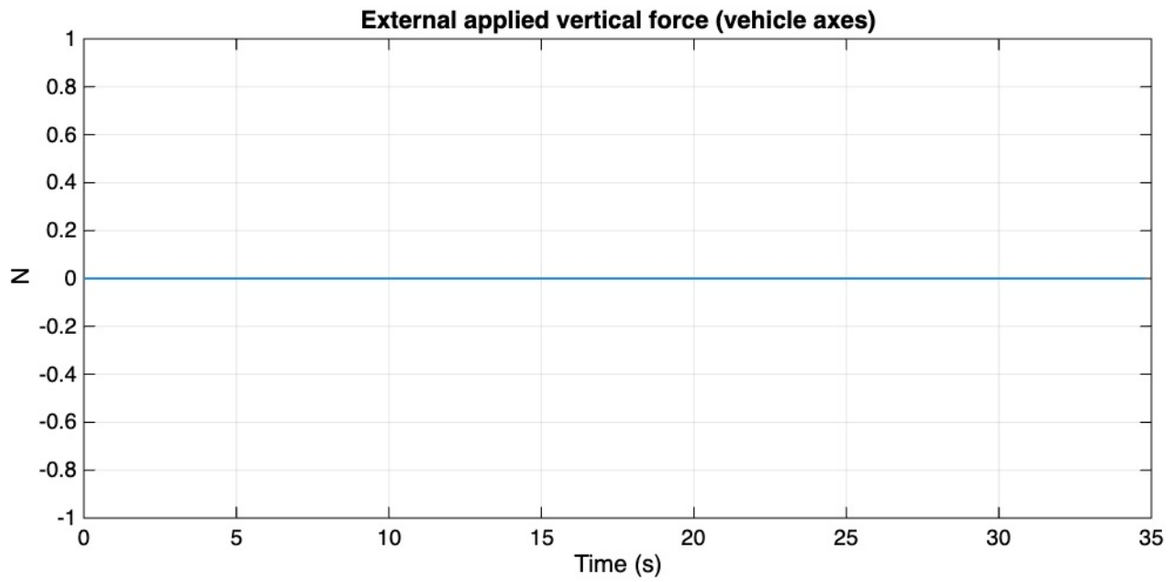

Total longitudinal force (vehicle axes)
N
Time (s)

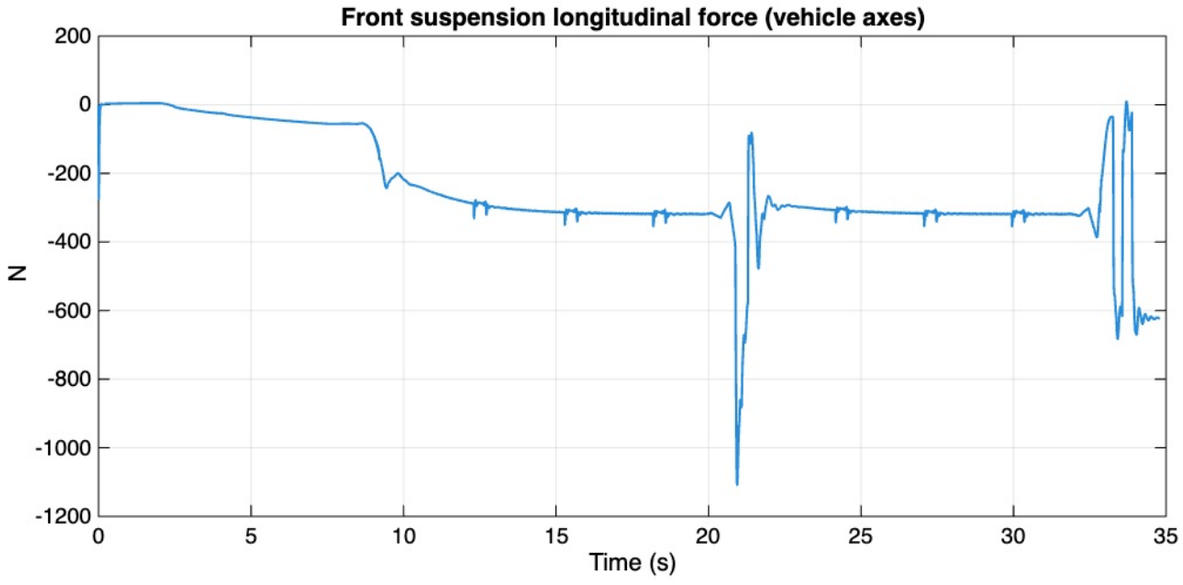

Total lateral force (vehicle axes)
N
Time (s)

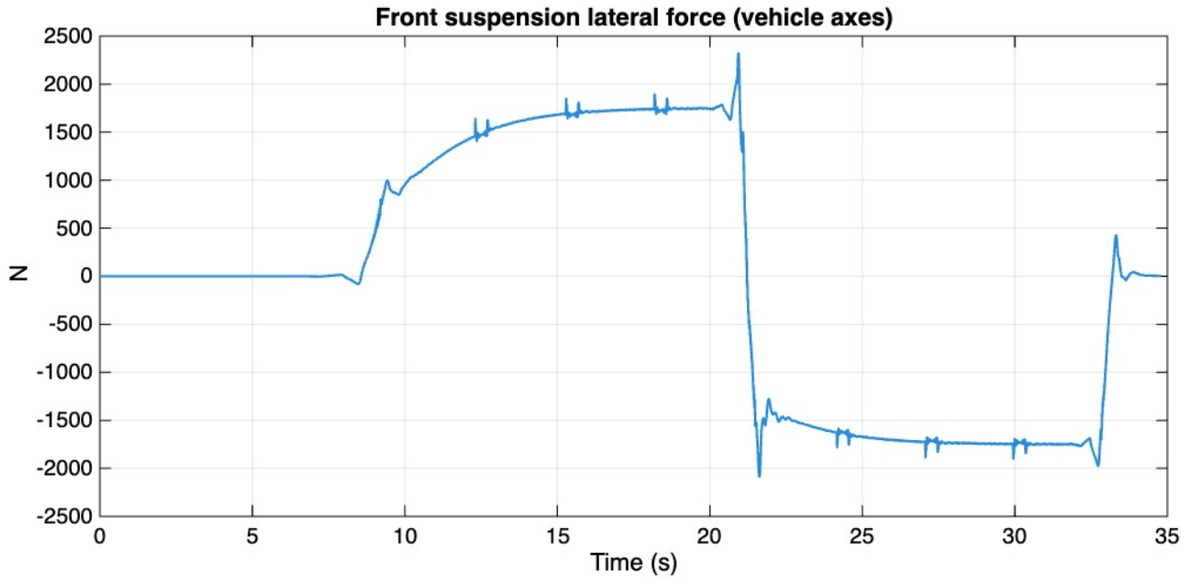

Total vertical force (vehicle axes)
N
Time (s)

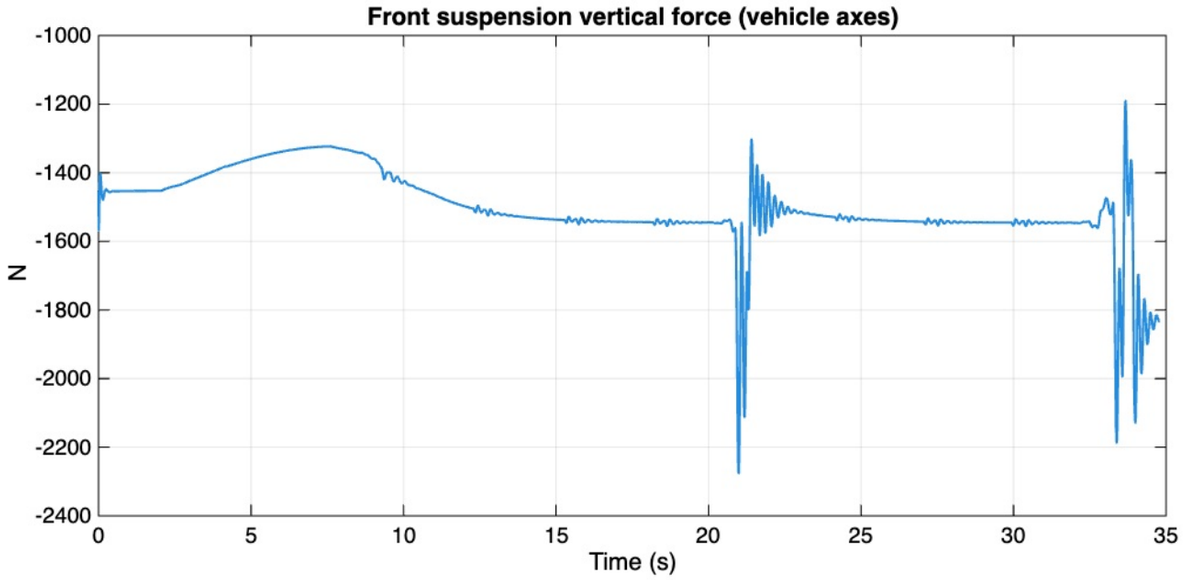

External applied longitudinal force (vehicle axes)
N
Time (s)

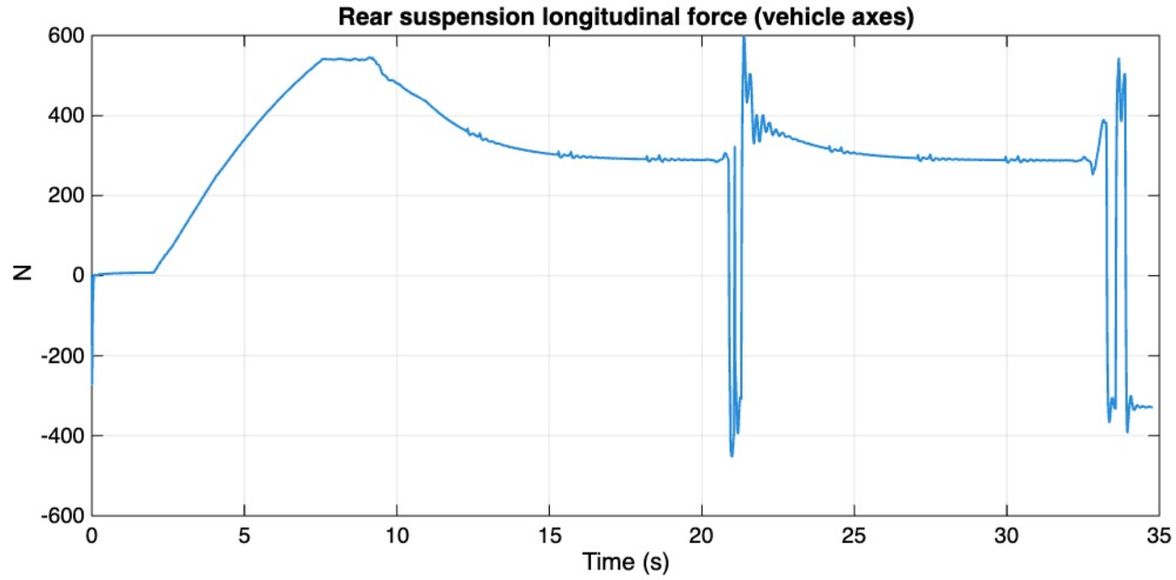

External applied lateral force (vehicle axes)
N
Time (s)

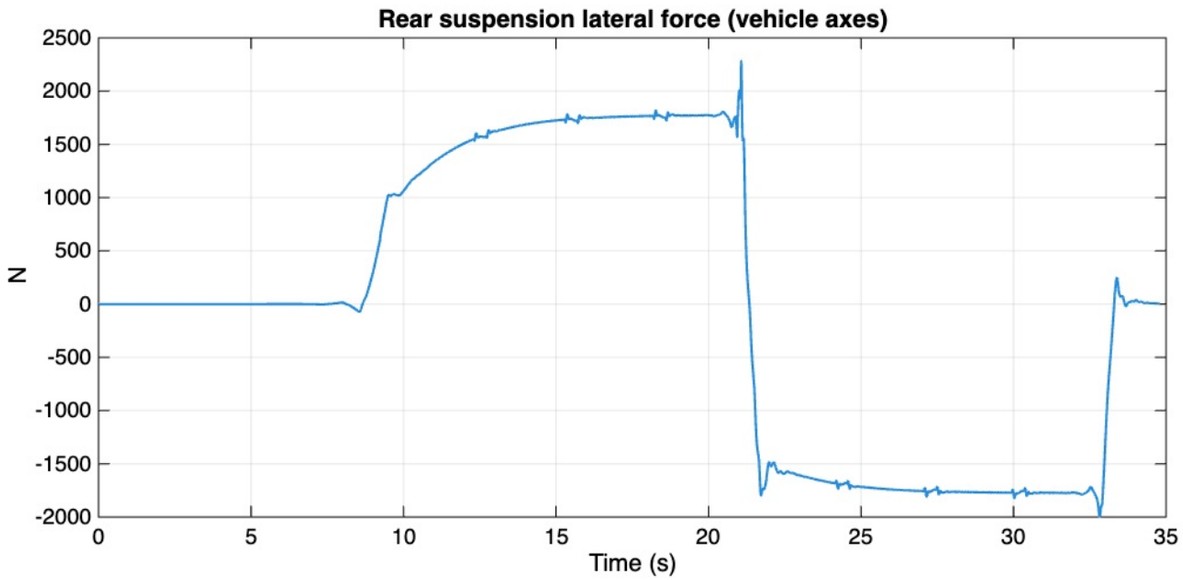

External applied vertical force (vehicle axes)
N
Time (s)

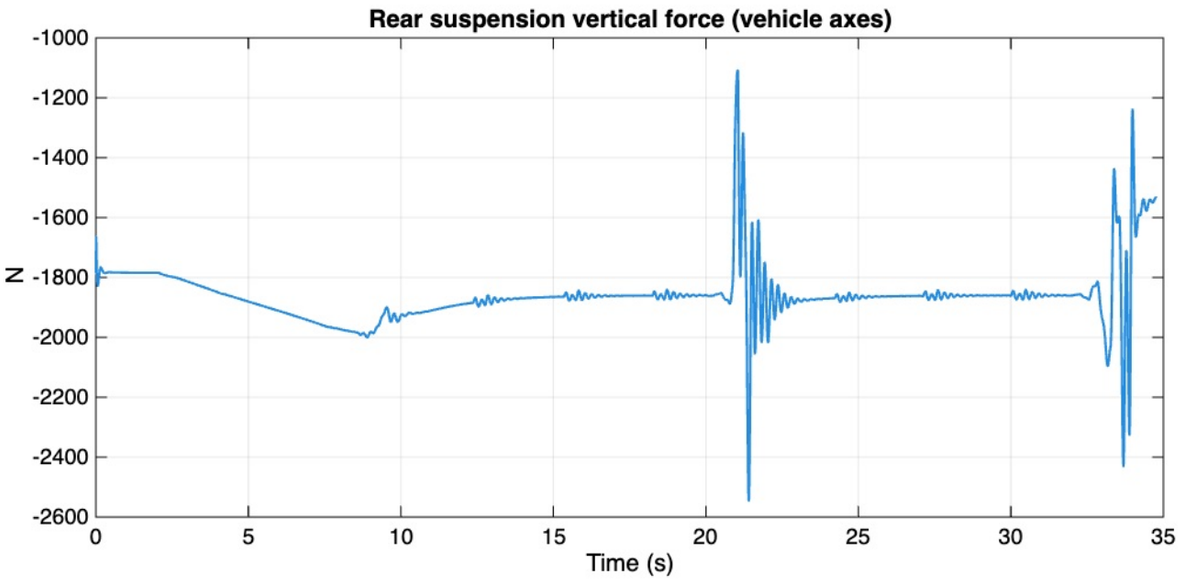

Front suspension longitudinal force (vehicle axes)
N
Time (s)

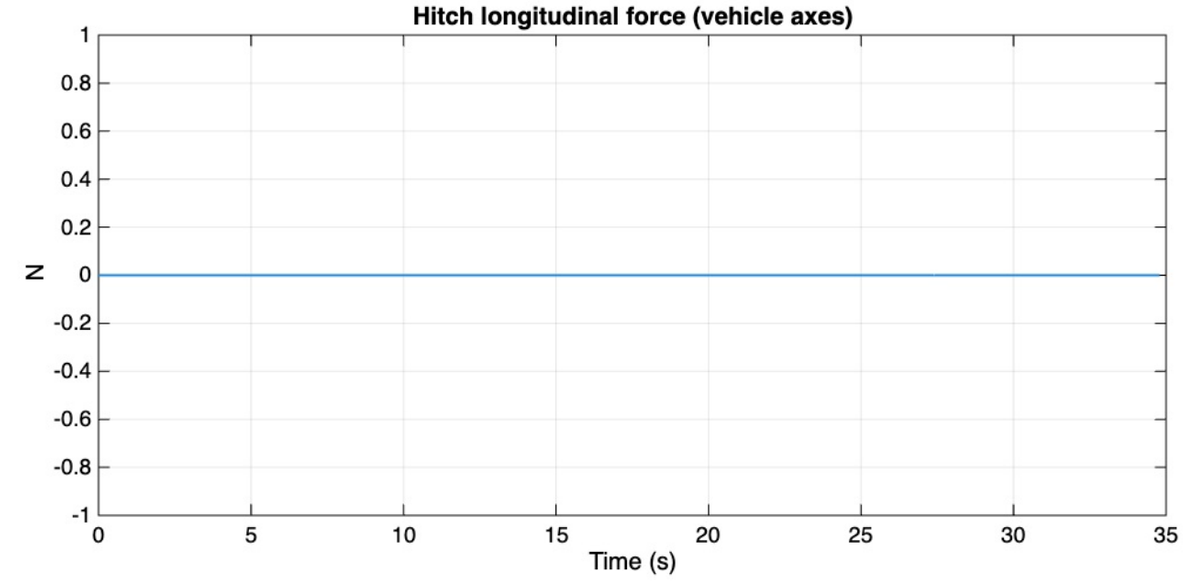

Front suspension lateral force (vehicle axes)
N
Time (s)

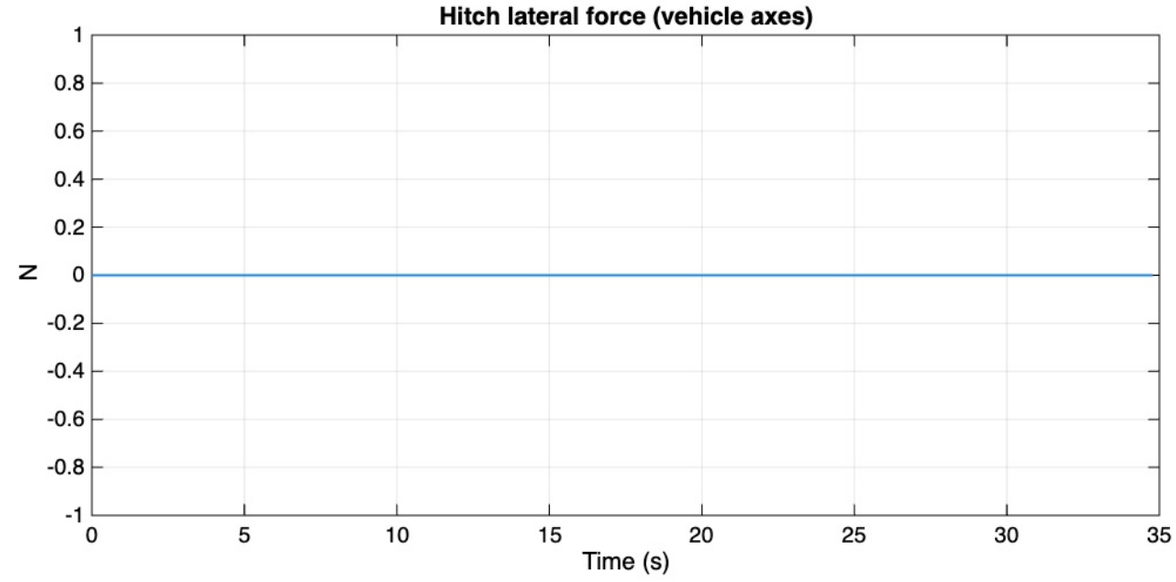

Front suspension vertical force (vehicle axes)
N
Time (s)

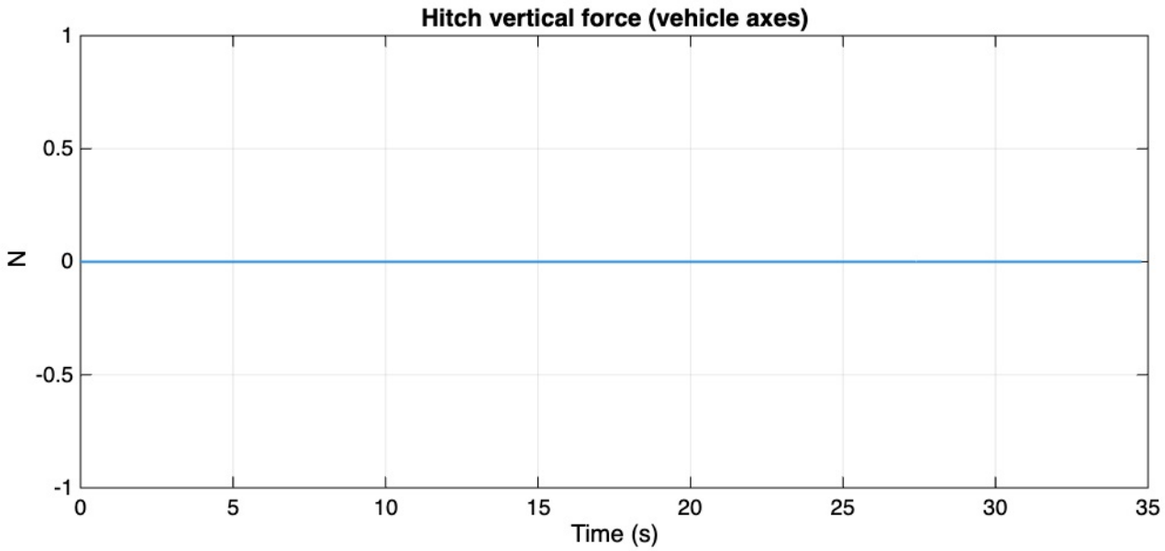

Rear suspension longitudinal force (vehicle axes)
N
Time (s)

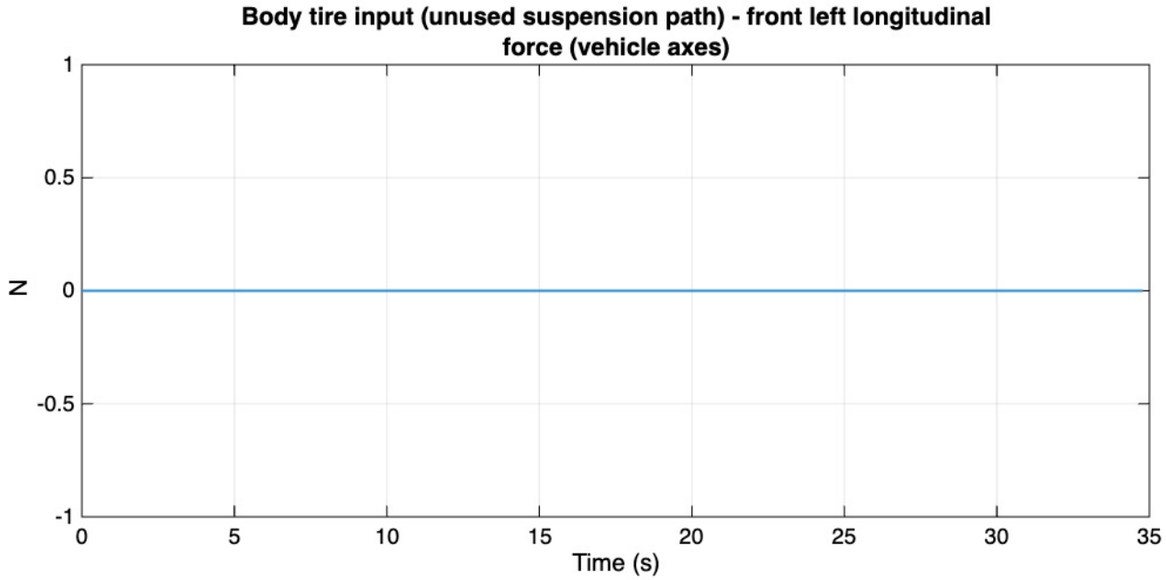

Rear suspension lateral force (vehicle axes)
N
Time (s)

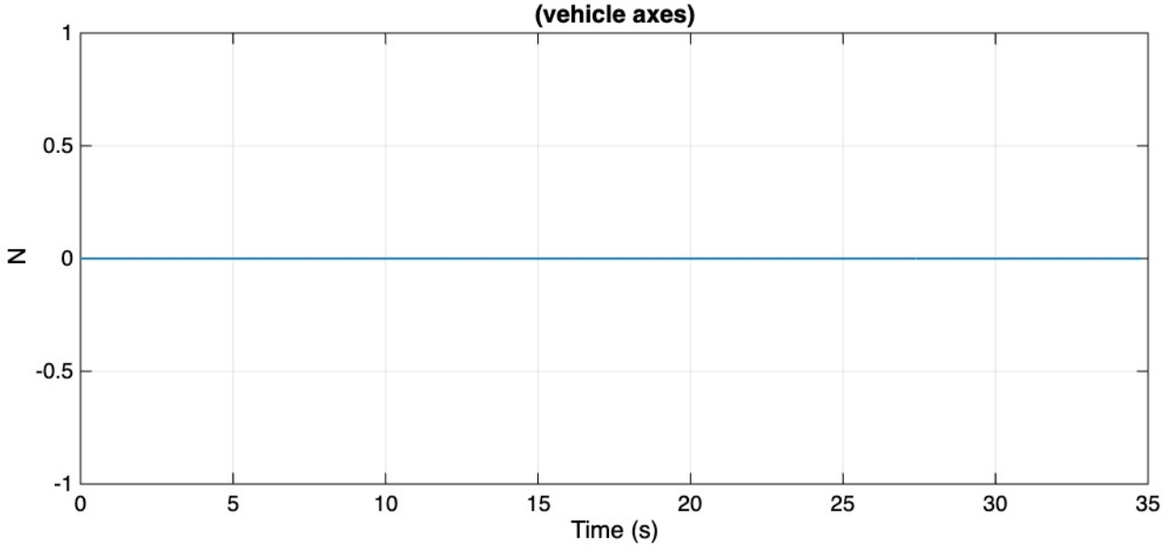

Rear suspension vertical force (vehicle axes)
N
Time (s)

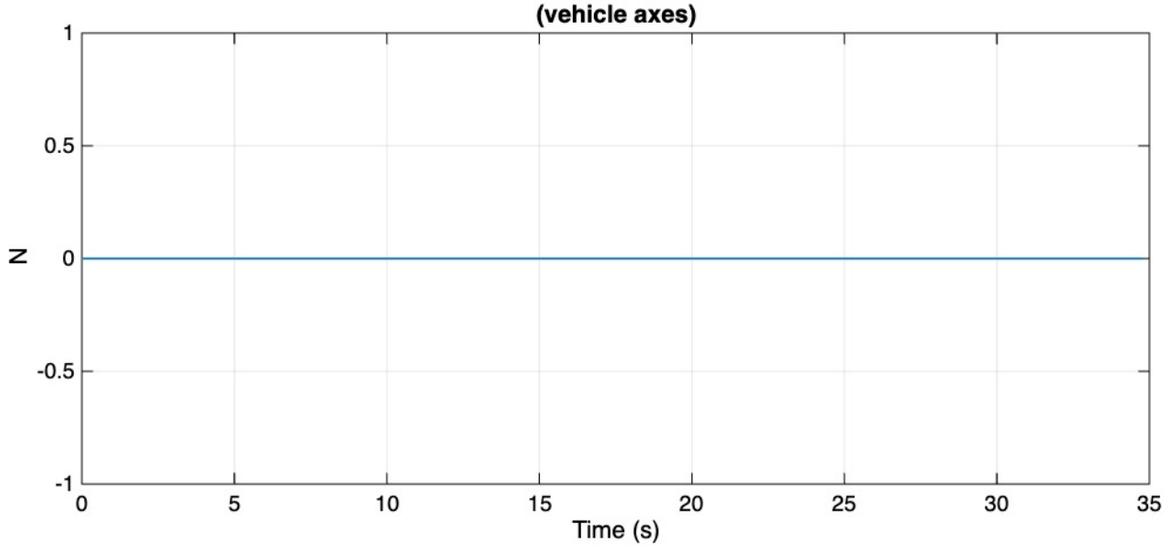

Hitch longitudinal force (vehicle axes)
N
Time (s)

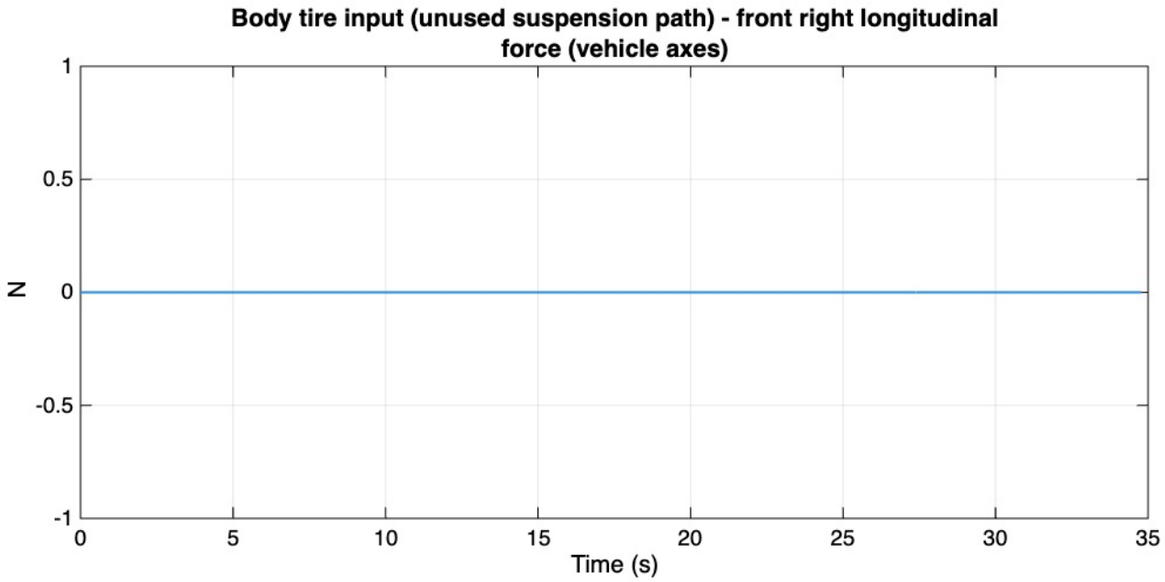

Hitch lateral force (vehicle axes)
N
Time (s)

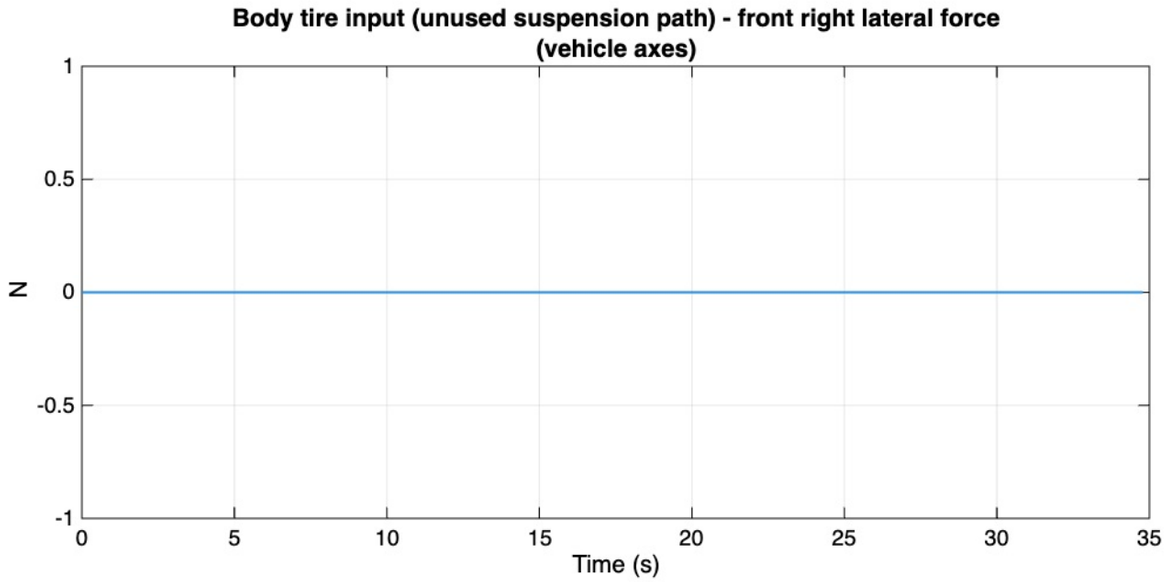

Hitch vertical force (vehicle axes)
N
Time (s)

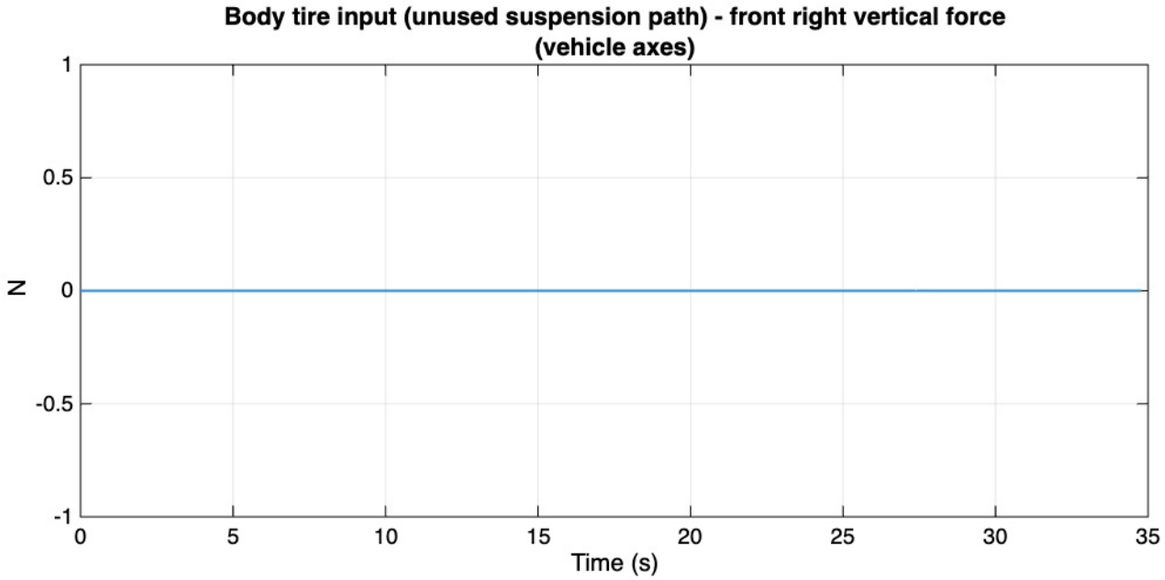

Body tire input (unused suspension path) - front left longitudinal force (vehicle axes)
N
Time (s)

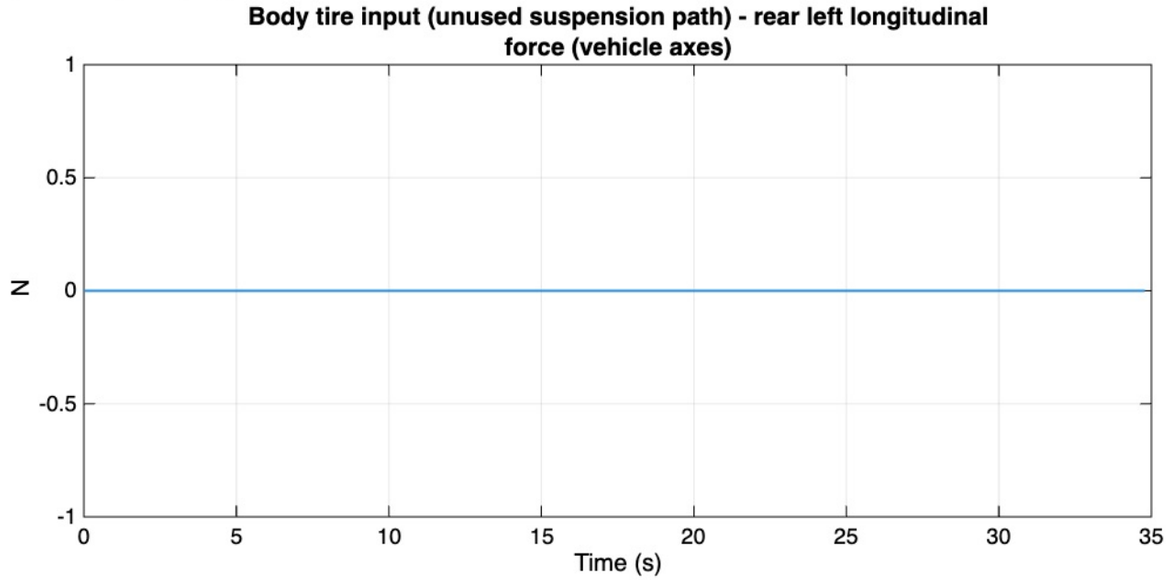

Body tire input (unused suspension path) - front left lateral force (vehicle axes)
N
Time (s)

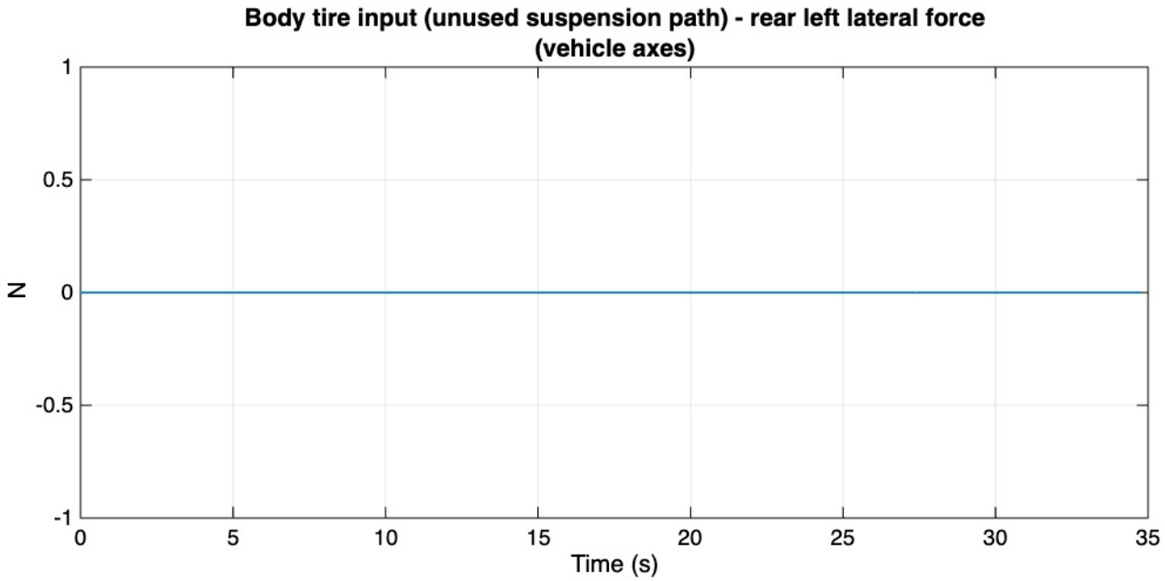

Body tire input (unused suspension path) - front left vertical force (vehicle axes)
N
Time (s)

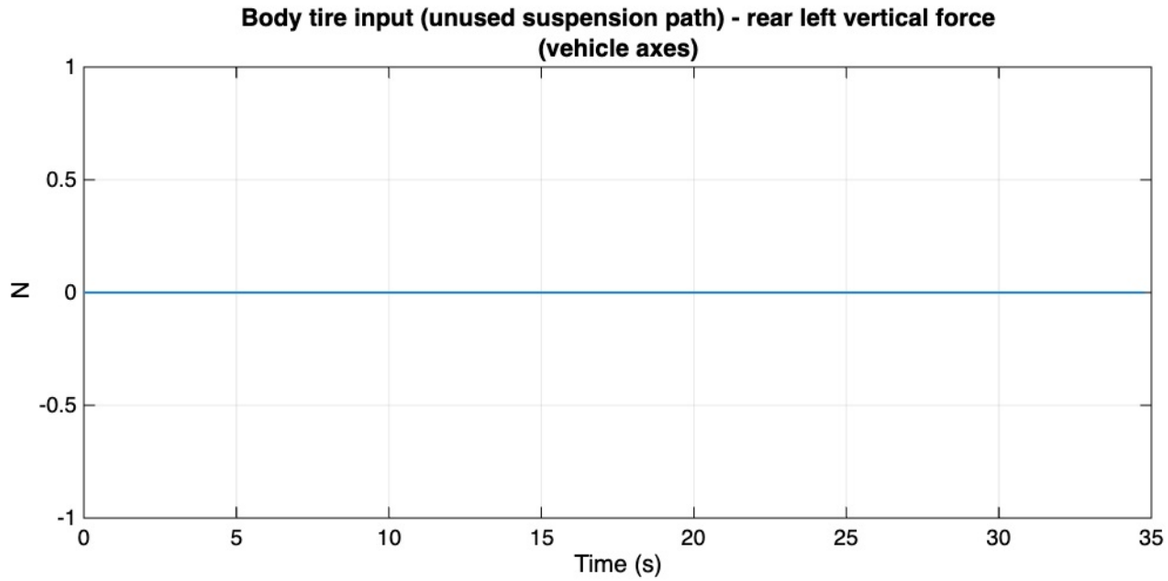

Body tire input (unused suspension path) - front right longitudinal force (vehicle axes)
N
Time (s)

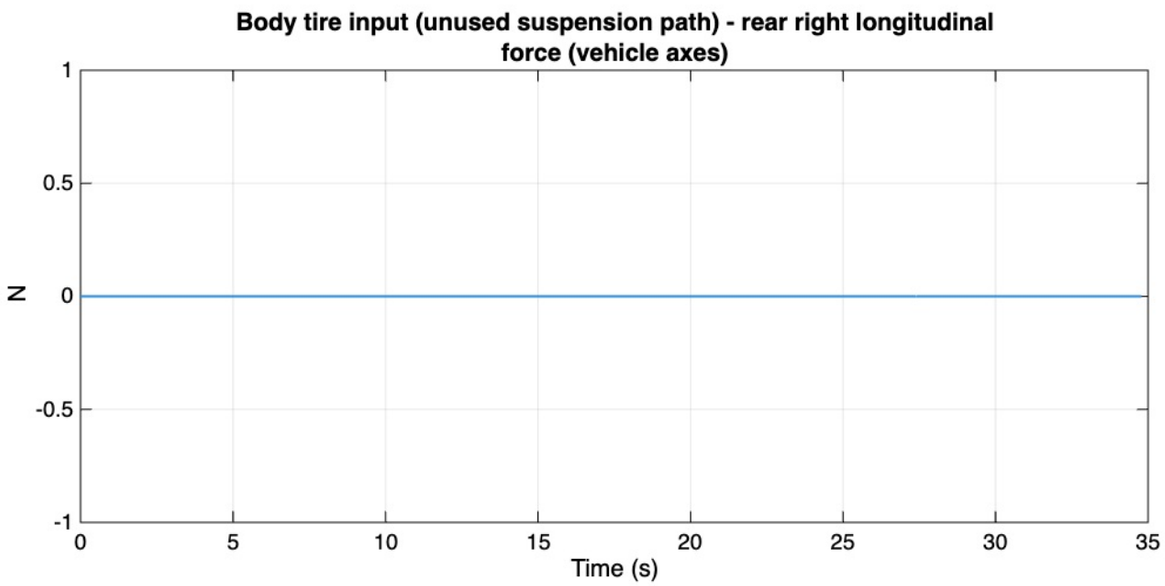

Body tire input (unused suspension path) - front right lateral force (vehicle axes)
N
Time (s)

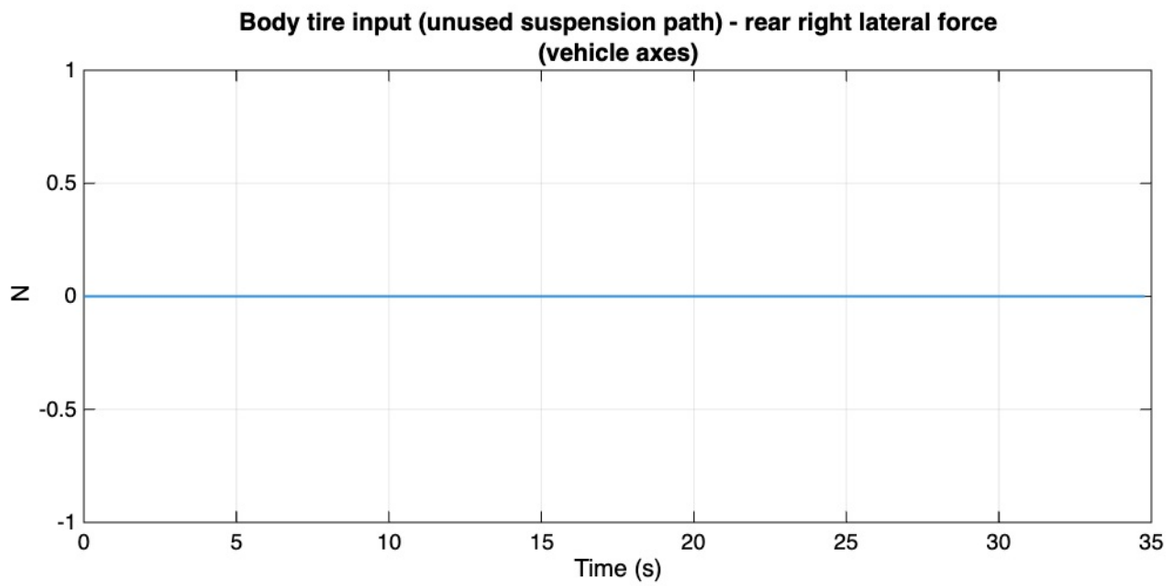

Body tire input (unused suspension path) - front right vertical force (vehicle axes)
N
Time (s)

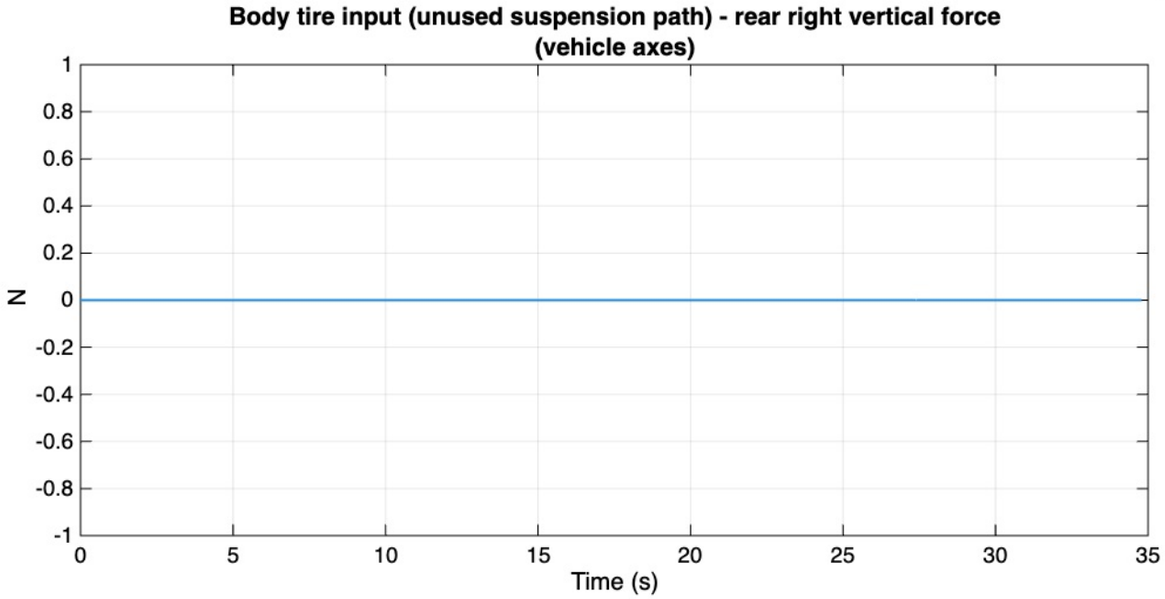

Body tire input (unused suspension path) - rear left longitudinal force (vehicle axes)
N
Time (s)

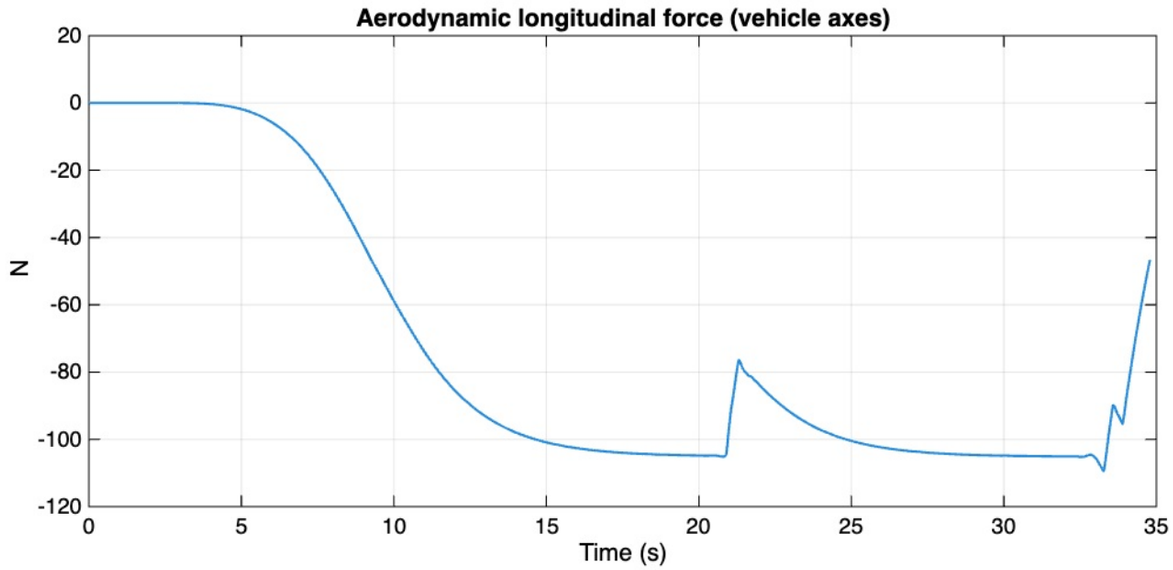

Body tire input (unused suspension path) - rear left lateral force (vehicle axes)
N
Time (s)

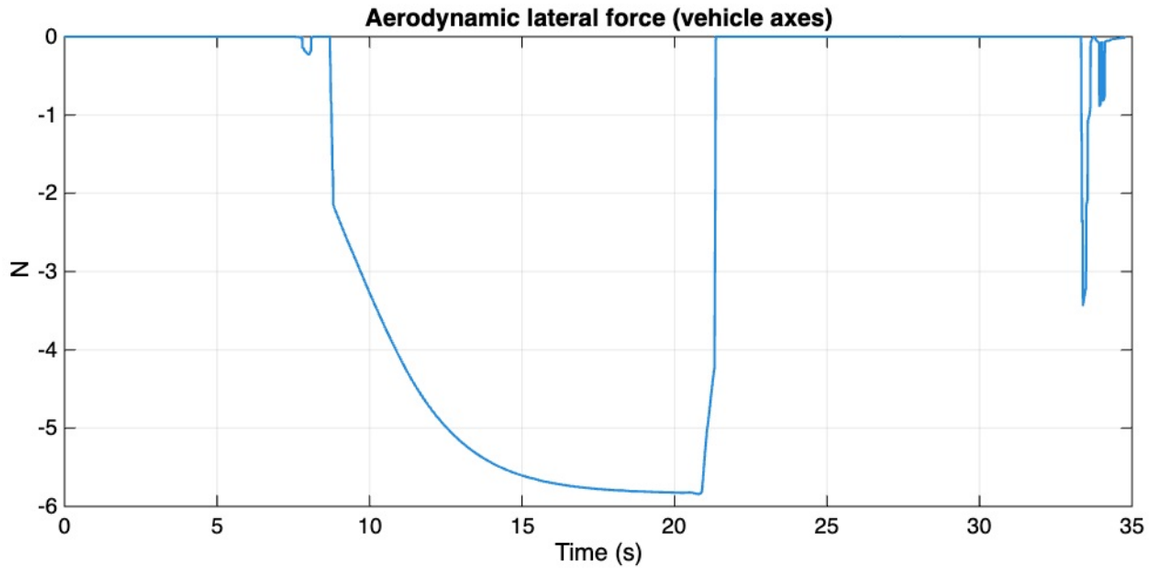

Body tire input (unused suspension path) - rear left vertical force (vehicle axes)
N
Time (s)

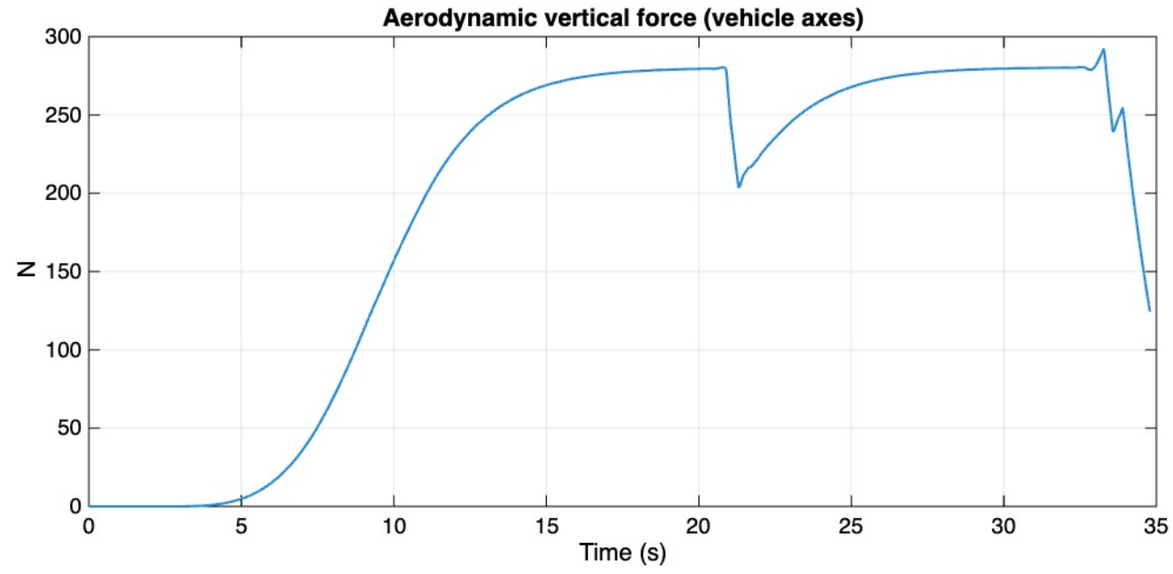

Body tire input (unused suspension path) - rear right longitudinal force (vehicle axes)
N
Time (s)

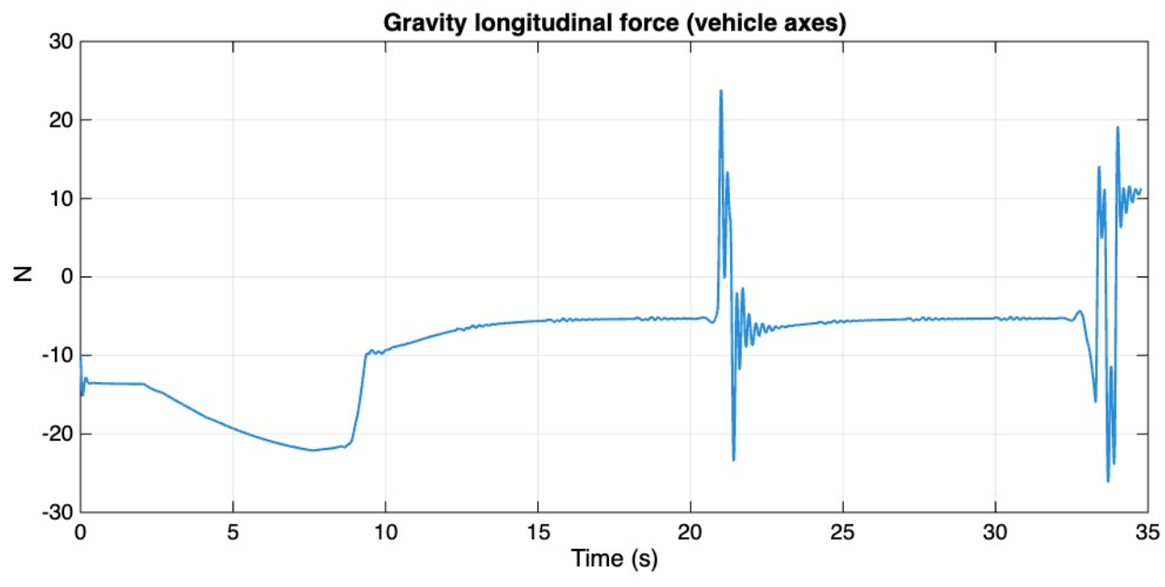

Body tire input (unused suspension path) - rear right lateral force (vehicle axes)
N
Time (s)

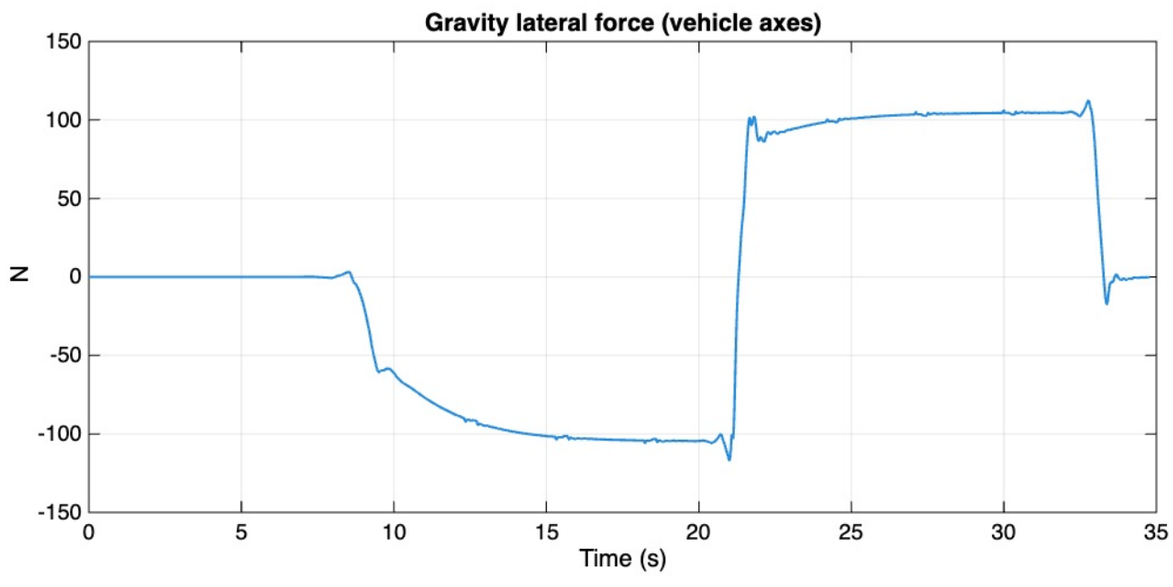

Body tire input (unused suspension path) - rear right vertical force (vehicle axes)
N
Time (s)

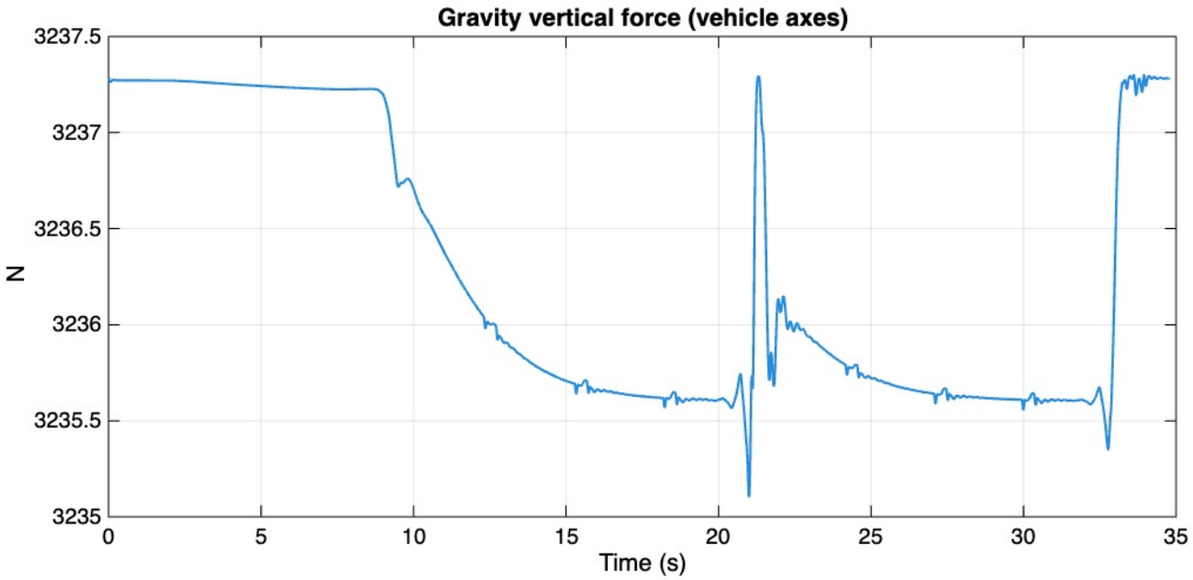

Aerodynamic longitudinal force (vehicle axes)
N
Time (s)

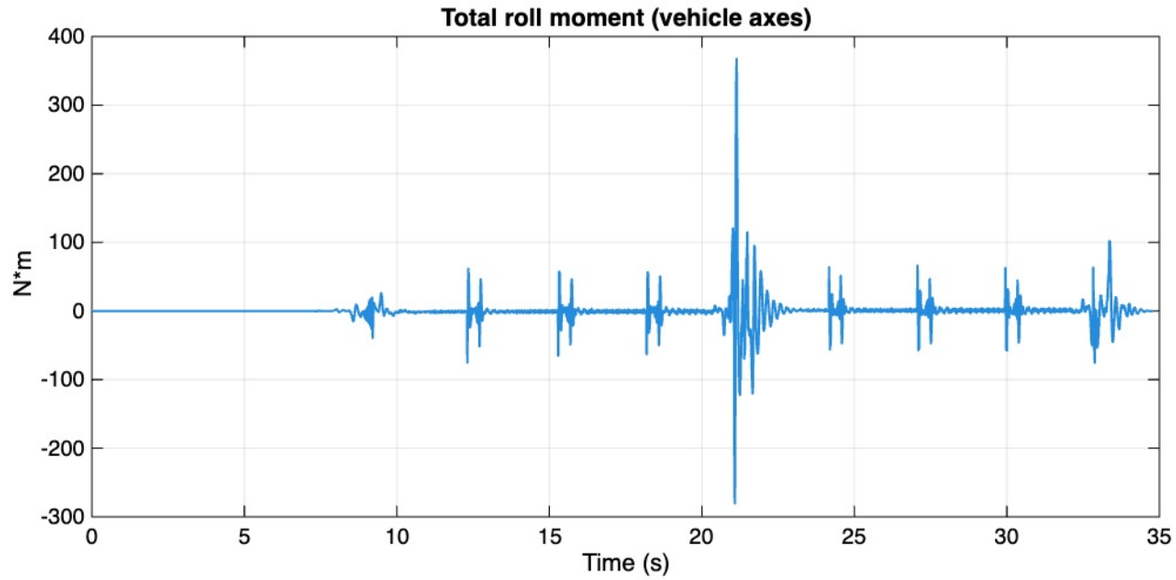

Aerodynamic lateral force (vehicle axes)
N
Time (s)

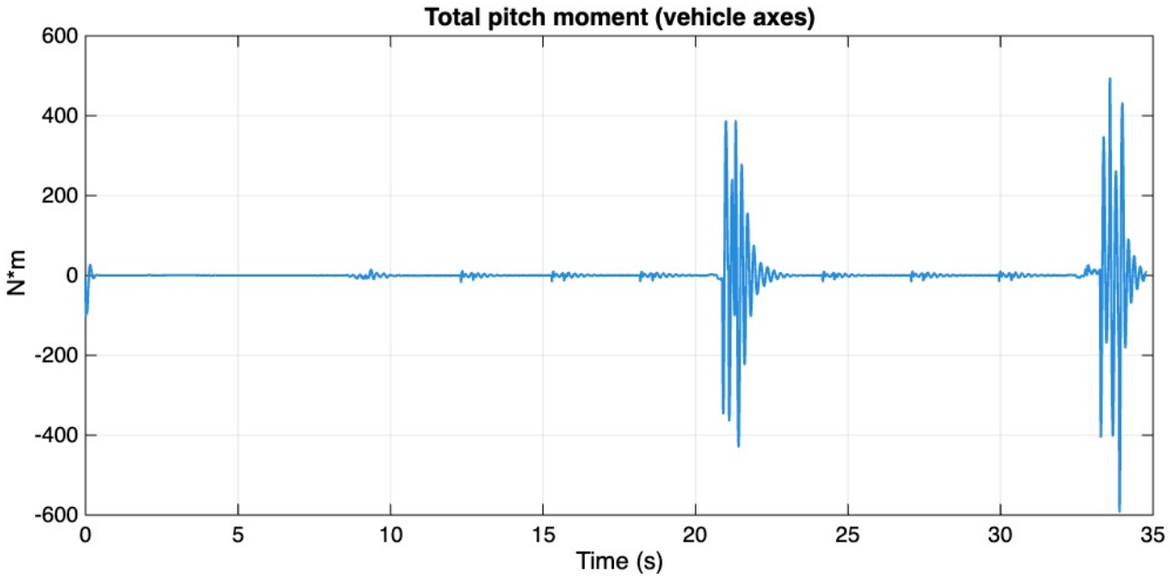

Aerodynamic vertical force (vehicle axes)
N
Time (s)


Gravity longitudinal force (vehicle axes)
N
Time (s)

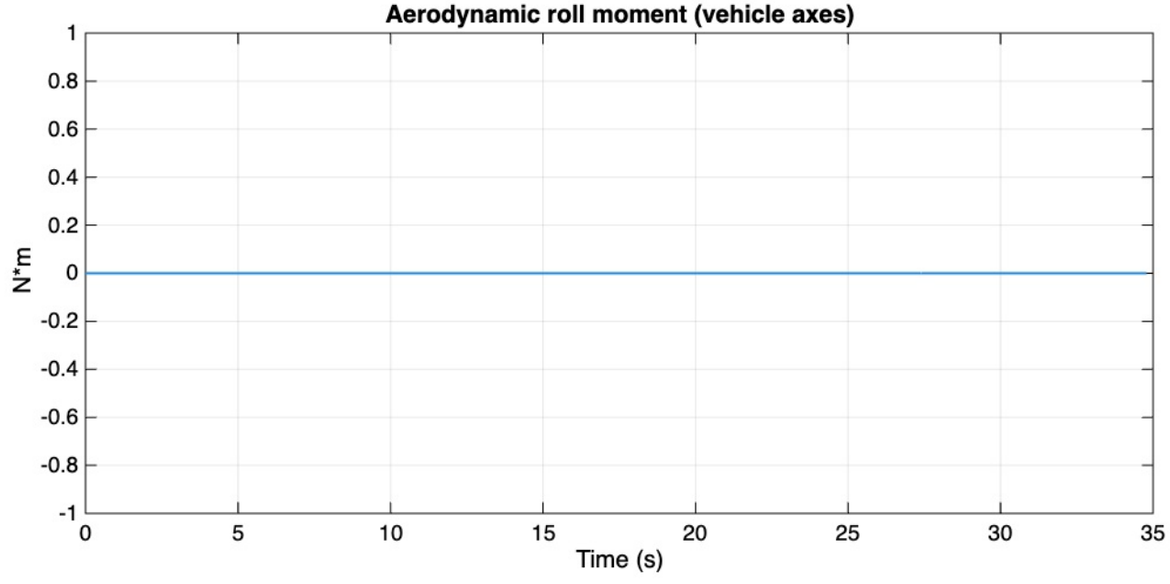

Gravity lateral force (vehicle axes)
N
Time (s)

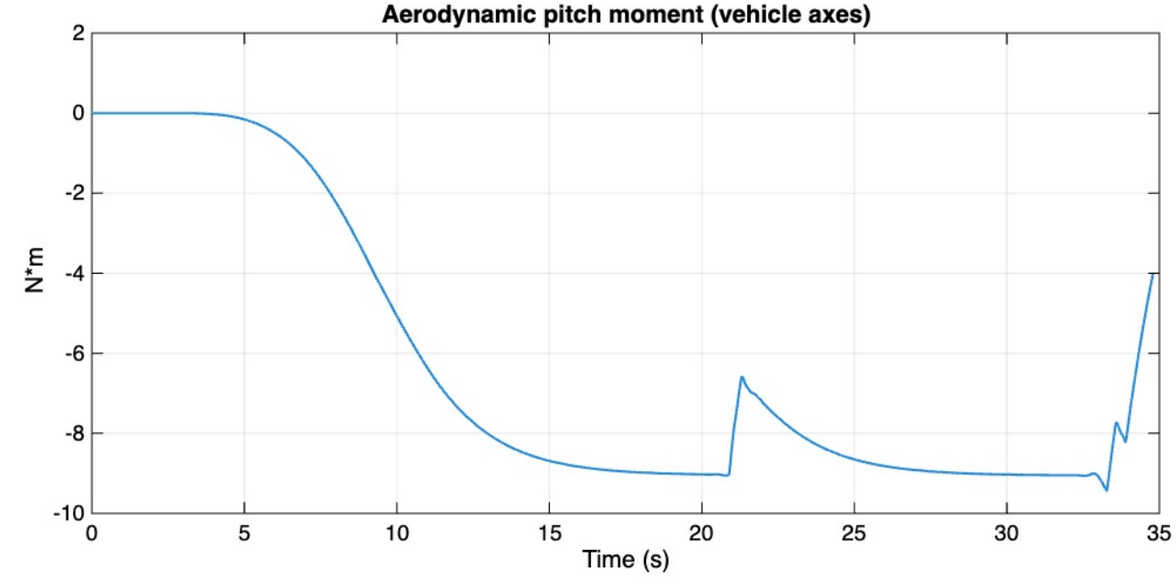

Gravity vertical force (vehicle axes)
N
Time (s)

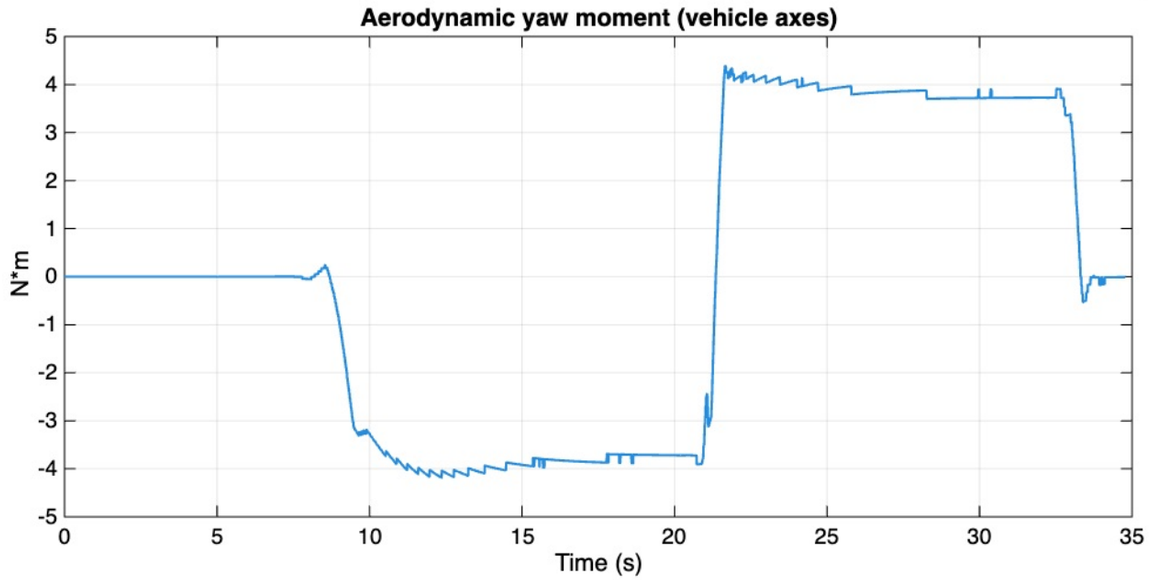

Total roll moment (vehicle axes)
N*m
Time (s)

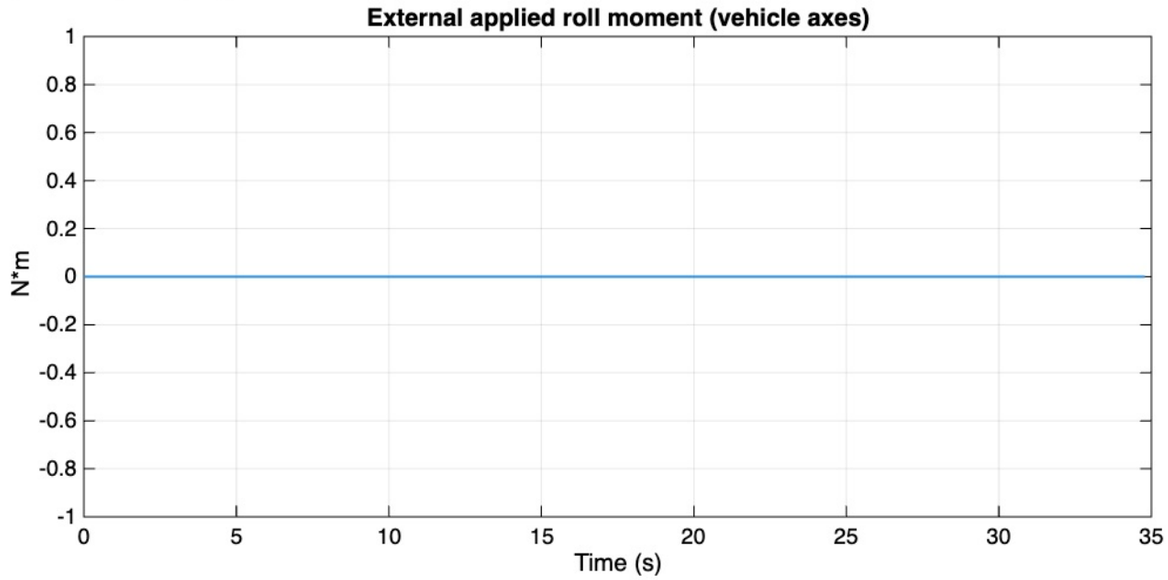

Total pitch moment (vehicle axes)
N*m
Time (s)

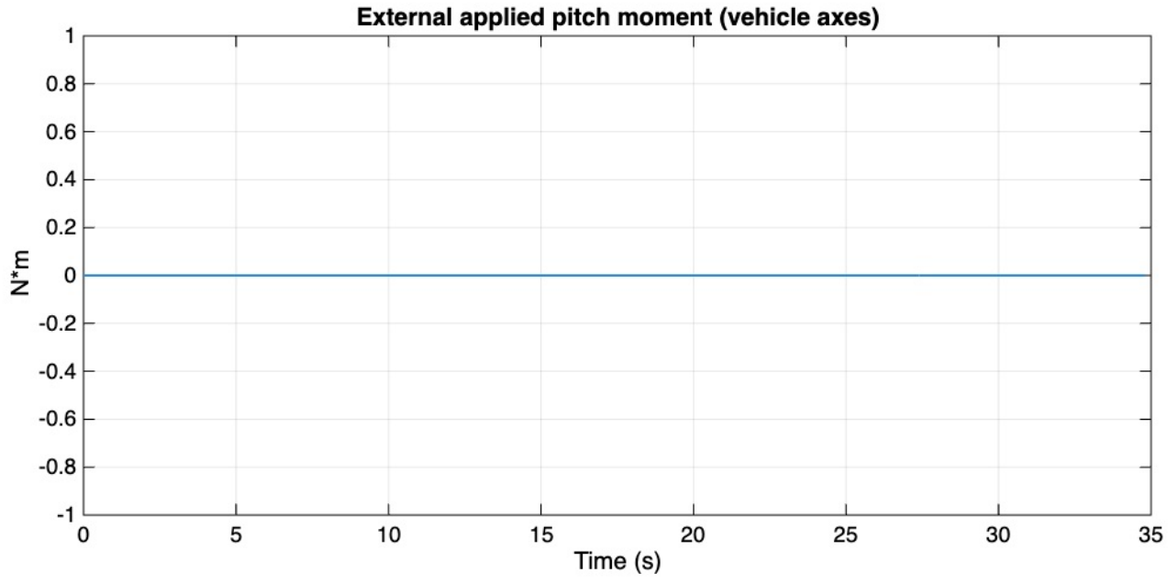

Total yaw moment (vehicle axes)
N*m
Time (s)

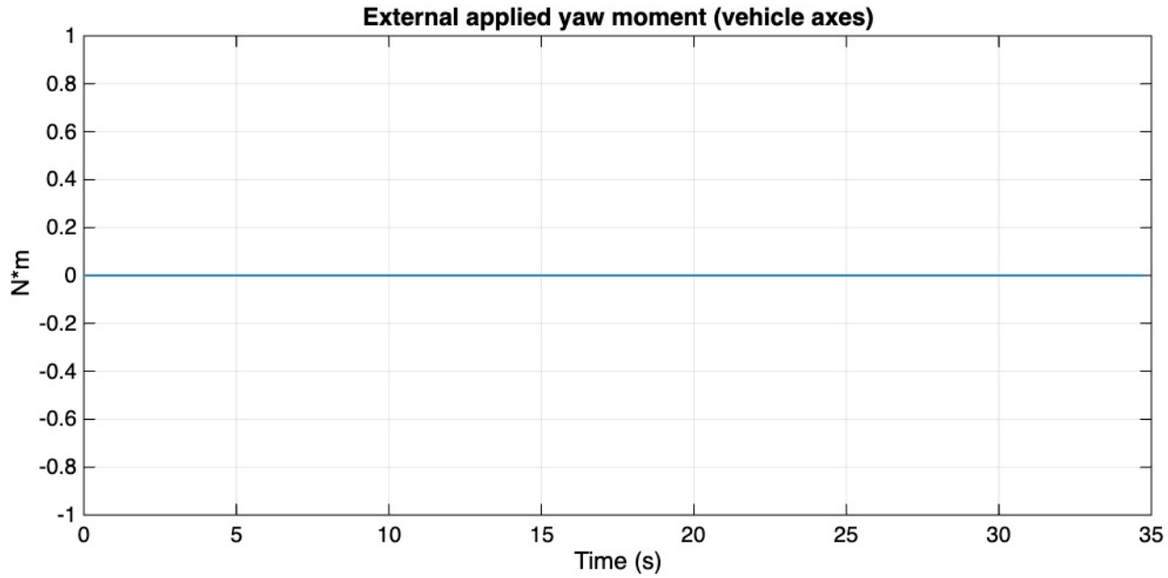

Aerodynamic roll moment (vehicle axes)
N*m
Time (s)

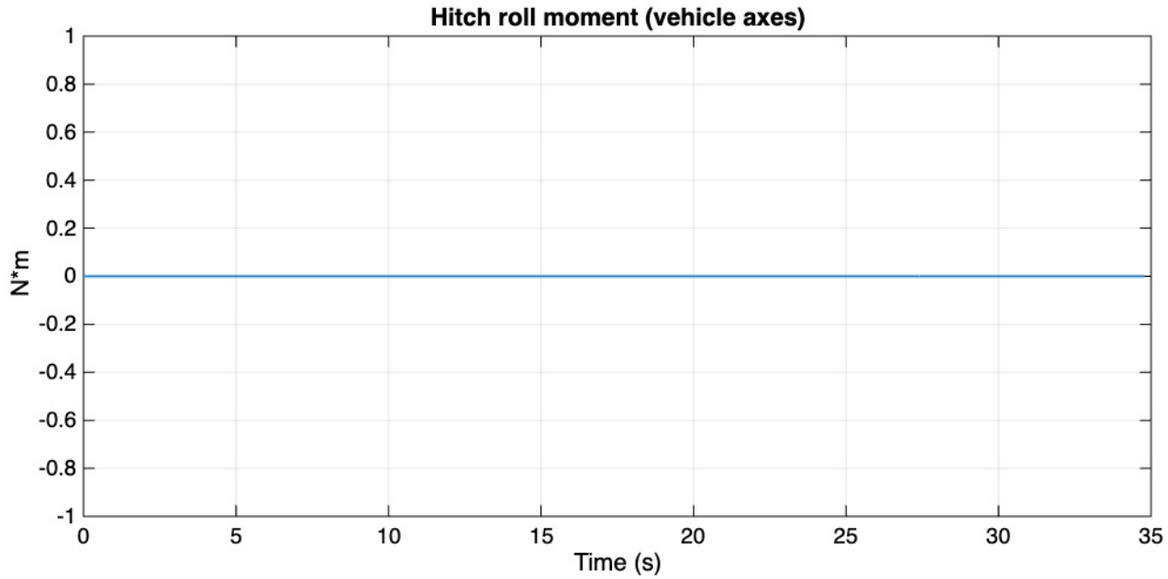

Aerodynamic pitch moment (vehicle axes)
N*m
Time (s)

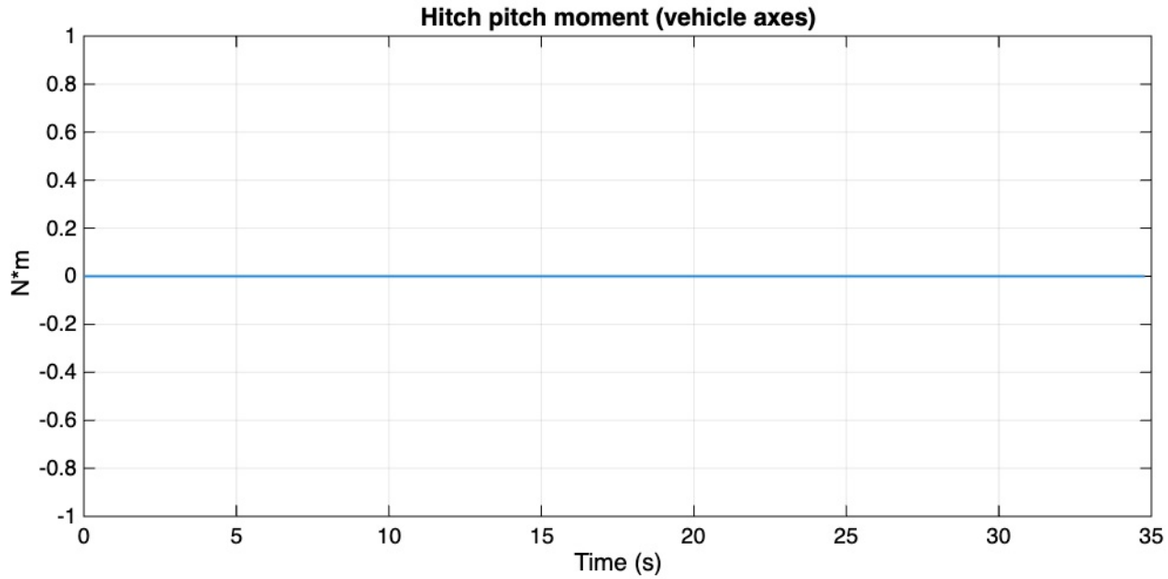

Aerodynamic yaw moment (vehicle axes)
N*m
Time (s)

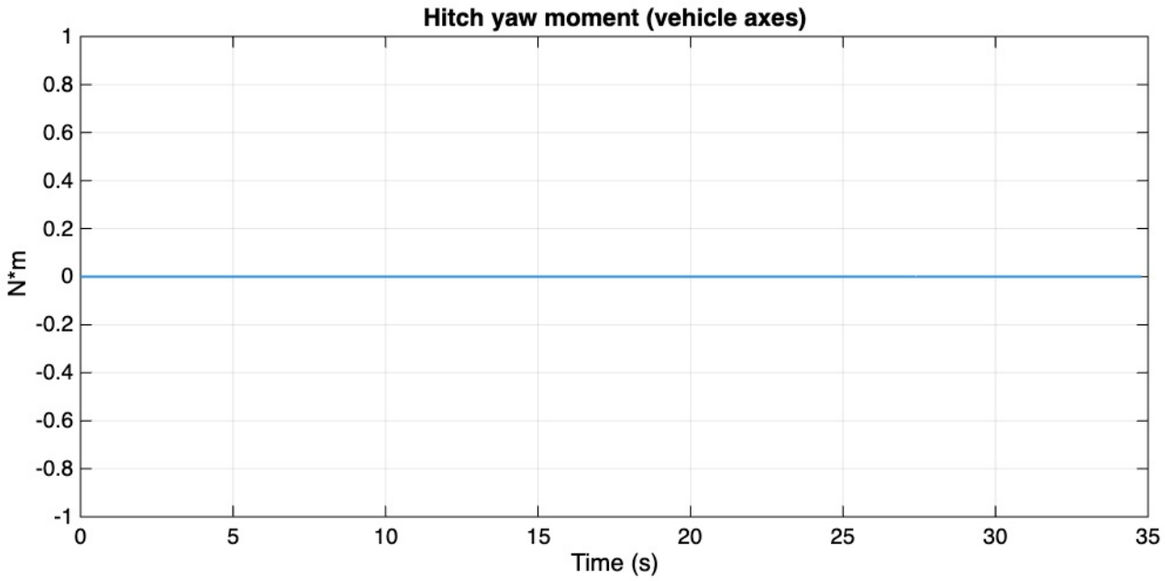

External applied roll moment (vehicle axes)
N*m
Time (s)

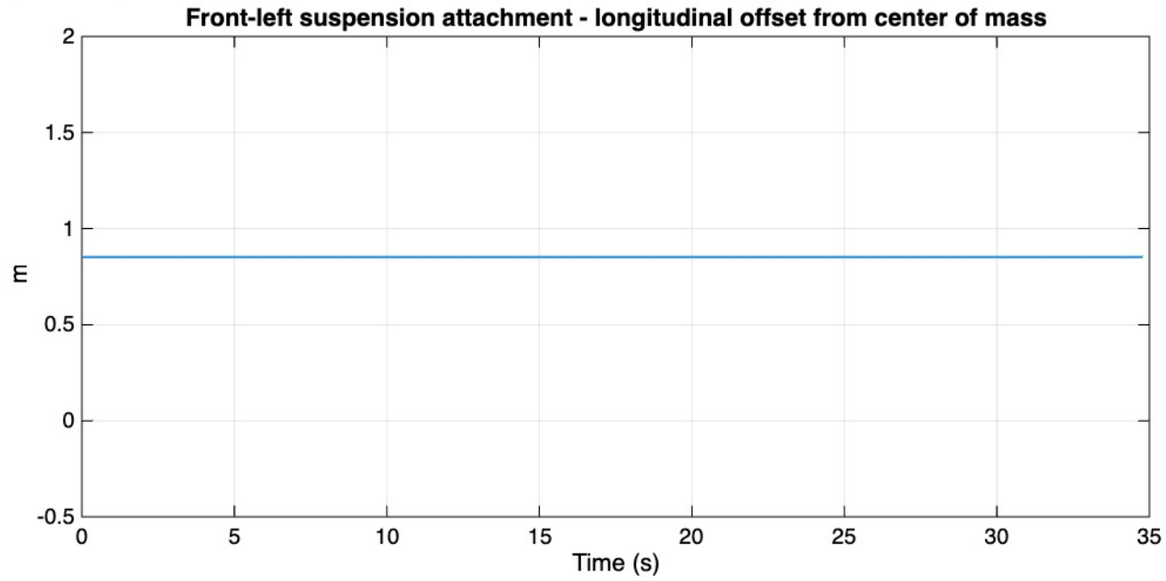

External applied pitch moment (vehicle axes)
N*m
Time (s)

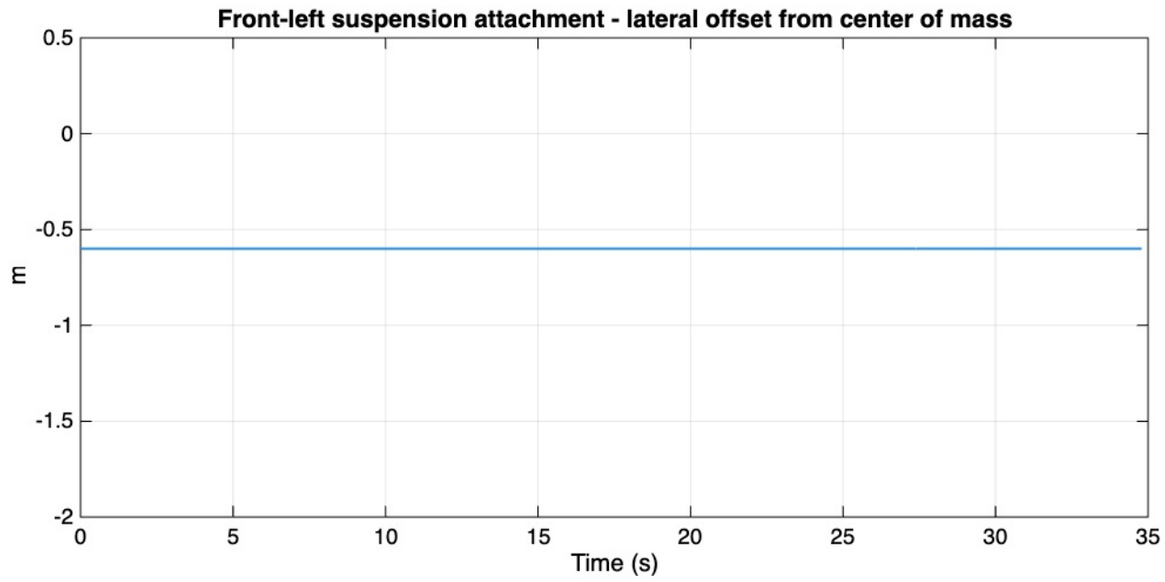

External applied yaw moment (vehicle axes)
N*m
Time (s)

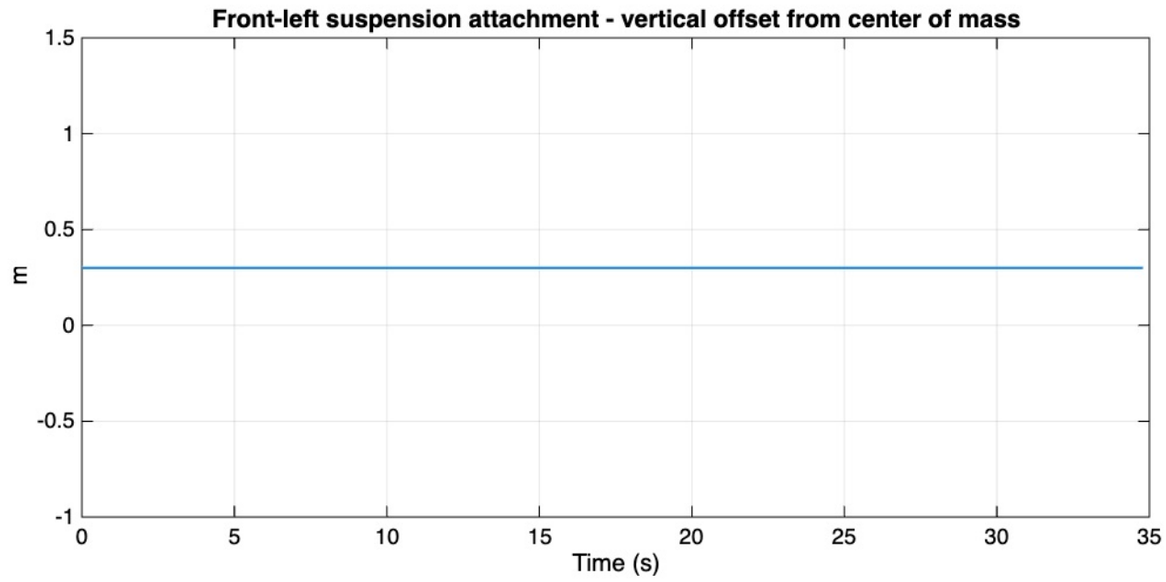

Hitch roll moment (vehicle axes)
N*m
Time (s)

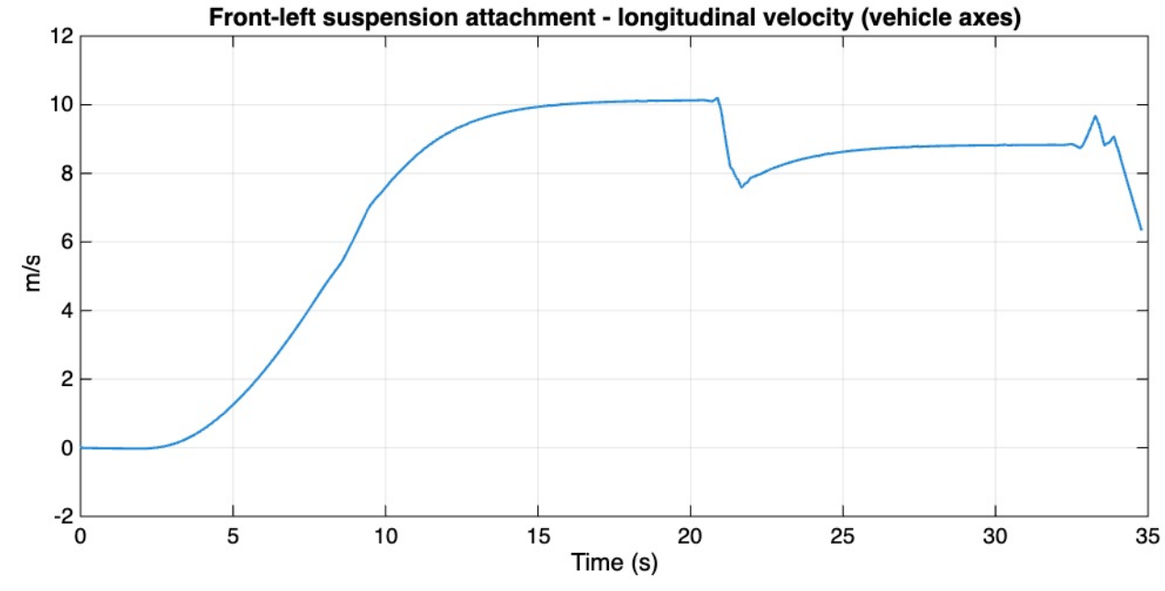

Hitch pitch moment (vehicle axes)
N*m
Time (s)

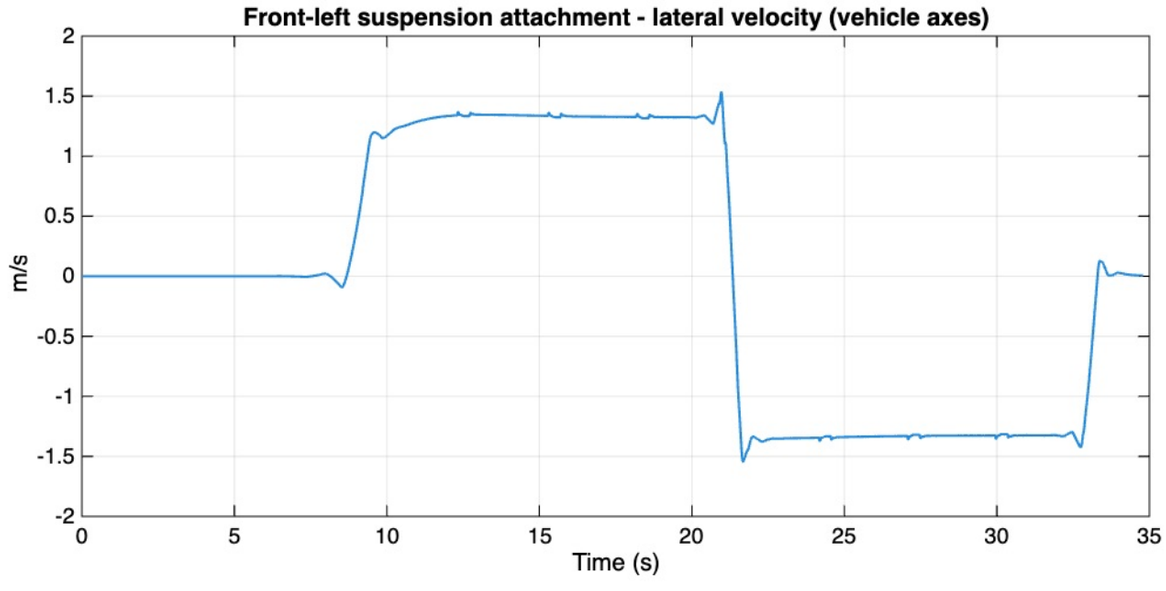

Hitch yaw moment (vehicle axes)
N*m
Time (s)

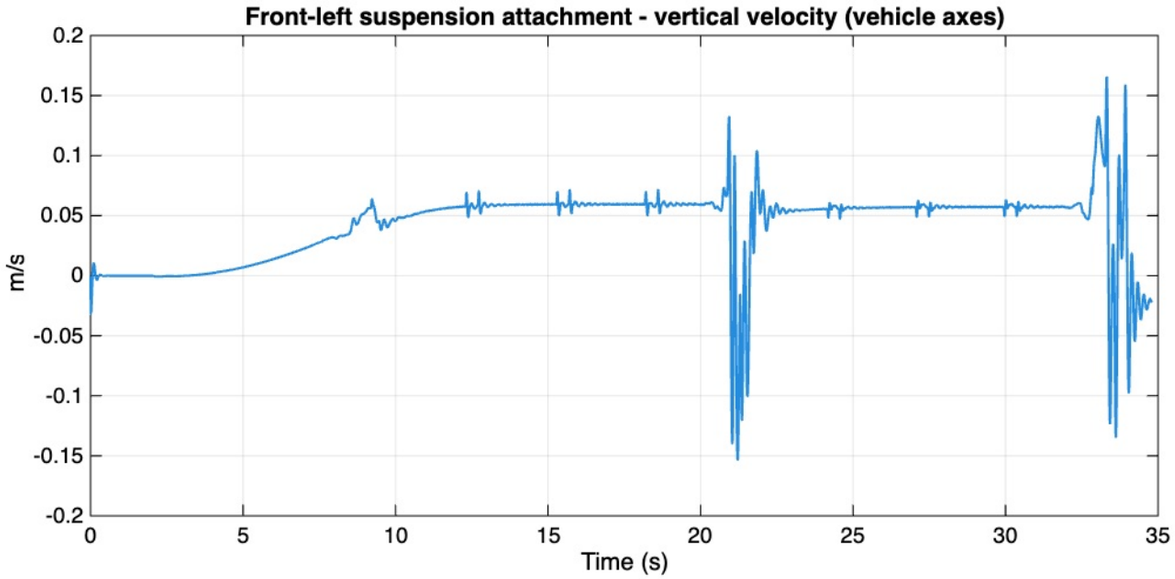

Front-left suspension attachment - longitudinal offset from center of mass
m
Time (s)

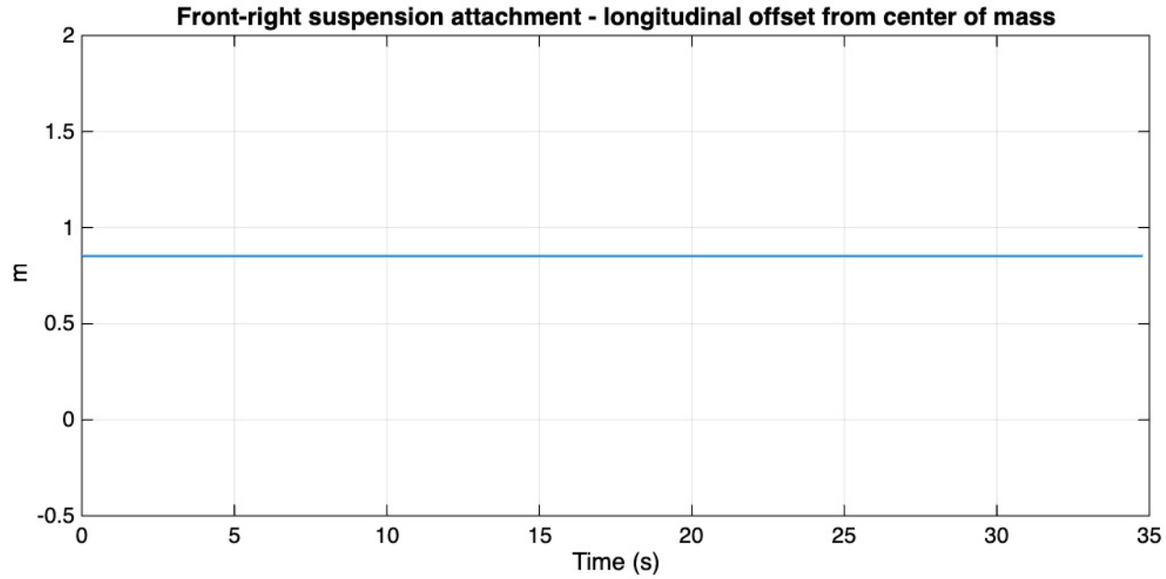

Front-left suspension attachment - lateral offset from center of mass
m
Time (s)

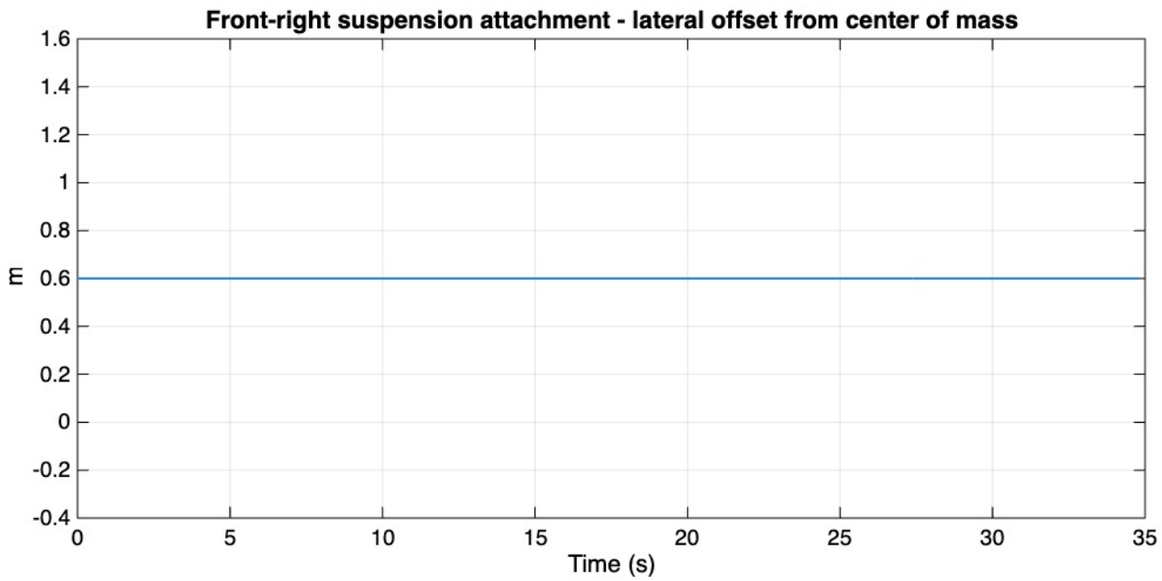

Front-left suspension attachment - vertical offset from center of mass
m
Time (s)

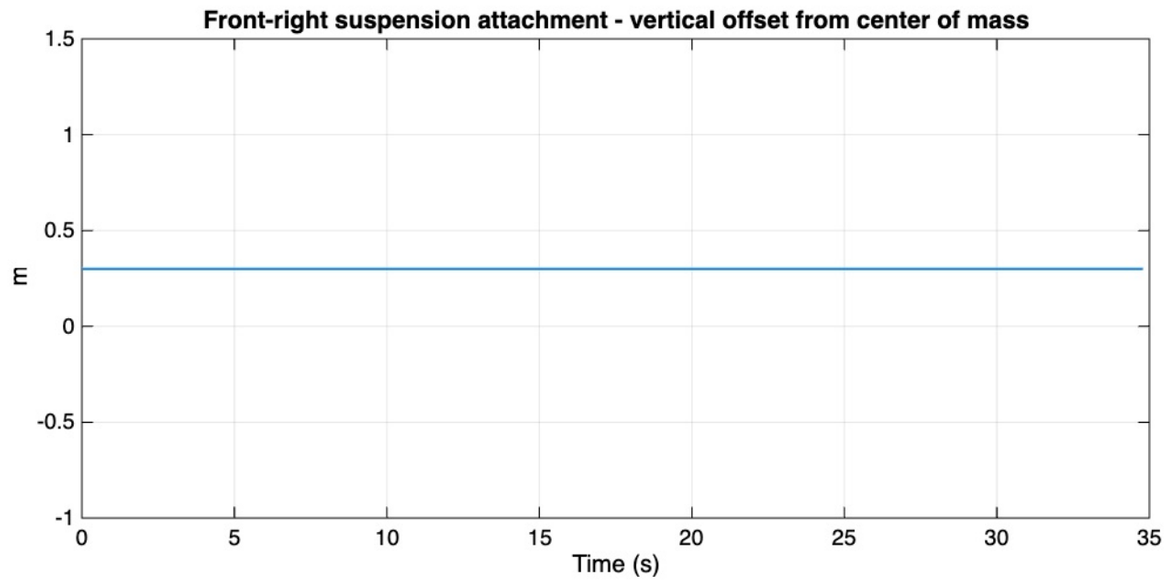

Front-left suspension attachment - longitudinal velocity (vehicle axes)
m/s
Time (s)

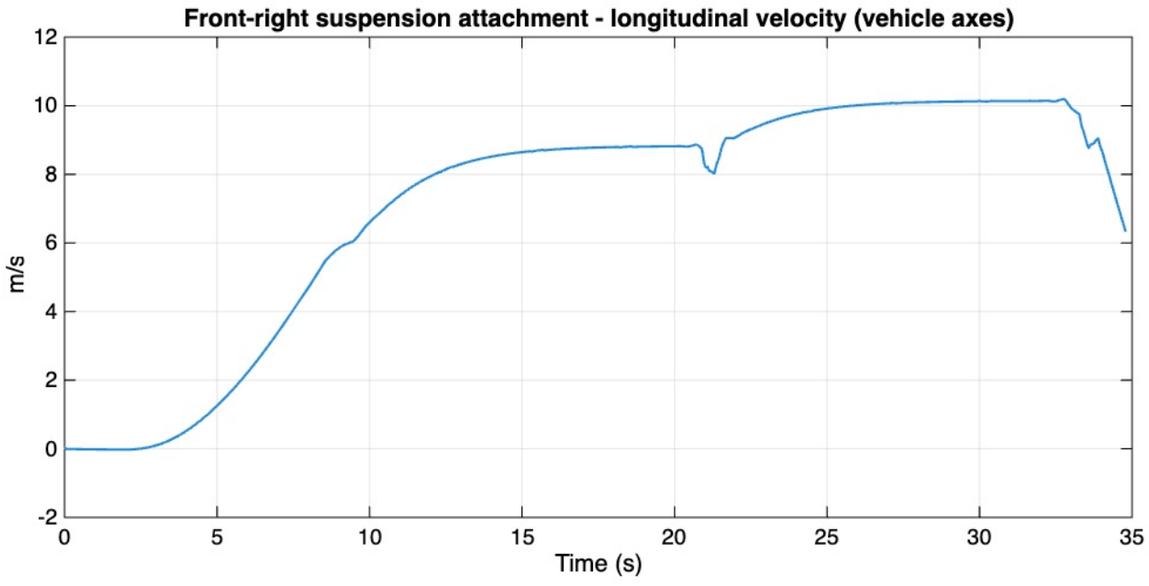

Front-left suspension attachment - lateral velocity (vehicle axes)
m/s
Time (s)

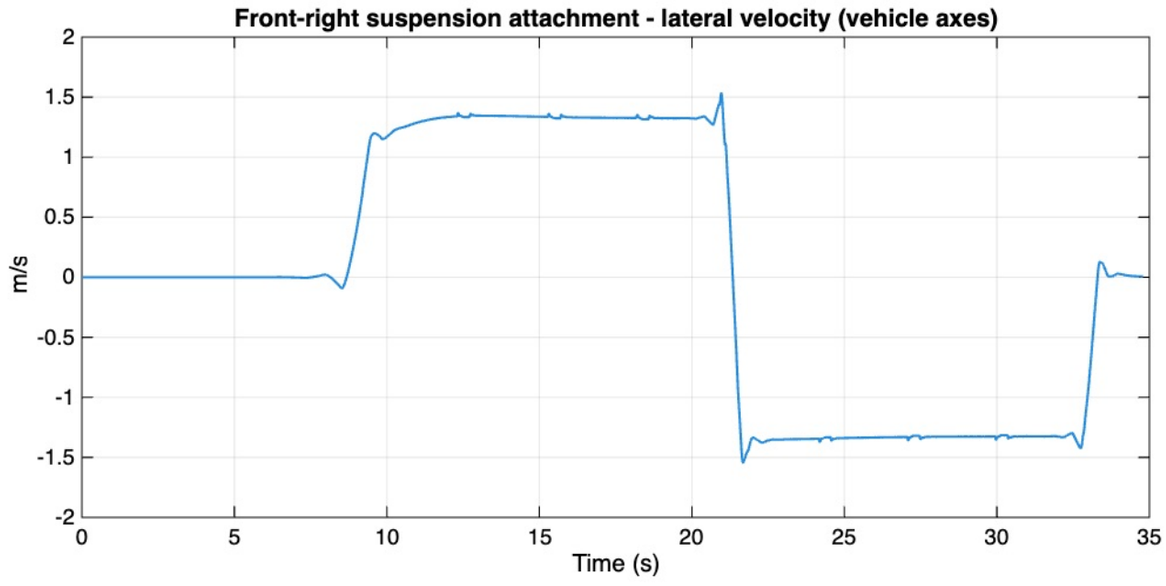

Front-left suspension attachment - vertical velocity (vehicle axes)
m/s
Time (s)


Front-right suspension attachment - longitudinal offset from center of mass
m
Time (s)

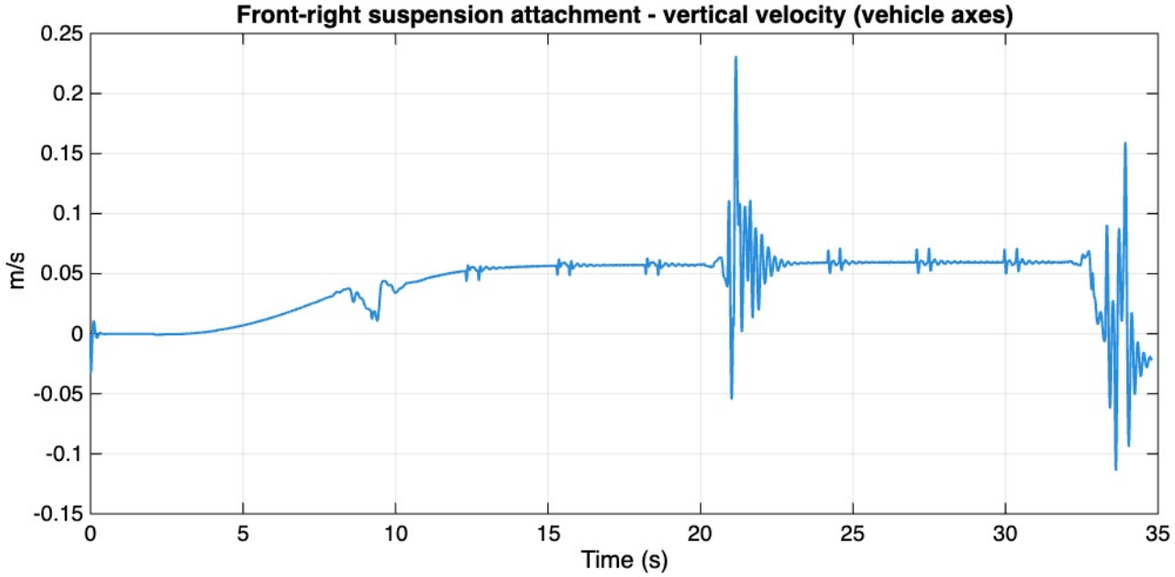

Front-right suspension attachment - lateral offset from center of mass
m
Time (s)


Front-right suspension attachment - vertical offset from center of mass
m
Time (s)

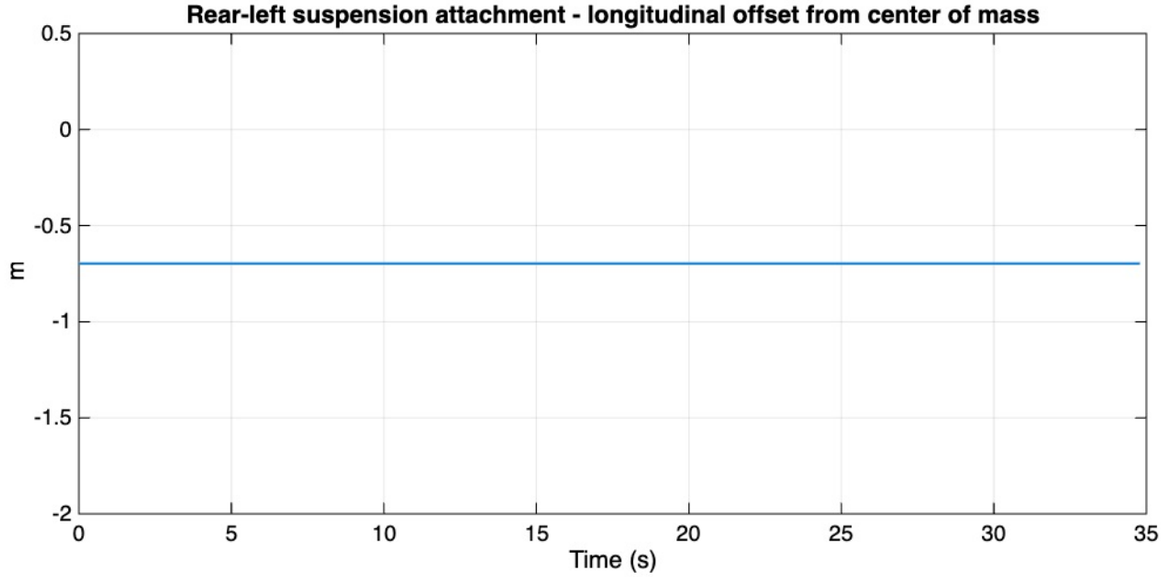

Front-right suspension attachment - longitudinal velocity (vehicle axes)
m/s
Time (s)


Front-right suspension attachment - lateral velocity (vehicle axes)
m/s
Time (s)

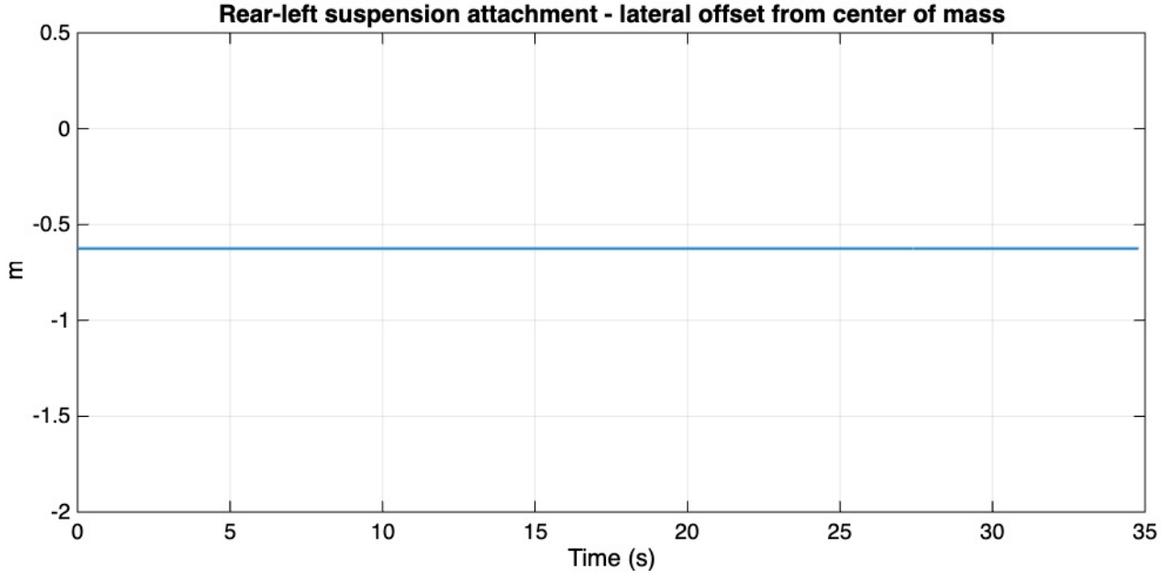
Front-right suspension attachment - vertical velocity (vehicle axes)



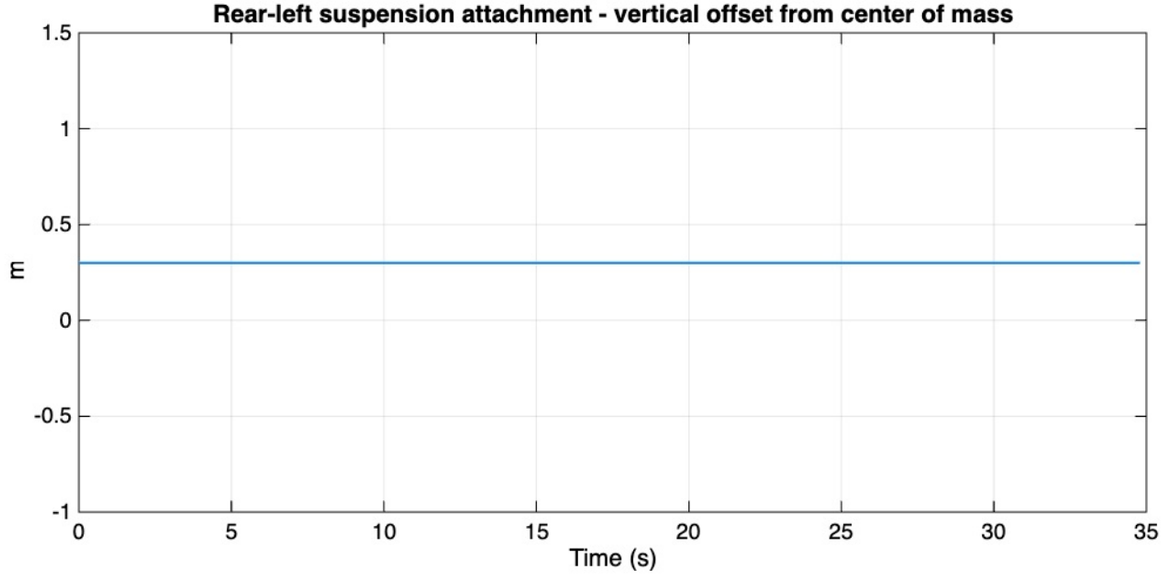
Rear-left suspension attachment - longitudinal offset from center of mass



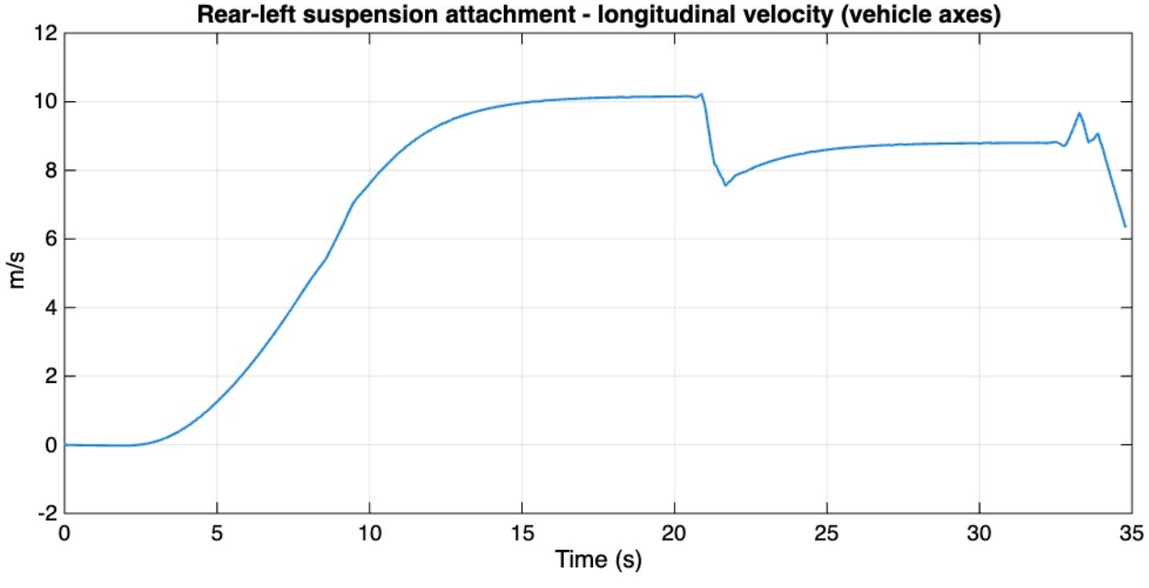
Rear-left suspension attachment - lateral offset from center of mass



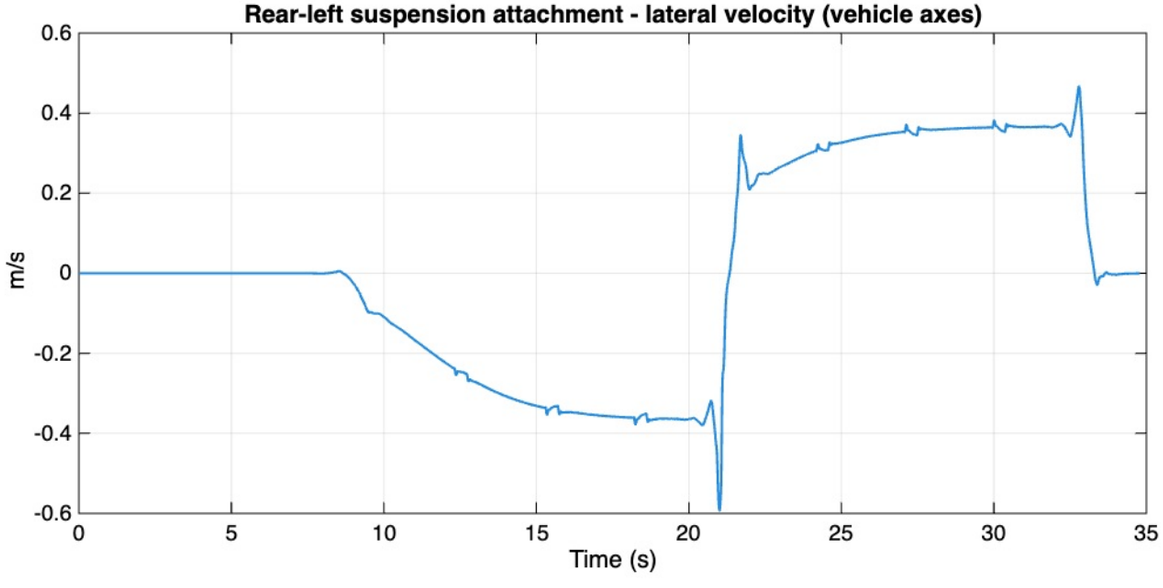
Rear-left suspension attachment - vertical offset from center of mass

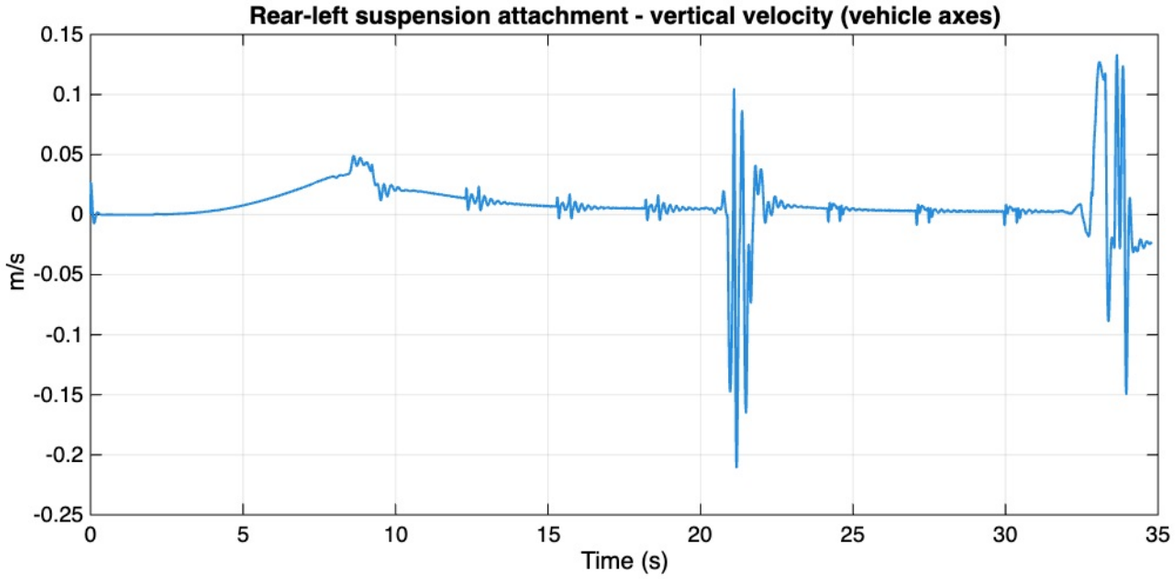


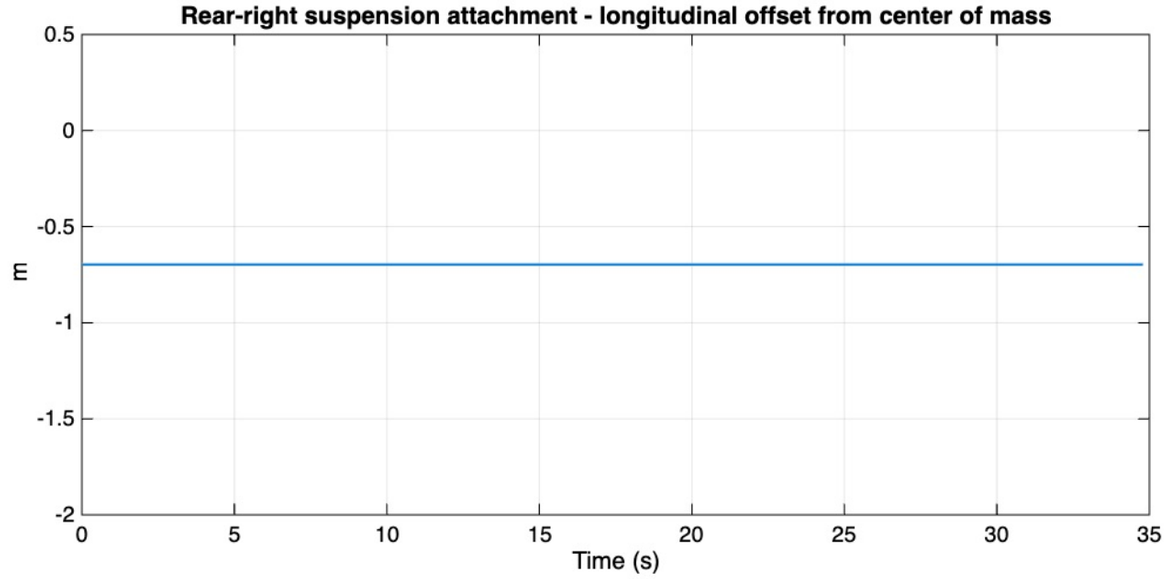
Rear-left suspension attachment - longitudinal velocity (vehicle axes)

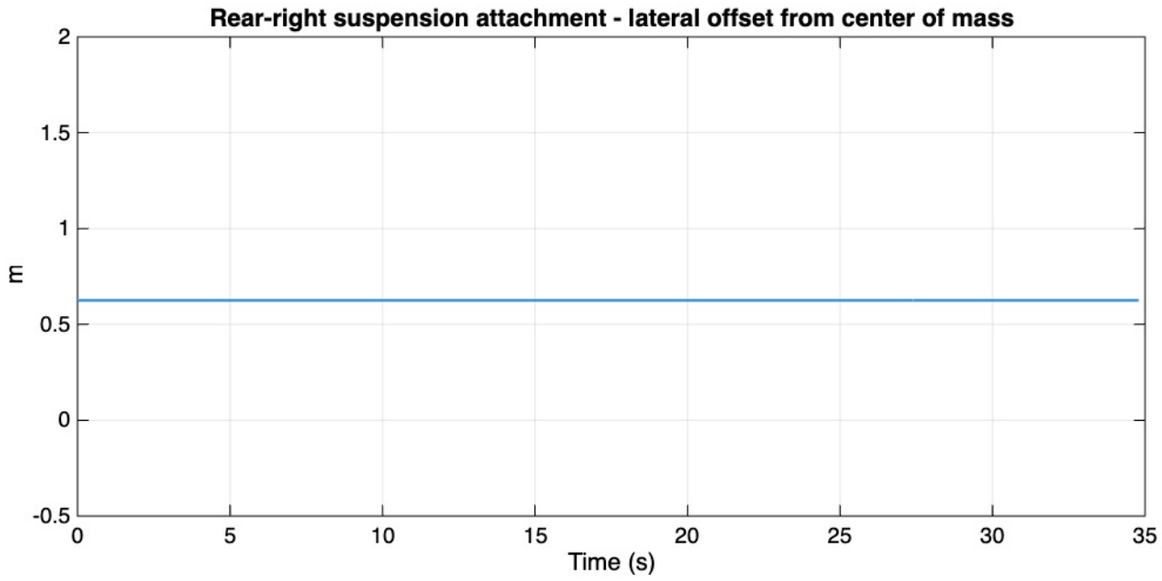


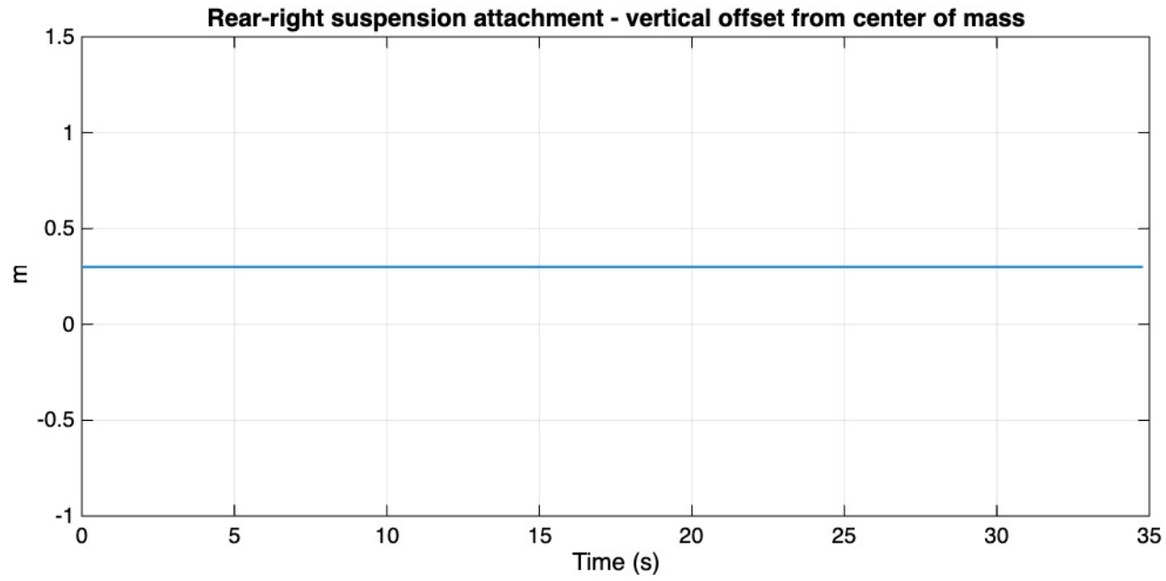
Rear-left suspension attachment - lateral velocity (vehicle axes)

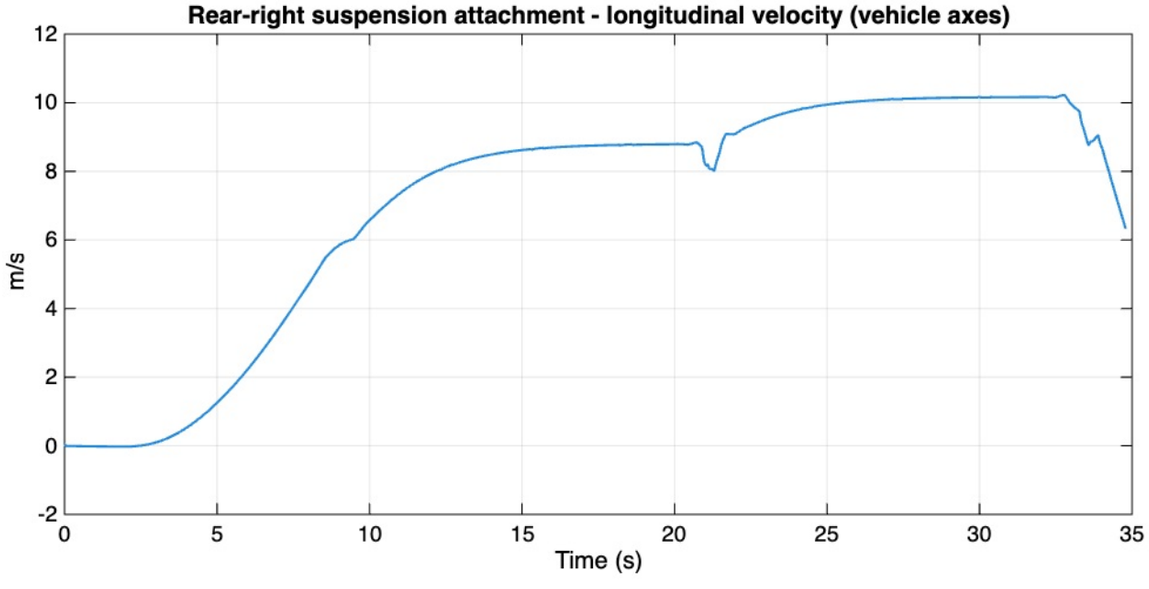


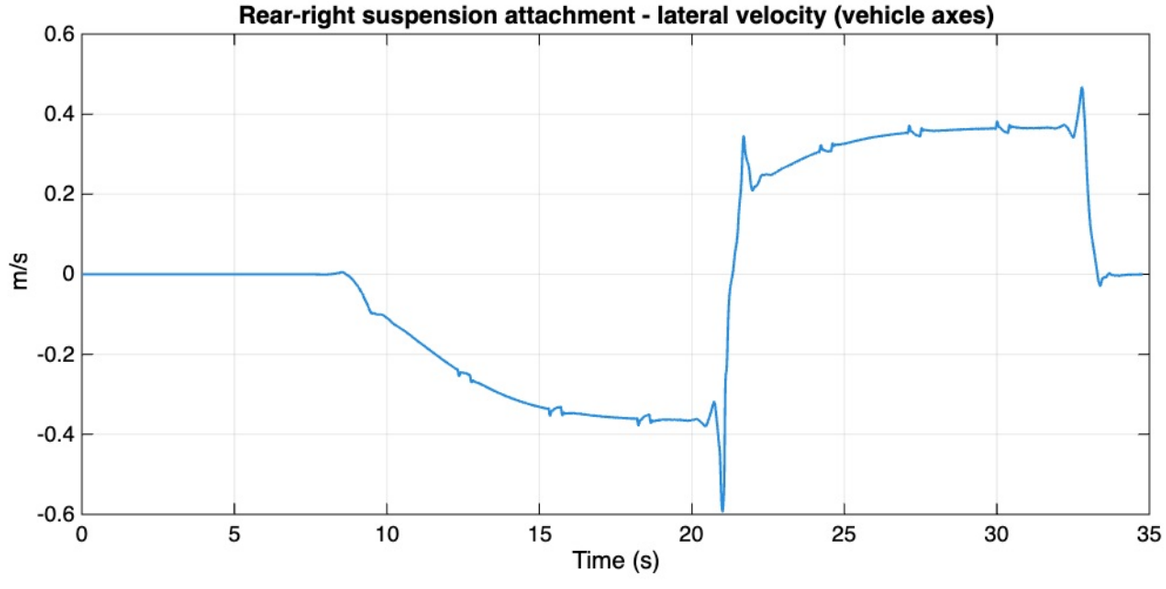

Rear-left suspension attachment - vertical velocity (vehicle axes)
m/s
Time (s)

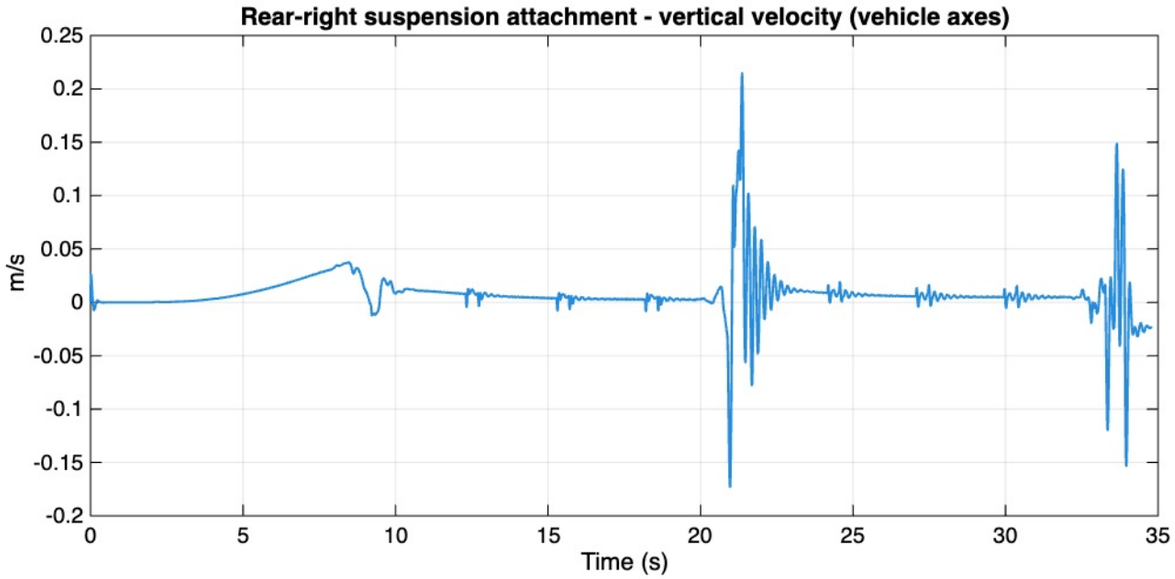

Rear-right suspension attachment - longitudinal offset from center of mass
m
Time (s)

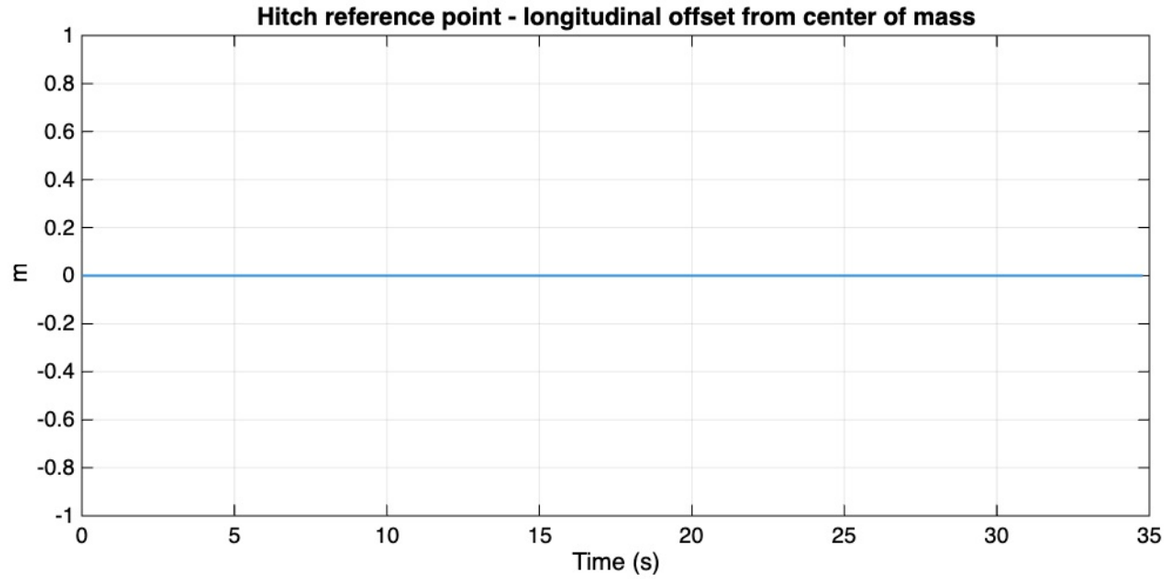

Rear-right suspension attachment - lateral offset from center of mass
m
Time (s)

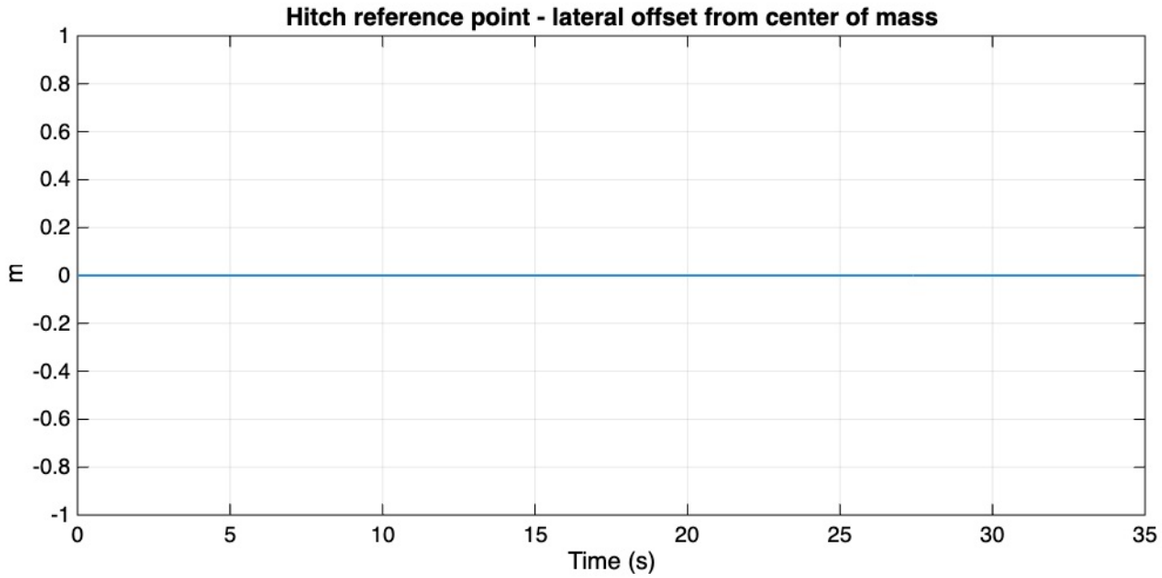

Rear-right suspension attachment - vertical offset from center of mass
m
Time (s)

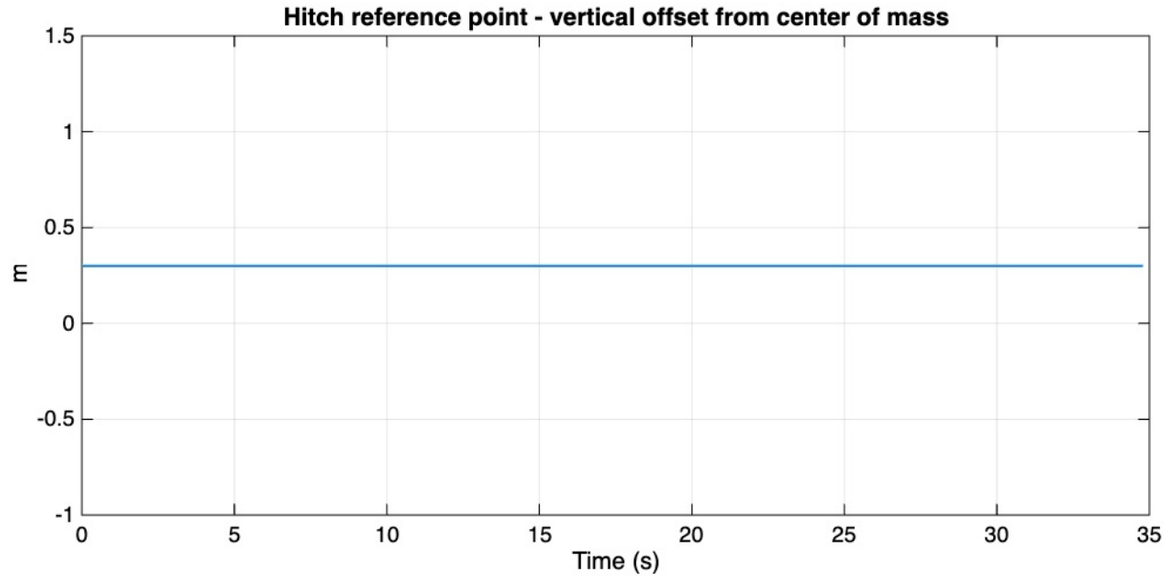

Rear-right suspension attachment - longitudinal velocity (vehicle axes)
m/s
Time (s)

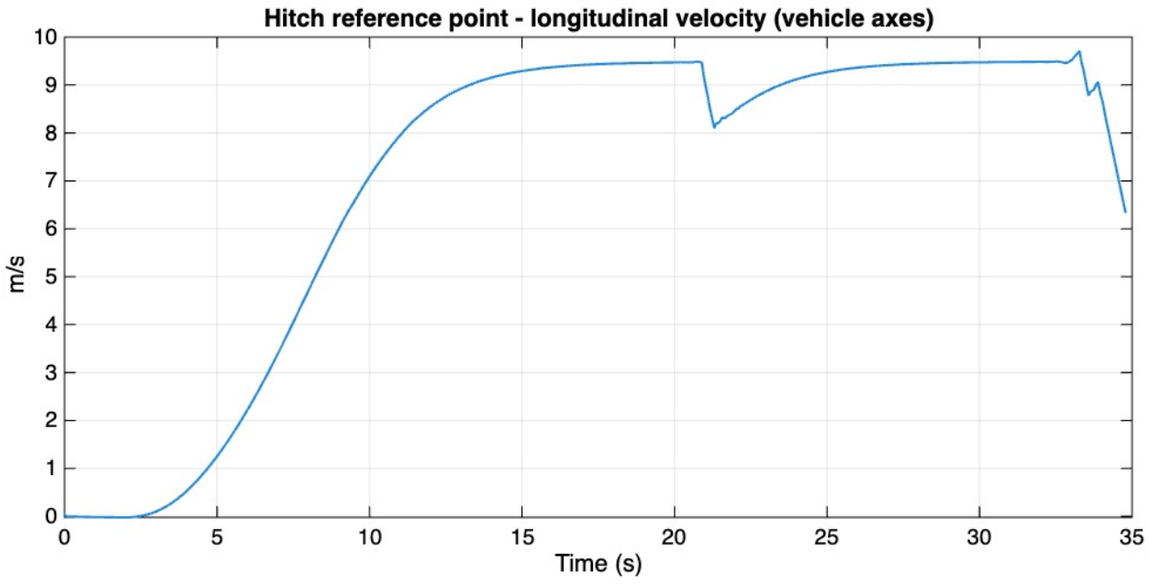

Rear-right suspension attachment - lateral velocity (vehicle axes)
m/s
Time (s)

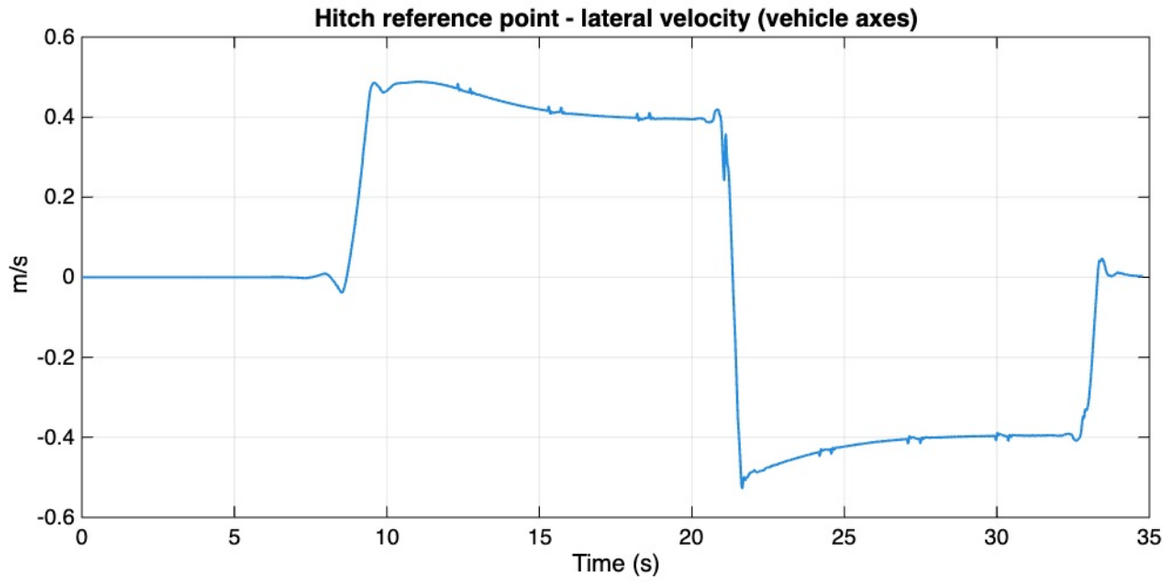

Rear-right suspension attachment - vertical velocity (vehicle axes)
m/s
Time (s)

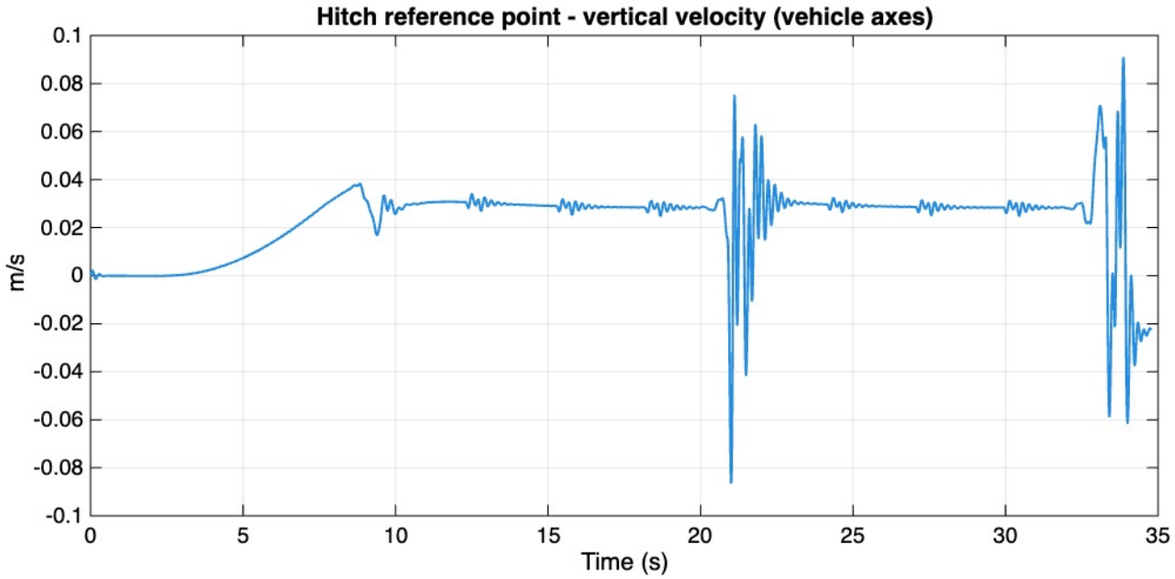

Hitch reference point - longitudinal offset from center of mass
m
Time (s)

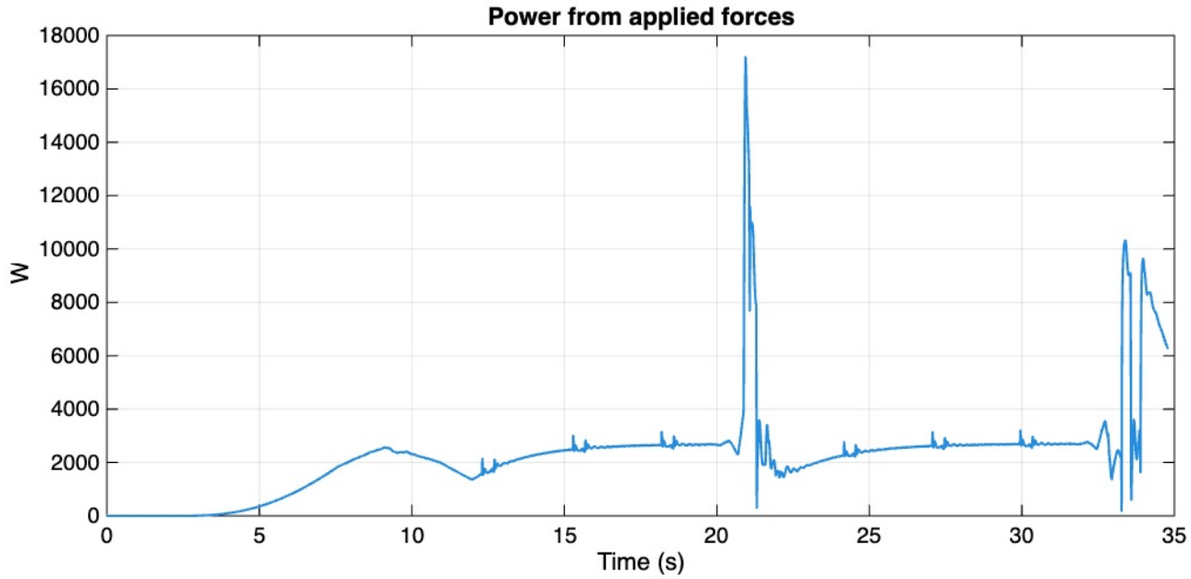

Hitch reference point - lateral offset from center of mass
m
Time (s)

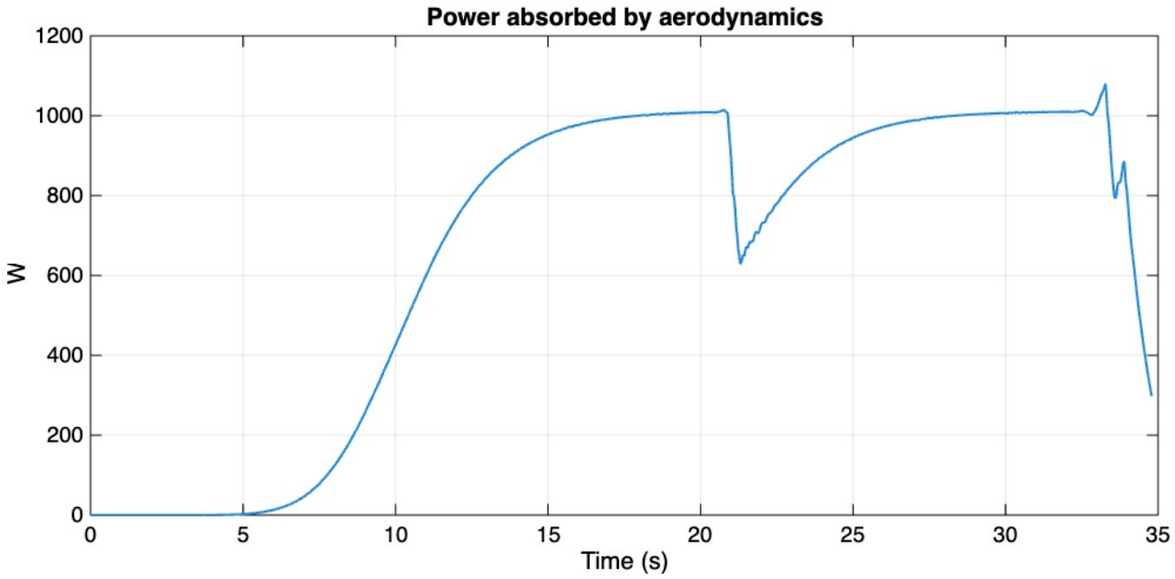

Hitch reference point - vertical offset from center of mass
m
Time (s)

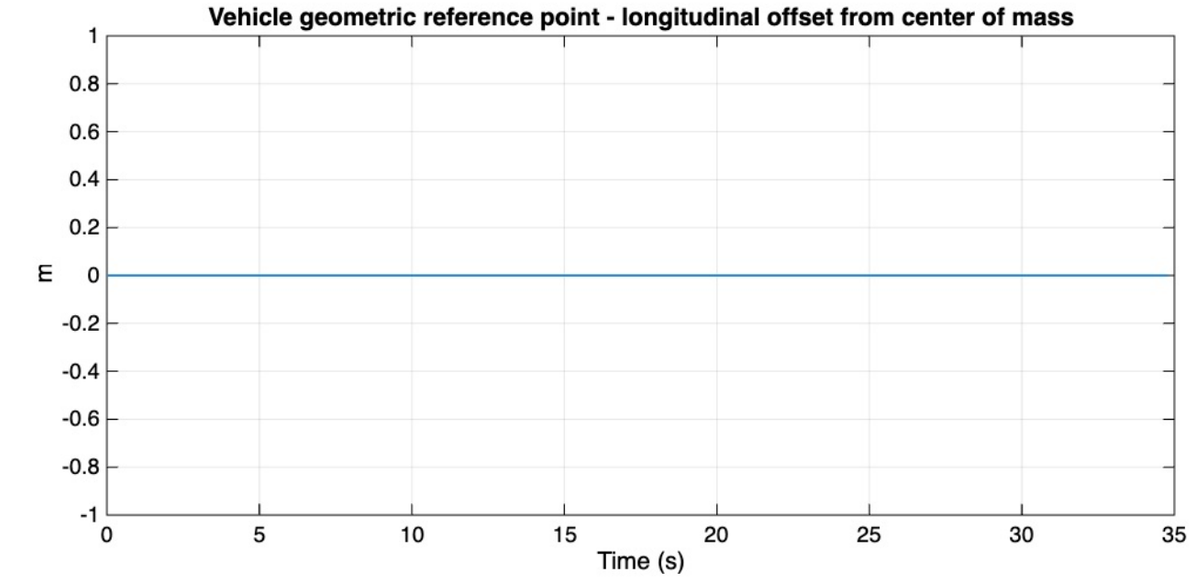

Hitch reference point - longitudinal velocity (vehicle axes)
m/s
Time (s)

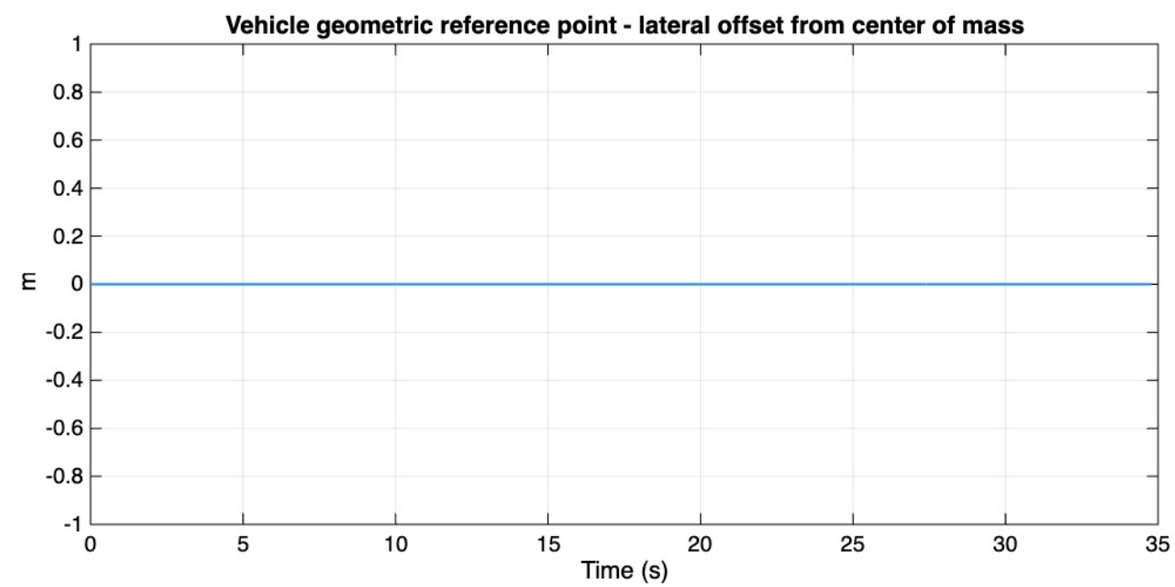

Hitch reference point - lateral velocity (vehicle axes)
m/s
Time (s)

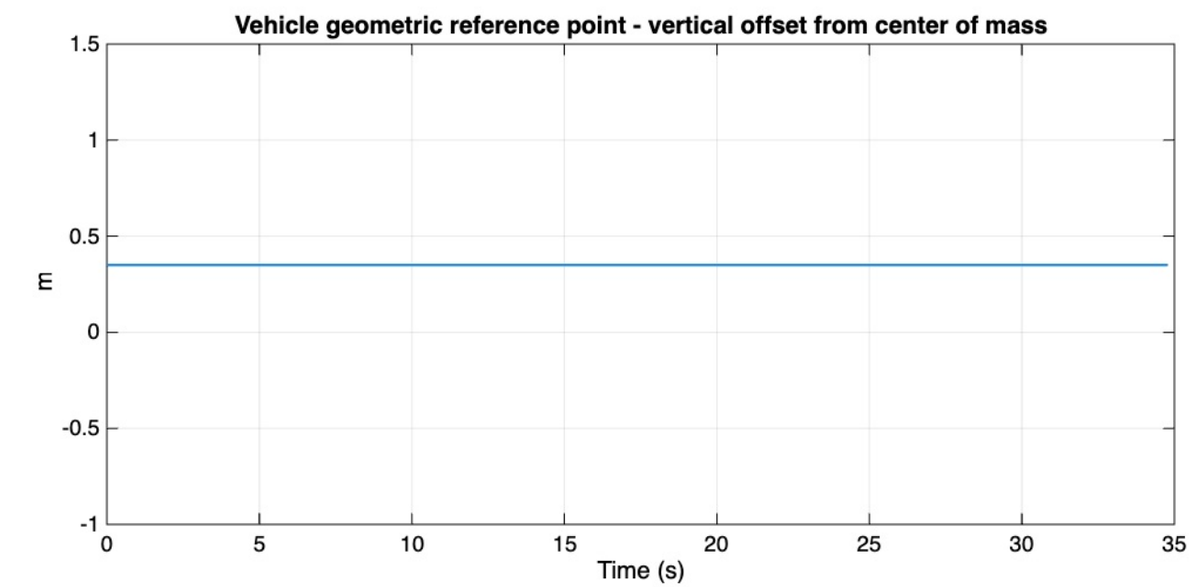

Hitch reference point - vertical velocity (vehicle axes)
m/s
Time (s)

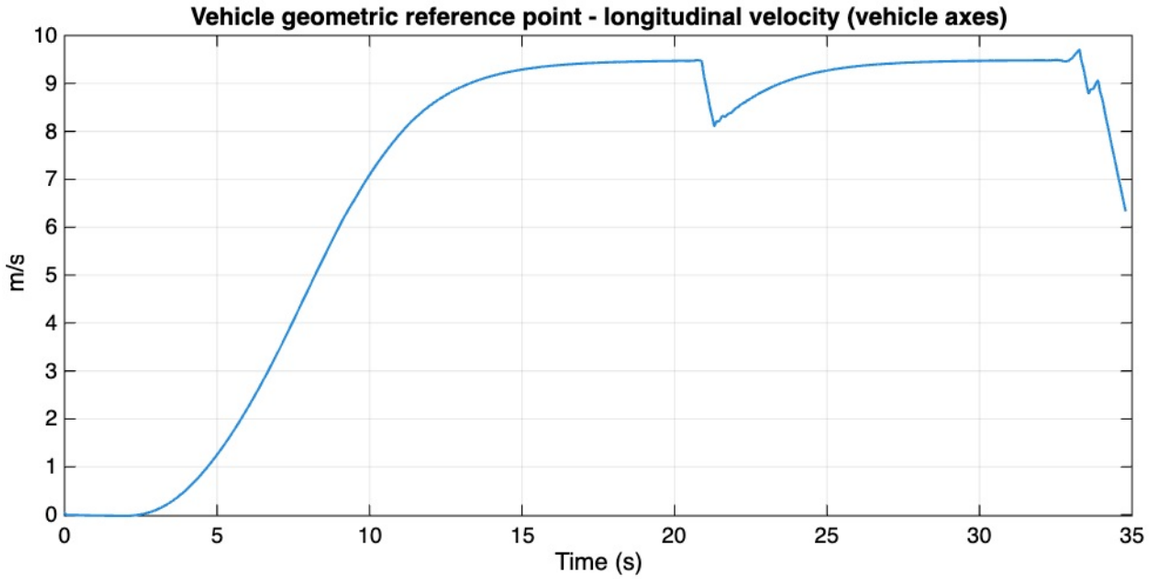

Power from applied forces
W
Time (s)

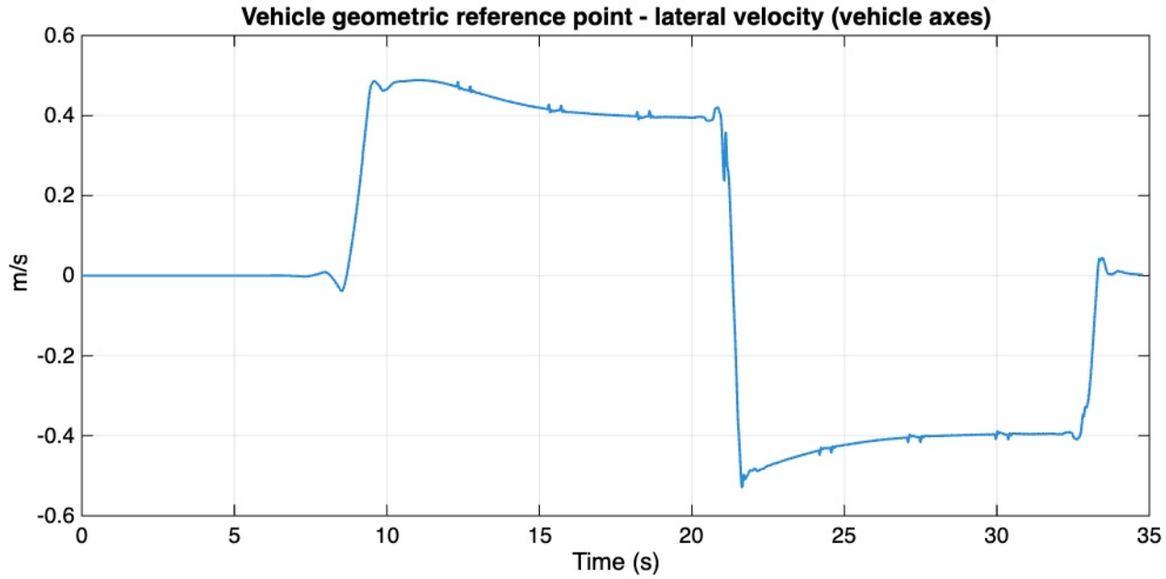

Power absorbed by aerodynamics
W
Time (s)

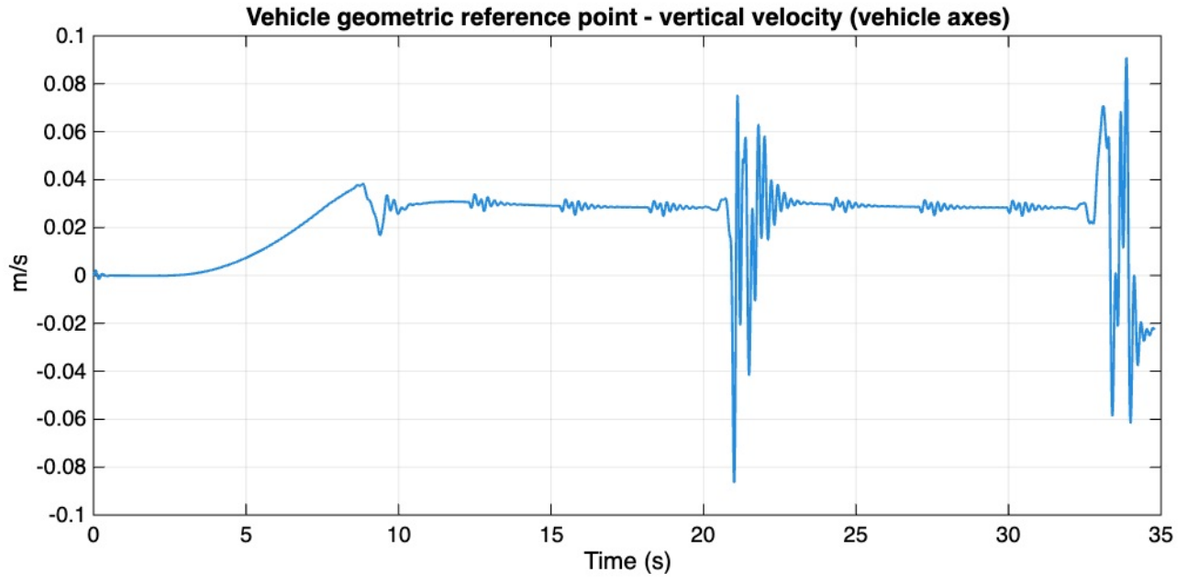

Vehicle geometric reference point - longitudinal offset from center of mass
m
Time (s)

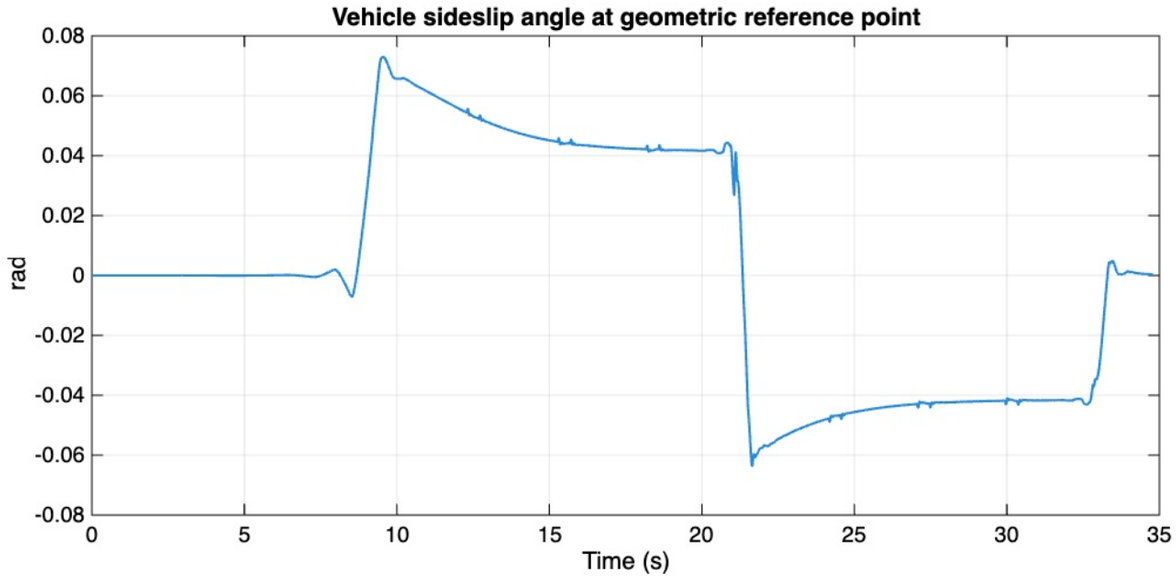

Vehicle geometric reference point - lateral offset from center of mass
m
Time (s)

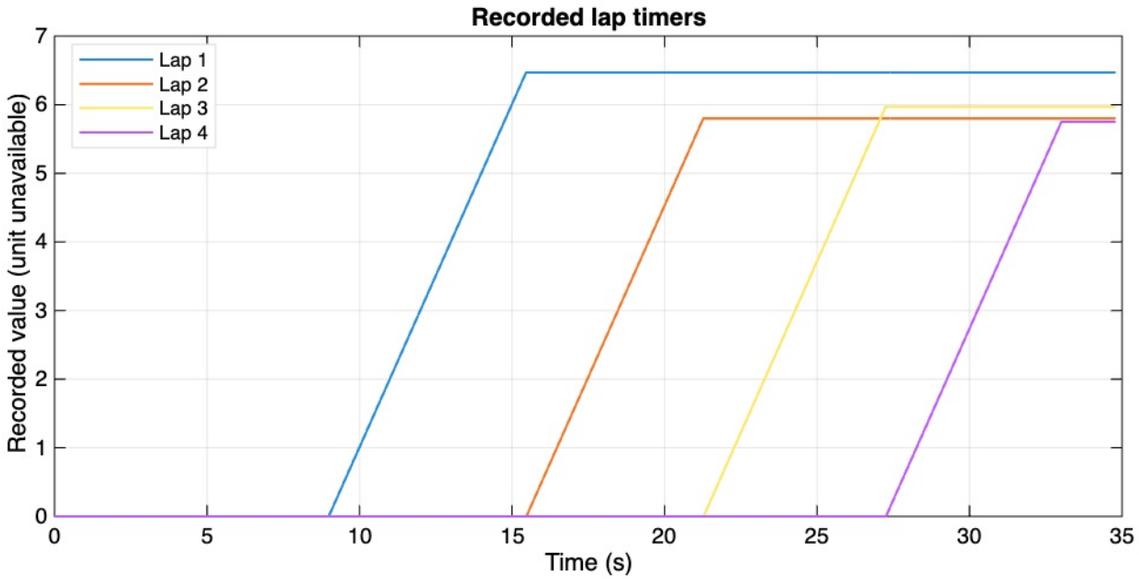

Vehicle geometric reference point - vertical offset from center of mass
m
Time (s)

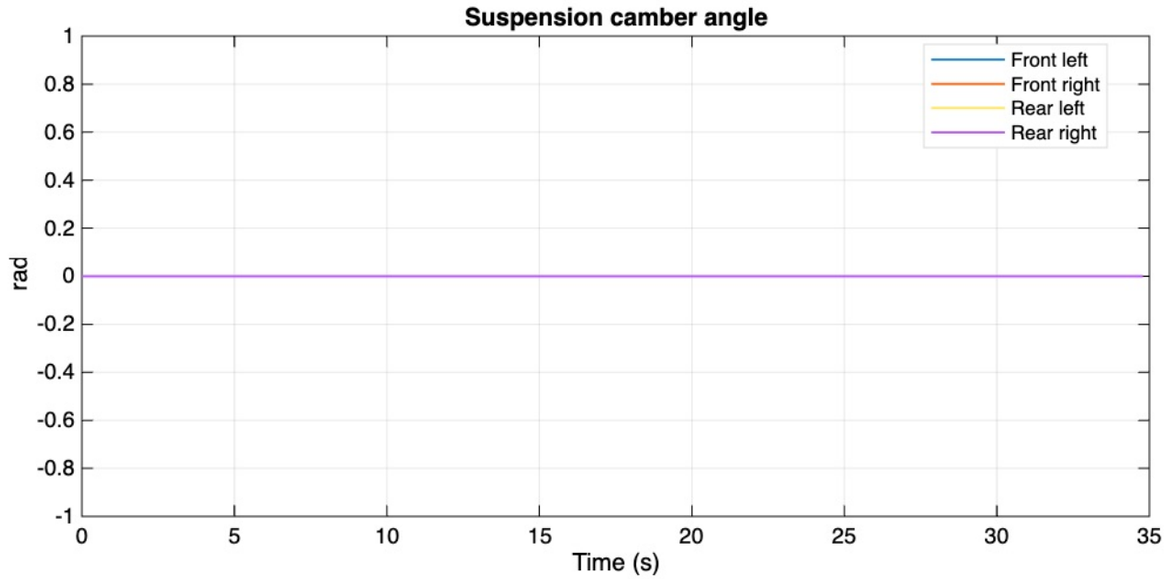

Vehicle geometric reference point - longitudinal velocity (vehicle axes)
m/s
Time (s)


Vehicle geometric reference point - lateral velocity (vehicle axes)
m/s
Time (s)

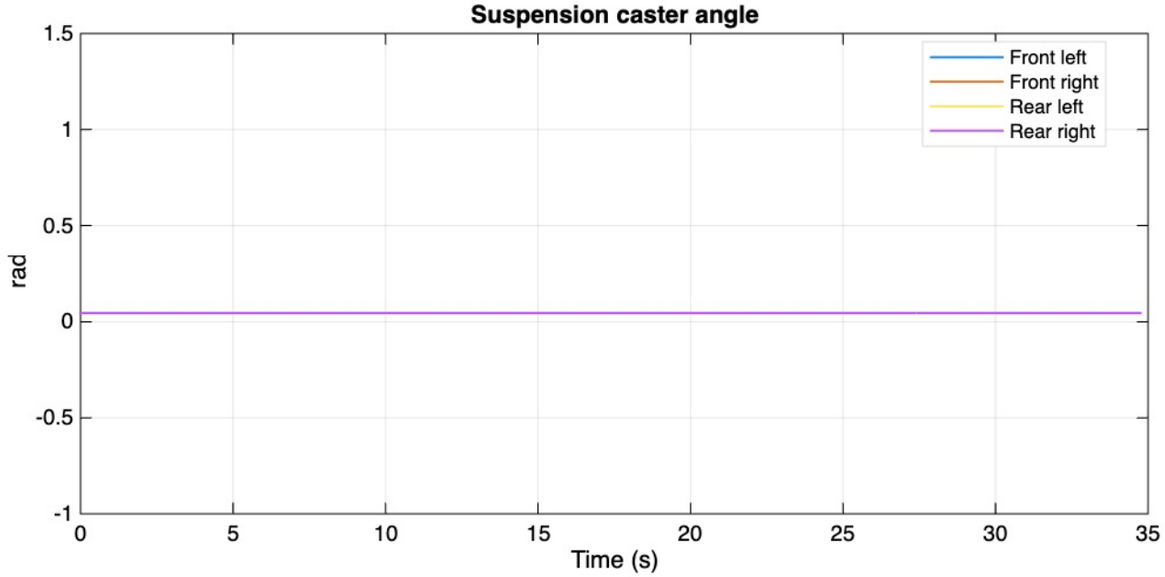

Vehicle geometric reference point - vertical velocity (vehicle axes)
m/s
Time (s)


Vehicle sideslip angle at geometric reference point
rad
Time (s)

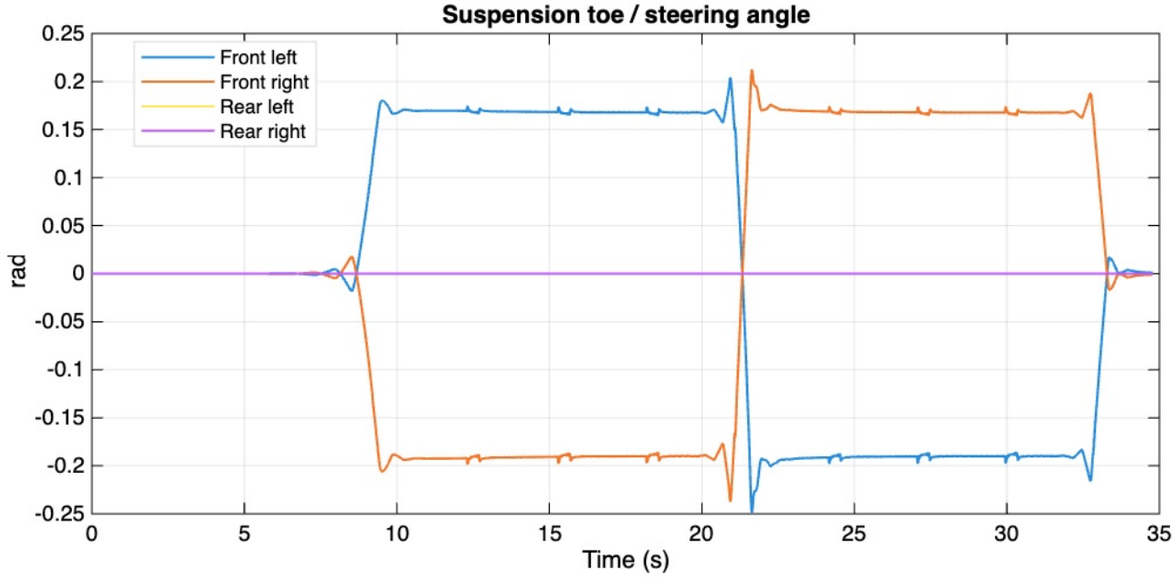

Recorded lap timers
Lap 1
Lap 2
Lap 3
Lap 4
Recorded value (unit unavailable)
Time (s)


Suspension camber angle
Front left
Front right
Rear left
Rear right
rad
Time (s)

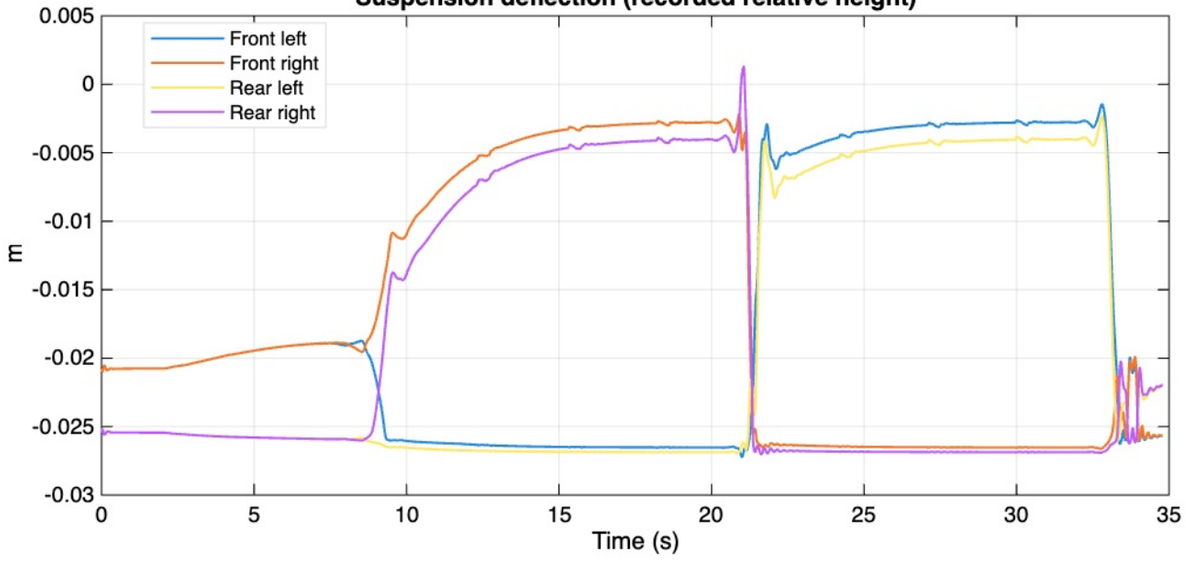
Suspension caster angle



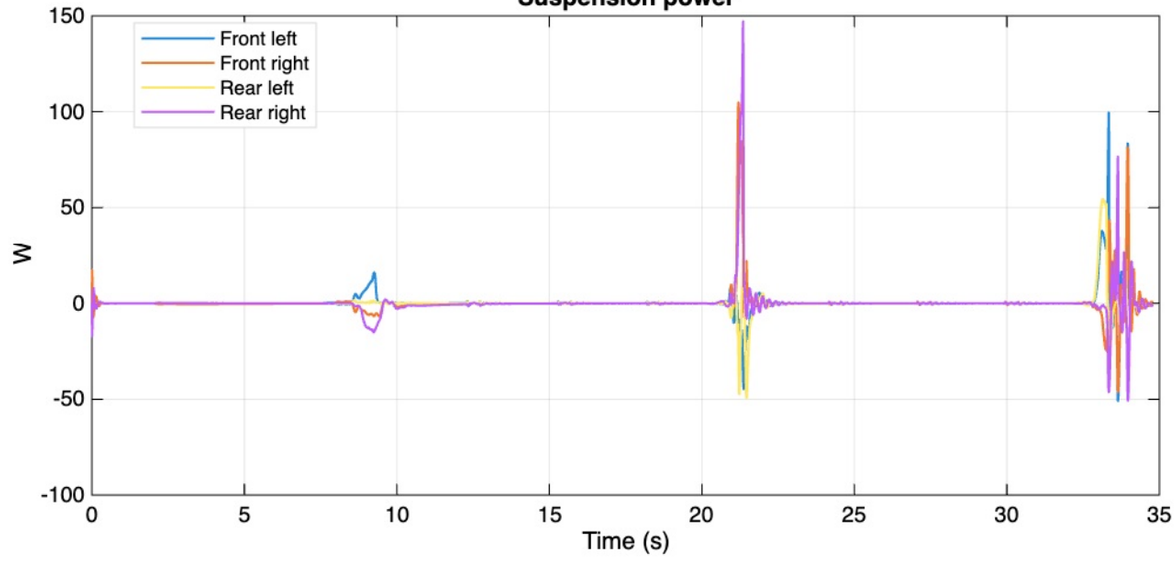
Suspension toe / steering angle



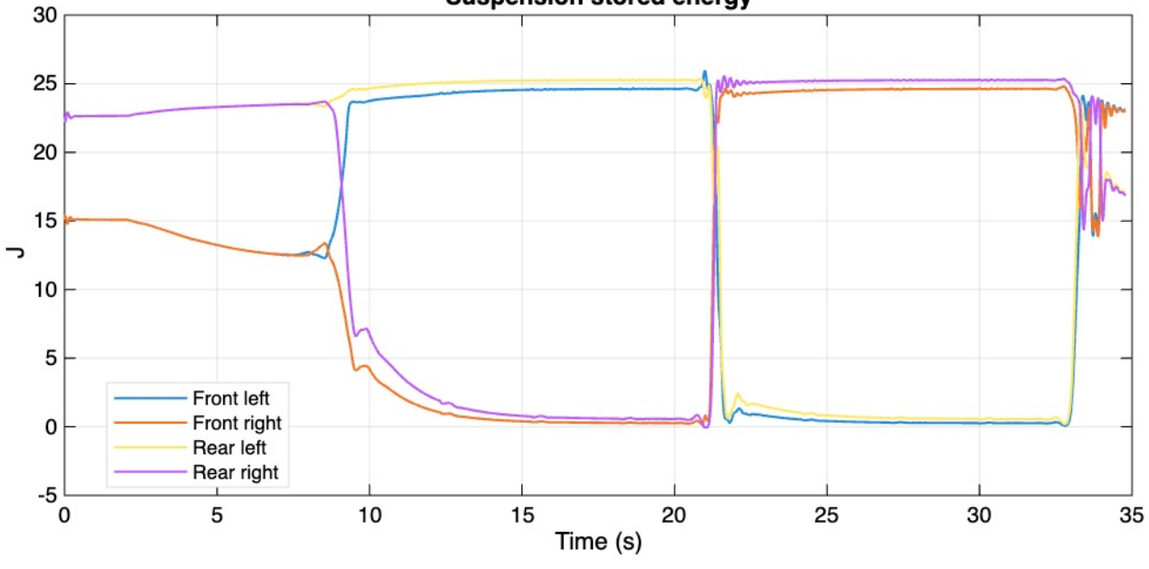
Suspension deflection (recorded relative height)



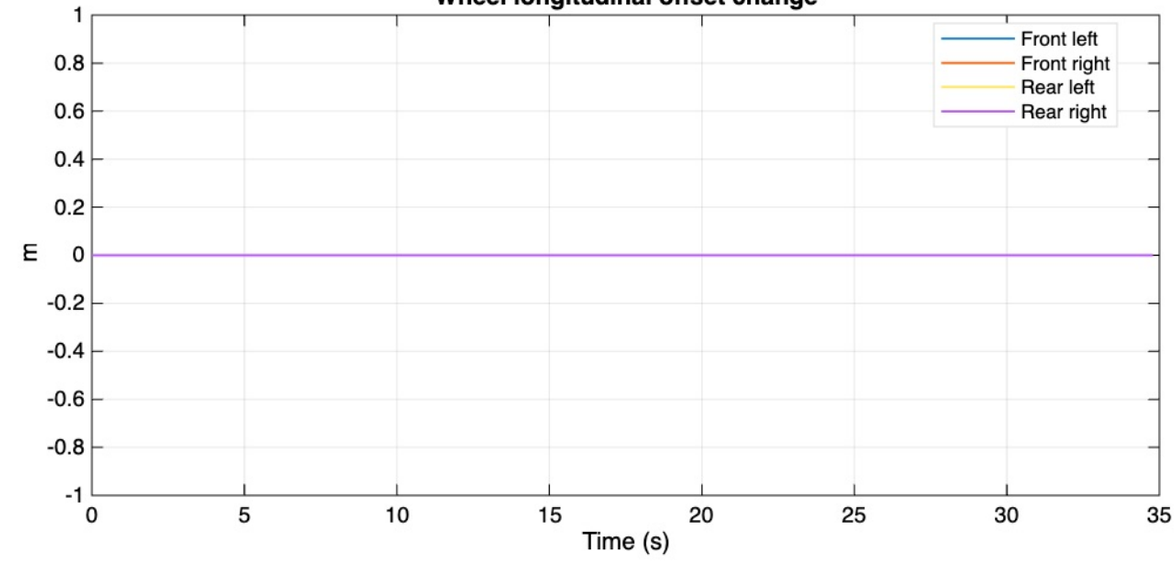
Suspension power



Suspension stored energy



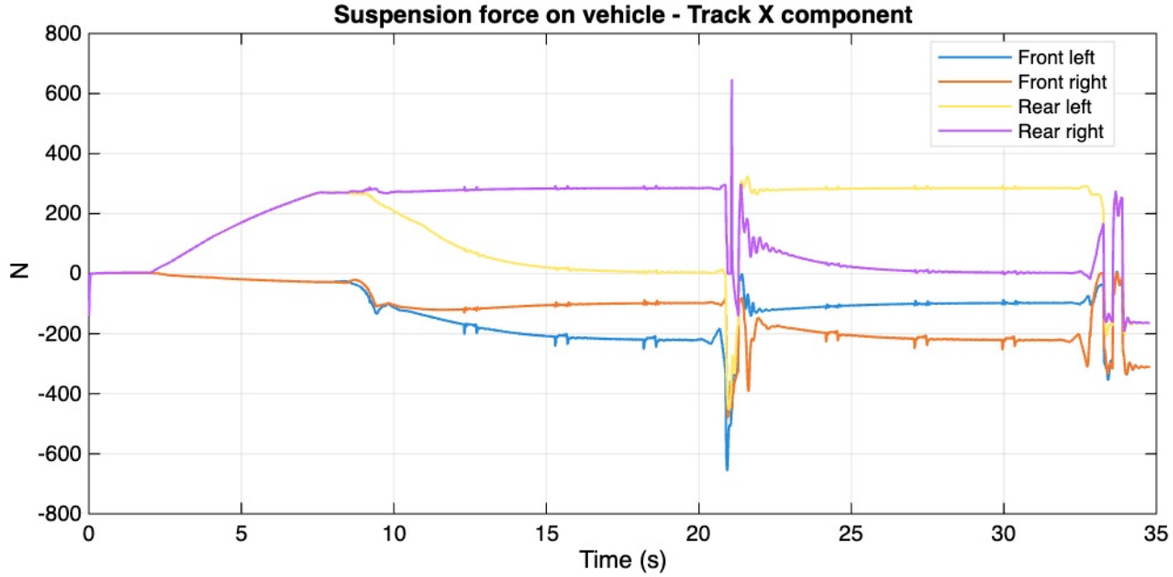
Wheel longitudinal offset change



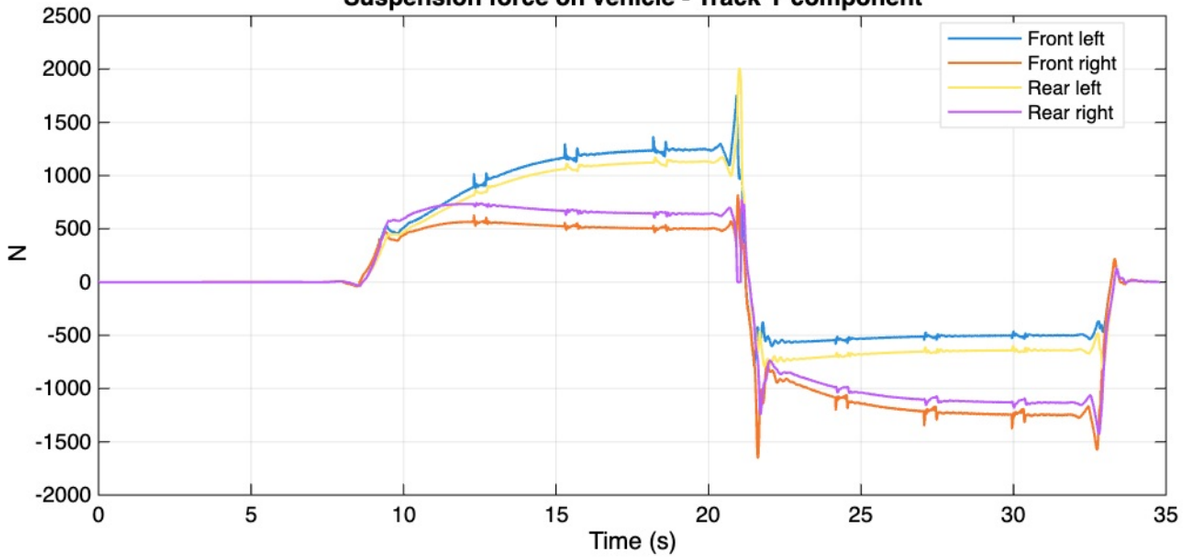
Wheel lateral offset change



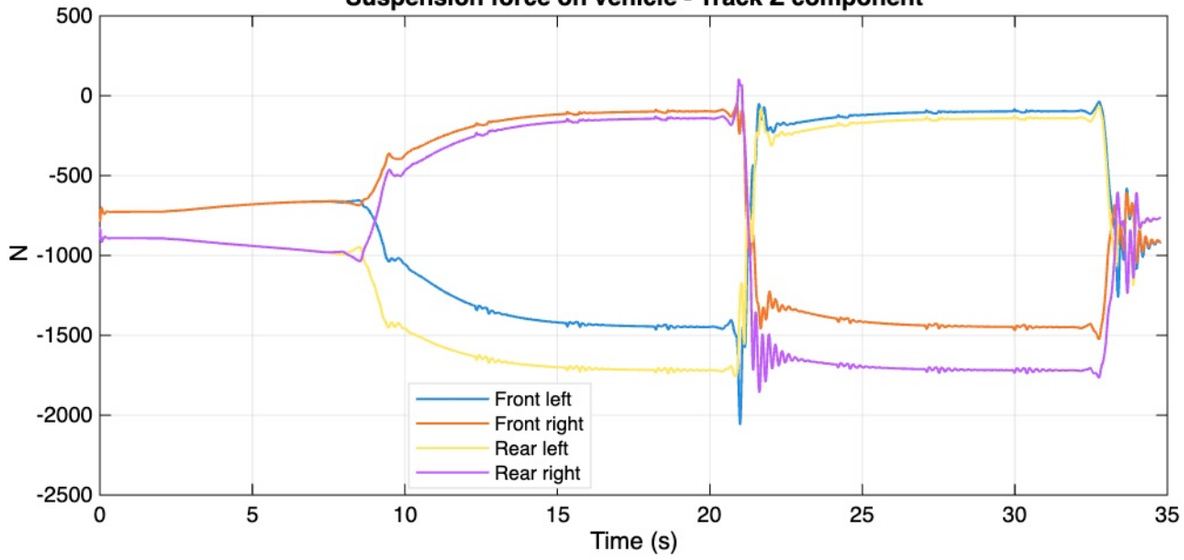
Suspension force on vehicle - Track X component



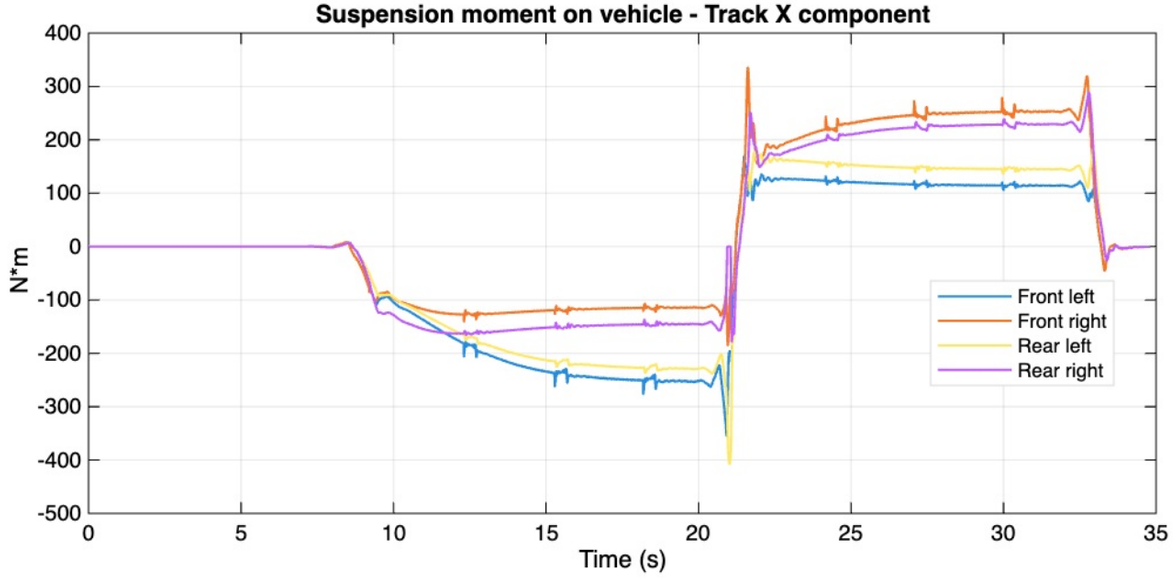
Suspension force on vehicle - Track Y component



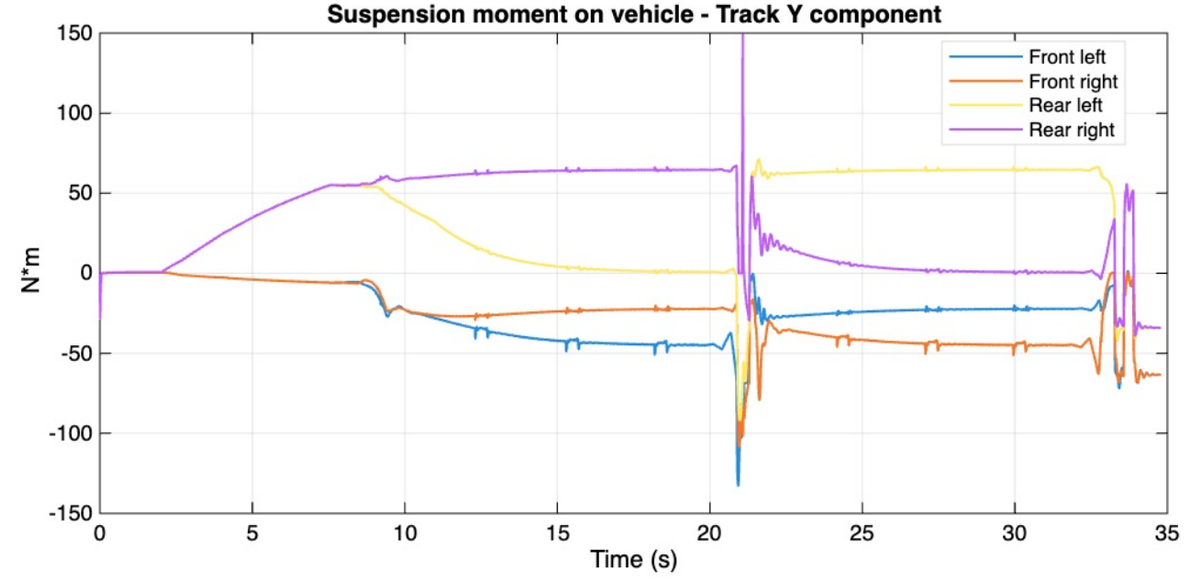
Suspension force on vehicle - Track Z component

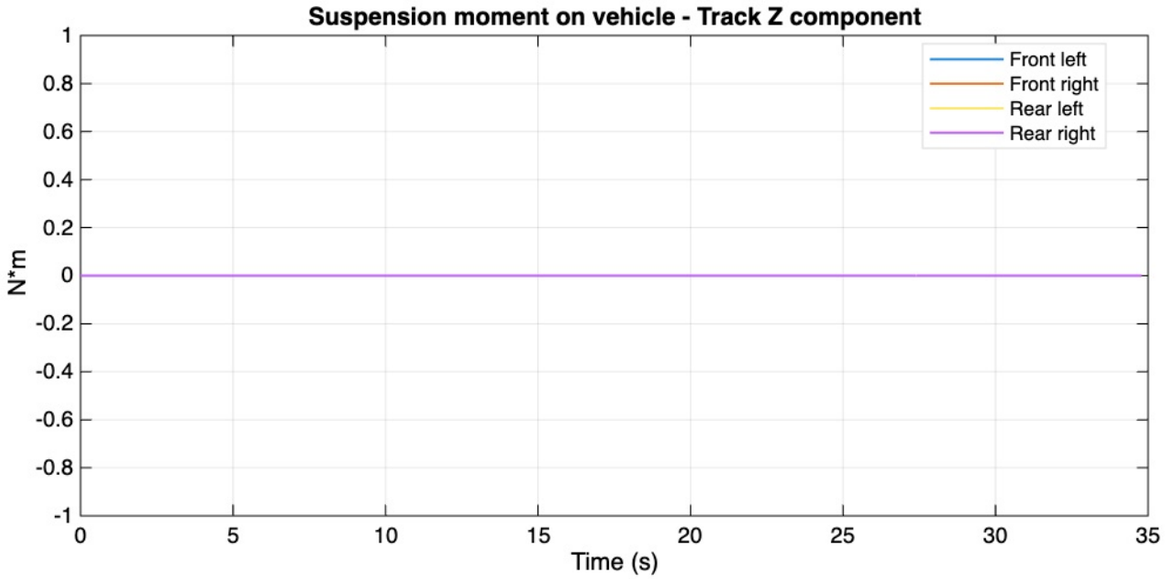


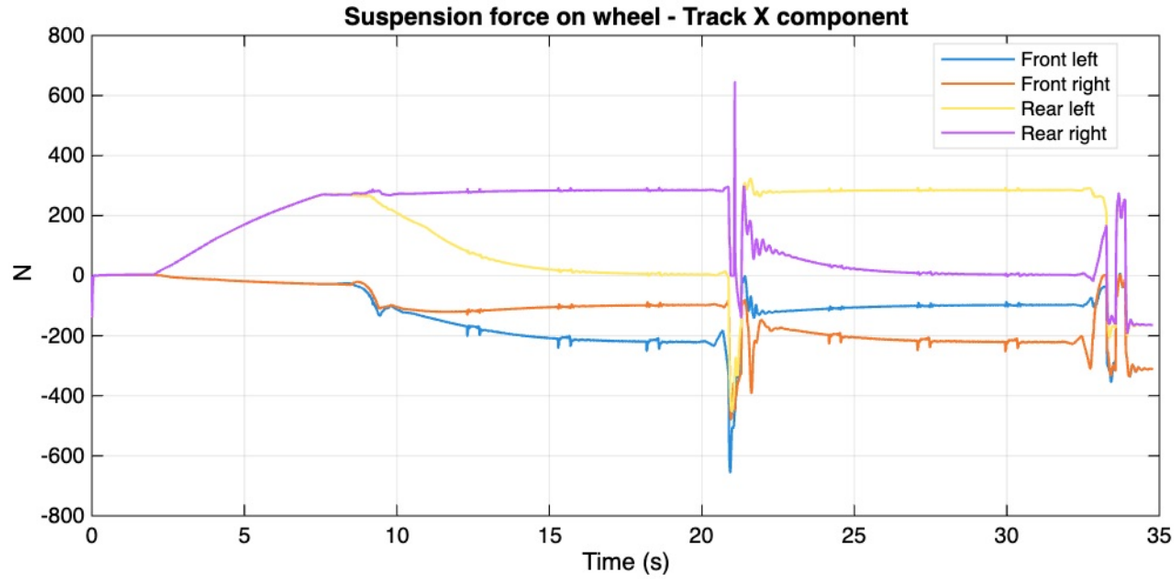
Suspension moment on vehicle - Track X component

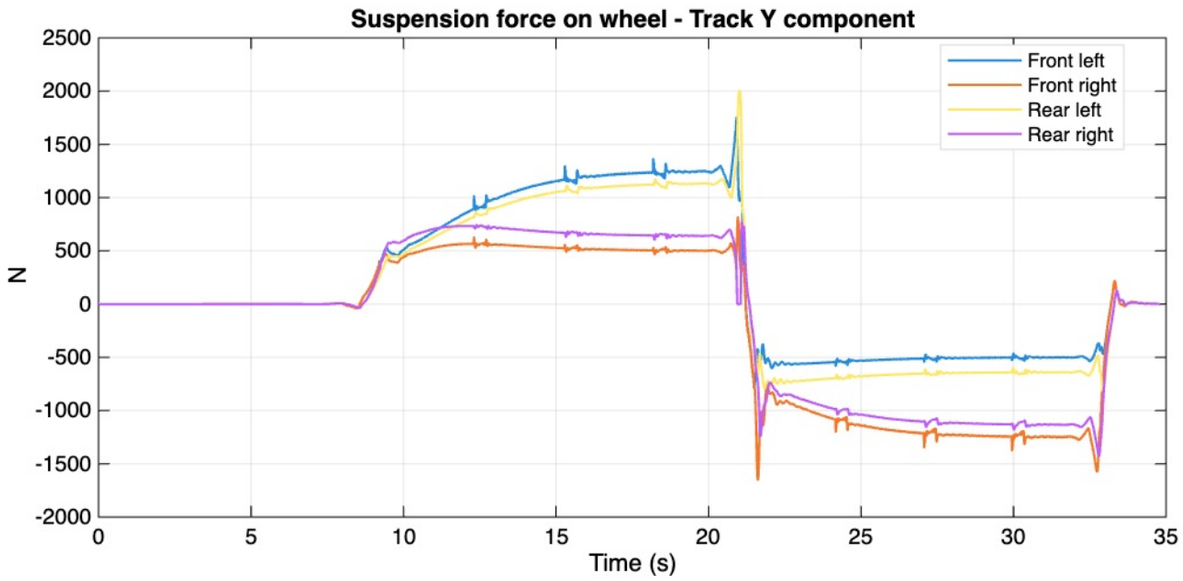


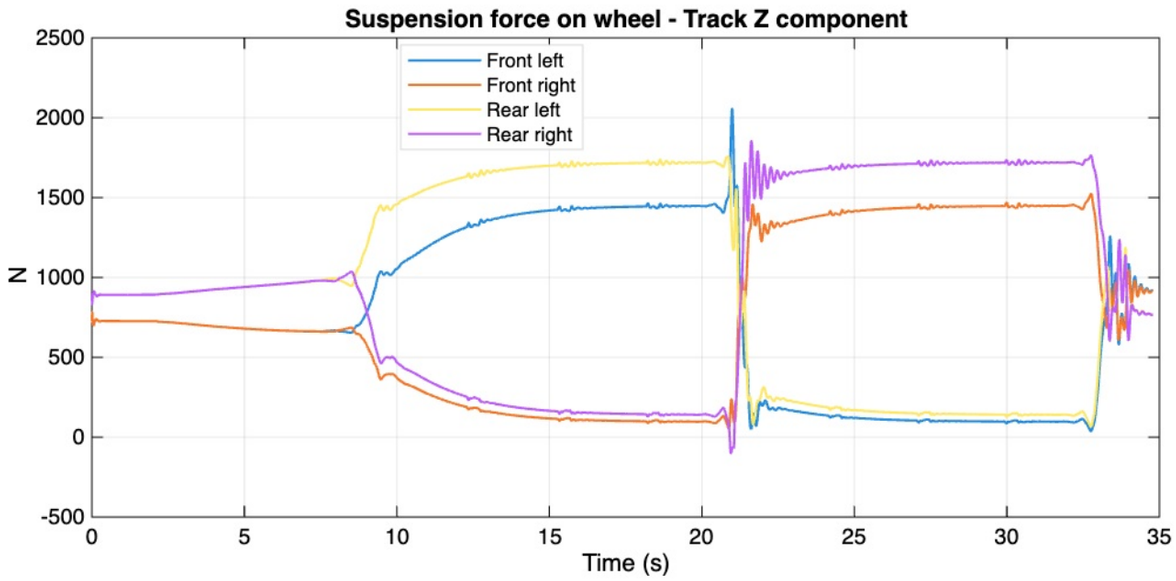
Suspension moment on vehicle - Track Y component

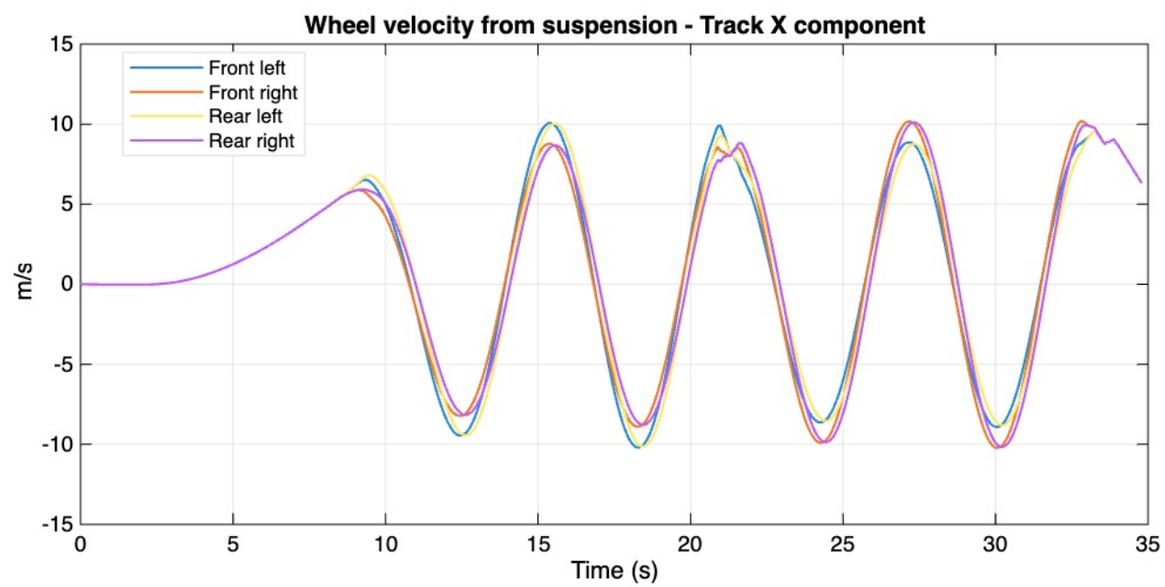


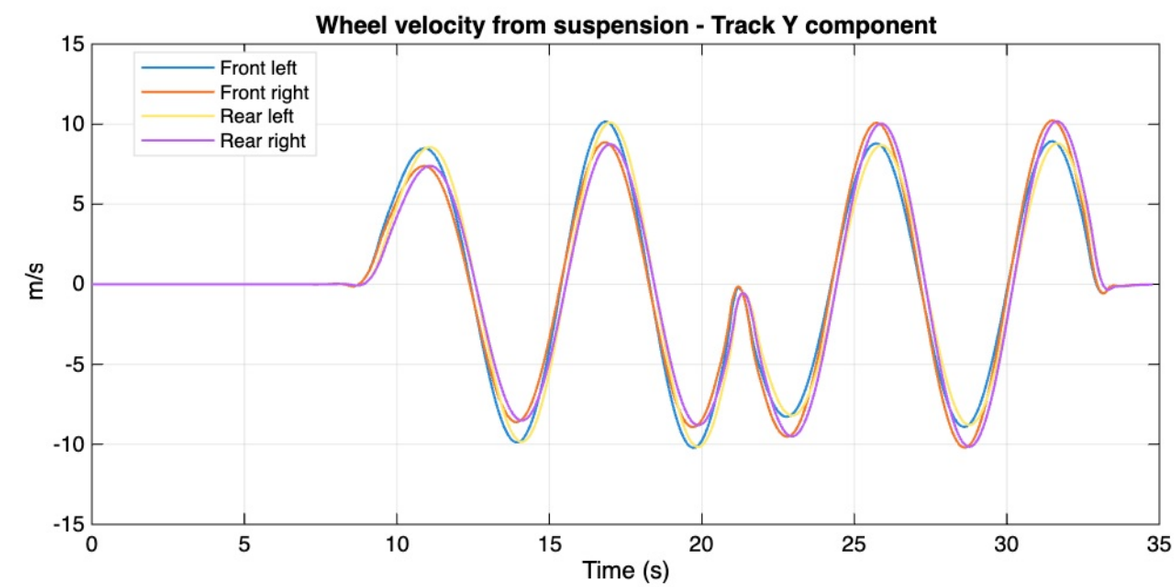

Suspension moment on vehicle - Track Z component
N*m
Time (s)
Front left
Front right
Rear left
Rear right

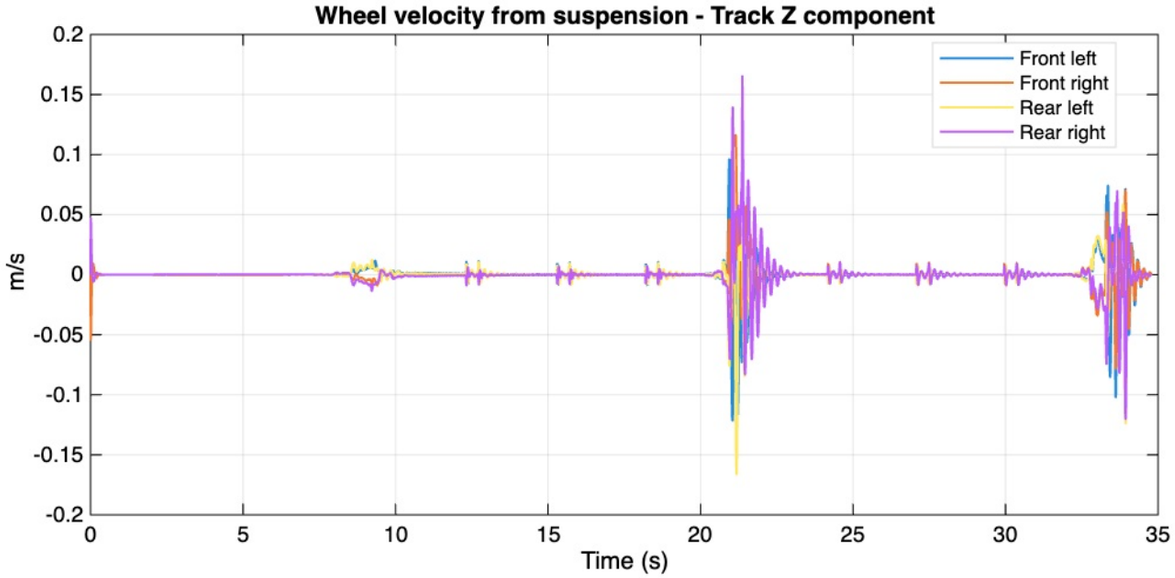

Suspension force on wheel - Track X component
N
Time (s)
Front left
Front right
Rear left
Rear right

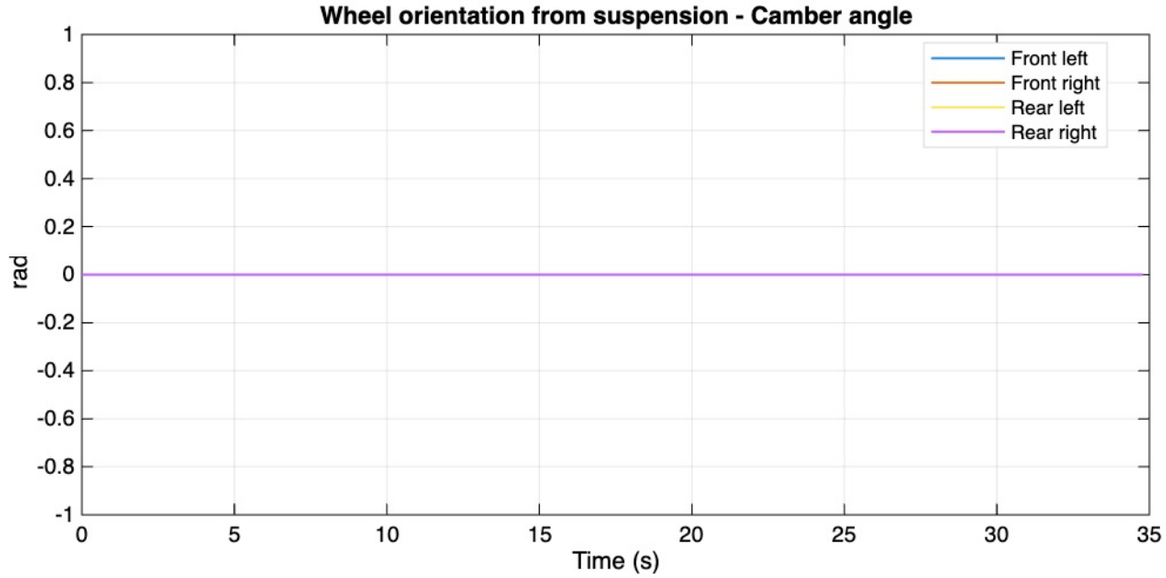

Suspension force on wheel - Track Y component
N
Time (s)
Front left
Front right
Rear left
Rear right

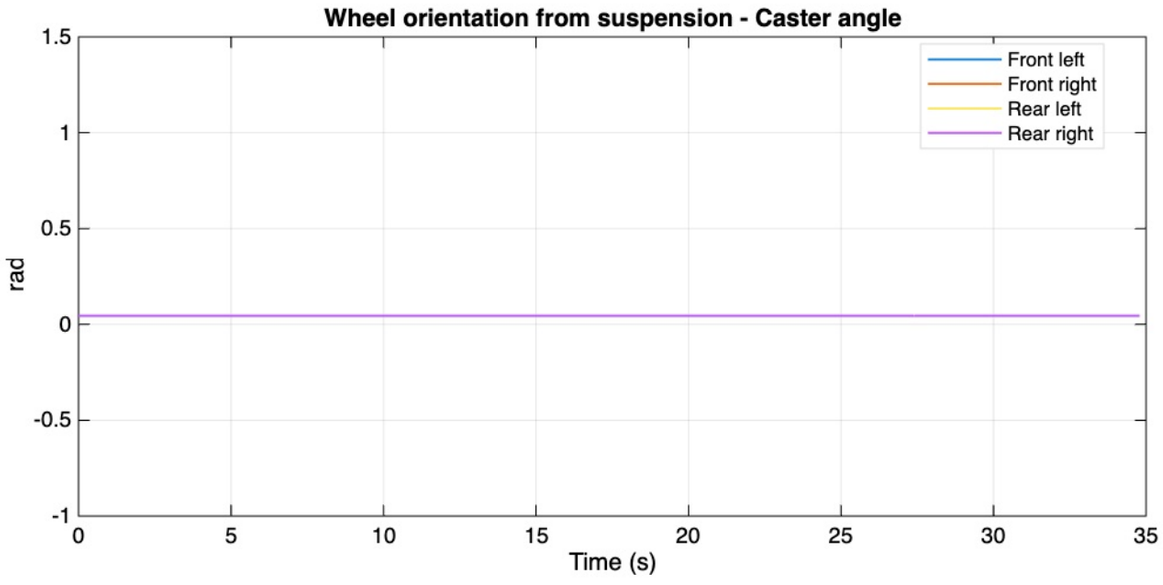

Suspension force on wheel - Track Z component
N
Time (s)
Front left
Front right
Rear left
Rear right

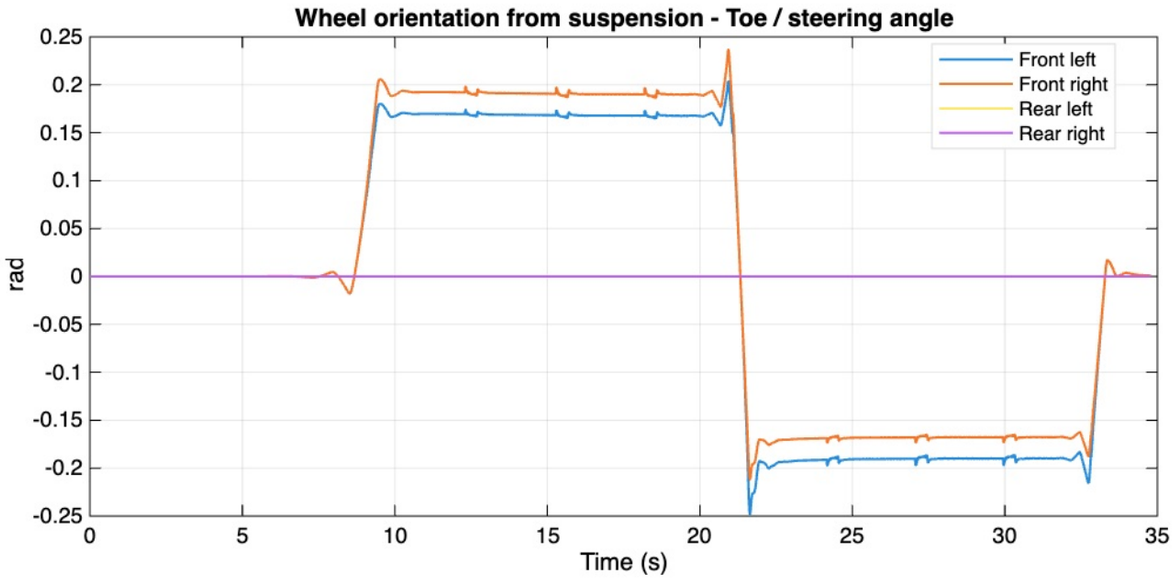

Wheel velocity from suspension - Track X component
m/s
Time (s)
Front left
Front right
Rear left
Rear right

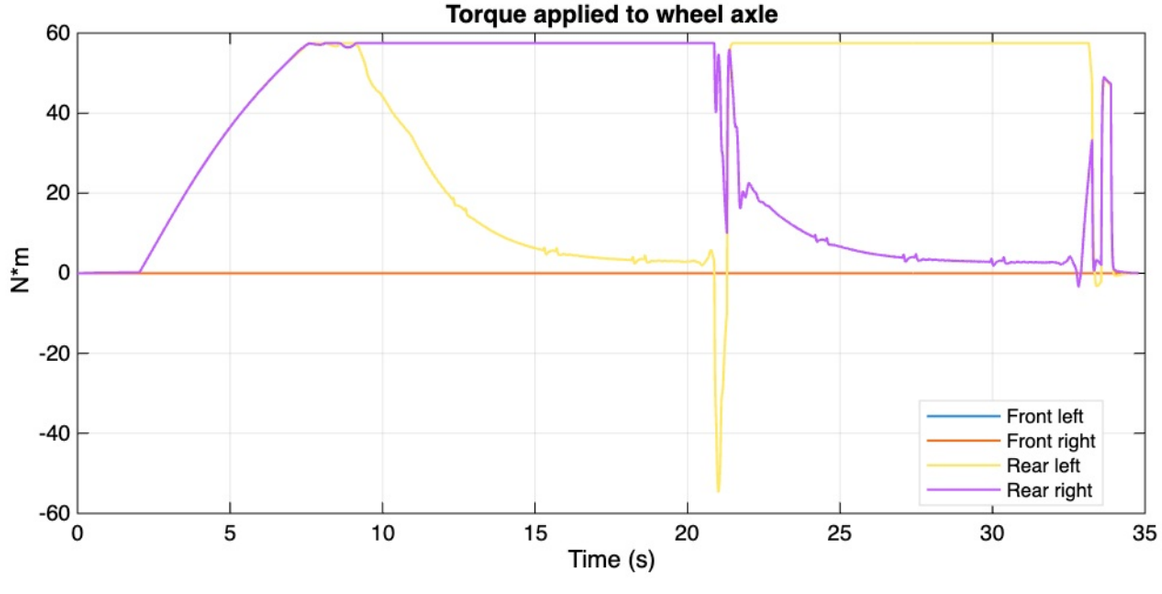

Wheel velocity from suspension - Track Y component
m/s
Time (s)
Front left
Front right
Rear left
Rear right

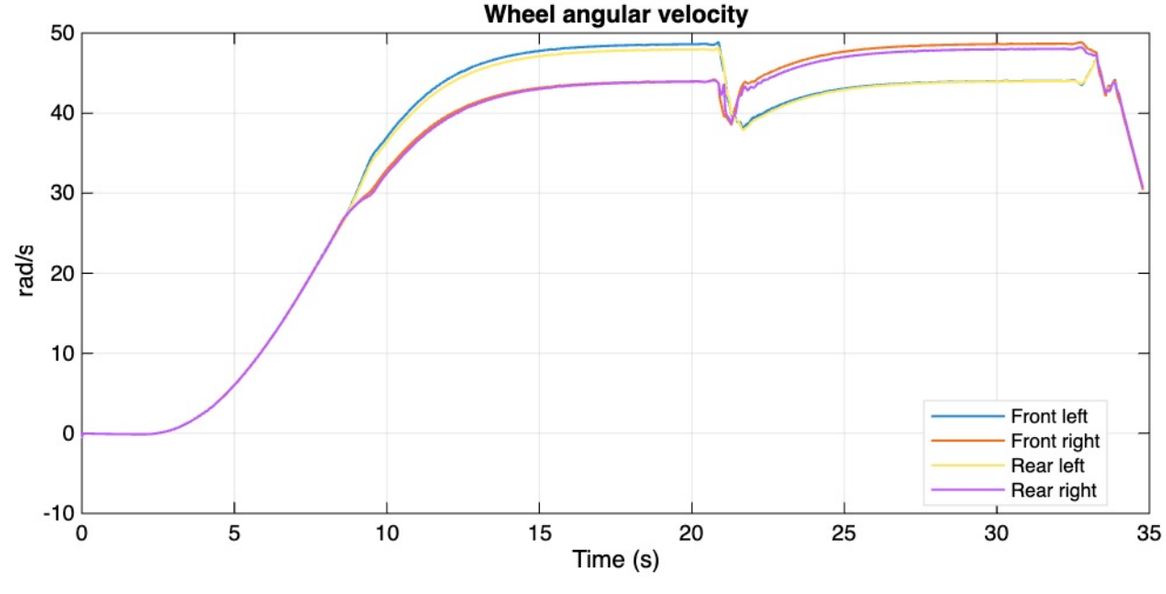

Wheel velocity from suspension - Track Z component
m/s
Time (s)
Front left
Front right
Rear left
Rear right

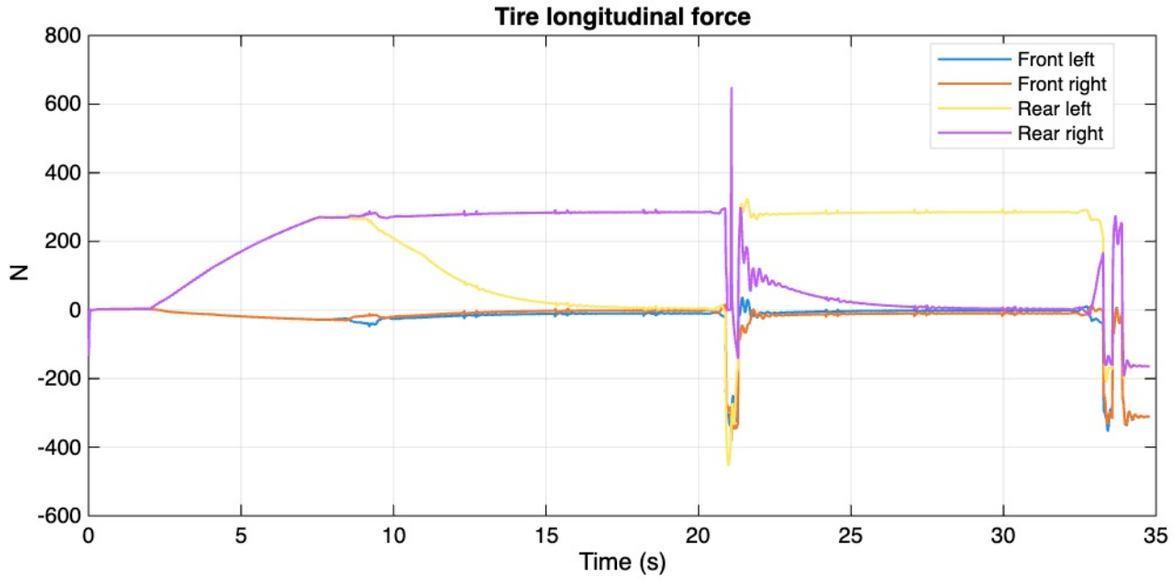

Wheel orientation from suspension - Camber angle
rad
Time (s)
Front left
Front right
Rear left
Rear right

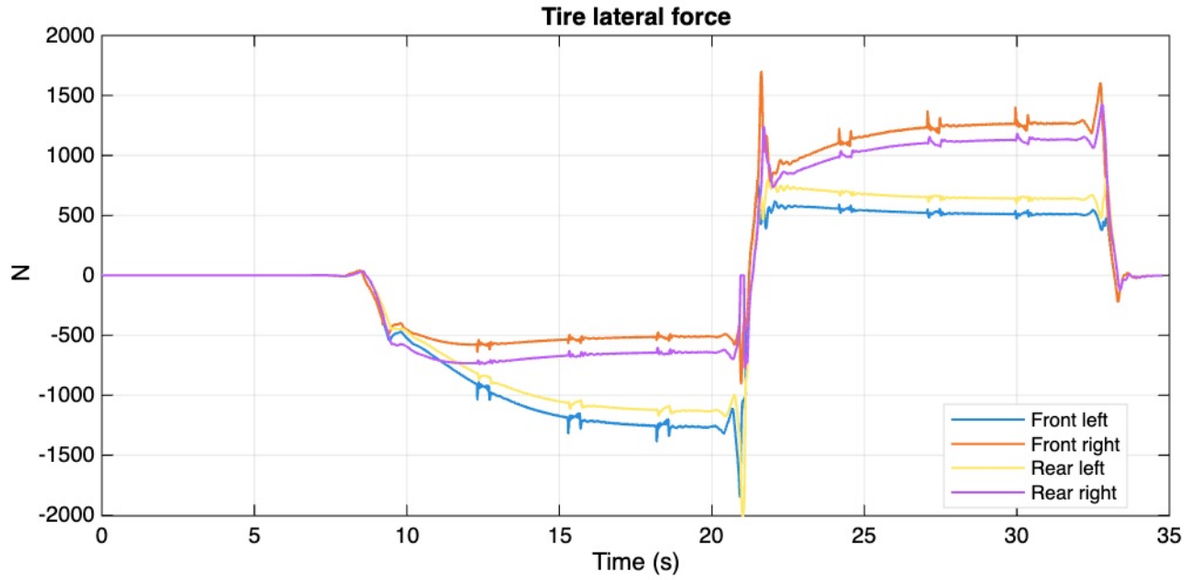

Wheel orientation from suspension - Caster angle
rad
Time (s)
Front left
Front right
Rear left
Rear right

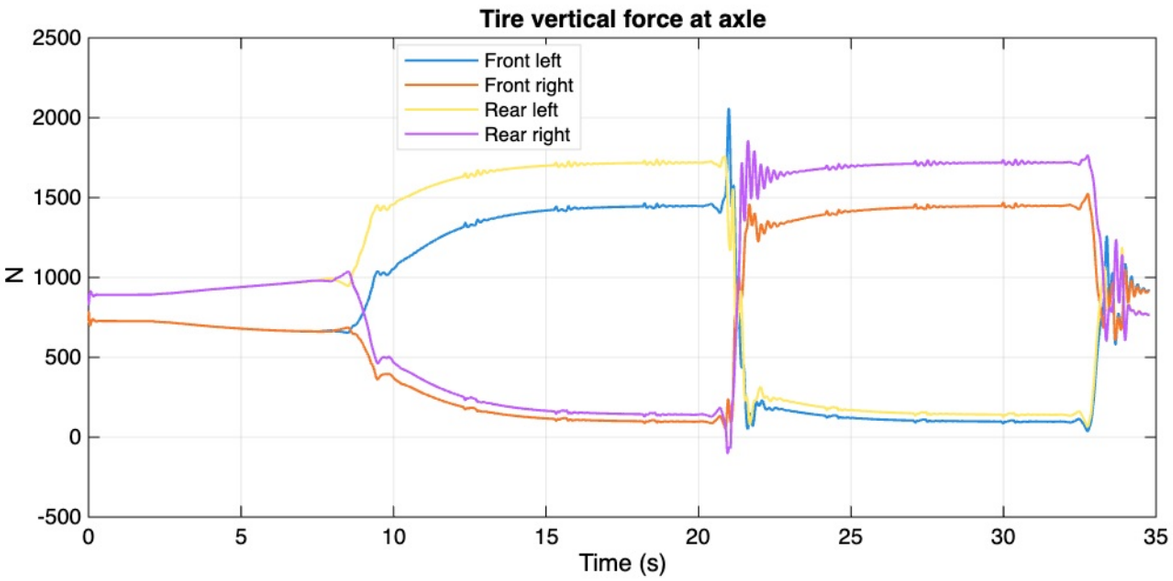

Wheel orientation from suspension - Toe / steering angle
rad
Time (s)
Front left
Front right
Rear left
Rear right

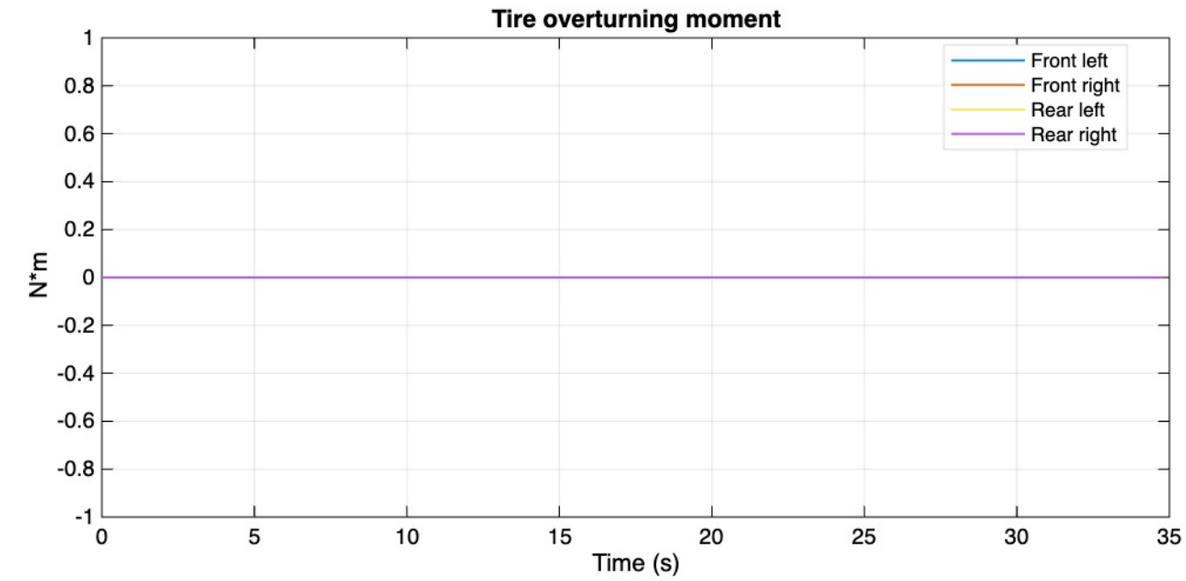

Torque applied to wheel axle
N*m
Time (s)
Front left
Front right
Rear left
Rear right

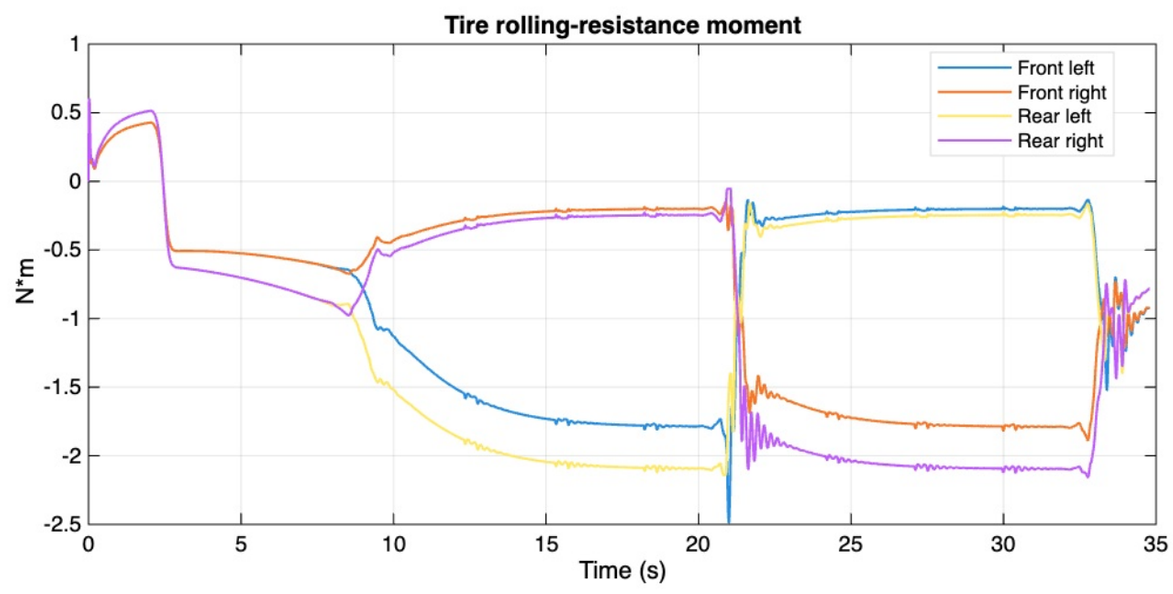

Wheel angular velocity
rad/s
Time (s)
Front left
Front right
Rear left
Rear right

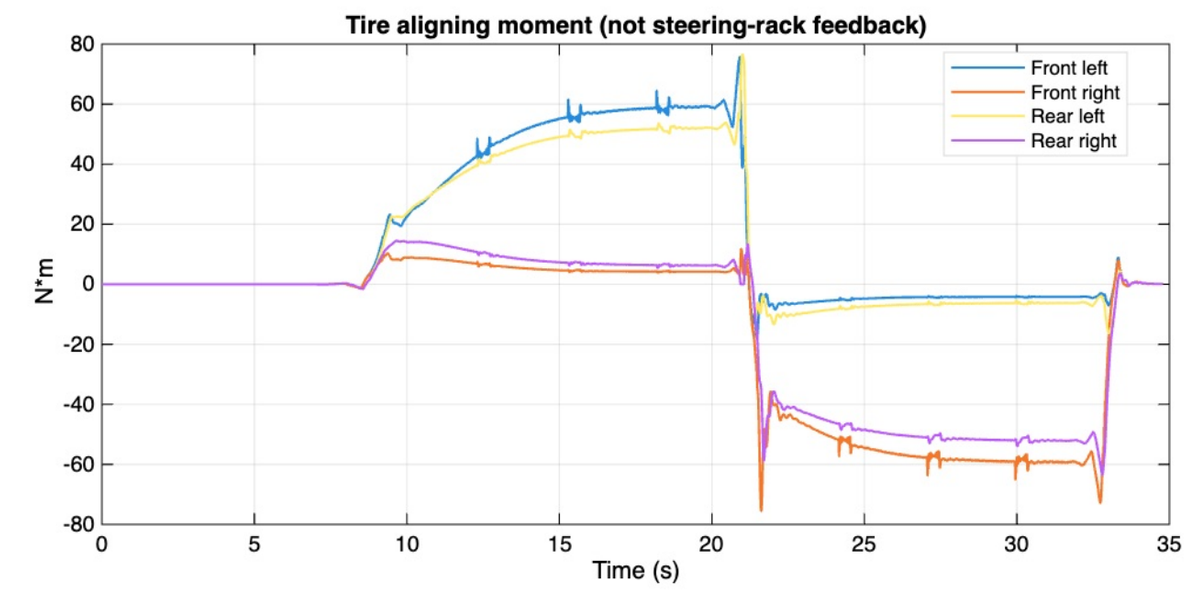

Tire longitudinal force
Front left
Front right
Rear left
Rear right
N
Time (s)


Tire lateral force
Front left
Front right
Rear left
Rear right
N
Time (s)

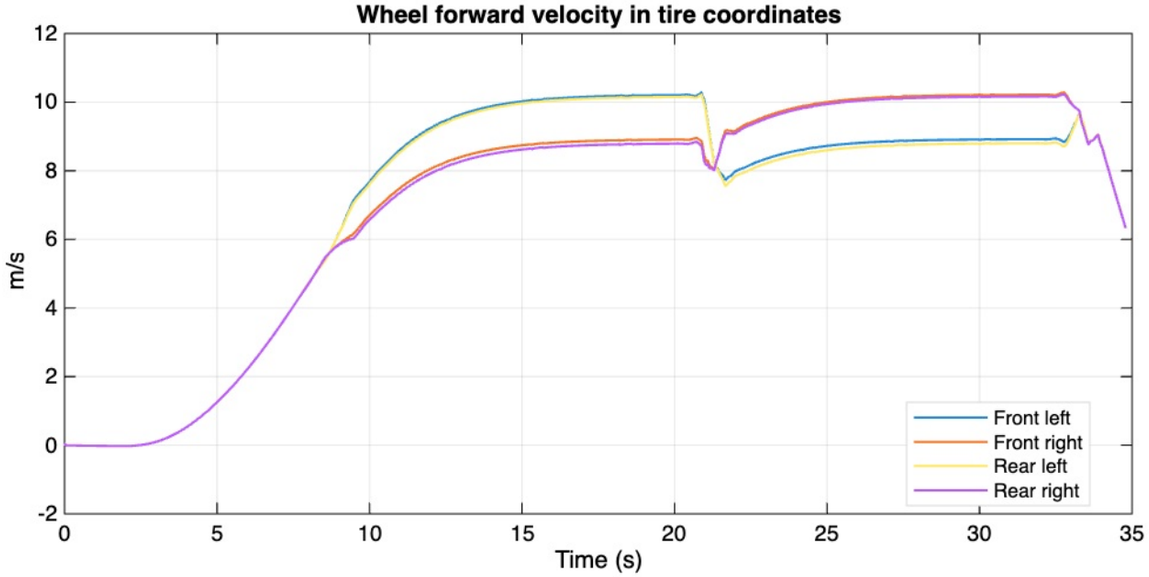

Tire vertical force at axle
Front left
Front right
Rear left
Rear right
N
Time (s)


Tire overturning moment
Front left
Front right
Rear left
Rear right
N*m
Time (s)

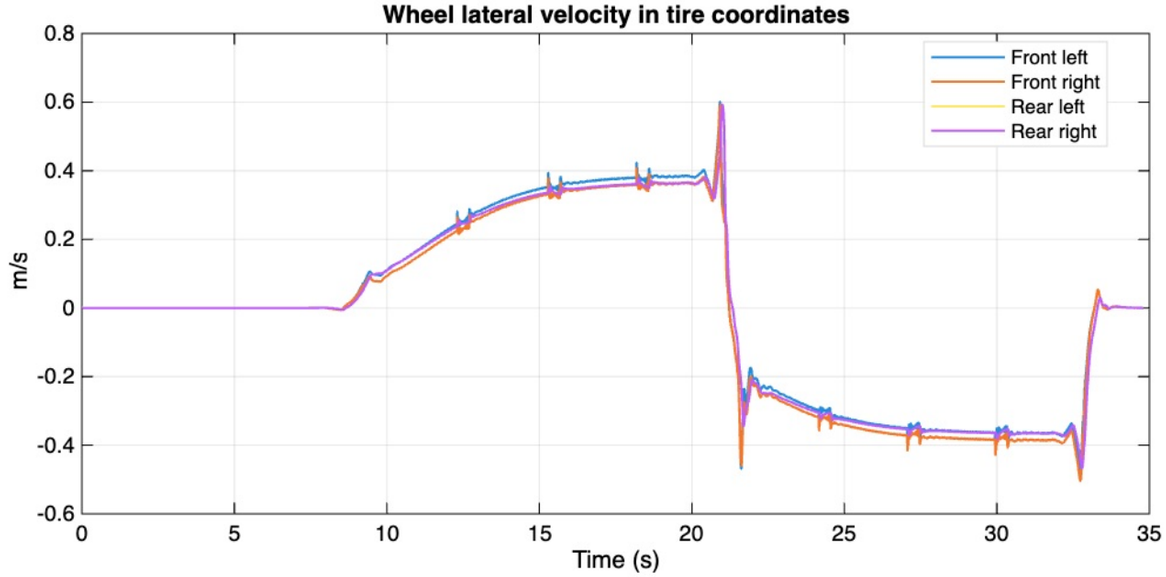

Tire rolling-resistance moment
Front left
Front right
Rear left
Rear right
N*m
Time (s)


Tire aligning moment (not steering-rack feedback)
Front left
Front right
Rear left
Rear right
N*m
Time (s)

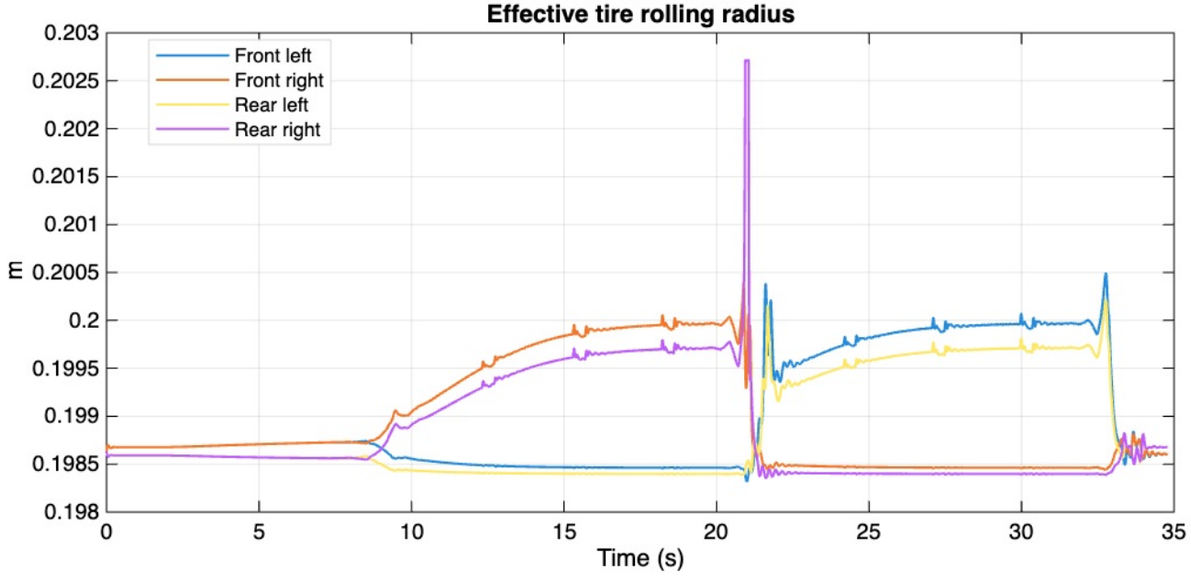
Wheel forward velocity in tire coordinates



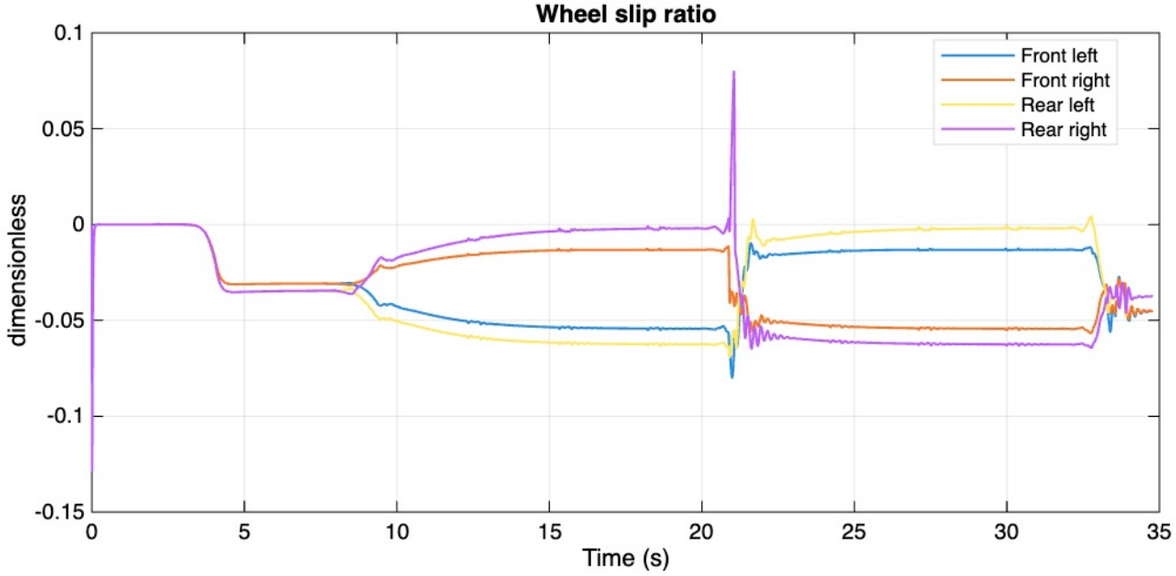
Wheel lateral velocity in tire coordinates



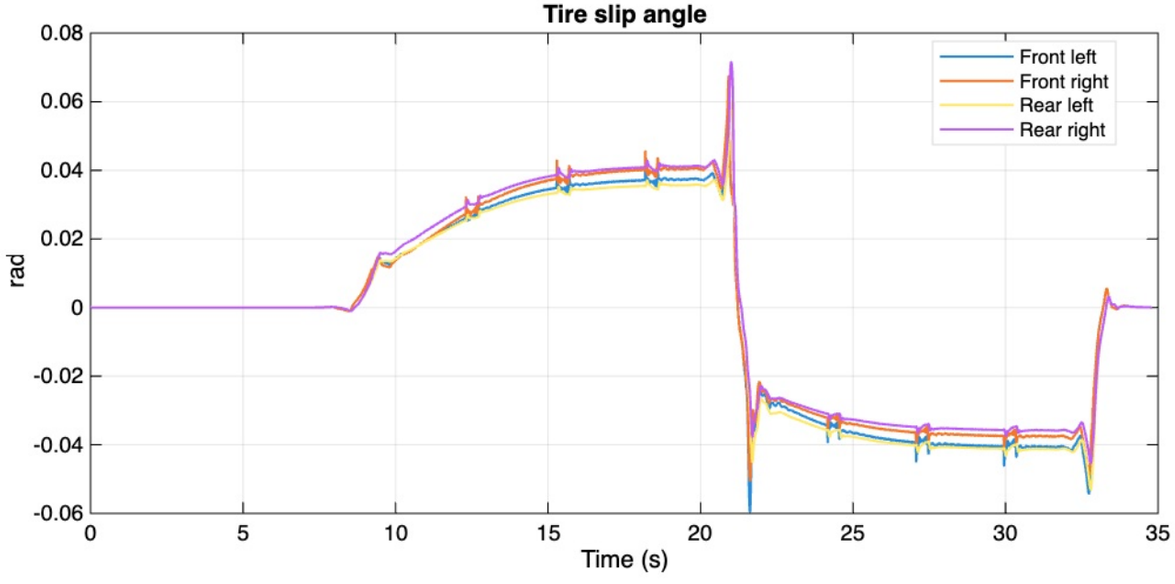
Effective tire rolling radius



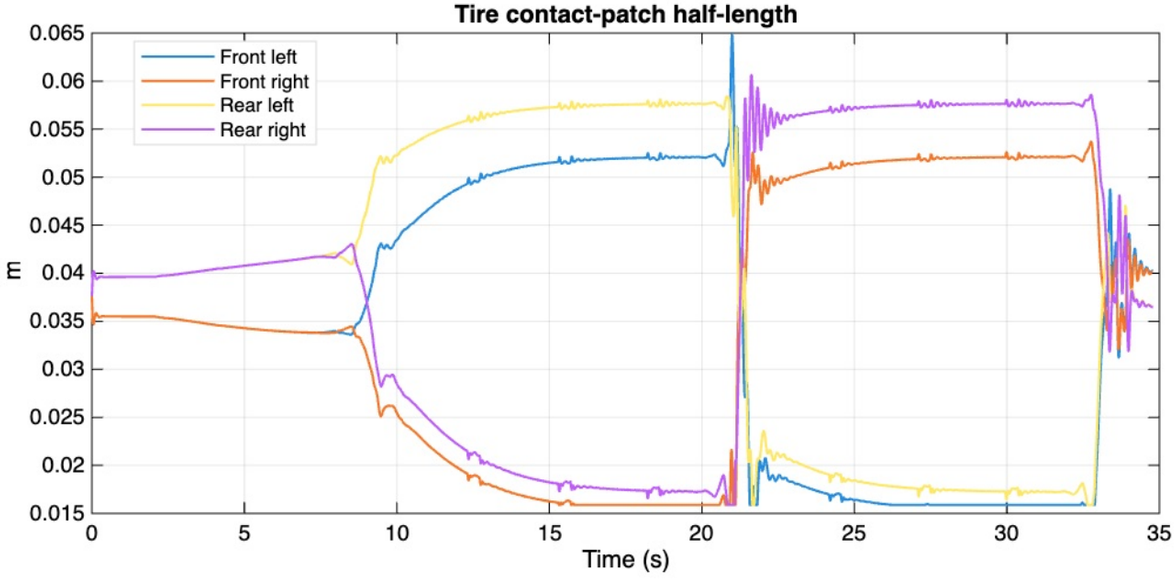
Wheel slip ratio

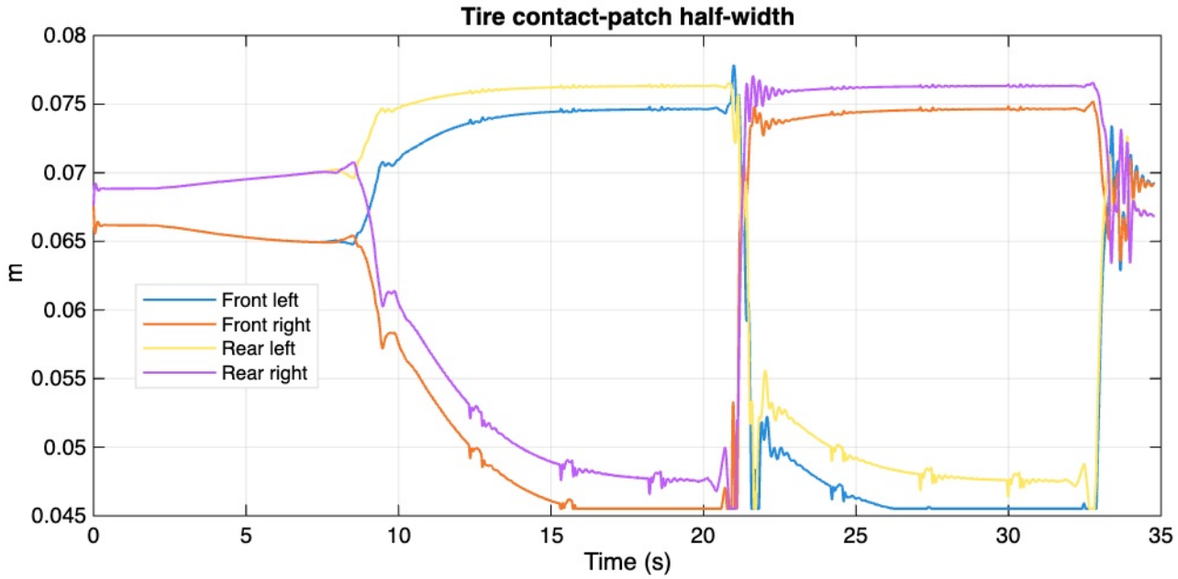


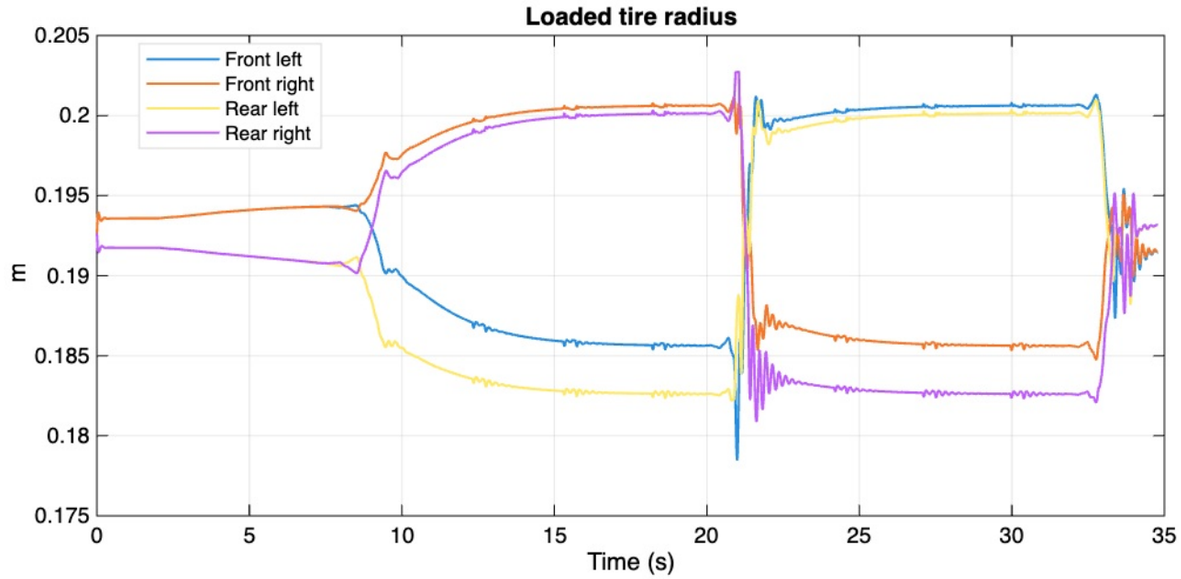
Tire slip angle

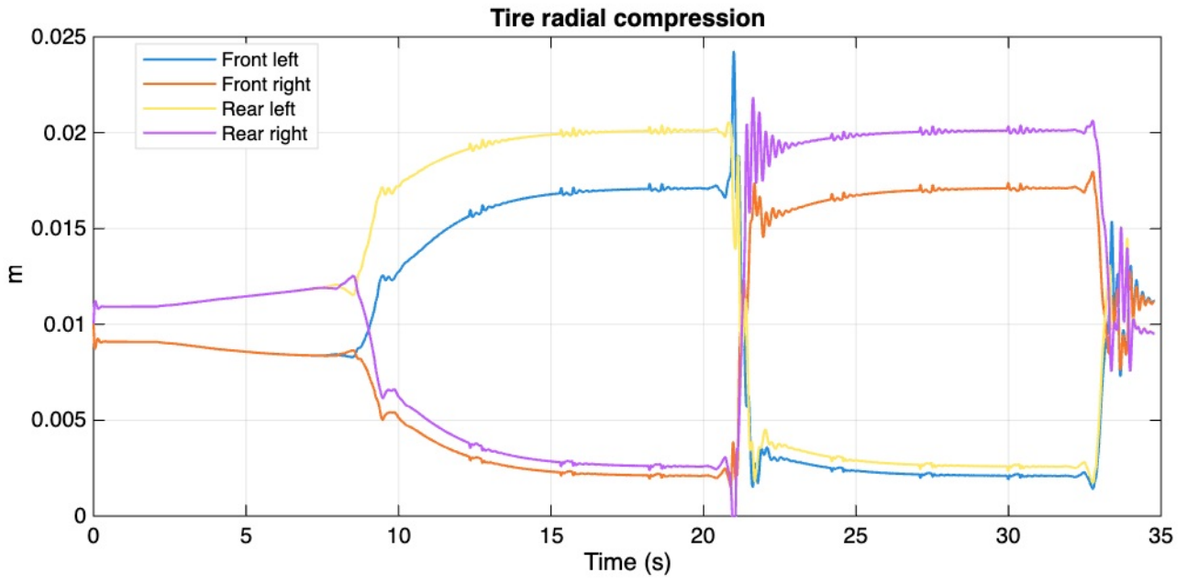


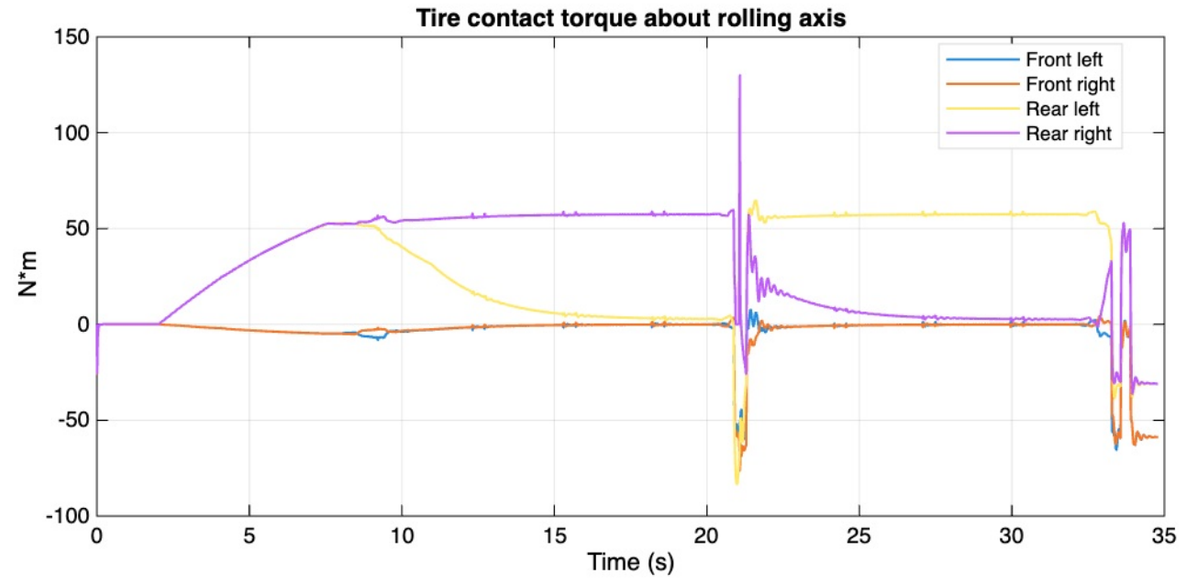
Tire contact-patch half-length

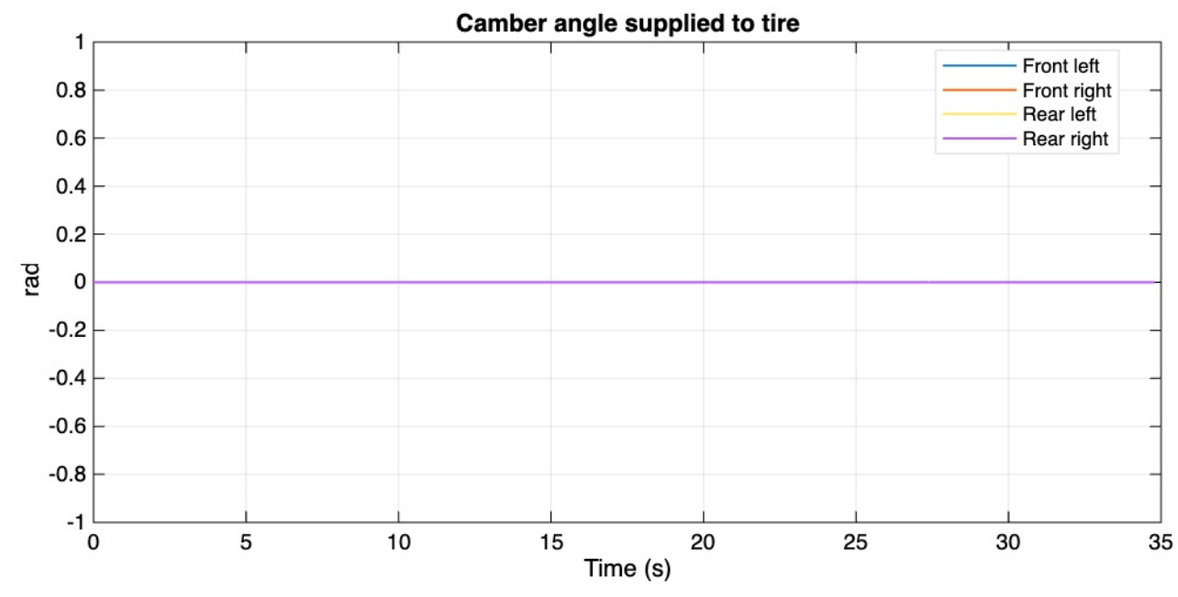


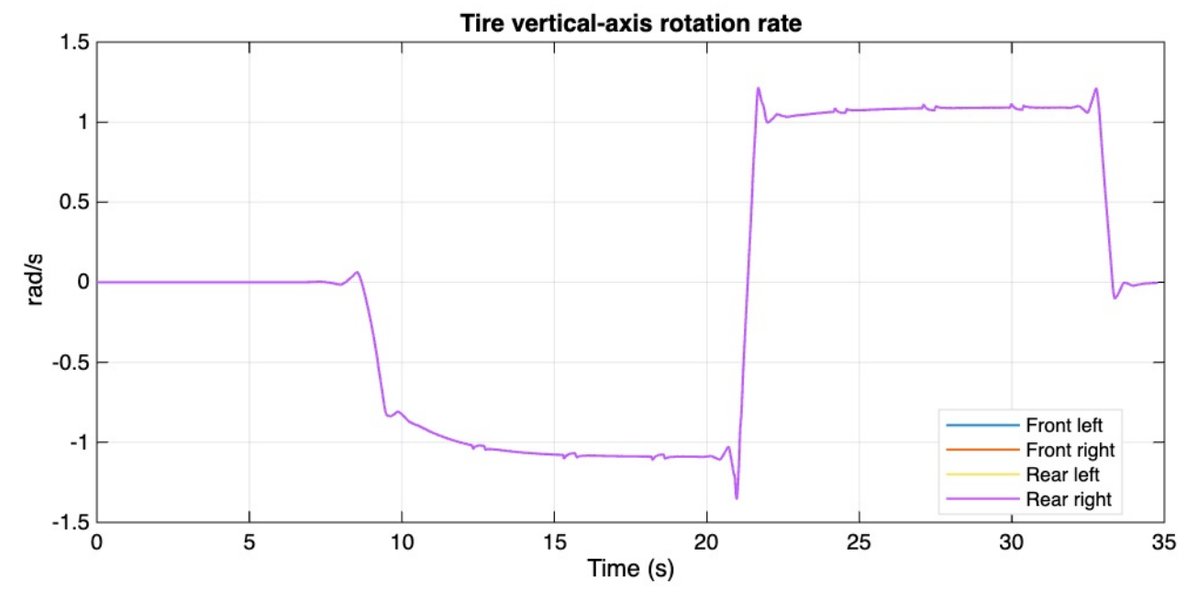

Tire contact-patch half-width
m
Time (s)
Front left
Front right
Rear left
Rear right

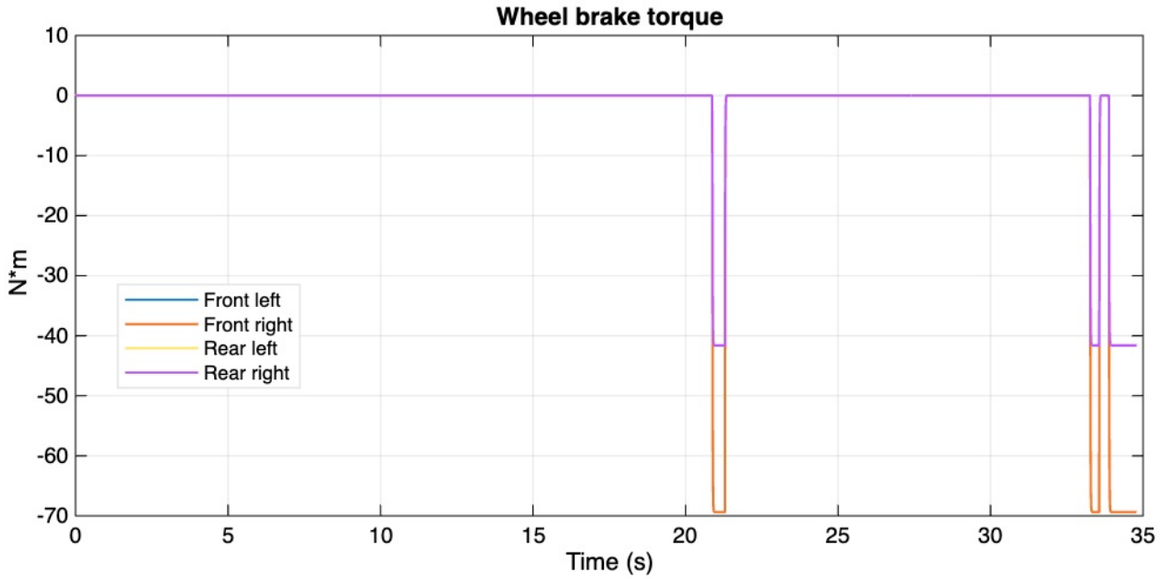

Loaded tire radius
m
Time (s)
Front left
Front right
Rear left
Rear right

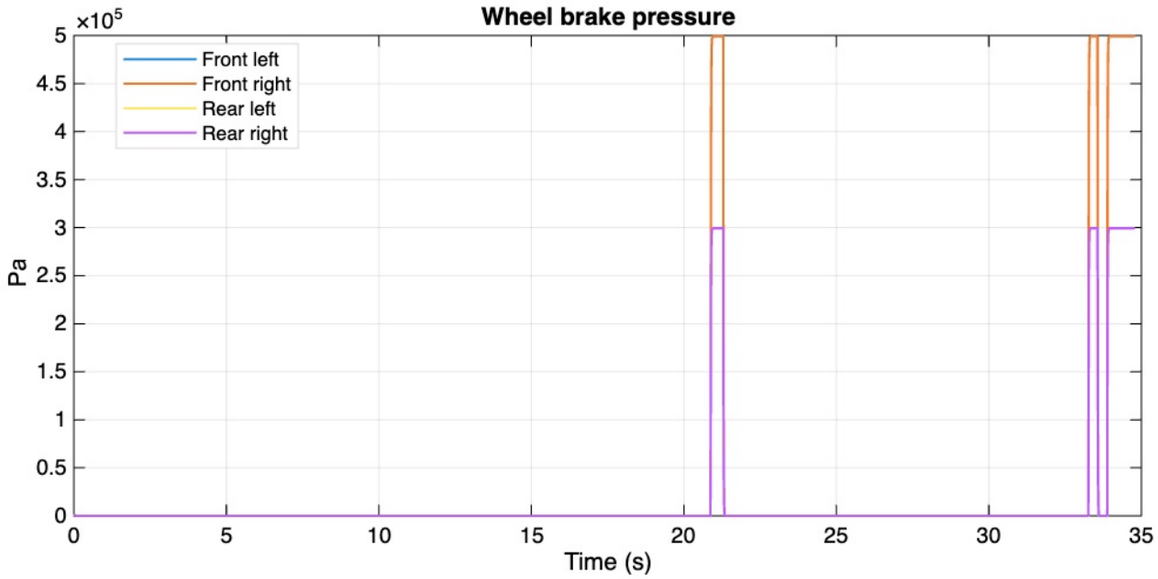

Tire radial compression
m
Time (s)
Front left
Front right
Rear left
Rear right

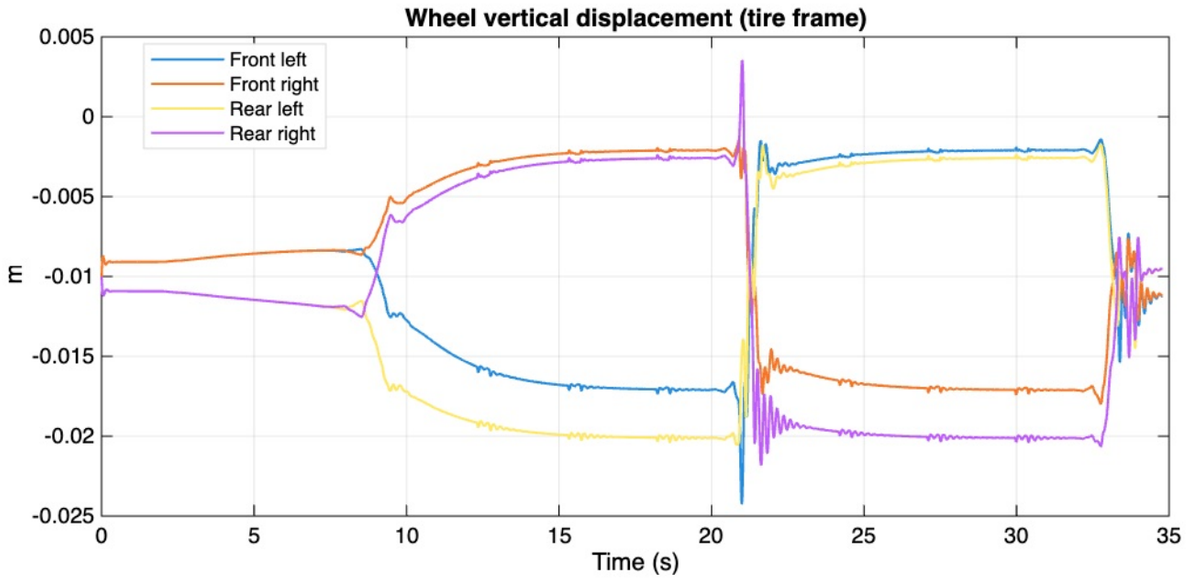

Tire contact torque about rolling axis
N*m
Time (s)
Front left
Front right
Rear left
Rear right

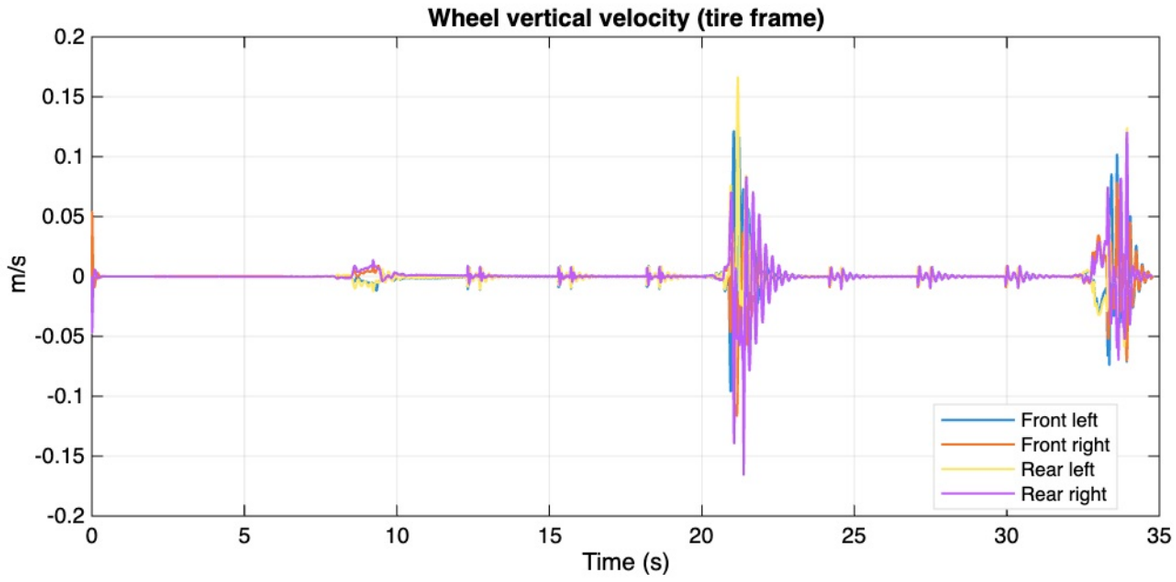

Camber angle supplied to tire
rad
Time (s)
Front left
Front right
Rear left
Rear right

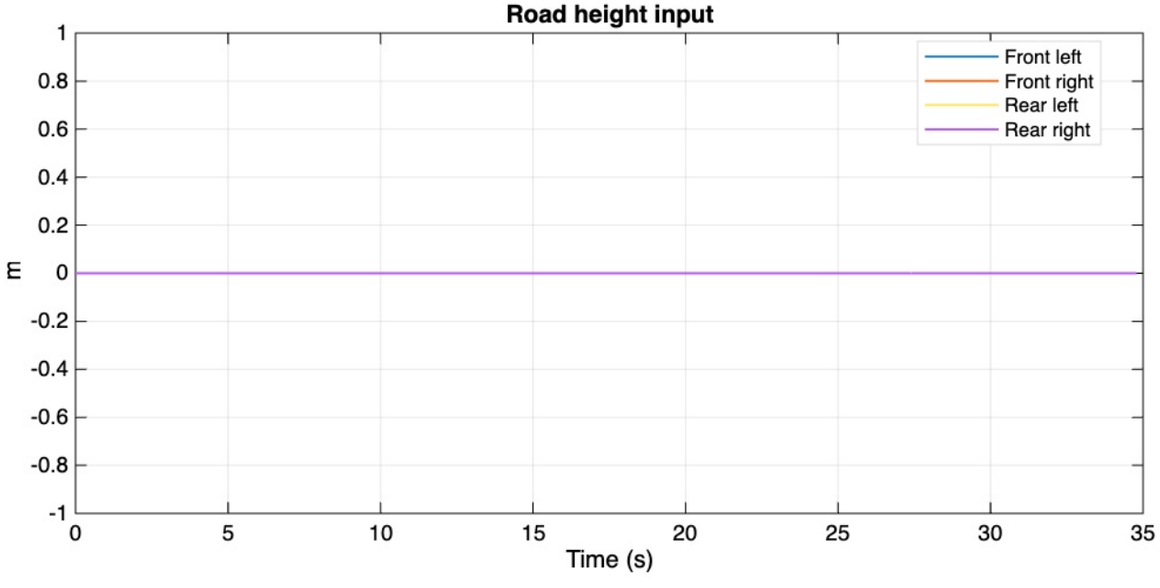

Tire vertical-axis rotation rate
rad/s
Time (s)
Front left
Front right
Rear left
Rear right

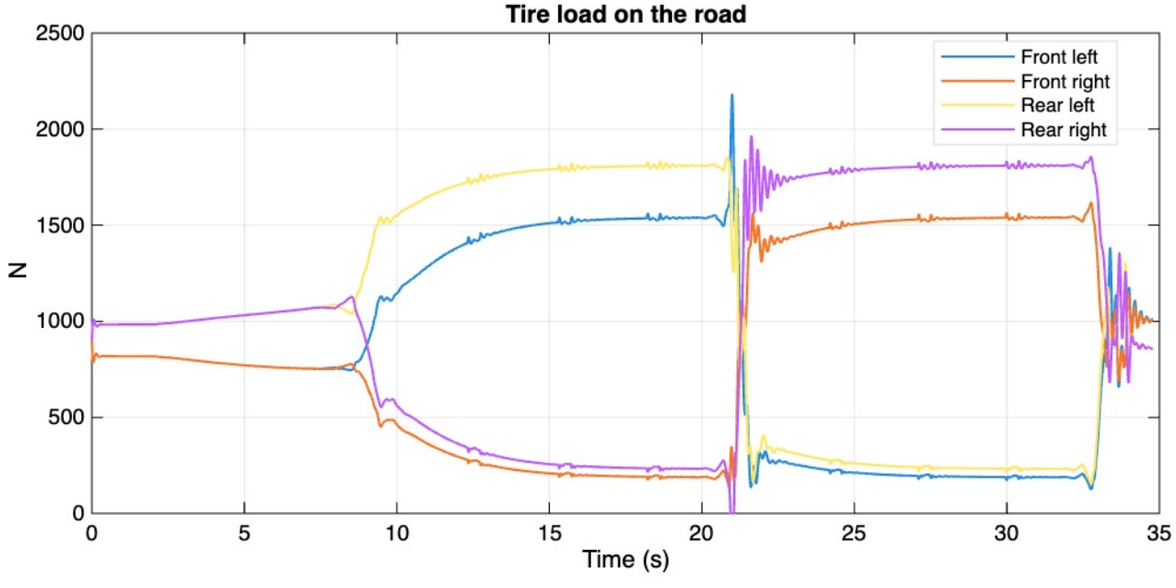

Wheel brake torque
N*m
Time (s)
Front left
Front right
Rear left
Rear right

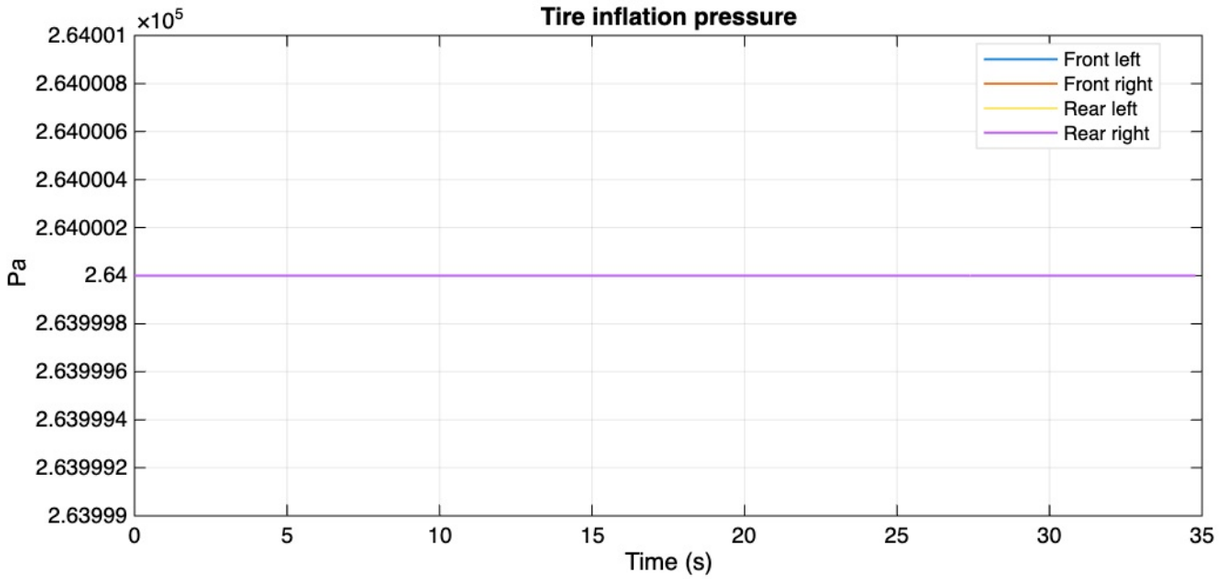

Wheel brake pressure
×10^5
Pa
Time (s)
Front left
Front right
Rear left
Rear right

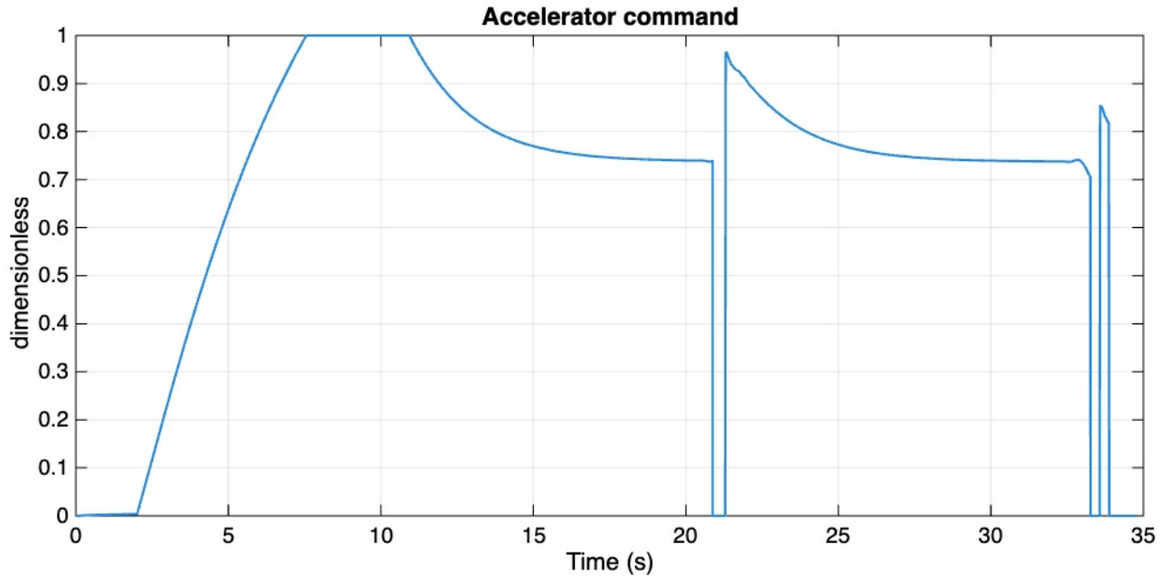

Wheel vertical displacement (tire frame)
m
Time (s)
Front left
Front right
Rear left
Rear right

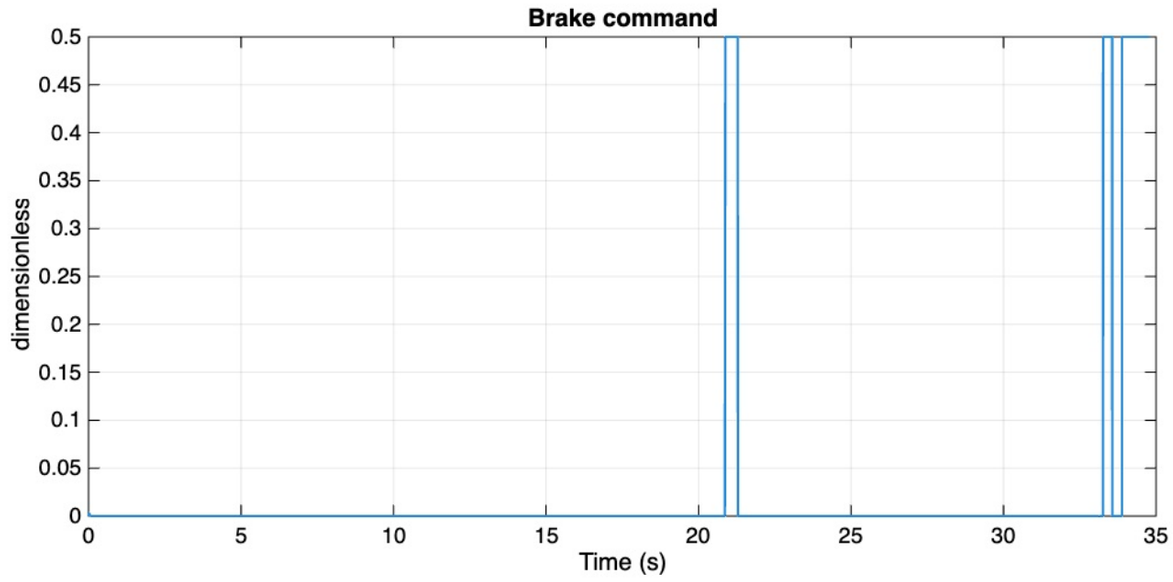

Wheel vertical velocity (tire frame)
m/s
Time (s)
Front left
Front right
Rear left
Rear right

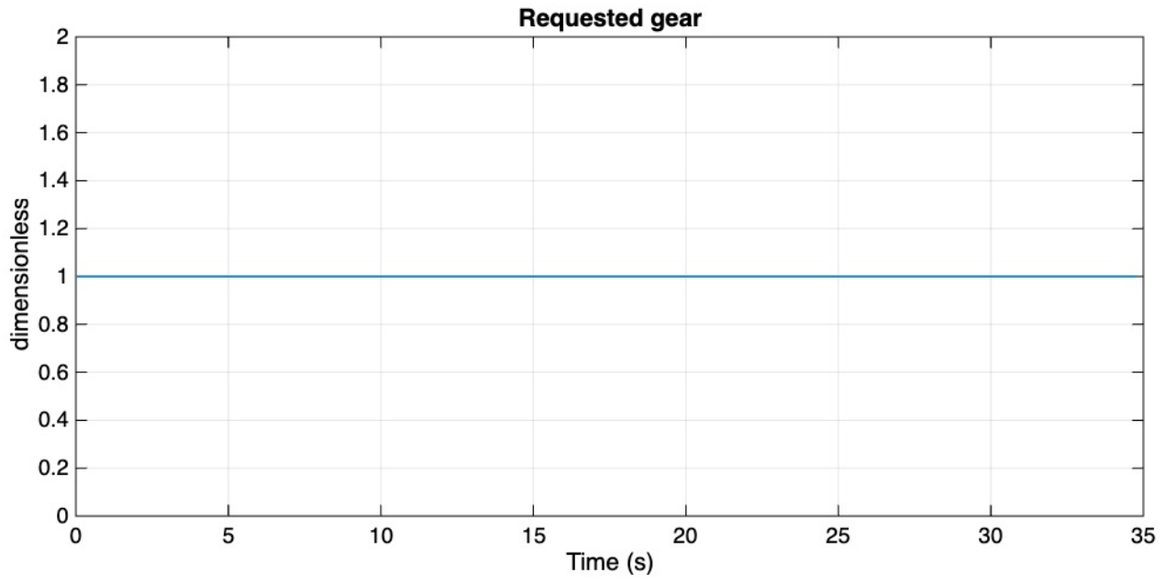

Road height input
m
Time (s)
Front left
Front right
Rear left
Rear right

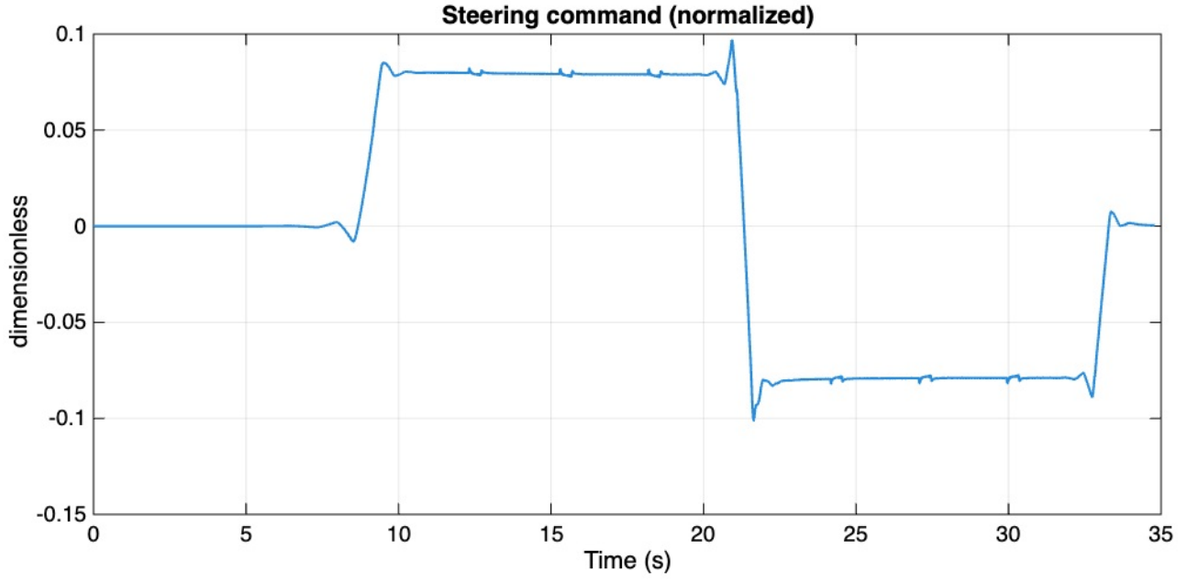

Tire load on the road
N
Time (s)
Front left
Front right
Rear left
Rear right

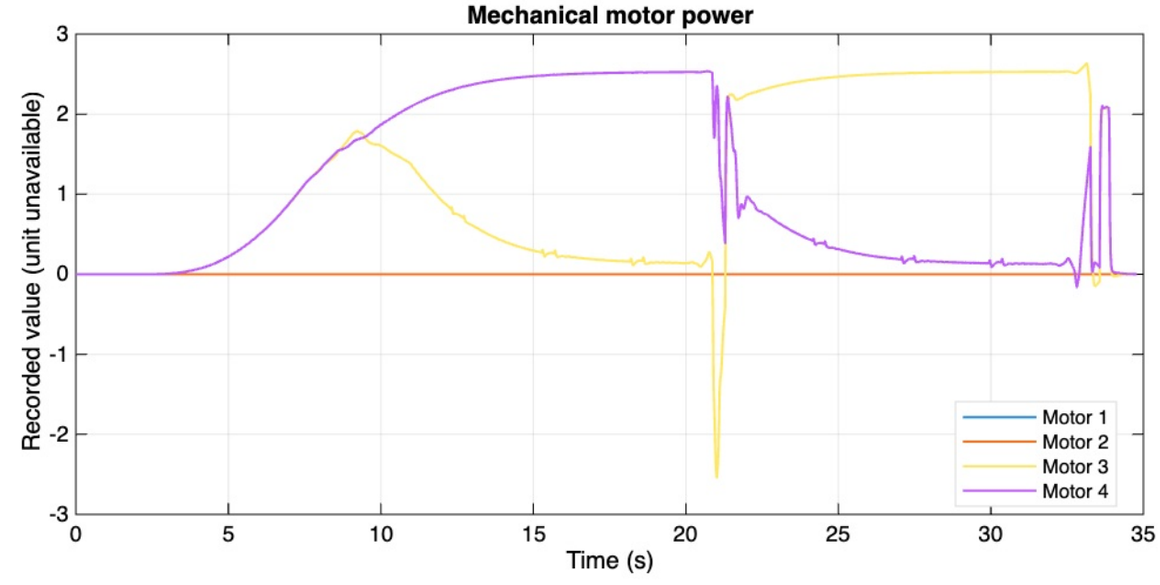

Tire inflation pressure
×10^5
2.64001
2.640008
2.640006
2.640004
2.640002
2.64
2.639998
2.639996
2.639994
2.639992
2.63999
Pa
0
5
10
15
20
25
30
35
Time (s)
Front left
Front right
Rear left
Rear right

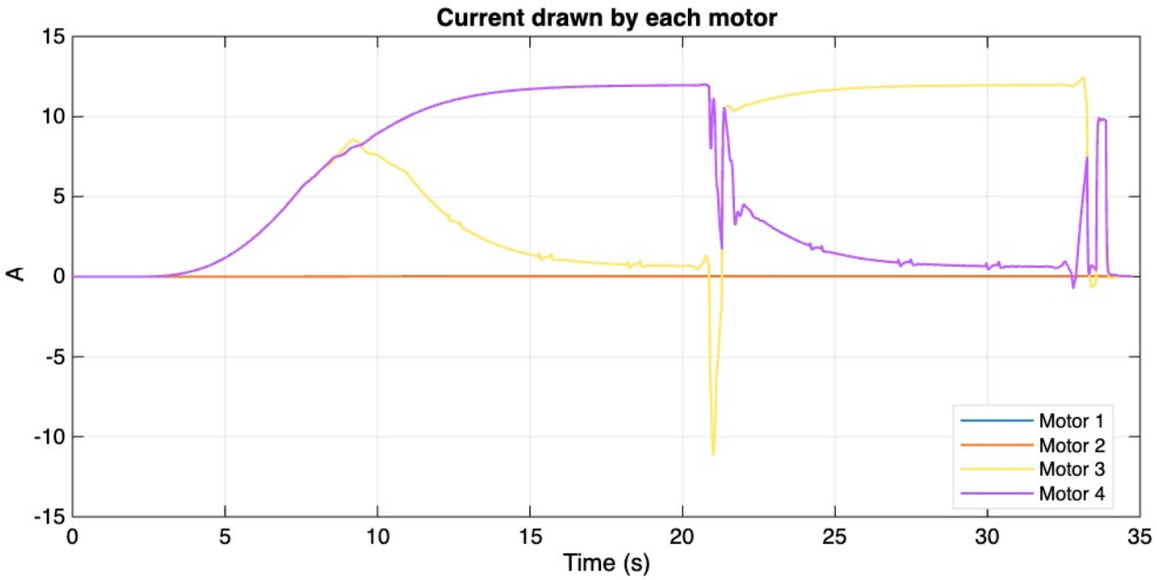

Accelerator command
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
dimensionless
0
5
10
15
20
25
30
35
Time (s)

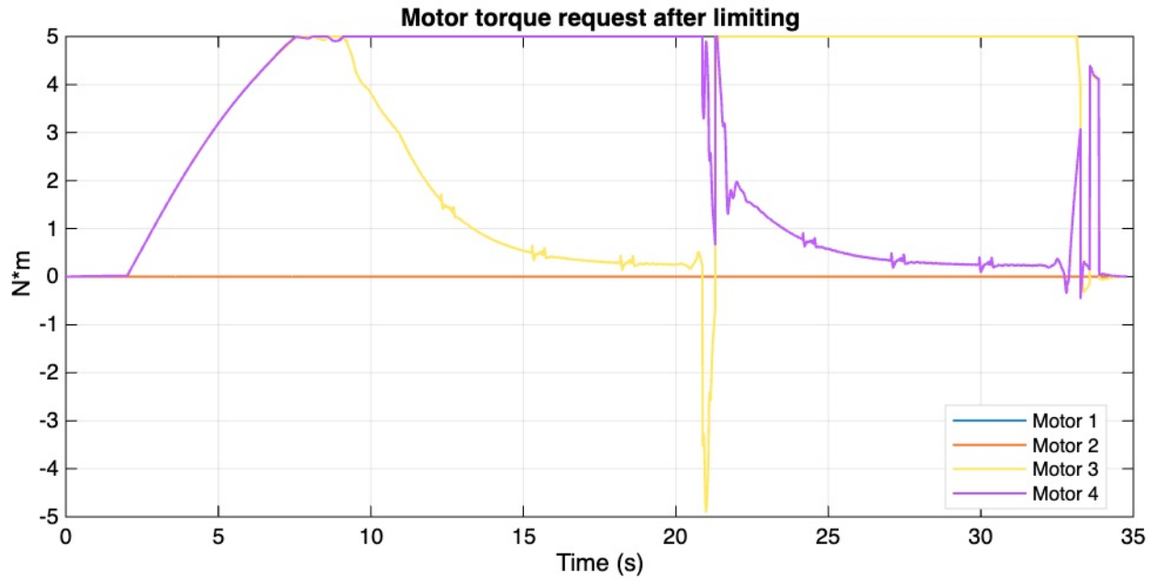

Brake command
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
dimensionless
0
5
10
15
20
25
30
35
Time (s)

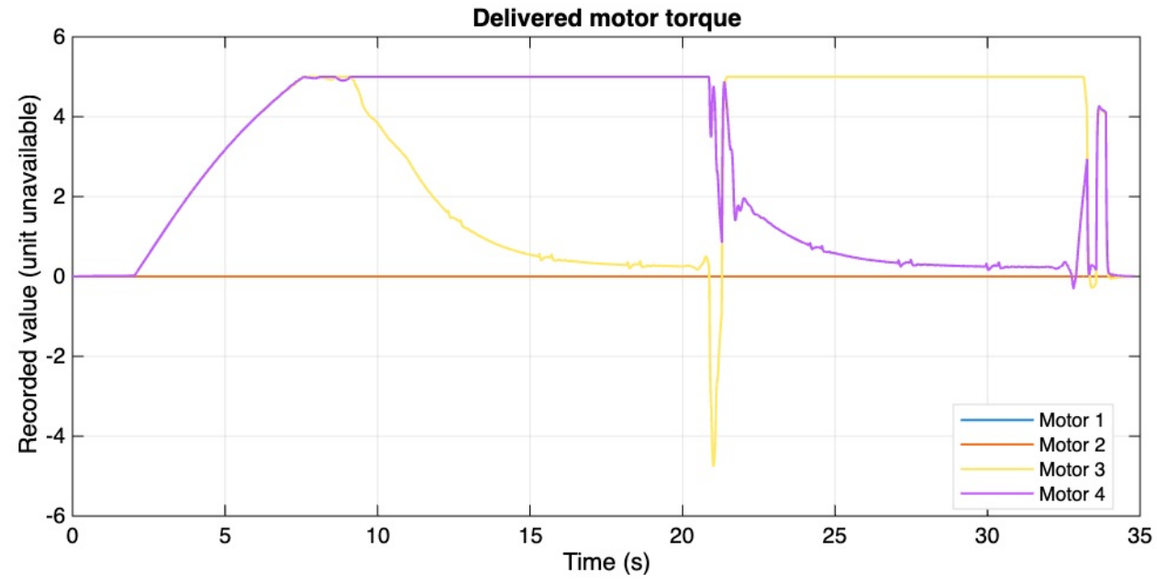

Requested gear
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
dimensionless
0
5
10
15
20
25
30
35
Time (s)

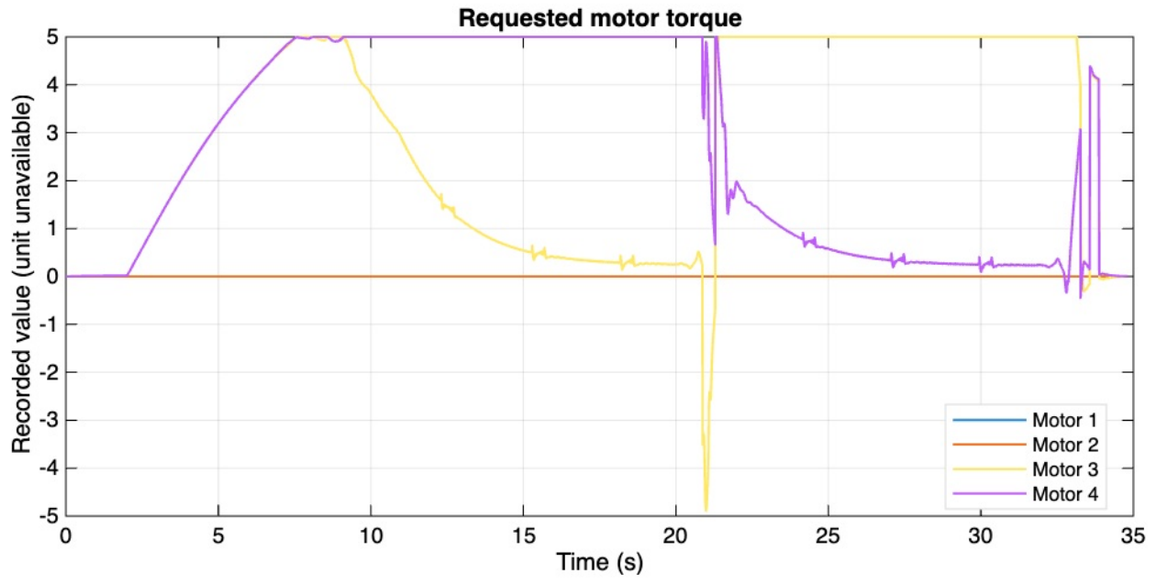

Steering command (normalized)
0.1
0.05
0
-0.05
-0.1
-0.15
dimensionless
0
5
10
15
20
25
30
35
Time (s)

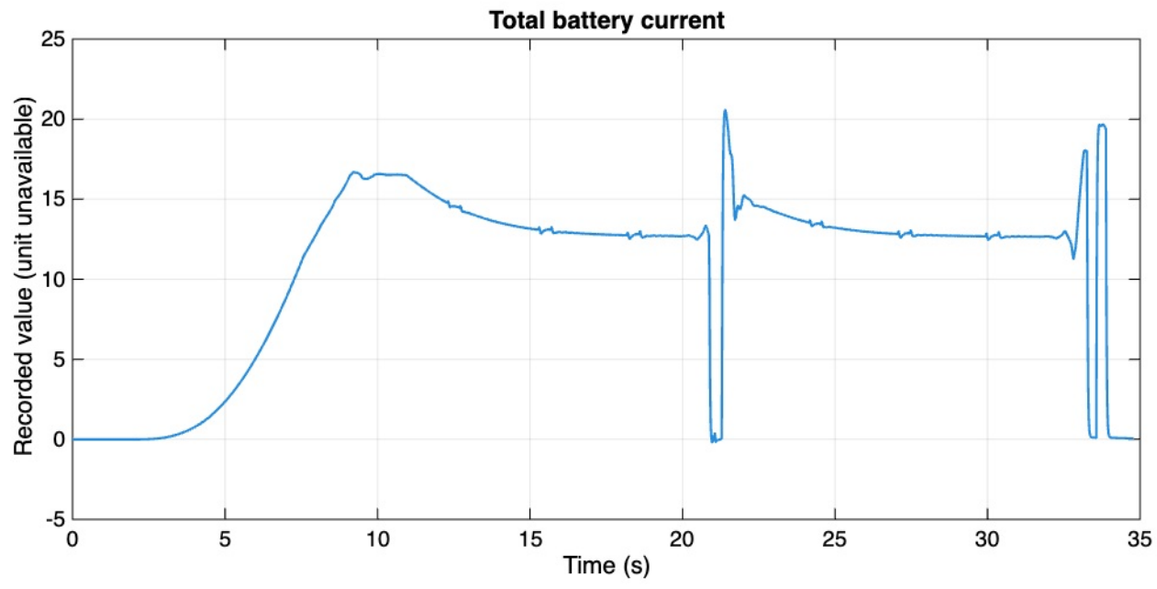

Mechanical motor power
3
2
1
0
-1
-2
-3
Recorded value (unit unavailable)
0
5
10
15
20
25
30
35
Time (s)
Motor 1
Motor 2
Motor 3
Motor 4

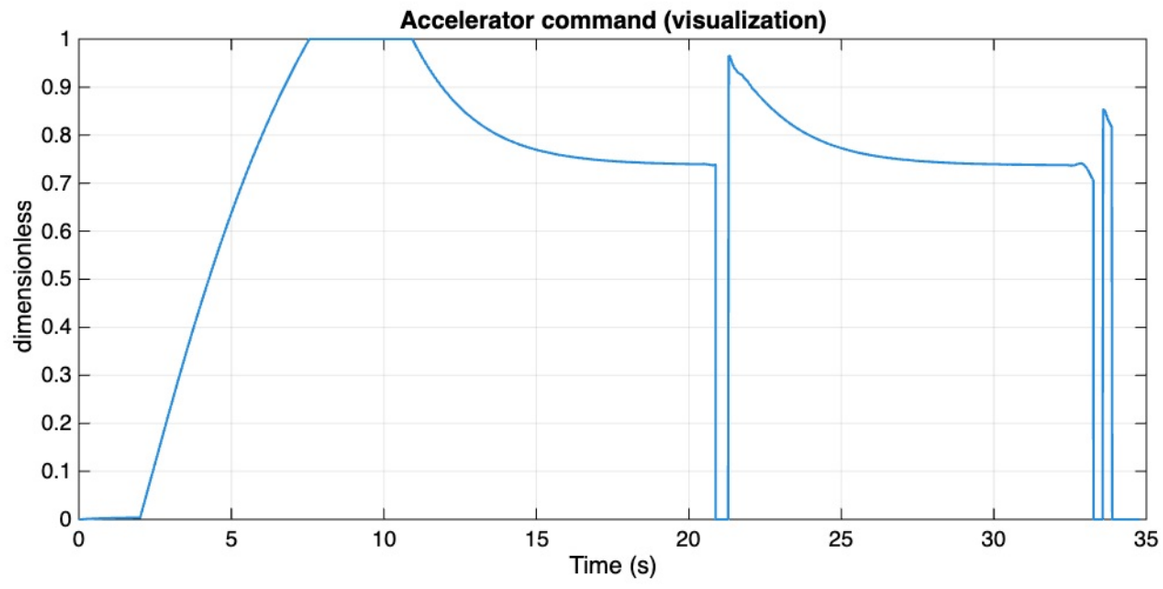

Current drawn by each motor
A
Time (s)
Motor 1
Motor 2
Motor 3
Motor 4

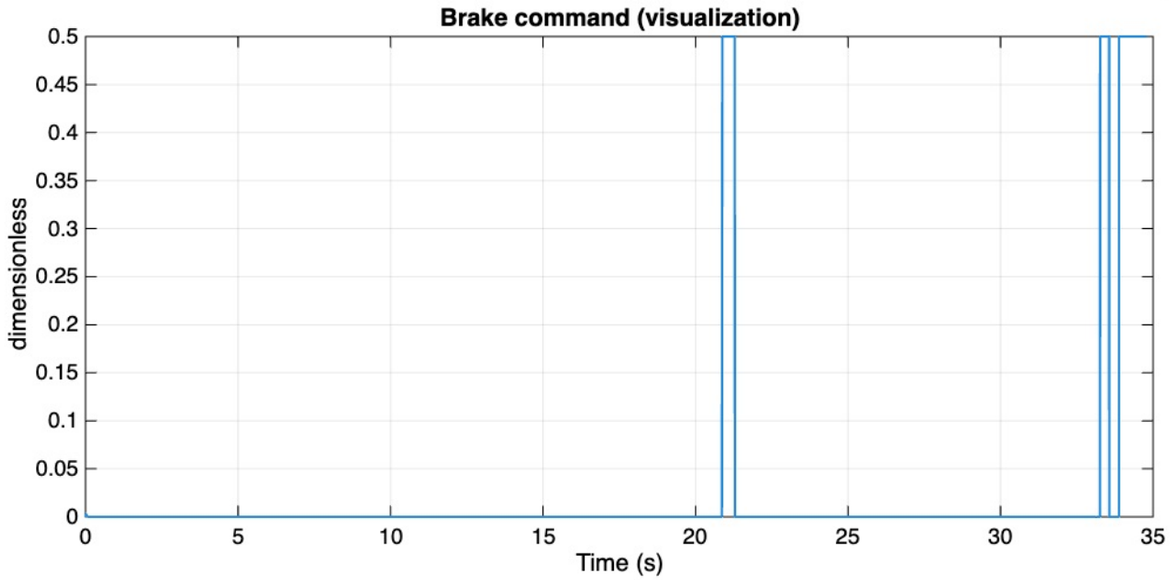

Motor torque request after limiting
N*m
Time (s)
Motor 1
Motor 2
Motor 3
Motor 4

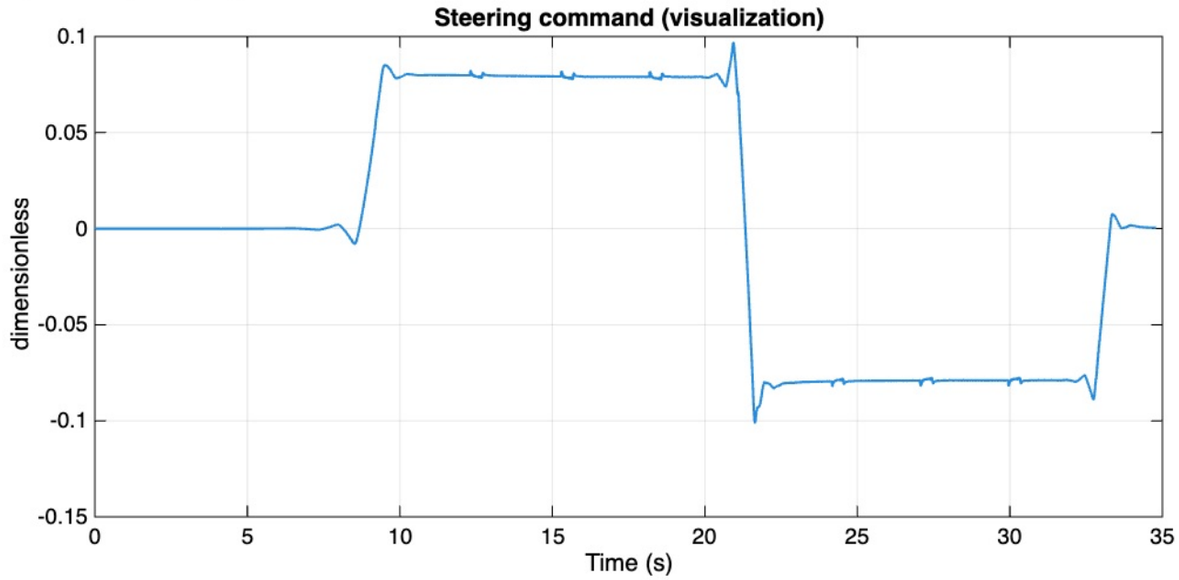

Delivered motor torque
Recorded value (unit unavailable)
Time (s)
Motor 1
Motor 2
Motor 3
Motor 4

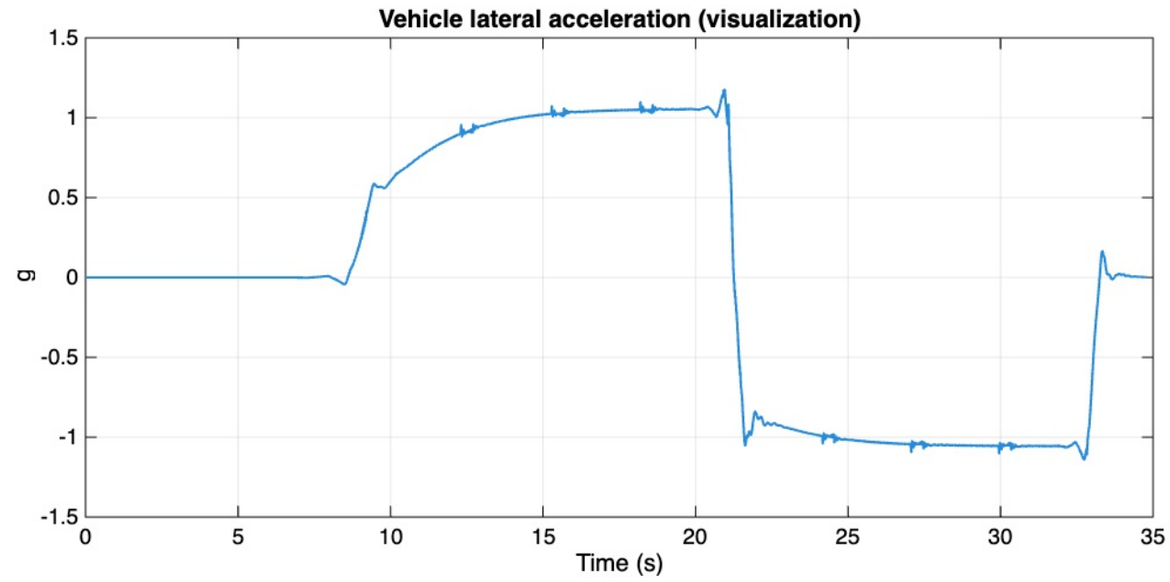

Requested motor torque
Recorded value (unit unavailable)
Time (s)
Motor 1
Motor 2
Motor 3
Motor 4

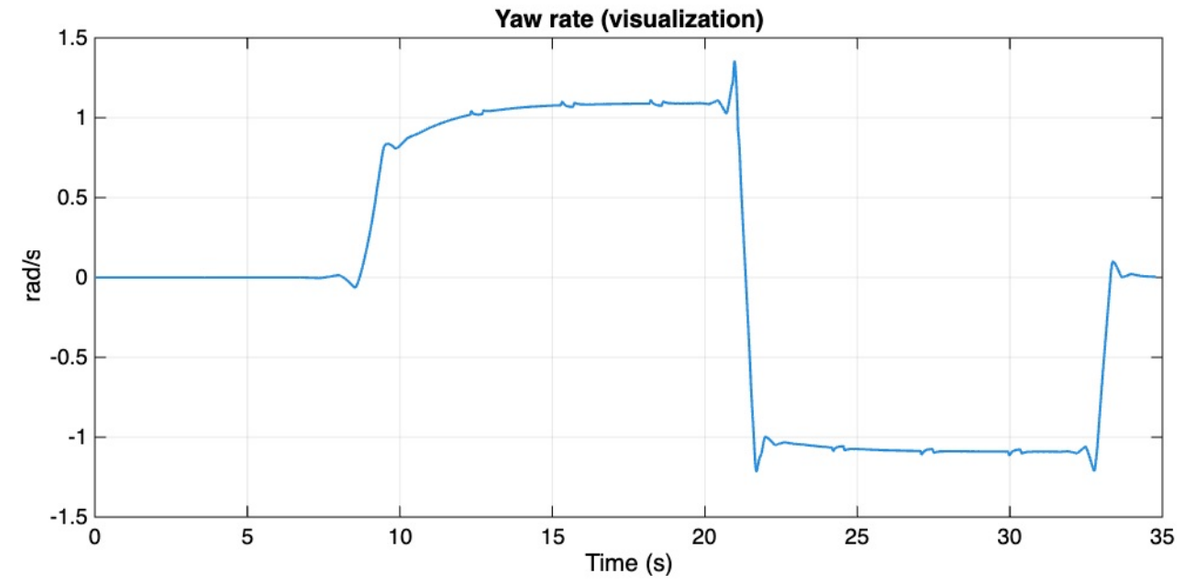

Total battery current
Recorded value (unit unavailable)
Time (s)

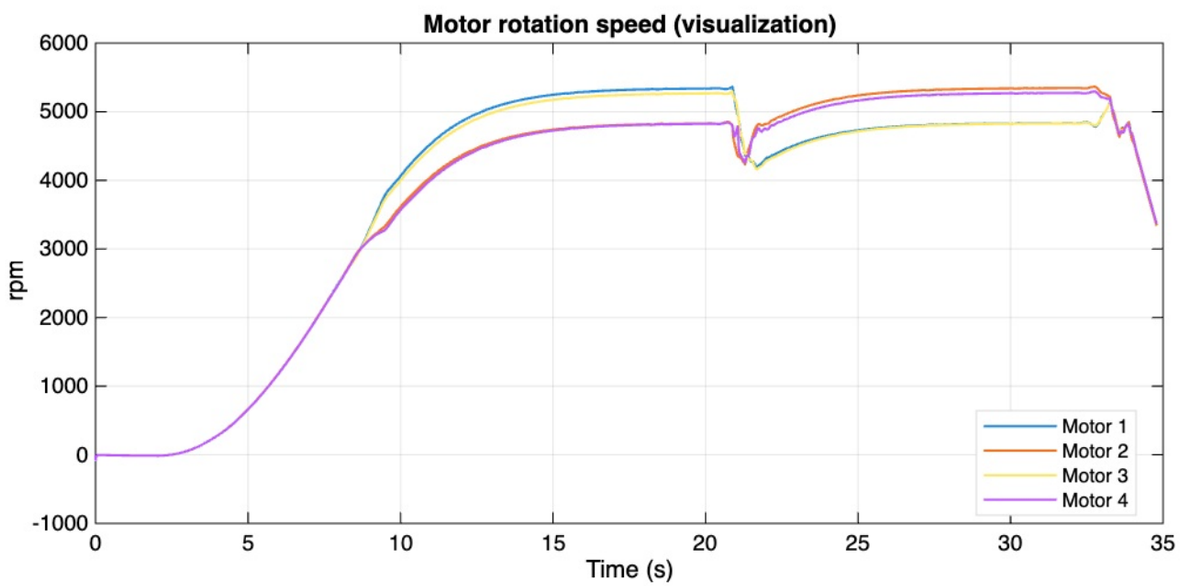

Accelerator command (visualization)
dimensionless
Time (s)

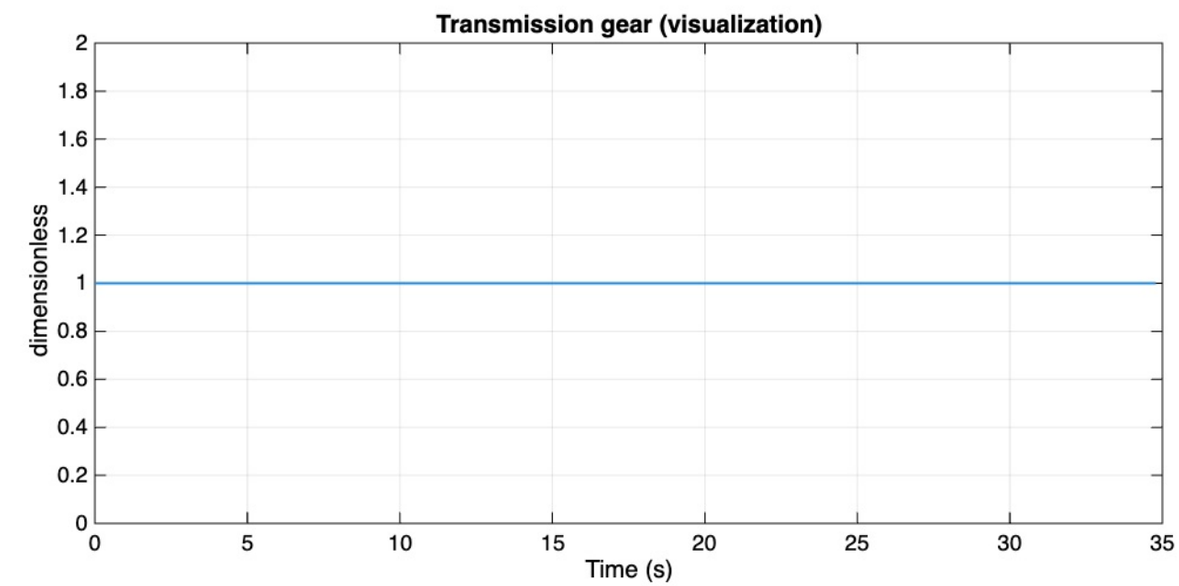

Brake command (visualization)
dimensionless
Time (s)

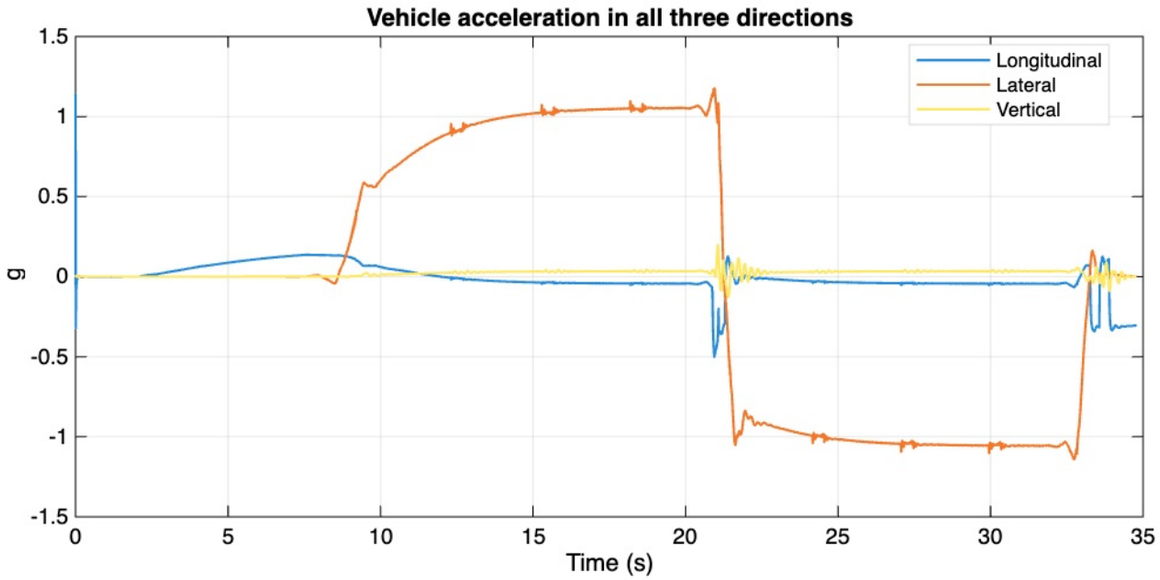

Steering command (visualization)
dimensionless
Time (s)

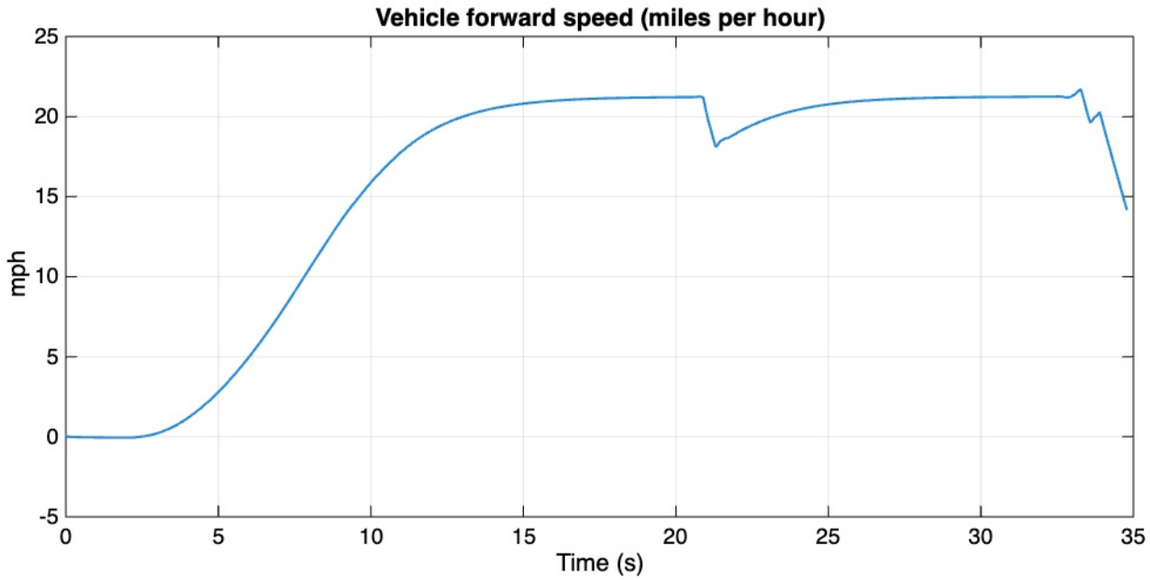

Vehicle lateral acceleration (visualization)
g
Time (s)

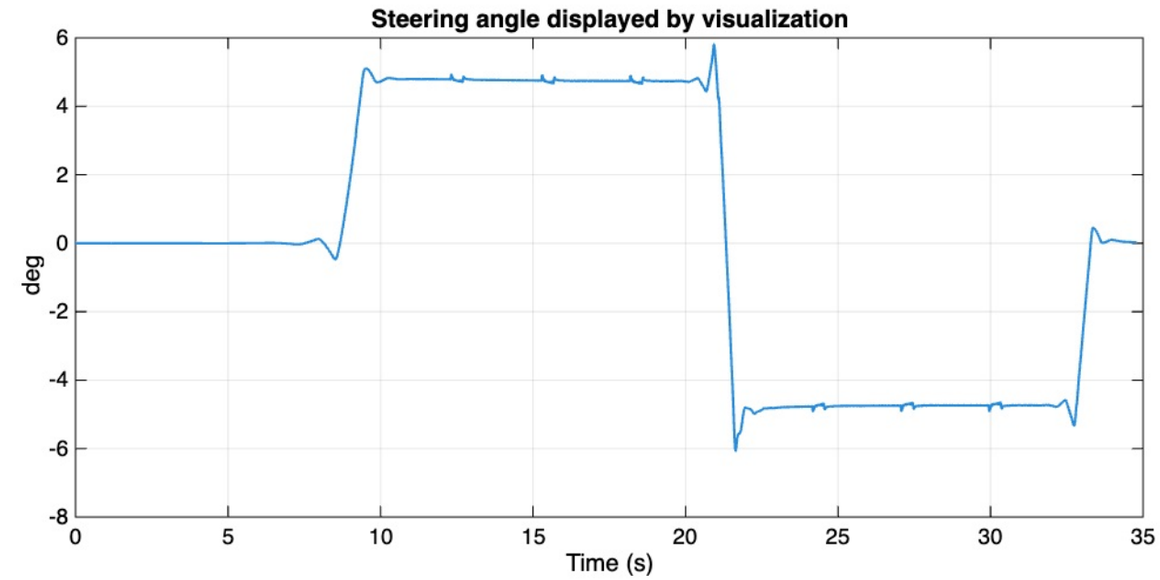

Yaw rate (visualization)
rad/s
Time (s)

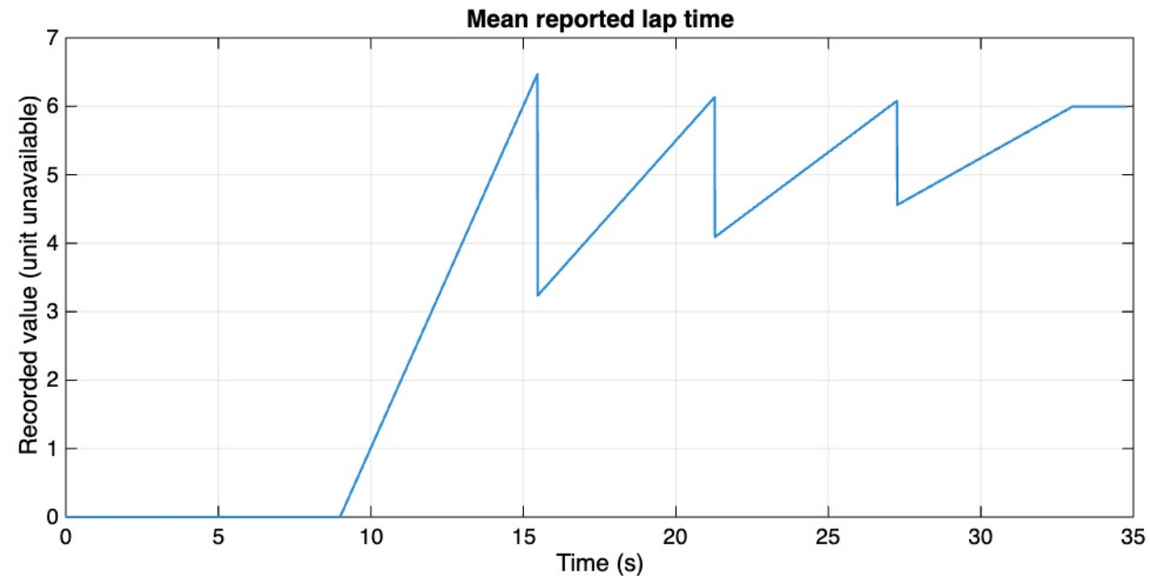

Motor rotation speed (visualization)
rpm
Time (s)
Motor 1
Motor 2
Motor 3
Motor 4

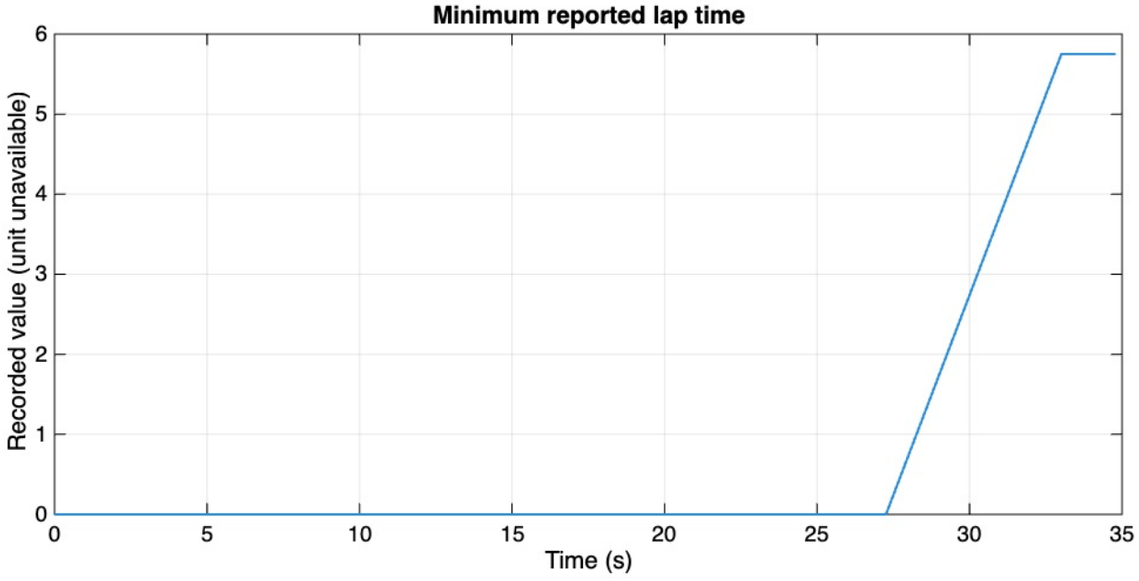

Transmission gear (visualization)
dimensionless
Time (s)


Vehicle acceleration in all three directions
Longitudinal
Lateral
Vertical
g
Time (s)


Vehicle forward speed (miles per hour)
mph
Time (s)


Steering angle displayed by visualization
deg
Time (s)


Mean reported lap time
Recorded value (unit unavailable)
Time (s)


Minimum reported lap time
Recorded value (unit unavailable)
Time (s)